

CyberLab Reference Guide

Field quick-reference for the kit
capability → tool → command → every option explained

Generated from the repository. Markdown is the source of truth.
github.com/project-cyberlab/cyberlab-reference-guide

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Alphabetical Tool Index

CyberLab Reference Guide

A field quick-reference for the kit. You open it mid-mission or mid-exercise when you need a **capability** and do not remember which tool provides it — it routes you to the tool, the exact command, and **every option, explained**.

Start here

You are...	Go to
Facing a problem — " <i>I need to carve deleted files</i> "	Capability Index — 46 capabilities
Asking what the kit even has	Kit Tool List — 1,006 tools, VM → category → tool
Wanting a finished page	fls · vol · yara
Carrying it into the field	PDF — 372 pages
Wanting to know how pages are built and why	The Format

Where it stands

Kit inventory	1,006 tools, 5 platforms, 54 categories
Captured from a real binary	945 commands
Capabilities	46 , across 10 investigation phases
Tool pages	135 , every one with a complete options table
Flags with usage guidance	153 across 20 reviewed tools
Mined invocations	360 real commands from cyberlab
Citations checked	255 URLs — 245 live
Linter	0 errors , 269 warnings (tracked debt)

What makes this different

Every DFIR quick-reference in the field either organises by tool (so you must already know the tool) or skips flag documentation. The most complete one in public circulation states outright that it "*assumes operator familiarity*."

This guide inverts that:

- **Capability-first.** You search by what you need to *do*, not what the binary is called.
- **Options completely listed out** — every flag, with *what it does* and *when you'd use it*.
- **Tools attached to commands**, each tagged with the VM that carries it, so you know whether you can run it on the box in front of you.

Scope is binding

Only tools in `catalog/kit-tools.json` may be documented. It is generated from the kit VMs' own upstream manifests:

Platform	Tools	Source of truth
Kali Linux	403	<code>kali-meta</code> <code>debian/control</code> (<code>kali-tools-*</code>)
REMnux	268	<code>REMnux/docs</code> <code>discover-the-tools</code>
SIFT Workstation	162	<code>teamdfir/sift-saltstack</code>
FLARE-VM	137	<code>mandiant/flare-vm</code> <code>config.xml</code>
Security Onion	36	kit tracker CSV

Every flag is real

Options are **captured from the running binary** (`--help` / `man`, inside the kit container), stored raw under `capture/`, and enforced by `scripts/lint.py`: a documented flag absent from a capture is a **build error**.

This answers a measured failure in the predecessor project, where fabricated CLI flags reached ~44 of 61 modules. Rationale in `docs/FORMAT.md`.

Citations are liveness-checked too — see `capture/SOURCES.md`.

Rebuild everything

```
python scripts/build_kit_list.py      # kit inventory from upstream manifests
python scripts/candidates.py          # candidate command names
# scripts/probe_container.sh runs inside the kit container on rick
python scripts/merge_coverage.py      # merge probe results -> coverage report
python scripts/mine_cyberlab.py       # candidate invocations (read-only)
python scripts/generate_pages.py      # tool pages (never overwrites hand-written)
python scripts/build_index.py         # capability index
python scripts/lint.py                # gate: 0 errors required
python scripts/validate_sources.py    # citation liveness
python scripts/build_pdf.py           # print render
```

Relationship to cyberlab

Separate repository, deliberately. The `cyberlab training lab` is **on hold** and receives no changes from this work; it is read-only input for mining.

Layout

catalog/	kit tool list – the binding scope (generated)
reference/	the guide: INDEX.md + one page per tool, grouped by capability
capture/	raw --help/man output, committed as evidence
docs/	FORMAT.md (how pages are built), PLAN.md (roadmap)
scripts/	build, probe, mine, generate, lint, validate, render

Capability Index

Start here when you have a problem. Find what you need to *do*; it points you at the tool and its page.

Phases follow [NIST SP 800-86](#) (collect → examine → analyse → report), ordered the way an investigation actually runs.

Legend: **bold** = captured from a real binary, page can be written · plain = in the kit but not captured yet (needs a booted VM or has no CLI).

Phases

- [Acquire & preserve](#)
- [Examine the filesystem](#)
- [Build the timeline](#)
- [Windows artifacts](#)
- [Memory forensics](#)
- [Network analysis](#)
- [Malware triage — static](#)
- [Malware triage — documents](#)
- [Reverse engineering](#)
- [Decode & deobfuscate](#)
- [Report & support](#)

Acquire & preserve

Image a disk, volume or device

[dc3dd](#), [dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#), [guymager](#)

Verify evidence integrity with hashes

[rahash2](#), [ssdeep](#), [sha256sum](#), [md5sum](#), [sigtool](#), [hashdeep](#), [md5deep](#), [sha256deep](#)

Inspect or mount a forensic image container

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#), [bdeinfo](#)

Capture live network traffic

[dumpcap](#), [tshark](#), [tcpdump](#)

Examine the filesystem

See the partition and volume layout

[mmls](#), [fsstat](#), [img_stat](#), [testdisk](#)

List files and directories, including deleted ones

[fls](#), [ffind](#), [ils](#), [tsk_recover](#)

Recover deleted or lost files

[tsk_recover](#), [icat](#), [photorec](#), [testdisk](#), [blkls](#), [ext4magic](#), [extundelete](#), [ext3grep](#)

Carve files out of unstructured data

[foremost](#), [scalpel](#), [binwalk](#), [bulk_extractor](#), [tcpxtract](#), [magicrescue](#)

Inspect metadata for one file or inode

[istat](#), [ils](#), [file](#), [stat](#), [exiftool](#), [trid](#), [magika](#)

Search raw data for a pattern

[rafind2](#), [strings](#), [grep](#), [xxd](#), [bulk_extractor](#), [lightgrep](#)

Build the timeline

Build a super-timeline from many artifact sources

[log2timeline.py](#), [psort.py](#), [pinfo.py](#), [psteal.py](#)

Build a filesystem MAC-time timeline

[fls](#), [mactime](#), [tsk_gettimes](#)

Windows artifacts

Parse registry hives

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#), [regipy-diff](#), [RECcmd](#), [hivexget](#), [hivexml](#)

Parse Windows event logs

[evtxexport](#), [evtxinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#), [evtx_dump](#)

Parse ESE / SRUM / Amcache databases

[esedbexport](#), [esedbinfo](#), [SrumsECmd](#), [AmcacheParser](#)

Parse execution and persistence artifacts

[PECmd](#), [AppCompatCacheParser](#), [MFTECmd](#), [AmcacheParser](#), [analyzeMFT](#)

Parse mail stores

[pffexport](#), [pffinfo](#), [readpst](#)

Memory forensics

Analyse a memory image

[vol](#), [volatility3](#), [volshell](#), [vol.py](#), [rekall](#)

Recover encryption keys from memory

[aeskeyfind](#), [rsakeyfind](#), [bulk_extractor](#)

Network analysis

Read and filter packet captures

[tshark](#), [capinfos](#), [ngrep](#), [tcpflow](#)

Split, merge or repair capture files

[editcap](#), [mergcap](#), [reordercap](#)

Extract files and payloads from traffic

[tcpxtract](#), [tcpflow](#), [foremost](#)

Detect intrusions in traffic

[suricata](#), [suricatasc](#), [zeek](#), [zeek-cut](#), [rita](#)

Probe or scan hosts and services

[nmap](#), [nping](#), [arp-scan](#), [ncat](#), [netdiscover](#)

Simulate network services for detonation

[inetsim](#), [fakedns](#), [fakenet](#)

Malware triage — static

Identify what a file actually is

[file](#), [die](#), [diec](#), [trid](#), [magika](#), [exiftool](#)

Scan with signatures for known-bad

[yara](#), [yarak](#), [clamscan](#), [freshclam](#), [sigtool](#), [clamscan](#)

Identify capabilities in a binary

[capa](#), [floss](#)

Extract strings, including obfuscated ones

[floss](#), [strings](#), [base64dump.py](#), [numbers-to-string.py](#)

Inspect PE / ELF structure

[rabin2](#), [readelf](#), [readelf.py](#), [objdump](#), [pescan](#), [pecheck.py](#), [peframe](#)

Detect and reverse packing

[upx](#), [die](#), [diec](#), [binwalk](#), [7za](#), [unzip](#)

Compare or cluster samples

[ssdeep](#), [radiff2](#), [bytehist](#)

Malware triage — documents

Analyse Office documents and macros

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#), [oleobj](#), [mraptor](#), [msodde](#), [pcodedmp](#), [pyxswf](#), [xlmdEOFfuscator](#), [runxlrd2.py](#), [vipermonkey](#)

Analyse RTF documents and embedded objects

[rtfobj](#), [oleobj](#), [rtfdump.py](#)

Analyse PDFs

[pdfid](#), [pdfid.py](#), [pdf-parser](#), [pdf-parser.py](#), [peepdf](#), [qpdf](#)

Analyse other container formats

[7za](#), [unzip](#), [msOFFcrypto-tool](#), [onenoteanalyzer](#)

Reverse engineering

Disassemble and explore a binary

[r2](#), [rabin2](#), [rasm2](#), [objdump](#), [vivbin](#), [vdbbin](#), [rizin](#), [ghidra](#), [cutter](#)

Diff two binaries

[radiff2](#), [bindiff](#)

Emulate or instrument execution

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#), [frida-ls-devices](#), [speakeasy](#), [qiling](#), [scdbg](#), [unicorn](#)

Analyse shellcode

[rasm2](#), [xortool](#), [scdbg](#), [shellcode2exe](#)

Decode & deobfuscate

Decode, decrypt or transform encoded data

[cyberchef](#), [base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

Break simple obfuscation

[xortool](#), [floss](#), [xlmdeobfuscator](#), [de4dot](#)

Crack passwords and hashes

[hashcat](#), [john](#), [hydra](#), [unshadow](#), [zip2john](#), [rar2john](#), [fcrackzip](#), [hashid](#)

Find hidden data

[binwalk](#), [ssdeep](#), [steghide](#), [stegosuite](#)

Report & support

Fetch and verify external references

[curl](#), [wget](#)

Inspect files by hand

[xxd](#), [ezhexviewer](#), [less](#), [hexdump](#)

Coverage of this index

- **46 capabilities** across 11 phases
- **147 captured tools** are reachable through a capability
- **5 named kit tools map to no capability** (below)
- **2 capabilities have no captured tool** behind them yet

Capabilities with nothing captured yet

These are real needs the kit may cover with GUI or Windows-only tools that a Linux container cannot capture.

- Windows artifacts / Parse mail stores
- Network analysis / Detect intrusions in traffic

Named kit tools not yet mapped to a capability

These are named in the kit manifests, captured from a real binary, and reach no capability in this index. Each is a taxonomy gap to close or an explicit out-of-scope decision. Listed rather than silently dropped, because a missing capability is otherwise invisible.

`aircrack-ng`, `cabextract`, `ccrypt`, `cryptsetup`, `dislocker`

Container-provided commands (not named in the kit manifests)

Discovered by walking the container's `PATH`. They ship in a kit image but no kit manifest names them, so the great majority is OS plumbing rather than investigative tooling. Kept for completeness only.

► 807 container-provided commands

The Format

How every page in this guide is built, and why. Derived from studying the reference works that already do this well, then fixing what they each get wrong for our use case: **a field guide you open mid-mission when you need a capability and do not remember the tool.**

1. What the research said

Source	What it does well	What it does not give us
tldr-pages style guide	Curated examples (≤ 8), imperative mood, <code>{{placeholder}}</code> syntax, machine-linted	Deliberately incomplete — omits most options by design. No capability index.
POSIX Utility Conventions ch.12 + man-pages(7)	Unambiguous synopsis grammar: <code>[]</code> optional, <code>\ </code> exclusive, <code>...</code> repeatable	Reference-shaped, not task-shaped. You must already know the tool's name.
SANS DFIR cheat sheets	Organised by investigative task — the right primary axis	Print-first, not greppable, not exhaustive on flags
Jai Minton DFIR cheatsheet	200+ scannable anchors; artifact paths and ATT&CK IDs as headings	<i>"Most entries lack detailed flag documentation... assumes operator familiarity"</i>
navi / cheat.sh	Machine-parseable, taggable, directly executable entries	No explanation layer at all
Progressive disclosure (Nielsen, 1995)	Layer information: essentials first, depth on demand	—

The gap we fill: every existing DFIR quick-reference assumes you already know the tool and its flags. None of them explains the options. That is exactly the layer this guide exists to provide, and it is why "completely listed out" is a hard requirement rather than a stretch goal.

2. Two entry points, one dataset

You arrive at this guide in one of two states:

1. **"I have a problem."** — *I need to recover deleted files from this image.* → **Capability Index:** capability → tool → command → options. This is the primary path and the reason the guide exists.
2. **"What do I have?"** — *Is there anything in the kit for Go binaries?* → **Kit Index:** VM → category → tool. This is `catalog/KIT-TOOLS.md`, generated from upstream manifests.

Both views are projections of the same JSON, so they can never disagree.

3. Scope is binding

A tool may appear in this guide **only** if it is in `catalog/kit-tools.json`, which is generated from the kit's own VM manifests. If it is not in the kit, it does not get a page — no matter how good the tool is.

Every command entry is tagged with the **VM(s) that carry it**, so in the field you instantly know whether you can actually run it on the box in front of you.

4. The page format

Every tool page has these sections, in this order. Sections are layered by progressive disclosure: a reader in a hurry stops after §3; a reader who needs precision continues to §4.

Header

```
# <tool>
**Kit:** REMnux · SIFT    **Category:** Filesystem analysis    **Version:** 4.11.1
**Docs:** <upstream URL>    **Verified:** 2026-07-29 from `cyberlab-aio:v1`
```

The version and the verification date are not decoration — they are the claim that the options below were read off the real binary on a known date.

1. Purpose — one line, imperative

List files and directory names in a disk image, including deleted entries.

2. Synopsis — POSIX notation, verbatim from the tool

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

3. Common invocations — ≤8, task-titled, copy-paste ready

Each example is titled by **the task it accomplishes**, not by the flag it uses, because the reader is searching by intent. Placeholders use tldr syntax so they are visually obvious and machine-checkable:

```
# List deleted files in a partition, recursively, with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}
```

4. Options — COMPLETE, one row per flag

Every flag the binary accepts. Not the useful ones — **all** of them.

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	Display deleted entries only	Narrowing a recovery hunt to what was removed

The fourth column is the one that makes this a guide rather than a man page dump. It answers *why would I reach for this*, which is what you actually need under time pressure.

5. Gotchas

Real operational traps, each one earned. Examples already known on this project: `timeout-guard` `vol`, `binwalk`, `clamscan` and `radare2` in any harness — a synthetic memory image once pinned `vol` at 99% CPU indefinitely.

6. See also

Sibling tools for the same capability, so a dead end reroutes instead of stalling.

5. Verification: why the options can be trusted

The predecessor project shipped **fabricated CLI flags in roughly 44 of its 61 modules** — invented options for `yara`, `clamscan`, `PE-bear`, `scdbg` and others — because content was generated from a language model rather than read off the tools. That failure is the reason this guide exists in its current form.

So the rule here is mechanical, not aspirational:

1. Run the tool's `--help` / `-h` / man page **inside the kit container or VM**.
2. Store the raw output under `capture/<tool>.help.txt`, committed.
3. Generate/check the options table **against that capture**.
4. `scripts/lint.py` fails the build if a documented flag is absent from the capture, or if a captured flag is missing from the page.

A flag that cannot be produced from a real binary cannot ship. This is the same discipline `tldr-lint` applies to formatting, applied instead to truth.

Worked proof. The captured `fls` usage line advertises `[-adDFlhpruvV]`, but the body of the same output documents `-P`, `-B`, `-S` and `-k` as well. Anyone transcribing the synopsis — or working from memory — silently loses four real options. Only a capture catches that.

6. Sources and citation validation

Two different claims need two different kinds of evidence, and they must not be confused:

- **What a flag is** → the binary. Captured, committed, linter-enforced (§5). No online source overrides a real `--help`.
- **What a tool is for, and how practitioners actually use it** → authoritative upstream documentation and reputable DFIR publishers.

Acceptable sources for the second kind, in preference order: the project's own docs/repo (sleuthkit.org, volatilityfoundation.org, docs.remnux.org, ericzimmerman.github.io, Microsoft Learn), then recognised practitioner sources (SANS, thedfirreport.com, Red Canary, Mandiant/Unit42), then MITRE ATT&CK for technique mapping.

Every cited URL is liveness-checked before it ships. The predecessor project shipped hallucinated citations that looked plausible and 404'd; it already has a hardened, allowlisted, deadline-guarded fetcher (`cyberlab_retrieval.py`, with `verify_url`) built precisely for this, and this guide reuses that discipline. A dead or unreachable citation fails the build rather than degrading quietly.

Version skew is a real hazard: upstream docs describe the *current* release, while the kit runs pinned versions (TSK 4.11.1 in the container vs 4.15.0 in the tracker, for example). Where docs and capture disagree, **the capture wins** and the discrepancy is worth a line in Gotchas.

7. Output

- **Markdown** is the source of truth: greppable, diffable, reviewable, renders on GitHub.
- **PDF/print render** is generated from the markdown for field use. It is a build artifact, never hand-edited, so the two can never drift.

Implementation Plan

Goal: a field quick-reference covering the kit's tools, capability-indexed, with every command's options completely listed and explained, verified against real binaries.

Status legend:  done ·  next ·  pending

Phase 0 — Foundations

-  **Scope fixed.** `catalog/kit-tools.json` + `KIT-TOOLS.md` : 1,006 tools, 5 platforms, 54 categories, each from an upstream machine-readable manifest.
-  **Format designed** from six reference works — `docs/FORMAT.md`.
-  **Verification proven.** `cyberlab-aio:v1` on rick has `fls`, `clamscan`, `binwalk`, `yara`, `vol`, `tshark`, `exiftool`, `log2timeline.py`.
-  **Worked example** — `reference/examine-the-filesystem/fls.md`, 22 options from a real capture.

Phase 1 — Capability taxonomy

The primary index. Draft ~40–60 capabilities phrased as **problems**, not tool names, e.g. "*Recover deleted files from an image*", "*Find what persisted on a host*", "*Decode an obfuscated payload*", "*Prove lateral movement*".

- Seed from the categories already carried by upstream: REMnux's 11 capability categories, Kali's 12 `kali-tools-*` groups, Security Onion's 9, and SANS investigative-task ordering.
- Map every one of the 1,006 kit tools to ≥ 1 capability. Tools that map to none are reported, not silently dropped — an unmapped tool is either a taxonomy gap or genuinely out of scope, and both need a decision.
- **Deliverable:** `reference/INDEX.md` — capability $\rightarrow$ tools $\rightarrow$ page links.

Open question for review: should the capability taxonomy follow an existing public framework (NIST/DFIR phases, MITRE ATT&CK tactics) for familiarity, or a kit-native one tuned to what we actually carry? Recommendation: kit-native primary, with ATT&CK IDs as secondary tags where they genuinely apply — forcing forensic tools into attacker tactics distorts both.

Phase 1b — Mine cyberlab as a source

The cyberlab training lab already contains a large body of commands and prior exhaustive research. It is **read-only input** here — cyberlab is on hold and receives no changes.

- `scripts/mine_cyberlab.py` : extract every fenced command block from the 61 modules, plus `docs/CURRICULUM_ROADMAP.md`, `docs/TEST_REPORT.md` and `docs/CONTENT_QUALITY_AUDIT.md`, and the `catalog/` research.
- Produce `capture/cyberlab-candidates.json` : tool $\rightarrow$ observed invocations, with the module each came from.
- These become **candidate invocations** for the "Common invocations" section — real workflows someone already thought through, which is exactly the part that is expensive to invent.

Mine the cleaned branch, not `main`. The content-quality audit found fabricated CLI flags in ~44 of 61 modules, concentrated in the generated "enrichment tail". Branch `audit/content-quality-p0-p1` (PR #1) already removed that tail — ~11,000 lines — so it is the higher-signal source. Modules 57 (acquisition) and 58 (eventlog) were rated the cleanest and are worth reading first.

Candidates are never trusted. Every mined invocation still has to pass the Phase 3 linter against a real capture. Mining saves the *thinking*, not the *verification* — a mined command with an invented flag must fail exactly as loudly as one written from scratch.

Phase 2 — Capture pipeline

`scripts/capture_help.py` : for each in-scope tool, run `--help / -h / man` in the right container, store raw output under `capture/<tool>.help.txt`.

- Containers: `cyberlab-aio:v1` and `dfir-aio:v4` on rick (Docker active, 830 GB free).
- **Always timeout-guard** — `vol` once pinned 99% CPU indefinitely on a synthetic image. Hard per-tool timeout, failure recorded not retried blindly.
- Emit a coverage report: which kit tools are covered by a container, which need a booted VM (Kali, FLARE-VM Windows guests), which are GUI-only and have no CLI surface to document.

Reality check: the 1,006-tool list includes GUI-only tools (Autopsy, Wireshark GUI, PE-bear) and appliance services (Elasticsearch, Kibana) with no meaningful command surface. Expect the genuinely CLI-documentable set to be substantially smaller. Phase 2's report turns that from a guess into a number before any page is written.

Phase 3 — Page generation + linter

- Generate a page skeleton per tool from its capture: header, synopsis, complete options table. Mechanical, deterministic — no model invention.
- Human/model adds only the *judgement* columns: purpose, "when you'd use it", invocations, gotchas.
- `scripts/lint.py` gates the build:
- every documented flag exists in the capture;
- every captured flag appears in the page (**completeness**, the hard requirement);
- every tool is in `kit-tools.json`;
- format rules hold (imperative mood, ≤8 examples, placeholder syntax).

Phase 4 — Print render

Markdown → PDF for field carry. Build artifact only, never hand-edited. Per-capability booklets rather than one 1,000-page brick.

Phase 5 — Prioritised content ► (20 of 135 tools reviewed)

Do not attempt all tools at once. Order by field value:

1. Core forensics: TSK suite, Volatility, Plaso, RegRipper, bulk_extractor

2. Network: Zeek, Suricata, tshark, tcpdump
 3. Malware triage: YARA, ClamAV, capa, FLOSS, Detect-It-Easy
 4. Live response / hunting: Velociraptor, osquery, Hayabusa, Chainsaw
 5. Offensive-side as needed for exercises
-

Risks

Risk	Mitigation
Fabricated flags (killed the predecessor project)	Capture-or-it-does-not-ship, enforced by linter
Scale — 1,006 tools is a lot of pages	Phase 2 coverage report right-sizes it; Phase 5 prioritises
Kit VMs powered down (<code>kali</code> , <code>range-linux-web</code> , <code>redinfra01</code>)	Containers cover most Linux tooling; boot VMs only for the remainder
Version drift between capture and deployed kit	Version + capture date in every page header
GUI-only tools with no CLI	Identify in Phase 2; document workflow, not flags

Kit Tool List

The binding scope for the quick-reference guide. A tool absent from this list is **not in the kit** and must not be documented in the guide.

Every entry is derived from an upstream machine-readable manifest; none is written from memory. Sources are listed at the end.

Contents

- [REMnux](#) — 268 tools, 31 categories
- [Kali Linux](#) — 403 tools, 12 categories
- [FLARE-VM](#) — 137 tools, 1 categories
- [SIFT Workstation](#) — 162 tools, 1 categories
- [Security Onion](#) — 36 tools, 9 categories

Total: 1006 tools across 5 platforms.

REMnux

Analyze Documents / Email Messages

Tool	Command(s)	Purpose
emldump.py	—	Parse and analyze EML files.
mail-parser	—	Parse raw SMTP and .MSG email messages and generate a parsed object from them.
msg-extractor	<code>extract_msg</code>	Extract emails and attachments from MSG files.
msgconvert	—	Convert MSG files to MBOX files.

Analyze Documents / General

Tool	Command(s)	Purpose
base64dump.py	—	Locate and decode strings encoded in Base64 and other common encodings.
Tesseract OCR	<code>tesseract</code>	Examine images to identify and extract text using optical character recognition (OCR).

Analyze Documents / Microsoft Office

Tool	Command(s)	Purpose
EvilClippy	—	Modify aspects of Microsoft Office documents.
Hachoir	<code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code>	View, edit, and carve contents of various binary file types.
libolecf	<code>olecfexport</code> , <code>olecfinfo</code> , <code>olecfmount</code> , <code>etc</code>	Microsoft Office OLE2 compound documents.

Tool	Command(s)	Purpose
msoffcrypto-crack.py	—	Recover the password of an encrypted Microsoft Office document.
msoffcrypto-tool	—	Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key.
msoffice-crypt	—	Encrypt and decrypt OOXML Microsoft Office documents.
oledump.py	—	Analyze OLE2 Structured Storage files.
olefile	—	Python package to parse, read and write MS OLE2 files.
oletools	mraptor , msodde , olebrowse , oledir , oleid , olemap , olemeta , oleobj , oletimes , olevba , pyxswf , rtfobj , ezhexviewer	Microsoft Office OLE2 compound documents.
onedump.py	—	Extract and analyze embedded files from OneNote documents.
pcode2code	—	Decompile VBA macro p-code from Microsoft Office documents.
pcodedmp	—	Disassemble VBA p-code.
rtfdump.py	—	Analyze a suspicious RTF file.
SSView	ssview	Analyze OLE2 Structured Storage files.
XLMMacroDeobfuscator	xlmdeobfuscator , runxlrd2.py	Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.
xmldump.py	—	Extract contents of XML files, in particular OOXML-formatted Microsoft Office documents.
zipdump.py	—	Analyze zip-compressed files.

Analyze Documents / Pdf

Tool	Command(s)	Purpose
Origamindee	pdfcop , pdfdecompress , pdfdecrypt , pdfextract , etc	Parse, modify, generate PDF files.
pdf-parser.py	—	Examine elements of the PDF file.
pdfid.py	—	Identify suspicious elements of the PDF file.
pdfresurrect	—	Extract previous versions of content from PDF files.
pdftk-java	pdftk	Edit, create, and examine PDF files.
pdftool.py	—	Analyze PDF files to identify incremental updates to the document.
peepdf-3	—	Examine elements of the PDF file.

Tool	Command(s)	Purpose
qpdf	—	Manipulate (merge, convert, transform) PDF files.

Dynamically Reverse-Engineer Code / Elf Files

Tool	Command(s)	Purpose
edb	—	An AArch32/x86/x86-64 debugger, well suited for debugging ELF files.
GNU Project Debugger	—	Multi-language debugger.
ltrace	—	Trace library calls and signals.
strace	—	Trace process' system calls and signals.

Dynamically Reverse-Engineer Code / General

Tool	Command(s)	Purpose
Frida	<code>frida</code> , <code>frida-ps</code> , <code>frida-trace</code> , <code>frida-discover</code> , <code>frida-ls-devices</code> , <code>frida-kill</code>	Trace the execution of a process to analyze its behavior.
r2pipe	—	Examine binary files, including disassembling and debugging.
radare2	<code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code>	Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A
Wine	<code>wine</code>	Run Windows applications.
x64dbg Automate MCP (OpenCode skills)	—	Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh

Dynamically Reverse-Engineer Code / Scripts

Tool	Command(s)	Purpose
box-js	<code>box-js</code> , <code>box-export</code>	Analyze suspicious JavaScript scripts.
JavaScript Deobfuscator	—	Deobfuscate JavaScript by removing common obfuscation techniques such as string arrays and proxy functions.
js_unshroud	—	Monitor and deobfuscate JavaScript behavior in a headless browser to analyze malicious web pages.
JStillery	<code>jstillery</code>	Deobfuscate JavaScript scripts using AST and Partial Evaluation techniques.
objects.js	—	Emulate common browser and PDF viewer objects, methods, and properties when deobfuscating JavaScript.
PowerShell Core	<code>pwsh</code>	Run PowerShell scripts and commands.
Rhino Debugger	<code>rhino-debugger</code>	GUI JavaScript debugger.
SpiderMonkey	<code>js</code>	Execute and deobfuscate JavaScript using Mozilla's standalone JavaScript engine.

Tool	Command(s)	Purpose
SpiderMonkey (Patched)	<code>js-ascii</code> , <code>js-file</code>	Execute and deobfuscate JavaScript using a patched version of Mozilla's standalone JavaScript engine.
STPyV8	—	Python3 and JavaScript interop engine, fork of the original PyV8 project.
Webcrack	—	Deobfuscate, unminify, and unpack bundled JavaScript, including scripts protected with obfuscator.io.

Dynamically Reverse-Engineer Code / Shellcode

Tool	Command(s)	Purpose
libemu	—	A library for x86 code emulation and shellcode detection.
Qiling	—	Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.
runsc	—	Run shellcode to trace and analyze its execution.
scdbg	—	Analyze shellcode by emulating its execution.
shcode2exe	—	Convert 32 and 64-bit shellcode to a Windows executable file.
shellcode2exe.bat	—	Convert 32 and 64-bit shellcode to a Windows executable file.
Speakeasy	—	Emulate code execution, including shellcode, Windows drivers, and Windows PE files.
XORSearch	<code>xorsearch</code>	Locate and decode strings obfuscated using common techniques.

Examine Static Properties / .Net

Tool	Command(s)	Purpose
dnfile	—	Analyze static properties of .NET files.
dotnetfile	—	Analyze static properties of .NET files.
monodis	—	Disassemble and extract resources from .NET assemblies.

Examine Static Properties / Deobfuscation

Tool	Command(s)	Purpose
1768.py	—	Analyze Cobalt Strike beacons.
Balbuzard	<code>balbuzard</code> , <code>bbcrack</code> , <code>bbharvest</code> , <code>bbtrans</code>	Extract and deobfuscate patterns from suspicious files.
base64dump.py	—	Locate and decode strings encoded in Base64 and other common encodings.
brxor.py	—	Bruteforce XOR'ed strings to find those that are English words.
Chepy	<code>chepy</code>	Decode and otherwise analyze data using this command-line tool and Python library.
Cobalt Strike Configuration Extractor (CSCE) and Parser	<code>csce</code> , <code>list-cs-settings</code>	Analyze Cobalt Strike beacons.
cs-analyze-processdump.py	—	Analyze Cobalt Strike beacon process dumps to detect sleep mask encoding.

Tool	Command(s)	Purpose
cs-decrypt-metadata.py	—	Decrypt Cobalt Strike metadata.
cs-extract-key.py	—	Extract AES and HMAC keys from Cobalt Strike beacon process memory.
cut-bytes.py	—	Cut out a part of a data stream.
CyberChef	cyberchef	Decode and otherwise analyze data using this browser app.
DC3-MWCP	mwcp	Parsing configuration information from malware.
ex_pe_xor.py	—	Search an XOR'ed file for indications of executable binaries.
FLOSS	floss	Extract and deobfuscate strings from PE executables.
format-bytes.py	—	Decompose structured binary data with format strings.
hex-to-bin.py	—	Convert hexadecimal text dumps to binary data.
Malchive	—	Perform static analysis of various aspects of malicious code.
NoMoreXOR.py	—	Help guess a file's 256-byte XOR by using frequency analysis.
numbers-to-string.py	—	Translate number sequences into ASCII characters.
re-search.py	—	Search files using regular expressions from a built-in library or custom patterns.
sets.py	—	Perform set operations on lines or bytes in text files.
strdeob.pl	—	Locate and decode stack strings in executable files.
translate.py	—	Translate bytes according to a Python expression.
unicode	—	Display Unicode character properties.
unXOR	—	Deobfuscate XOR'ed files.
xor-kpa.py	—	Implement a XOR known plaintext attack.
xorBruteForcer.py	—	Bruteforce an XOR-encoded file.
XORSearch	xorsearch	Locate and decode strings obfuscated using common techniques.
xorsearch.py	—	Search for XOR, ROL, ROT, and SHIFT encoded strings with YARA and regex support.
XORStrings	—	Search for XOR encoded strings in a file.
xortool	—	Analyze XOR-encoded data.

Examine Static Properties / Elf Files

Tool	Command(s)	Purpose
pyelftools	readelf.py	Python library for parsing and analyzing ELF files and DWARF debugging information.

Examine Static Properties / General

Tool	Command(s)	Purpose
7-Zip	<code>7za</code>	Compress and decompress files using a variety of algorithms.
binwalk	—	Extract and analyze firmware images.
bulk_extractor	—	Extract interesting strings from binary files.
ClamAV	<code>clamscan</code> , <code>freshclam</code>	Scan files for malware signatures.
Detect-It-Easy	—	Determine types of files and examine file properties.
disitool	<code>disitool.py</code>	Manipulate embedded digital signatures.
DroidLysis	<code>droidlysis</code>	Perform static analysis of Android applications.
ExifTool	<code>exiftool</code>	Tool to read from, write to, and edit EXIF metadata of various file types.
file	—	Identify file type using "magic" numbers.
file-magic.py	—	Identify file types using the Python magic module.
Hachoir	<code>hachoir-grep</code> , <code>hachoir-metadata</code> , <code>hachoir-strip</code> , <code>hachoir-wx</code>	View, edit, and carve contents of various binary file types.
Hash ID	<code>hash-id.py</code>	Identify different types of hashes.
LIEF	—	Parse and analyze PE, ELF, MachO, DEX, OAT, VDEX, ART, and DWARF executable formats.
Magika	—	Identify file type using signatures.
Malcat Lite	—	Analyze binary files using a hex editor, disassembler, and file dissector.
msitools	—	Create, inspect and extract Windows Installer (.msi) files.
Name-That-Hash	<code>nth</code>	Identify different types of hashes.
numbers-to-string.py	—	Convert decimal numbers to strings.
re-search.py	—	Search the file for built-in regular expressions of common suspicious artifacts.
signsrch	—	Find patterns of common encryption, compression, or encoding algorithms.
Sleuth Kit	—	Analyze disk images and recover files from them.
ssdeep	—	Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.
strings.py	—	Extract ASCII and Unicode strings from binary files with length sorting and filtering.
thefuzz	—	Fuzzy String Matching in Python.
TrID	<code>trid</code> , <code>tridupdate</code>	Identify file type using signatures.
wxHexEditor	—	Hex editor.

Tool	Command(s)	Purpose
Yara Rules	—	Scan a file with YARA rules to identify capabilities and behaviors (packer detection, anti-debug, networking).
YARA-Forge Rules	—	Scan files with curated YARA rules from 45+ sources for malware family identification.

Examine Static Properties / Go

Tool	Command(s)	Purpose
GoReSym	<code>GoReSym</code>	Extract metadata and symbols from Go binaries, including stripped ones.
Redress	<code>redress</code>	Analyze stripped Go binaries to recover symbols, types, source structure, and integrate with Radare2.

Examine Static Properties / Pe Files

Tool	Command(s)	Purpose
bearparser	<code>bearcommander</code>	Parse PE file contents.
debloat	—	Remove junk contents from bloated Windows executables.
disitool.py	—	Extract, delete, copy, and inject digital signatures in PE files.
dllcharacteristics.py	—	Read and set DLL characteristics of a PE file.
Manalyze	—	Perform static analysis of suspicious PE files.
PE Tree	<code>pe-tree</code>	Examine contents and structure of PE files.
pecheck.py	—	Analyze static properties of PE files.
pedump	—	Statically analyze PE files and extract their components (e.g., resources).
pefile	—	Python library for analyzing static properties of PE files.
PEframe	<code>peframe</code>	Statically analyze PE and Microsoft Office files.
pev	<code>pestr</code> , <code>readpe</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>peldd</code> , <code>peres</code>	Analyze PE files and extract strings from them.
PortEx	<code>portex</code>	Statically analyze PE files.
readpe (formerly pev)	<code>readpe</code> , <code>pestr</code> , <code>pedis</code> , <code>pehash</code> , <code>pescan</code> , <code>pesec</code> , <code>peldd</code> , <code>pepack</code> , <code>peres</code> , <code>ofs2rva</code> , <code>rva2ofs</code>	Analyze PE files and extract strings from them.

Explore Network Interactions / Connecting

Tool	Command(s)	Purpose
Anomy	<code>anomy</code>	A wrapper around wget, ssh, sftp, ftp, and telnet to route these connections through Tor to anonymize your traffic.
cURL	<code>curl</code>	Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.
EPIC IRC Client	<code>epic5</code>	Examine IRC activities with this IRC client.
GNU Wget	<code>wget</code>	Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.
netcat	<code>nc</code>	Read and write data across network connections.
thug	—	Examine suspicious website using this low-interaction honeyclient.
tor	—	Obfuscate your origins by routing traffic through a network of anonymizing nodes.
Unfurl	—	Deconstruct and decode data from a URL.
zbarimg	<code>zbarimg</code>	Decode QR codes and barcodes from image files.

Explore Network Interactions / Monitoring

Tool	Command(s)	Purpose
Burp Suite Community Edition	<code>burpsuite</code>	Investigate website interactions using this web proxy.
cs-parse-traffic.py	—	Decrypt and parse Cobalt Strike beacon network traffic.
mitmproxy	<code>mitmproxy</code> , <code>mitmdump</code> , <code>mitmweb</code>	Investigate website interactions using this web proxy.
monitor-network	—	Monitor traffic on the first active network interface using tshark, printing a live summary to the screen or saving it t
Network Miner Free Edition	<code>networkminer</code>	Examine network traffic and carve PCAP capture files.
ngrep	—	Look for patterns in network traffic.
PolarProxy	<code>polarproxy</code>	Intercept and decrypt TLS traffic.
tcpdump	—	Capture and analyze network traffic with this command-line sniffer.
tcpflow	—	Analyze the flow of network traffic.
tcpick	—	Capture and analyze network traffic with this command-line sniffer.
tcpxtract	—	Extract files from network traffic.
tshark	—	Capture and analyze network traffic with this console-based sniffer.
wireshark	—	Capture and analyze network traffic with this sniffer.

Explore Network Interactions / Services

Tool	Command(s)	Purpose
accept-all-ips	—	Accept connections to all IPv4 and IPv6 addresses and redirect it to the corresponding local port.
dnsresolver.py	—	DNS resolver tool for dynamic analysis with wildcard and tracking support.

Tool	Command(s)	Purpose
fakedns	—	Respond to DNS queries with the specified IP address.
fakemail	—	Intercept and examine SMTP email activity with this fake SMTP server.
FakeNet-NG	—	Emulate common network services and interact with malware.
INetSim	<code>inetsim</code>	Emulate common network services and interact with malware.
inspired 3	—	Examine IRC activity with this IRC server.
netcat	<code>nc</code>	Read and write data across network connections.
Nginx	—	Web server.

Gather And Analyze Data / General

Tool	Command(s)	Purpose
DeXRAY	<code>dexray</code>	Extract and decode data from antivirus quarantine files.
dissect	—	Perform a variety of forensics and incident response tasks using this DFIR framework and toolset.
dnslib	—	Python library to encode/decode DNS wire-format packets.
ioc_parser	—	Extract IOCs from security report PDFs.
ipwhois	—	Retrieve and parse whois data for IP addresses.
malwoverview	<code>malwoverview</code>	Query public repositories of malware data (e.g., VirusTotal, HybridAnalysis).
nsrlookup	—	Look up MD5 file hashes in the NIST National Software Reference Library (NSRL).
pdnstool	—	Query passive DNS databases for DNS data.
Scalpel	—	Carve contents out of binary files, such as partitions.
time-decode	—	Decode and encode date and timestamps.
virustotal-search	<code>virustotal-search.py</code>	Search VirusTotal for file hashes.
virustotal-submit	<code>virustotal-submit.py</code>	Submit files to VirusTotal.
Yara	<code>yara</code>	Identify and classify malware samples using Yara rules.
YARA-X	—	Scan files using YARA rules, the next generation of YARA written in Rust.

General Utilities / General

Tool	Command(s)	Purpose
7-Zip	<code>7za</code>	Compress and decompress files using a variety of algorithms.
cabextract	—	Extract Microsoft cabinet (cab) files.
cURL	<code>curl</code>	Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.
Docker	—	Run and manage containers.
Firefox	<code>firefox</code>	Web browser.

Tool	Command(s)	Purpose
GNOME Calculator	<code>galculator</code>	Calculator.
IBus	<code>ibus-setup</code>	Adjust input methods for the GUI.
Info-ZIP	<code>zip</code> , <code>unzip</code>	Compress and decompress files using the zip algorithm.
myip	—	Determine the IP address of the default network interface.
myjson-filter.py	—	Filter data formatted using the JSON format used by Didier Stevens' tools.
nasm	—	An x86-64 assembler.
Nautilus	—	Graphical file manager.
OpenSSH	<code>sftp</code> , <code>ssh</code> , <code>etc</code>	Initiate and receive SSH and SFTP connections.
PowerShell Core	<code>pwsh</code>	Run PowerShell scripts and commands.
RAR	<code>rar</code>	Compress and decompress files using a variety of algorithms.
REMnux Installer	—	Install and update the REMnux distro.
restrict-egress	—	Restrict outbound network access to an allowlist of domains and CIDRs using an nftables default-deny egress policy. It i
sortcanon.py	—	Sort text files using canonicalization functions built into this tool.
SQLite	<code>sqlite3</code>	Manage and interact with SQL database files.
sshpas	<code>sshpas</code>	Supply a password to SSH non-interactively for automated logins.
texteditor.py	—	Edit text files from the command line using search-and-replace commands.
unrar-free	<code>unrar</code>	Decompress files using a variety of algorithms.
Wine	<code>wine</code>	Run Windows applications.

Investigate System Interactions / General

Tool	Command(s)	Purpose
ProcDOT	<code>procdot</code>	Visualize and examine the output of Process Monitor.
ProcmonMCP	—	MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.
sandfly-processdecloak	—	Find hidden processes on the local Linux system.
Unhide	—	Find hidden processes or connections on the local Linux system.

Perform Memory Forensics / General

Tool	Command(s)	Purpose
AESKeyFinder	<code>aeskeyfind</code>	Find 128-bit and 256-bit AES keys in a memory image.
<code>bulk_extractor</code>	—	Extract interesting strings from binary files.
RSAKeyFinder	<code>rsakeyfind</code>	Find BER-encoded RSA private keys in a memory image.
Volatility Framework	—	Memory forensics tool and framework.

Statically Analyze Code / .Net

Tool	Command(s)	Purpose
de4dot	—	Deobfuscate and unpack .NET programs.
ILSpy	—	Examine and decompile .NET programs.

Statically Analyze Code / Android

Tool	Command(s)	Purpose
androguard	—	Examine Android files.
AndroidProjectCreator	—	Convert an Android APK application file into an Android Studio project for easier analysis.
APKiD	<code>apkid</code>	Identify compilers, packers, and obfuscators used to protect Android APK and DEX files.
apktool	—	Reverse-engineer Android APK files.
baksmali	—	Disassembler for the dex format used by Dalvik, Android's Java VM implementation.
dex2jar	<code>dex-tools</code> , <code>d2j-dex2jar</code>	Examine Dalvik Executable (dex) files.
DroidLysis	<code>droidlysis</code>	Perform static analysis of Android applications.
JADX	<code>jadx</code> , <code>jadx-gui</code>	Generate Java source code from Dalvik Executable (dex) and Android APK files.

Statically Analyze Code / General

Tool	Command(s)	Purpose
Cutter	—	Reverse engineering platform powered by Rizin.
Detect-It-Easy	—	Determine types of files and examine file properties.
Ghidra	—	Software reverse engineering tool suite.
<code>objdump</code>	—	Disassemble binary files.
Qiling	—	Emulate code execution of PE files, shellcode, etc. for a variety of OS and hardware platforms.
radare2	<code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code>	Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A
Vivisect	<code>vivbin</code> , <code>vdbbin</code>	Statically examine and emulate binary files.

Statically Analyze Code / Java

Tool	Command(s)	Purpose
cfr	—	Java decompiler.
Java IDX Parser	<code>idx_parser.py</code>	Analyze Java IDX files.
Javassist	—	Java bytecode engineering toolkit/library.

Tool	Command(s)	Purpose
JD-GUI Java Decompiler	<code>jd-gui</code>	Java decompiler with GUI.
Procyon	<code>procyon</code>	Java decompiler.

Statically Analyze Code / Pe-Files

Tool	Command(s)	Purpose
binee (Binary Emulation Environment)	—	Analyze I/O operations of a suspicious PE file by emulating its execution.
capa	—	Detect suspicious capabilities in PE files.
Malchive	—	Perform static analysis of various aspects of malicious code.
mbcscan	<code>mbcscan.py</code>	Scan a PE file to list the associated Malware Behavior Catalog (MBC) details.
Speakeasy	—	Emulate code execution, including shellcode, Windows drivers, and Windows PE files.

Statically Analyze Code / Python

Tool	Command(s)	Purpose
Decompyle++	<code>pycdas</code> , <code>pycdc</code>	Python bytecode disassembler and decompiler.
PyInstaller Extractor	<code>pyinstxtractor.py</code>	Extract contents of a PyInstaller-generated PE files.
pyinstxtractor-ng	<code>pyinstxtractor-ng</code>	Extract contents of PyInstaller-generated executables without requiring a matching Python version.
uncompyle6	<code>uncompyle6</code>	Python cross-version bytecode decompiler for Python 1.0 through 3.8.

Statically Analyze Code / Scripts

Tool	Command(s)	Purpose
Autolt-Ripper	<code>autoit-ripper</code>	Extract Autolt scripts embedded in PE binaries.
decode-vbe.py	—	Decode encoded VBS scripts (VBE).
GootLoaderAutoJsDecode.py	—	Statically deobfuscate GootLoader (GOOTLOADER) malicious JScript to recover the payload and extract C2 domains.
JS Beautifier	<code>js-beautify</code>	Reformat JavaScript scripts for easier analysis.

Statically Analyze Code / Unpacking

Tool	Command(s)	Purpose
binwalk	—	Extract and analyze firmware images.
Bytehist	<code>bytehist</code>	Generate byte-usage-histograms for all types of files with a focus on PE files.
ClamAV	<code>clamscan</code> , <code>freshclam</code>	Scan files for malware signatures.
TrID	<code>trid</code> , <code>tridupdate</code>	Identify file type using signatures.

Tool	Command(s)	Purpose
UPX	<code>upx</code>	Pack and unpack PE files.

Use Artificial Intelligence / General

Tool	Command(s)	Purpose
GhidraAssistMCP	—	MCP server for AI-assisted reverse engineering in Ghidra.
OpenCode	—	Open-source AI coding agent for the terminal.
ProcmonMCP	—	MCP server that lets AI assistants analyze Process Monitor (Procmon) XML captures.
radare2	<code>r2</code> , <code>rasm2</code> , <code>rabin2</code> , <code>rahash2</code> , <code>rafind2</code> , <code>r2ai</code> , <code>decai</code> , <code>pdg</code>	Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (A
REMnux MCP Server	<code>remnux-mcp-server</code>	MCP server for using the REMnux malware analysis toolkit via AI assistants.
x64dbg Automate MCP (OpenCode skills)	—	Drive x64dbg on a remote Windows VM from OpenCode on REMnux, with eight AI commands for tracing, unpacking, state snapsh

View Or Edit Files / General

Tool	Command(s)	Purpose
dos2unix	<code>dos2unix</code> , <code>mac2unix</code> , <code>unix2dos</code> , <code>unix2mac</code>	Convert text files with Windows or macOS line breaks to Unix line breaks and vice versa.
Evince	<code>evince</code>	View documents in a variety of formats, including PDF.
feh	—	View images.
ImageMagick	<code>display</code> , <code>convert</code> , <code>mogrify</code> , <code>etc</code>	View and manipulate image and related files.
Mermaid Viewer	—	View Mermaid diagrams, such as AI-generated code-analysis workflow diagrams, in a local browser.
SciTE	—	Edit text files.
VBinDiff	<code>vbindiff</code>	Compare binary files.
Visual Studio Code	<code>code</code>	Powerful source code editor.
wxHexEditor	—	Hex editor.

Kali Linux

Crypto Stego

Tool	Command(s)	Purpose
aesfix	<code>aesfix</code>	
aeskeyfind	<code>aeskeyfind</code>	

Tool	Command(s)	Purpose
ccrypt	ccrypt	
steghide	steghide	
stegosuite	stegosuite	
stegsnow	stegsnow	

Database

Tool	Command(s)	Purpose
jsql-injection	jsql-injection	
mdbtools	mdbtools	
osscanner	osscanner	
sidguesser	sidguesser	
sqldict	sqldict	
sqlitebrowser	sqlitebrowser	
sqlmap	sqlmap	
sqlninja	sqlninja	
sqlsus	sqlsus	
tnscmd10g	tnscmd10g	

Exploitation

Tool	Command(s)	Purpose
armitage	armitage	
beef-xss	beef-xss	
exploitdb	exploitdb	
metasploit-framework	metasploit-framework	
msfpc	msfpc	
set	set	
shellnoob	shellnoob	
sqlmap	sqlmap	
termineter	termineter	

Forensics

Tool	Command(s)	Purpose
7zip	7zip	
afflib-tools	afflib-tools	
apktool	apktool	

Tool	Command(s)	Purpose
autopsy	autopsy	
binwalk	binwalk	
binwalk3	binwalk3	
bulk-extractor	bulk-extractor	
bytecode-viewer	bytecode-viewer	
cabextract	cabextract	
chkrootkit	chkrootkit	
creddump7	creddump7	
dc3dd	dc3dd	
dcfldd	dcfldd	
ddrescue	ddrescue	
dumpzilla	dumpzilla	
edb-debugger	edb-debugger	
ewf-tools	ewf-tools	
exifprobe	exifprobe	
exiv2	exiv2	
ext3grep	ext3grep	
ext4magic	ext4magic	
extundelete	extundelete	
fcrackzip	fcrackzip	
firmware-mod-kit	firmware-mod-kit	
foremost	foremost	
forensic-artifacts	forensic-artifacts	
forensics-colorize	forensics-colorize	
galleta	galleta	
gdb	gdb	
gpart	gpart	
gparted	gparted	
grokevt	grokevt	
guymager	guymager	
hashdeep	hashdeep	
inetsim	inetsim	
jadx	jadx	
jasvasnoop	jasvasnoop	

Tool	Command(s)	Purpose
libhivex-bin	libhivex-bin	
libsmali-java	libsmali-java	
lvm2	lvm2	
lynis	lynis	
mac-robber	mac-robber	
magicrescue	magicrescue	
md5deep	md5deep	
mdbtools	mdbtools	
memdump	memdump	
metacam	metacam	
missidentify	missidentify	
myrescue	myrescue	
nasm	nasm	
nasty	nasty	
ollydbg	ollydbg	
parted	parted	
pasco	pasco	
pdf-parser	pdf-parser	
pdfid	pdfid	
plaso	plaso	
polenum	polenum	
pst-utils	pst-utils	
python3-capstone	python3-capstone	
python3-dfdatetime	python3-dfdatetime	
python3-dfvfs	python3-dfvfs	
python3-dfwinreg	python3-dfwinreg	
python3-distorm3	python3-distorm3	
radare2	radare2	
readpe	readpe	
recoverdm	recoverdm	
recoverjpeg	recoverjpeg	
recstudio	recstudio	
reglookup	reglookup	
regripper	regripper	

Tool	Command(s)	Purpose
rephrase	rephrase	
rifiuti	rifiuti	
rifiuti2	rifiuti2	
rizin-cutter	rizin-cutter	
rkhunter	rkhunter	
rsakeyfind	rsakeyfind	
rz-ghidra	rz-ghidra	
safecopy	safecopy	
samdump2	samdump2	
scalpel	scalpel	
scrounge-ntfs	scrounge-ntfs	
sleuthkit	sleuthkit	
sqlitebrowser	sqlitebrowser	
ssdeep	ssdeep	
tcpdump	tcpdump	
tcpflow	tcpflow	
tcpick	tcpick	
tcpreplay	tcpreplay	
truecrack	truecrack	
undbx	undbx	
unhide	unhide	
unrar	unrar	
upx-ucl	upx-ucl	
vinetto	vinetto	
wce	wce	
winregfs	winregfs	
wireshark	wireshark	
xmount	xmount	
xplico	xplico	
yara	yara	

Fuzzing

Tool	Command(s)	Purpose
afl++	afl++	
sfuzz	sfuzz	

Tool	Command(s)	Purpose
spike	spike	
wfuzz	wfuzz	

Information Gathering

Tool	Command(s)	Purpose
Otrace	Otrace	
arping	arping	
braa	braa	
dmitry	dmitry	
dnsenum	dnsenum	
dnsmap	dnsmap	
dnsrecon	dnsrecon	
dnstracer	dnstracer	
dnswalk	dnswalk	
enum4linux	enum4linux	
fierce	fierce	
firewalk	firewalk	
fping	fping	
fragrouter	fragrouter	
ftester	ftester	
hping3	hping3	
ike-scan	ike-scan	
intrace	intrace	
irpas	irpas	
lbd	lbd	
legion	legion	
maltego	maltego	
masscan	masscan	
metagoofil	metagoofil	
nbtscan	nbtscan	
ncat	ncat	
netdiscover	netdiscover	
netmask	netmask	
nmap	nmap	
onesixtyone	onesixtyone	

Tool	Command(s)	Purpose
p0f	p0f	
qsslcaudit	qsslcaudit	
recon-ng	recon-ng	
smbmap	smbmap	
smtp-user-enum	smtp-user-enum	
snmpcheck	snmpcheck	
ssldump	ssldump	
sslh	sslh	
sslscan	sslscan	
sslyze	sslyze	
swaks	swaks	
thc-ipv6	thc-ipv6	
theharvester	theharvester	
tlssled	tlssled	
twofi	twofi	
unicornscan	unicornscan	
urlcrazy	urlcrazy	
wafw00f	wafw00f	
zenmap	zenmap	

Passwords

Tool	Command(s)	Purpose
cewl	cewl	
chntpw	chntpw	
cisco-auditing-tool	cisco-auditing-tool	
cmospwd	cmospwd	
crackmapexec	crackmapexec	
creddump7	creddump7	
crunch	crunch	
fcrackzip	fcrackzip	
freerdp3-x11	freerdp3-x11	
gpp-decrypt	gpp-decrypt	
hash-identifier	hash-identifier	
hashcat	hashcat	
hashcat-tils	hashcat-tils	

Tool	Command(s)	Purpose
hashid	hashid	
hydra	hydra	
hydra-gtk	hydra-gtk	
john	john	
johnny	johnny	
maskprocessor	maskprocessor	
medusa	medusa	
mimikatz	mimikatz	
ncrack	ncrack	
onesixtyone	onesixtyone	
ophcrack	ophcrack	
ophcrack-cli	ophcrack-cli	
pack	pack	
pack2	pack2	
passing-the-hash	passing-the-hash	
patator	patator	
pdfcrack	pdfcrack	
pipal	pipal	
polenum	polenum	
rainbowcrack	rainbowcrack	
rarcrack	rarcrack	
rcracki-mt	rcracki-mt	
rsmangler	rsmangler	
samdump2	samdump2	
seclists	seclists	
sipcrack	sipcrack	
sipvicious	sipvicious	
smbmap	smbmap	
sqldict	sqldict	
statsprocessor	statsprocessor	
sucrack	sucrack	
thc-pptp-bruter	thc-pptp-bruter	
truecrack	truecrack	
twofi	twofi	

Tool	Command(s)	Purpose
wordlists	wordlists	

Post Exploitation

Tool	Command(s)	Purpose
cymothoa	cymothoa	
dbd	dbd	
dns2tcp	dns2tcp	
exe2hexbat	exe2hexbat	
iodine	iodine	
laudanum	laudanum	
mimikatz	mimikatz	
miredo	miredo	
nishang	nishang	
powersploit	powersploit	
proxychains4	proxychains4	
proxytunnel	proxytunnel	
ptunnel	ptunnel	
pwnat	pwnat	
sbd	sbd	
shellter	shellter	
sslh	sslh	
stunnel4	stunnel4	
udptunnel	udptunnel	
veil	veil	
webacoo	webacoo	
weeveily	weeveily	

Reverse Engineering

Tool	Command(s)	Purpose
apktool	apktool	
bytecode-viewer	bytecode-viewer	
clang	clang	
dex2jar	dex2jar	
edb-debugger	edb-debugger	
jadx	jadx	

Tool	Command(s)	Purpose
javasnoop	javasnoop	
jd-gui	jd-gui	
metasploit-framework	metasploit-framework	
ollydbg	ollydbg	
radare2	radare2	
recstudio	recstudio	
rizin	rizin	
rizin-cutter	rizin-cutter	
rz-ghidra	rz-ghidra	

Sniffing Spoofing

Tool	Command(s)	Purpose
above	above	
bettercap	bettercap	
darkstat	darkstat	
dnscraf	dnscraf	
driftnet	driftnet	
dsniff	dsniff	
ettercap-graphical	ettercap-graphical	
ferret-sidejack	ferret-sidejack	
fiked	fiked	
hamster-sidejack	hamster-sidejack	
hexinject	hexinject	
isr-evilgrade	isr-evilgrade	
macchanger	macchanger	
mitmproxy	mitmproxy	
netsniff-ng	netsniff-ng	
rebind	rebind	
responder	responder	
sniffjoke	sniffjoke	
sslsniff	sslsniff	
sslsplit	sslsplit	
tcpflow	tcpflow	
tcpreplay	tcpreplay	
wifi-honey	wifi-honey	

Tool	Command(s)	Purpose
wireshark	wireshark	

Vulnerability

Tool	Command(s)	Purpose
afl++	afl++	
bed	bed	
cisco-auditing-tool	cisco-auditing-tool	
cisco-global-exploiter	cisco-global-exploiter	
cisco-ocs	cisco-ocs	
cisco-torch	cisco-torch	
copy-router-config	copy-router-config	
dhcpig	dhcpig	
enumiax	enumiax	
gvm	gvm	
iaxflood	iaxflood	
inviteflood	inviteflood	
legion	legion	
lynis	lynis	
nikto	nikto	
nmap	nmap	
ohrwurm	ohrwurm	
peass	peass	
protos-sip	protos-sip	
rtpbreak	rtpbreak	
rtpflood	rtpflood	
rtpinsertsound	rtpinsertsound	
rtpmixsound	rtpmixsound	
sctpscan	sctpscan	
sfuzz	sfuzz	
siege	siege	
siparmyknife	siparmyknife	
sipp	sipp	
sipsak	sipsak	
sipvicious	sipvicious	
slowhttpptest	slowhttpptest	

Tool	Command(s)	Purpose
spike	spike	
t50	t50	
thc-ssl-dos	thc-ssl-dos	
unix-privesc-check	unix-privesc-check	
voiphopper	voiphopper	
yersinia	yersinia	

Web

Tool	Command(s)	Purpose
apache-users	apache-users	
apache2	apache2	
beef-xss	beef-xss	
burpsuite	burpsuite	
cadaver	cadaver	
commix	commix	
cutycapt	cutycapt	
davtest	davtest	
default-mysql-server	default-mysql-server	
dirb	dirb	
dirbuster	dirbuster	
dotdotpwn	dotdotpwn	
eyewitness	eyewitness	
ferret-sidejack	ferret-sidejack	
ftester	ftester	
hakrawler	hakrawler	
hamster-sidejack	hamster-sidejack	
heartleech	heartleech	
httpprint	httpprint	
httrack	httrack	
hydra	hydra	
hydra-gtk	hydra-gtk	
jboss-autopwn	jboss-autopwn	
joomscan	joomscan	
jsql-injection	jsql-injection	
laudanum	laudanum	

Tool	Command(s)	Purpose
lbd	<code>lbd</code>	
maltego	<code>maltego</code>	
medusa	<code>medusa</code>	
mitmproxy	<code>mitmproxy</code>	
ncrack	<code>ncrack</code>	
nikto	<code>nikto</code>	
nishang	<code>nishang</code>	
nmap	<code>nmap</code>	
osscanner	<code>osscanner</code>	
owasp-mantra-ff	<code>owasp-mantra-ff</code>	
padbuster	<code>padbuster</code>	
paros	<code>paros</code>	
patator	<code>patator</code>	
php	<code>php</code>	
php-mysql	<code>php-mysql</code>	
proxychains4	<code>proxychains4</code>	
proxytunnel	<code>proxytunnel</code>	
qsslcaudit	<code>qsslcaudit</code>	
redsocks	<code>redsocks</code>	
sidguesser	<code>sidguesser</code>	
siege	<code>siege</code>	
skipfish	<code>skipfish</code>	
slowhttptest	<code>slowhttptest</code>	
sqldict	<code>sqldict</code>	
sqlitebrowser	<code>sqlitebrowser</code>	
sqlmap	<code>sqlmap</code>	
sqlninja	<code>sqlninja</code>	
sqlsus	<code>sqlsus</code>	
ssldump	<code>ssldump</code>	
sslh	<code>sslh</code>	
sslscan	<code>sslscan</code>	
sslsniff	<code>sslsniff</code>	
sslsplit	<code>sslsplit</code>	
sslyze	<code>sslyze</code>	

Tool	Command(s)	Purpose
stunnel4	stunnel4	
thc-ssl-dos	thc-ssl-dos	
tlssled	tlssled	
tnscmd10g	tnscmd10g	
uniscan	uniscan	
wafw00f	wafw00f	
wapiti	wapiti	
watobo	watobo	
webacoo	webacoo	
webscarab	webscarab	
webshells	webshells	
weevely	weevely	
wfuzz	wfuzz	
whatweb	whatweb	
wireshark	wireshark	
wpscan	wpscan	
xsser	xsser	
zaproxy	zaproxy	

FLARE-VM

All Packages (choco)

Tool	Command(s)	Purpose
010editor	010editor	
7zip	7zip	
advanced-installer	advanced-installer	
angr	angr	
apimonitor	apimonitor	
apktool	apktool	
asar	asar	
autoit-ripper	autoit-ripper	
binaryninja	binaryninja	
bindiff	bindiff	
blobrunner	blobrunner	

Tool	Command(s)	Purpose
blobrunner64	blobrunner64	
bytecodeviewer	bytecodeviewer	
capa	capa	
capa-explorer-web	capa-explorer-web	
chrome.extensions	chrome.extensions	
cmdr	cmdr	
codetrack	codetrack	
cryptotester	cryptotester	
cutter	cutter	
cyberchef	cyberchef	
cygwin	cygwin	
de4dot-cex	de4dot-cex	
dependencywalker	dependencywalker	
dex2jar	dex2jar	
didier-stevens-beta	didier-stevens-beta	
didier-stevens-suite	didier-stevens-suite	
die	die	
dll-to-exe	dll-to-exe	
dnlib	dnlib	
dnspyex	dnspyex	
dotdumper	dotdumper	
dotnet3.5	dotnet3.5	
exeinfope	exeinfope	
explorersuite	explorersuite	
extreme_dumper	extreme_dumper	
ezviewer	ezviewer	
fakenet-ng	fakenet-ng	
fiddler	fiddler	
file	file	
floss	floss	
garbageman	garbageman	
ghidra	ghidra	
goresym	goresym	
gostringungarbler	gostringungarbler	

Tool	Command(s)	Purpose
hashmyfiles	hashmyfiles	
hollowshunter	hollowshunter	
hxd	hxd	
ida.plugin.capa	ida.plugin.capa	
ida.plugin.comida	ida.plugin.comida	
ida.plugin.delphihelper	ida.plugin.delphihelper	
ida.plugin.dereferencing	ida.plugin.dereferencing	
ida.plugin.diaphora	ida.plugin.diaphora	
ida.plugin.flare	ida.plugin.flare	
ida.plugin.flare-emu	ida.plugin.flare-emu	
ida.plugin.hashdb	ida.plugin.hashdb	
ida.plugin.hrtng	ida.plugin.hrtng	
ida.plugin.ifl	ida.plugin.ifl	
ida.plugin.xray	ida.plugin.xray	
ida.plugin.xrefer	ida.plugin.xrefer	
idafree	idafree	
idr	idr	
ifpstools	ifpstools	
ilspy	ilspy	
innoextract	innoextract	
innounp	innounp	
internet_detector	internet_detector	
ipython	ipython	
isd	isd	
js-beautify	js-beautify	
js-deobfuscator	js-deobfuscator	
keystone	keystone	
libraries.python3	libraries.python3	
magika	magika	
malware-jail	malware-jail	
map	map	
microsoft-office	microsoft-office	
nasm	nasm	
net-reactor-slayer	net-reactor-slayer	

Tool	Command(s)	Purpose
nmap	<code>nmap</code>	
notepadplusplus	<code>notepadplusplus</code>	
notepadpp.plugin.compare	<code>notepadpp.plugin.compare</code>	
notepadpp.plugin.jstool	<code>notepadpp.plugin.jstool</code>	
notepadpp.plugin.xmltools	<code>notepadpp.plugin.xmltools</code>	
obfuscator-io-deobfuscator	<code>obfuscator-io-deobfuscator</code>	
offvis	<code>offvis</code>	
onenoteanalyzer	<code>onenoteanalyzer</code>	
pdbresym	<code>pdbresym</code>	
pdfstreamdumper	<code>pdfstreamdumper</code>	
pe_unmapper	<code>pe_unmapper</code>	
pebear	<code>pebear</code>	
peid	<code>peid</code>	
pesieve	<code>pesieve</code>	
pestudio	<code>pestudio</code>	
pkg-unpacker	<code>pkg-unpacker</code>	
pma-labs	<code>pma-labs</code>	
procdot	<code>procdot</code>	
processdump	<code>processdump</code>	
psnotify	<code>psnotify</code>	
pycdas	<code>pycdas</code>	
pycdc	<code>pycdc</code>	
pylingual	<code>pylingual</code>	
rat-king-parser	<code>rat-king-parser</code>	
recaf	<code>recaf</code>	
reg_export	<code>reg_export</code>	
regcool	<code>regcool</code>	
regshot	<code>regshot</code>	
resourcehacker	<code>resourcehacker</code>	
rundotnetdll	<code>rundotnetdll</code>	
sctdbg	<code>sctdbg</code>	
sclauncher	<code>sclauncher</code>	
sclauncher64	<code>sclauncher64</code>	
sfextract	<code>sfextract</code>	

Tool	Command(s)	Purpose
shellcode_launcher	shellcode_launcher	
sysinternals	sysinternals	
systeminformer	systeminformer	
ttd	ttd	
uncompyle6	uncompyle6	
uniextract2	uniextract2	
unpyc3	unpyc3	
upx	upx	
vb-decompiler-lite	vb-decompiler-lite	
vbdec	vbdec	
vcbuildtools	vcbuildtools	
vcredist-all	vcredist-all	
vscode	vscode	
vscode.extension.jupyter	vscode.extension.jupyter	
vscode.extension.python	vscode.extension.python	
windbg	windbg	
windows-terminal	windows-terminal	
wireshark	wireshark	
x64dbg	x64dbg	
x64dbg.plugin.dbgchild	x64dbg.plugin.dbgchild	
x64dbg.plugin.ollydumpex	x64dbg.plugin.ollydumpex	
x64dbg.plugin.scyllahide	x64dbg.plugin.scyllahide	
x64dbg.plugin.x64dbgpy	x64dbg.plugin.x64dbgpy	
yara	yara	

SIFT Workstation

Forensic Packages

Tool	Command(s)	Purpose
absent	absent	
aeskeyfind	aeskeyfind	
afflib-tools	afflib-tools	
aircrack-ng	aircrack-ng	
android-sdk-platform-tools	android-sdk-platform-tools	

Tool	Command(s)	Purpose
arp-scan	<code>arp-scan</code>	
autopsy	<code>autopsy</code>	
avfs	<code>avfs</code>	
aws-cli	<code>aws-cli</code>	
bless	<code>bless</code>	
blt	<code>blt</code>	
bulk-extractor	<code>bulk-extractor</code>	
cabextract	<code>cabextract</code>	
ccrypt	<code>ccrypt</code>	
chromium-browser	<code>chromium-browser</code>	
clamav	<code>clamav</code>	
claude-code	<code>claude-code</code>	
cmospwd	<code>cmospwd</code>	
cryptcat	<code>cryptcat</code>	
cryptsetup	<code>cryptsetup</code>	
dc3dd	<code>dc3dd</code>	
dcfldd	<code>dcfldd</code>	
default-jre	<code>default-jre</code>	
disktype	<code>disktype</code>	
dislocker	<code>dislocker</code>	
docker	<code>docker</code>	
dos2unix	<code>dos2unix</code>	
dotnet	<code>dotnet</code>	
driftnet	<code>driftnet</code>	
dsniff	<code>dsniff</code>	
e2fsprogs	<code>e2fsprogs</code>	
ent	<code>ent</code>	
epic5	<code>epic5</code>	
etherape	<code>etherape</code>	
ettercap-graphical	<code>ettercap-graphical</code>	
ewf-tools	<code>ewf-tools</code>	
exfat-extras	<code>exfat-extras</code>	
exfat-fuse	<code>exfat-fuse</code>	
exif	<code>exif</code>	

Tool	Command(s)	Purpose
extundelete	extundelete	
fdupes	fdupes	
feh	feh	
file	file	
flex	flex	
foremost	foremost	
gawk	gawk	
gdb	gdb	
gddrescue	gddrescue	
ghex	ghex	
graphviz	graphviz	
grepcidr	grepcidr	
gthumb	gthumb	
gzrt	gzrt	
hashdeep	hashdeep	
hexedit	hexedit	
hydra	hydra	
hydra-gtk	hydra-gtk	
init	init	
ipython3	ipython3	
jq	jq	
kdiff3	kdiff3	
kpartx	kpartx	
lft	lft	
lvm2	lvm2	
magnus	magnus	
mdadm	mdadm	
mtd-utils	mtd-utils	
nbd-client	nbd-client	
nbtscan	nbtscan	
netcat	netcat	
netpbm	netpbm	
netsed	netsed	
netwox	netwox	

Tool	Command(s)	Purpose
nfdump	nfdump	
ngrep	ngrep	
nikto	nikto	
ntfs-3g	ntfs-3g	
okular	okular	
onboard	onboard	
open-iscsi	open-iscsi	
openjdk	openjdk	
ophcrack	ophcrack	
ophcrack-cli	ophcrack-cli	
orca	orca	
outguess	outguess	
p0f	p0f	
p7zip-full	p7zip-full	
patch	patch	
pdftk-java	pdftk-java	
perl	perl	
pev	pev	
pff-tools	pff-tools	
phonon	phonon	
pkg-config	pkg-config	
plaso-tools	plaso-tools	
powershell	powershell	
pst-utils	pst-utils	
pv	pv	
python-flowgrep	python-flowgrep	
python3-debian	python3-debian	
python3-dfvfs	python3-dfvfs	
python3-fuse	python3-fuse	
python3-keyrings-alt	python3-keyrings-alt	
python3-m2crypto	python3-m2crypto	
python3-magic	python3-magic	
python3-pefile	python3-pefile	
python3-plaso	python3-plaso	

Tool	Command(s)	Purpose
python3-pypff	python3-pypff	
python3-pyqt5	python3-pyqt5	
python3-pytsk3	python3-pytsk3	
python3-redis	python3-redis	
python3-setuptools-rust	python3-setuptools-rust	
python3-tk	python3-tk	
python3-tsk	python3-tsk	
python3-virtualenv	python3-virtualenv	
python3-wxgtk4	python3-wxgtk4	
python3-xlswriter	python3-xlswriter	
python3-yara	python3-yara	
qemu	qemu	
qemu-utils	qemu-utils	
radare2	radare2	
rar	rar	
rsakeyfind	rsakeyfind	
safecopy	safecopy	
samdump2	samdump2	
scalpel	scalpel	
silversearcher-ag	silversearcher-ag	
sleuthkit	sleuthkit	
socat	socat	
squashfs-tools	squashfs-tools	
ssdeep	ssdeep	
ssldump	ssldump	
sslsniff	sslsniff	
stunnel4	stunnel4	
swig	swig	
tcl	tcl	
tcpflow	tcpflow	
tcpick	tcpick	
tcpreplay	tcpreplay	
tcpslice	tcpslice	
tcpstat	tcpstat	

Tool	Command(s)	Purpose
tcptrace	tcptrace	
tcptrack	tcptrack	
tcpxtract	tcpxtract	
testdisk	testdisk	
tofrodo	tofrodo	
transmission	transmission	
ugrep	ugrep	
unity-control-center	unity-control-center	
unrar	unrar	
upx-ucl	upx-ucl	
vbindiff	vbindiff	
virtuoso-minimal	virtuoso-minimal	
vmfs-tools	vmfs-tools	
winbind	winbind	
wine	wine	
wireshark	wireshark	
xdot	xdot	
xfsprogs	xfsprogs	
xmount	xmount	
zenity	zenity	
zlib1g-dev	zlib1g-dev	

Security Onion

DB/Search

Tool	Command(s)	Purpose
Elasticsearch	—	NoSQL index for logs/metadata/alerts
ILM	—	Index Lifecycle Management (aging/rolling/deletion)
Kibana	—	Search analytics & dashboards

Detection

Tool	Command(s)	Purpose
ATT&CK Navigator	—	Visual defensive-coverage matrix
ElastAlert 2	—	Continuous alert-indicator scanner
Playbook	—	Custom detections mapped to MITRE ATT&CK

Tool	Command(s)	Purpose
Sigma / Sigma-CLI	—	Translate YAML detection rules to Elastic queries
so-idstools	—	Manage/update Suricata ET rules

Diagnostics

Tool	Command(s)	Purpose
Grafana	—	Infra/node-health dashboards
InfluxDB	—	Time-series store for hardware metrics
Telegraf	—	Hardware health/CPU/mem/disk agent

File Analysis

Tool	Command(s)	Purpose
DomainStats	—	Domain-age checks to flag malicious sites
FreqServer	—	Entropy analysis to detect DGAs
Strelka	—	Real-time file extraction & YARA/file carving

Host/EDR

Tool	Command(s)	Purpose
Elastic Agent	—	Host log/metric/audit collector
Elastic Fleet	—	Central agent config & deployment
OpenCanary	—	Network honeypot / decoy services
Osquery	—	Live endpoint queries via SQL
Wazuh	—	HIDS - File Integrity Monitoring & rootkit checks

IR/AI

Tool	Command(s)	Purpose
CyberChef	—	Visual data decode/parse/analysis
Onion AI Assistant	—	AI triage assistance for analysts
Security Onion Console (SOC)	—	Web UI for threat hunting & grid management
TheHive	—	Collaborative IR & case management

Identity/Infra

Tool	Command(s)	Purpose
ManagerHype	—	Hypervisor integration for VM mapping
Nginx (so-nginx)	—	Reverse proxy for internal web traffic
Ory Kratos & Dex	—	IAM - SSO/auth/sessions

Tool	Command(s)	Purpose
so-apt-cacher-ng	—	Local package caching proxy
so-firewall	—	iptables/nftables wrapper for inter-node comms

Log Routing

Tool	Command(s)	Purpose
Filebeat	—	Lightweight log shipper
Logstash	—	Data pipeline - ECS normalize & enrich
Redis	—	Log ingestion queue/broker
Rsyslog / Syslog-ng	—	Ingest traditional syslog from appliances

Network Visibility

Tool	Command(s)	Purpose
Sensoroni	—	Backend API agent retrieving/extracting PCAPs
Stenographer	—	High-speed raw full packet capture daemon
Suricata	—	Signature-based IDS/IPS & packet capture
Zeek	—	Network traffic analyzer & protocol metadata logging

Sources

- `remnux_docs` — <https://raw.githubusercontent.com/REMnux/docs/master/discover-the-tools>
- `kali_control` — <https://gitlab.com/kalilinux/packages/kali-meta/-/raw/kali/master/debian/control>
- `flare_config` — <https://raw.githubusercontent.com/mandiant/flare-vm/main/config.xml>

Tool Reference

One page per tool. Every option below was read off the real binary and is checked by the linter.

affcat

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affcat version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TODO: one-line imperative purpose.

Synopsis

```
affcat [options] infile [... more infiles]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-q	—	--- quiet; don't print to STDERR if a page is skipped	
-n	—	--- noisy; tell when pages are skipped.	
-l	—	--- List all of the segment names	
-L	—	--- List segment names, lengths, and args	
-d	—	--- debug. Print the page numbers to stderr as data goes to stdout	
-b	—	--- Output BADFALG for bad blocks (default is NULLs)	
-v	—	--- Just print the version number and exit.	

Gotchas

TODO: operational traps.

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

affconvert

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** affconvert version 3.7.20

Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Please, see more info in manpage.

Synopsis

```
affconvert [options] file1 [... files]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[dc3dd](#) , [dcfldd](#) , [dd](#) , [ewfacquire](#)

affinfo

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** affinfo version 3.7.20 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

(may be repeated)

Synopsis

```
affinfo [options] infile
-a = print ALL segments (normally data segments are suppressed)
-b = print how many bad blocks in each segment (implies -a)
-i = identify the files, don't do info on them.
-w = wide output; print more than 1 line if necessary.
-s segment = Just print information about 'segment'.
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

bdeinfo

Kit: SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: bdeinfo

Synopsis

```
bdeinfo [ -k keys ] [ -o offset ] [ -p password ]  
[ -r password ] [ -s filename ] [ -hvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs  
bdeinfo -V  
# from cyberlab 06-windows-artifact-libs  
bdeinfo exercise/bitlocker.raw
```

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	shows this help	
<code>-k</code>	—	the full volume encryption key and tweak key formatted in base16 and separated by a : character e.g. FVEK:TWEAK	
<code>-o</code>	—	specify the volume offset in bytes	
<code>-p</code>	—	specify the password/passphrase	
<code>-r</code>	—	specify the recovery password	
<code>-s</code>	—	specify the file containing the startup key. typically this file has the extension .BEK	

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	

Gotchas

TODO: operational traps.

See also

`ewfinfo`, `ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`

dc3dd

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dc3dd (dc3dd) 7.2.646
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

dc3dd [OPTION 1] [OPTION 2] ... [OPTION N]

Synopsis

Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo dc3dd if=/dev/sdX of=/evidence/case01.dd hash=sha256 log=/evidence/case01.log
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	display this help and exit	
--version	—	output version information and exit	
--flags	—	display compile-time flags and exit	

Gotchas

TODO: operational traps.

See also

[dcfldd](#), [dd](#), [ewfacquire](#), [affconvert](#)

dcfldd

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Version:** dcfldd (dcfldd) 1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Enhanced version of dd for forensics and security.

Synopsis

```
dcfldd [OPTION]...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	display this help and exit	
<code>--version</code>	—	output version information and exit	

Gotchas

TODO: operational traps.

See also

`dc3dd`, `dd`, `ewfacquire`, `affconvert`

dd

Kit: Base OS — present on every Linux image **Capability:** Image a disk, volume or device **Version:** dd (coreutils) 9.1
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

or: dd OPTION

Synopsis

```
dd [OPERAND]...  
or: dd OPTION
```

Common invocations

```
# from cyberlab 01-disk-forensics  
dd if=/dev/zero of=exercise/sample.dd bs=1M count=10  
# from cyberlab 51-linux-triage-workflow  
dd if=/dev/zero of=trriage_sample.raw bs=1M count=8
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	display this help and exit	
--version	—	output version information and exit	

Gotchas

TODO: operational traps.

See also

[dc3dd](#), [dcfldd](#), [ewfacquire](#), [affconvert](#)

dumpcap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs: https://www.wireshark.org](#)

← [Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
dumpcap [options] ...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 48 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-i</code>	interface	name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: rpcap:/// TCP@:	
<code>--interface</code>	interface	name or idx of interface (def: first non-loopback), or for remote capturing, use one of these formats: rpcap:/// TCP@:	
<code>--ifname</code>	name	name to use in the capture file for a pipe from which we're capturing	
<code>--ifdescr</code>	description	description to use in the capture file for a pipe from which we're capturing	
<code>-f</code>	capture filter	packet filter in libpcap filter syntax	
<code>-s</code>	snaplen	packet snapshot length (def: appropriate maximum)	
<code>--snapshot-length</code>	snaplen	packet snapshot length (def: appropriate maximum)	
<code>-p</code>	—	don't capture in promiscuous mode	
<code>--no-promiscuous-mode</code>	—	don't capture in promiscuous mode	

Flag	Argument	What it does	When you would use it
<code>-I</code>	—	capture in monitor mode, if available	
<code>--monitor-mode</code>	—	capture in monitor mode, if available	
<code>-B</code>	buffer size	size of kernel buffer in MiB (def: 2MiB)	
<code>--buffer-size</code>	buffer size	size of kernel buffer in MiB (def: 2MiB)	
<code>-y</code>	link type	link layer type (def: first appropriate)	
<code>--linktype</code>	link type	link layer type (def: first appropriate)	
<code>-D</code>	—	print list of interfaces and exit	
<code>--list-interfaces</code>	—	print list of interfaces and exit	
<code>-L</code>	—	print list of link-layer types of iface and exit	
<code>--list-data-link-types</code>	—	print list of link-layer types of iface and exit	
<code>--list-time-stamp-types</code>	—	print list of timestamp types for iface and exit	
<code>-d</code>	—	print generated BPF code for capture filter	
<code>-S</code>	—	print statistics for each interface once per second	
<code>-M</code>	—	for -D, -L, and -S, produce machine-readable output	
<code>-c</code>	packet count	stop after n packets (def: infinite)	
<code>-a</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	
<code>--autostop</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM kB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	
<code>-w</code>	filename	name of file to save (def: tempfile)	
<code>-g</code>	—	enable group read access on the output file(s)	
<code>-b</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in	
<code>--ring-buffer</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM kB files:NUM - ringbuffer: replace after NUM files packets:NUM - ringbuffer: replace after NUM packets in	
<code>-n</code>	—	use pcapng format instead of pcap (default)	
<code>-P</code>	—	use libpcap format instead of pcapng	
<code>--capture-comment</code>	comment	add a capture comment to the output file (only for pcapng)	
<code>--temp-dir</code>	directory	write temporary files to this directory (default: /tmp)	

Flag	Argument	What it does	When you would use it
<code>--log-level</code>	level	sets the active log level ("critical", "warning", etc.)	
<code>--log-fatal</code>	level	sets level to abort the program ("critical" or "warning")	
<code>--log-domains</code>	[!]list	comma separated list of the active log domains	
<code>--log-debug</code>	[!]list	comma separated list of domains with "debug" level	
<code>--log-noisy</code>	[!]list	comma separated list of domains with "noisy" level	
<code>--log-file</code>	path	file to output messages to (in addition to stderr)	
<code>-N</code>	packet_limit	maximum number of packets buffered within dumpcap	
<code>-C</code>	byte_limit	maximum number of bytes used for buffering packets within dumpcap	
<code>-t</code>	—	use a separate thread per interface	
<code>-q</code>	—	don't report packet capture counts	
<code>-v</code>	—	print version information and exit	
<code>--version</code>	—	print version information and exit	
<code>-h</code>	—	display this help and exit	
<code>--help</code>	—	display this help and exit	

Gotchas

TODO: operational traps.

See also

[tshark](#)

ewfacquire

Kit: Kali Linux · SIFT Workstation **Capability:** Image a disk, volume or device **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Acquire a disk, volume or device into an EWF/E01 evidence container, with the case metadata and hashes stored inside the image itself.

Synopsis

```
ewfacquire [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -C case_number ] [ -d digest_type ] [ -D description ]
[ -e examiner_name ] [ -E evidence_number ] [ -f format ]
[ -g number_of_sectors ] [ -l log_filename ]
[ -m media_type ] [ -M media_flags ] [ -N notes ]
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo ewfacquire -t /evidence/case01 -f encase6 -c deflate -S 2G /dev/sdX
```

Options

All 32 options parsed from the captured help text; 30 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-A</code>	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage, for non-ASCII metadata.
<code>-b</code>	—	specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768	Sectors per chunk. Larger chunks read faster but lose more data to each unreadable sector.
<code>-B</code>	—	specify the number of bytes to acquire (default is all bytes)	Acquire a fixed number of bytes rather than the whole device — with <code>-o</code> , the way to image one region.

Flag	Argument	What it does	When you would use it
<code>-c</code>	—	specify the compression values as: level or method:level compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty	Trade size against time. <code>best</code> on a slow USB source is often faster overall than <code>none</code> , because the bottleneck is the read, not the CPU.
<code>-C</code>	—	specify the case number (default is case_number).	Case number, stored in the image header.
<code>-d</code>	—	calculate additional digest (hash) types besides md5, options: sha1, sha256	Add sha1 or sha256 alongside the default md5. Worth doing once, at acquisition: md5 alone is increasingly challenged, and rehashing later means reading the whole image again.
<code>-D</code>	—	specify the description (default is description).	Description, stored in the image header.
<code>-e</code>	—	specify the examiner name (default is examiner_name).	Examiner name, stored in the image header.
<code>-E</code>	—	specify the evidence number (default is evidence_number).	Evidence number, stored in the image header.
<code>-f</code>	—	specify the EWF file format to write to, options: ewf, smart, ftk, encase2, encase3, encase4, encase5, encase6 (default), encase7, encase7-v2, linen5, linen6, linen7, ewfx	Choose the EWF variant. <code>encase6</code> is the safe default for interoperability; pick the format the tool that will read it expects, not the newest one available.
<code>-g</code>	—	specify the number of sectors to be used as error granularity	Error granularity — how much data around a bad sector is discarded. Lower values preserve more of a failing disk at the cost of speed.
<code>-h</code>	—	shows this help	
<code>-l</code>	—	logs acquiry errors and the digest (hash) to the log_filename	Write the acquisition log, including errors and hashes, to a file. This is the record you cite later; do not skip it.
<code>-m</code>	—	specify the media type, options: fixed (default), removable, optical, memory	Record the media type (fixed, removable, optical, memory) in the header.
<code>-M</code>	—	specify the media flags, options: logical, physical (default)	Record whether this is a physical device or a logical volume.
<code>-N</code>	—	specify the notes (default is notes).	Free-text notes, stored in the image header. These five metadata flags are the reason to choose E01 over a raw <code>dd</code> image: the container describes its own provenance.
<code>-o</code>	—	specify the offset to start to acquire (default is 0)	Start at an offset rather than sector 0.
<code>-p</code>	—	specify the process buffer size (default is the chunk size)	Process buffer size — a throughput tuning knob.
<code>-P</code>	—	specify the number of bytes per sector (default is 512) (use this to override the automatic bytes per sector detection)	Override bytes-per-sector. Needed on 4Kn drives where the 512-byte assumption is wrong.
<code>-q</code>	—	quiet shows minimal status information	Minimal status output, for logs.
<code>-r</code>	—	specify the number of retries when a read error occurs (default is 2)	Read retries before giving up on a sector. Raise it for a dying drive; lower it when retries are heating a drive that may not survive the acquisition.
<code>-R</code>	—	resume acquiry at a safe point	Resume an interrupted acquisition at a safe point, instead of restarting a multi-hour read.

Flag	Argument	What it does	When you would use it
<code>-s</code>	—	swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)	Swap byte pairs for a big-endian source. Rare, and wrong unless you know the source endianness differs.
<code>-S</code>	—	specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for encase6 and encase7 format and 1.9 GiB for other formats)	Segment size. The 1.4 GiB default suits FAT32 and optical media; raise it on a modern filesystem to avoid hundreds of fragments.
<code>-t</code>	—	specify the target file (without extension) to write to	Set the output name, without extension — <code>ewfacquire</code> appends <code>.E01</code> , <code>.E02</code> ... itself. The one flag you always pass.
<code>-T</code>	—	specify the file containing the table of contents (TOC) of an optical disc. The TOC file must be in the CUE format.	Supply a CUE file when imaging optical media, so the track layout is preserved.
<code>-u</code>	—	unattended mode (disables user interaction)	Unattended — suppresses the interactive prompts. Required for any scripted or headless acquisition.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr — worth capturing alongside <code>-l</code> when a source is throwing read errors.
<code>-V</code>	—	print version	
<code>-w</code>	—	zero sectors on read error (mimic EnCase like behavior)	Zero unreadable sectors instead of aborting, the way EnCase does. Keeps offsets aligned so the filesystem still parses.
<code>-x</code>	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.
<code>-2</code>	—	specify the secondary target file (without extension) to write to	Write a second copy in the same pass. Two independent copies for the cost of one read, which matters when the source is failing and may not survive a second acquisition.

Gotchas

- The case metadata flags default to literal placeholder strings — `case_number`, `examiner_name`, `evidence_number`. Omit them and the image ships with those placeholders recorded as fact, which is worse than an empty field because it looks filled in.
- `-t` takes the name **without** an extension. Passing `image.E01` produces `image.E01.E01`.
- Acquisition is not verification. `ewfacquire` records hashes; proving the image still matches them later is `ewfverify`.

See also

`dc3dd`, `dcfldd`, `dd`, `affconvert`

ewfexport

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Convert an EWF/E01 image to raw, or to another EWF format, including extracting a subset of it.

Synopsis

```
ewfexport [ -A codepage ] [ -b number_of_sectors ]
[ -B number_of_bytes ] [ -c compression_values ]
[ -d digest_type ] [ -f format ] [ -l log_filename ]
[ -o offset ] [ -p process_buffer_size ]
[ -S segment_file_size ] [ -t target ] [ -hqsuvVwx ] ewf_files
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 19 options parsed from the captured help text; 17 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-A</code>	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage.
<code>-b</code>	—	specify the number of sectors to read at once (per chunk), options: 16, 32, 64 (default), 128, 256, 512, 1024, 2048, 4096, 8192, 16384 or 32768 (not used for raw and files formats)	Sectors per chunk.
<code>-B</code>	—	specify the number of bytes to export (default is all bytes)	Number of bytes to export — with <code>-o</code> , extracts one partition.
<code>-c</code>	—	specify the compression values as: level or method:level compression method options: deflate (default), bzip2 (bzip2 is only supported by EWF2 formats) compression level options: none (default), empty	Compression for an EWF target.

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	calculate additional digest (hash) types besides md5, options: sha1, sha256 (not used for raw and files format)	Calculate an additional digest over the exported data.
<code>-f</code>	—	specify the output format to write to, options: raw (default), files (restricted to logical volume files), ewf, smart, encase1, encase2, encase3, encase4, encase5, encase6, encase7, encase7-v2, linen5	Output format — <code>raw</code> for tools that cannot read EWF, or another EWF variant for compatibility.
<code>-h</code>	—	shows this help	
<code>-l</code>	—	logs export errors and the digest (hash) to the log_filename	Log the export.
<code>-o</code>	—	specify the offset to start the export (default is 0)	Start offset, to export a region rather than the whole image.
<code>-p</code>	—	specify the process buffer size (default is the chunk size)	Process buffer size.
<code>-q</code>	—	quiet shows minimal status information	Minimal output.
<code>-s</code>	—	swap byte pairs of the media data (from AB to BA) (use this for big to little endian conversion and vice versa)	Swap byte pairs.
<code>-S</code>	—	specify the segment file size in bytes (default is 1.4 GiB) (minimum is 1.0 MiB, maximum is 7.9 EiB for raw, encase6 and encase7 format and 1.9 GiB for other formats) (not used for files format)	Segment size for the output.
<code>-t</code>	—	specify the target file to export to, use - for stdout (default is export) stdout is only supported for the raw format	Target name. <code>-</code> writes to stdout, which lets the image be piped straight into another tool without staging a full raw copy on disk.
<code>-u</code>	—	unattended mode (disables user interaction)	Unattended, for scripted conversion.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	print version	
<code>-w</code>	—	zero sectors on checksum error (mimic EnCase like behavior)	Zero sectors that cannot be read.
<code>-x</code>	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.

Gotchas

- Exporting to raw discards the metadata and hashes that justified using E01. Keep the original: the raw copy is a working artefact, not the evidence.
- A raw export needs the full uncompressed size in free space. E01 compression routinely hides a 2 TB image inside 700 GB.

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Show the metadata, hashes and acquisition details recorded inside an EWF/E01 image.

Synopsis

```
ewfinfo [ -A codepage ] [ -d date_format ] [ -f format ]
[ -himvVx ] ewf_files
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfinfo /evidence/case01.E01
```

Options

All 9 options parsed from the captured help text; 7 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage, for images with non-ASCII metadata.
-d	—	specify the date format, options: ctime (default), dm (day/month), md (month/day), iso8601	Date format. iso8601 is the unambiguous choice for a report.
-e	—	only show EWF read error information	Read errors recorded at acquisition. Check this before trusting a clean-looking image: unreadable sectors are noted here, not in the filesystem.
-f	—	specify the output format, options: text (default), dfxml	Emit DFXML instead of text, when the output feeds a tool rather than a person.
-h	—	shows this help	

Flag	Argument	What it does	When you would use it
<code>-i</code>	—	only show EWF acquiry information	Acquisition details only — who imaged it, when, with what.
<code>-m</code>	—	only show EWF media information	Media details only — geometry, sector size, media type.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	print version	

Gotchas

- This reads the header only. It reports the hash that was recorded at acquisition; it does not recompute it, so it cannot tell you the image is still intact. Use `ewfverify` for that.

See also

`ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

ewfmount

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: ewfmount

Synopsis

```
ewfmount [ -f format ] [ -X extended_options ] [ -hvV ] image mount_point
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
sudo ewfmount /evidence/case01.E01 /mnt/ewf
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	—	specify the input format, options: raw (default), files (restricted to logical volume files)	
-h	—	shows this help	
-v	—	verbose output to stderr, while ewfmount will remain running in the foreground	
-V	—	print version	
-X	—	extended options to pass to sub system	

Gotchas

TODO: operational traps.

See also

[ewfinfo](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [img_stat](#), [ntfs-3g](#), [vshadowinfo](#)

ewfverify

Kit: Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Recompute an EWF/E01 image's hashes and check them against the values stored at acquisition.

Synopsis

```
ewfverify [ -A codepage ] [ -d digest_type ] [ -f format ]
[ -l log_filename ] [ -p process_buffer_size ]
[ -hqVWwx ] ewf_files
```

Common invocations

```
# from cyberlab 57-forensic-acquisition
ewfverify /evidence/case01.E01
```

Options

All 11 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-A	—	codepage of header section, options: ascii (default), windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252, windows-1253, windows-1254, windows-125	Header codepage.
-d	—	calculate additional digest (hash) types besides md5, options: sha1, sha256	Also verify an additional digest such as sha256, when one was recorded at acquisition.
-f	—	specify the input format, options: raw (default), files (restricted to logical volume files)	Output format.
-h	—	shows this help	
-l	—	logs verification errors and the digest (hash) to the log_filename	Write the verification result to a log file — the artefact worth keeping with the case, not just terminal output.

Flag	Argument	What it does	When you would use it
<code>-p</code>	—	specify the process buffer size (default is the chunk size)	Process buffer size.
<code>-q</code>	—	quiet shows minimal status information	Minimal output, for scripted checks.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	print version	
<code>-w</code>	—	zero sectors on checksum error (mimic EnCase like behavior)	Wipe sectors that could not be read.
<code>-x</code>	—	use the chunk data instead of the buffered read and write functions.	Bypass the buffered read/write path.

Gotchas

- This reads every byte, so it takes as long as the acquisition did. Budget for that rather than discovering it mid-deadline.
- A pass proves the image matches what was recorded **at acquisition**. It says nothing about whether the acquisition captured the device correctly — a disk failing mid-image produces a verifiable image of incomplete data.

See also

`ewfinfo`, `ewfmount`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `ntfs-3g`, `vshadowinfo`

img_stat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect or mount a forensic image container; See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Analyze disk images and recover files from them.

Synopsis

```
img_stat [-tvV] [-i imgtype] [-b dev_sector_size] image
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-t	—	display type only	
-i	imgtype	The format of the image file (use '-i list' for list of supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-v	—	verbose output to stderr	
-V	—	Print version	

Gotchas

TODO: operational traps.

See also

[ewfinfo](#), [ewfmount](#), [ewfverify](#), [ewfexport](#), [affinfo](#), [affcat](#), [ntfs-3g](#), [vshadowinfo](#)

md5sum

Kit: Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** md5sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print or check MD5 (128-bit) checksums.

Synopsis

```
md5sum [OPTION]... [FILE]...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	read in binary mode	
--binary	—	read in binary mode	
-c	—	read checksums from the FILEs and check them	
--check	—	read checksums from the FILEs and check them	
--tag	—	create a BSD-style checksum	
-t	—	read in text mode (default)	
--text	—	read in text mode (default)	
-z	—	end each output line with NUL, not newline, and disable file name escaping	
--zero	—	end each output line with NUL, not newline, and disable file name escaping	
--ignore-missing	—	don't fail or report status for missing files	
--quiet	—	don't print OK for each successfully verified file	

Flag	Argument	What it does	When you would use it
<code>--status</code>	—	don't output anything, status code shows success	
<code>--strict</code>	—	exit non-zero for improperly formatted checksum lines	
<code>-w</code>	—	warn about improperly formatted checksum lines	
<code>--warn</code>	—	warn about improperly formatted checksum lines	
<code>--help</code>	—	display this help and exit	
<code>--version</code>	—	output version information and exit	

Gotchas

TODO: operational traps.

See also

`rahash2`, `ssdeep`, `sha256sum`, `sigtool`

ntfs-3g

Kit: SIFT Workstation **Capability:** Inspect or mount a forensic image container **Version:** ntfs-3g 2022.10.3

Captured: `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

ntfs-3g 2022.10.3 integrated FUSE 28 - Third Generation NTFS Driver

Synopsis

```
ntfs-3g [-o option[,...]] <device|image_file> <mount_point>
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`ewfinfo`, `ewfmount`, `ewfverify`, `ewfexport`, `affinfo`, `affcat`, `img_stat`, `vshadowinfo`

rahash2

Kit: REMnux **Capability:** Verify evidence integrity with hashes **Version:** rahash2 6.1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rahash2 [-BehjkLqRrvX] [-b S] [-a A] [-c H] [-E A] [-s S] [-f O] [-t O] [file] ...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 23 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-a</code>	algo	comma separated list of algorithms (default is 'sha256')	
<code>-b</code>	bsize	specify the size of the block (instead of full file)	
<code>-B</code>	—	show per-block hash	
<code>-c</code>	hash	compare with this hash	
<code>-e</code>	—	swap endian (use little endian)	
<code>-E</code>	algo	encrypt. Use -S to set key and -I to set IV	
<code>-D</code>	algo	decrypt. Use -S to set key and -I to set IV	

Flag	Argument	What it does	When you would use it
<code>-f</code>	from	start hashing at given address	
<code>-i</code>	num	repeat hash N iterations (f.ex: 3DES)	
<code>-I</code>	iv	use give initialization vector (IV) (hexa or s:string)	
<code>-j</code>	—	output in json	
<code>-J</code>	—	new simplified json output (same as -jj)	
<code>-S</code>	seed	use given seed (hexa or s:string) use ^ to prefix (key for -E) (- will slurp the key from stdin, the @ prefix points to a file)	
<code>-k</code>	—	show hash using the openssh's randomkey algorithm	
<code>-q</code>	—	run in quiet mode (-qq to show only the hash)	
<code>-L</code>	—	list muta plugins (combines with -q, used by -a, -E and -D)	
<code>-r</code>	—	output radare commands	
<code>-R</code>	—	output radare2 sdb commands (k file.=...)	
<code>-s</code>	string	hash this string instead of files	
<code>-t</code>	to	stop hashing at given address	
<code>-x</code>	hexstr	hash this hexpair string instead of files	
<code>-X</code>	—	output in hexpairs instead of binary/plain	
<code>-v</code>	—	show version information	

Gotchas

TODO: operational traps.

See also

`ssdeep`, `sha256sum`, `md5sum`, `sigtool`

sha256sum

Kit: Base OS — present on every Linux image **Capability:** Verify evidence integrity with hashes **Version:** sha256sum (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print or check SHA256 (256-bit) checksums.

Synopsis

```
sha256sum [OPTION]... [FILE]...
```

Common invocations

```
# from cyberlab 01-disk-forensics
sha256sum /tmp/recovered.txt
# from cyberlab 04-registry-analysis
sha256sum exercise/SYSTEM_sample.hive
# from cyberlab 05-file-carving
sha256sum /tmp/ak_tcp/*.jpg
# from cyberlab 10-malicious-documents
sha256sum exercise/sample.doc
# from cyberlab 21-yara-authoring
sha256sum exercise/eicar_sample.txt
# from cyberlab 26-metasploit-workflow
sha256sum exercise/scan.xml
# from cyberlab 32-remnux-static-triage
sha256sum exercise/sample.exe
# from cyberlab 33-binwalk-firmware
sha256sum firmware.bin
```

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	read in binary mode	

Flag	Argument	What it does	When you would use it
<code>--binary</code>	—	read in binary mode	
<code>-c</code>	—	read checksums from the FILEs and check them	
<code>--check</code>	—	read checksums from the FILEs and check them	
<code>--tag</code>	—	create a BSD-style checksum	
<code>-t</code>	—	read in text mode (default)	
<code>--text</code>	—	read in text mode (default)	
<code>-z</code>	—	end each output line with NUL, not newline, and disable file name escaping	
<code>--zero</code>	—	end each output line with NUL, not newline, and disable file name escaping	
<code>--ignore-missing</code>	—	don't fail or report status for missing files	
<code>--quiet</code>	—	don't print OK for each successfully verified file	
<code>--status</code>	—	don't output anything, status code shows success	
<code>--strict</code>	—	exit non-zero for improperly formatted checksum lines	
<code>-w</code>	—	warn about improperly formatted checksum lines	
<code>--warn</code>	—	warn about improperly formatted checksum lines	
<code>--help</code>	—	display this help and exit	
<code>--version</code>	—	output version information and exit	

Gotchas

TODO: operational traps.

See also

`rahash2`, `ssdeep`, `md5sum`, `sigtool`

sigtool

Kit: REMnux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.clamav.net>

← [Capability index](#) · [Kit tool list](#)

Purpose

Scan files for malware signatures.

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 35 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--help	—	Show this help	
-h	—	Show this help	
--version	—	Print version number and exit	
-V	—	Print version number and exit	
--quiet	—	Be quiet, output only error messages	
--debug	—	Enable debug messages	
--stdout	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
--hex-dump	—	Convert data from stdin to a hex string and print it on stdout	
--md5	FILES	Generate MD5 checksum from stdin or MD5 sigs for FILES	
--sha1	FILES	Generate SHA1 checksum from stdin or SHA1 sigs for FILES	
--sha256	FILES	Generate SHA256 checksum from stdin or SHA256 sigs for FILES	
--mdb	FILES	Generate .mdb (section hash) sigs	
--imp	FILES	Generate .imp (import table hash) sigs	
--html-normalise	FILE	Create normalised parts of HTML file	
--ascii-normalise	FILE	Create normalised text file from ascii source	

Flag	Argument	What it does	When you would use it
<code>--utf16-decode</code>	FILE	Decode UTF16 encoded files	
<code>--info</code>	FILE	-i FILE Print database information	
<code>--max-bad-sigs</code>	NUMBER	Maximum number of mismatched signatures When building a CVD. Default: 3000	
<code>--flevel</code>	FLEVEL	Specify a custom flevel. Default: 213	
<code>--cvd-version</code>	NUMBER	Specify the version number to use for the build. Default is to use the value+1 from the current CVD in --datadir. If no datafile is found the default behaviour is to prompt for a version number, this	
<code>--no-cdiff</code>	—	Don't generate .cdiff file	
<code>--unsigned</code>	—	Create unsigned database file (.cud)	
<code>--hybrid</code>	—	Create a hybrid (standard and bytecode) database file	
<code>--print-certs</code>	FILE	Print Authenticode details from a PE	
<code>--server</code>	ADDR	ClamAV Signing Service address	
<code>--datadir</code>	DIR	Use DIR as default database directory	
<code>--unpack</code>	FILE	-u FILE Unpack a CVD/CLD file	
<code>--unpack-current</code>	SHORTNAME	Unpack local CVD/CLD into cwd	
<code>--list-sigs</code>	FILE (optional)	-l[FILE] List signature names	
<code>--find-sigs</code>	REGEX	-fREGEX Find signatures matching REGEX	
<code>--decode-sigs</code>	—	Decode signatures from stdin	
<code>--vba</code>	FILE	Extract VBA/Word6 macro code	
<code>--vba-hex</code>	FILE	Extract Word6 macro code with hex values	
<code>--run-cdiff</code>	FILE	-r FILE Execute update script FILE in cwd	
<code>--tempdir</code>	DIRECTORY	Create temporary files in DIRECTORY	

Gotchas

TODO: operational traps.

See also

[rahash2](#), [ssdeep](#), [sha256sum](#), [md5sum](#), [yara](#), [yarak](#), [clamscan](#), [freshclam](#)

ssdeep

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Verify evidence integrity with hashes; Compare or cluster samples; Find hidden data **Version:** 2.14.1 **Captured:** cyberlab-aio via -h on 2026-08-02 [raw Docs:](https://ssdeep-project.github.io/ssdeep/index.html) <https://ssdeep-project.github.io/ssdeep/index.html>

← [Capability index](#) · [Kit tool list](#)

Purpose

Compute Context Triggered Piecewise Hashes (CTPH), also known as fuzzy hashes.

Synopsis

```
ssdeep [-m file] [-k file] [-dpgvrsblcxa] [-t val] [-h|-V] [FILES]
```

Common invocations

```
# from cyberlab 08-malware-static-triage
ssdeep -V
# from cyberlab 08-malware-static-triage
ssdeep exercise/sample.bin
# from cyberlab 32-remnux-static-triage
ssdeep sample.exe > baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m baseline.txt sample_mod.exe
# from cyberlab 32-remnux-static-triage
ssdeep exercise/sample.exe > exercise/baseline.txt
# from cyberlab 32-remnux-static-triage
ssdeep -m exercise/baseline.txt exercise/sample_mod.exe
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[rahash2](#), [sha256sum](#), [md5sum](#), [sigtool](#), [radiff2](#), [binwalk](#)

tshark

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Capture live network traffic; Read and filter packet captures **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Read, filter and dissect network captures from the command line.

Synopsis

```
tshark [options] ...
```

Common invocations

```
# from cyberlab 07-network-pcap
tshark --version | head -n 1
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -q -z io,phs | head -n 30
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e ip.dst -e http.host -e http.request.uri
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'dns.flags.response == 0' -T fields -e dns.qry.name | sort -u
# from cyberlab 07-network-pcap
tshark -r exercise/sample.pcap -Y 'http.request' -T fields -e http.host -e http.request.uri
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -c 20
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "dns.flags.response == 0" -T fields -e dns.qry.name
# from cyberlab 24-wireshark-deep
tshark -r exercise/sample.pcap -Y "http.request" -T fields -e http.host -e http.request.uri
```

Options

All 75 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-i</code>	interface	name or idx of interface (def: first non-loopback)	Capture live from an interface instead of reading a file.
<code>--interface</code>	interface	name or idx of interface (def: first non-loopback)	
<code>-f</code>	capture filter	packet filter in libpcap filter syntax	Apply a <i>capture</i> filter (BPF syntax) before packets are stored.
<code>-s</code>	snaplen	packet snapshot length (def: appropriate maximum)	
<code>--snapshot-length</code>	snaplen	packet snapshot length (def: appropriate maximum)	
<code>-p</code>	—	don't capture in promiscuous mode	
<code>--no-promiscuous-mode</code>	—	don't capture in promiscuous mode	
<code>-I</code>	—	capture in monitor mode, if available	
<code>--monitor-mode</code>	—	capture in monitor mode, if available	
<code>-B</code>	buffer size	size of kernel buffer (def: 2MB)	
<code>--buffer-size</code>	buffer size	size of kernel buffer (def: 2MB)	
<code>-y</code>	link type	link layer type (def: first appropriate)	
<code>--linktype</code>	link type	link layer type (def: first appropriate)	
<code>-D</code>	—	print list of interfaces and exit	
<code>--list-interfaces</code>	—	print list of interfaces and exit	
<code>-L</code>	—	print list of link-layer types of iface and exit	
<code>--list-data-link-types</code>	—	print list of link-layer types of iface and exit	
<code>--list-time-stamp-types</code>	—	print list of timestamp types for iface and exit	
<code>-c</code>	packet count	stop after n packets (def: infinite)	Stop after N packets — a fast way to sample a huge capture.
<code>-a</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM KB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	
<code>--autostop</code>	autostop cond.	duration:NUM - stop after NUM seconds filesize:NUM - stop this file after NUM KB files:NUM - stop after NUM files packets:NUM - stop after NUM packets	
<code>-b</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM KB files:NUM - ringbuffer: replace after NUM files packets:NUM - switch to next file after NUM packets in	

Flag	Argument	What it does	When you would use it
<code>--ring-buffer</code>	ringbuffer opt.	duration:NUM - switch to next file after NUM secs filesize:NUM - switch to next file after NUM KB files:NUM - ringbuffer: replace after NUM files packets:NUM - switch to next file after NUM packets in	
<code>-r</code>	infile	set the filename to read from (or '-' for stdin)	Read from a capture file — the normal forensic mode.
<code>--read-file</code>	infile	set the filename to read from (or '-' for stdin)	
<code>-2</code>	—	perform a two-pass analysis	Two-pass analysis, so fields needing later context resolve.
<code>-M</code>	packet count	perform session auto reset	
<code>-R</code>	read filter	packet Read filter in Wireshark display filter syntax (requires -2)	
<code>--read-filter</code>	read filter	packet Read filter in Wireshark display filter syntax (requires -2)	
<code>-Y</code>	display filter	packet display filter in Wireshark display filter syntax	Apply a <i>display</i> filter (Wireshark syntax) after dissection.
<code>--display-filter</code>	display filter	packet display filter in Wireshark display filter syntax	
<code>-n</code>	—	disable all name resolutions (def: "mNd" enabled, or as set in preferences)	Disable name resolution. Also stops DNS lookups leaking from an evidence host — use it by default on an investigation.
<code>-N</code>	name resolve flags	enable specific name resolution(s): "mnNtdv"	
<code>-H</code>	hosts file	read a list of entries from a hosts file, which will then be written to a capture file. (Implies -W n)	
<code>--enable-protocol</code>	proto_name	enable dissection of proto_name	
<code>--disable-protocol</code>	proto_name	disable dissection of proto_name	
<code>--enable-heuristic</code>	short_name	enable dissection of heuristic protocol	
<code>--disable-heuristic</code>	short_name	disable dissection of heuristic protocol	
<code>-w</code>	outfile -	write packets to a pcapng-format file named "outfile" (or '-' for stdout)	Write the (filtered) packets to a new capture file.
<code>--capture-comment</code>	comment	add a capture file comment, if supported	
<code>-C</code>	config profile	start with specified configuration profile	
<code>-F</code>	output file type	set the output file type, default is pcapng an empty "-F" option will list the file types	

Flag	Argument	What it does	When you would use it
<code>-V</code>	—	add output of packet tree (Packet Details)	
<code>-O</code>	protocols	Only show packet details of these protocols, comma separated	
<code>-P</code>	—	print packet summary even when writing to a file	
<code>--print</code>	—	print packet summary even when writing to a file	
<code>-S</code>	separator	the line separator to print between packets	
<code>-x</code>	—	add output of hex and ASCII dump (Packet Bytes)	Hex and ASCII dump of packet contents.
<code>--hexdump</code>	hexoption	add hexdump, set options for data source and ASCII dump all dump all data sources (-x default) frames dump only frame data source ascii include ASCII dump text (-x default) delimit delimit ASCII dump	
<code>-j</code>	protocolfilter	protocols layers filter if -T ek pdml json selected (e.g. "ip ip.flags text", filter does not expand child nodes, unless child is specified also in the filter)	
<code>-J</code>	protocolfilter	top level protocol filter if -T ek pdml json selected (e.g. "http tcp", filter which expands all child nodes)	
<code>-e</code>	field	field to print if -Tfields selected (e.g. tcp.port, _ws.col.info) this option can be repeated to print multiple fields	With <code>-T fields</code> , name each field to print. Repeatable.
<code>-l</code>	—	flush standard output after each packet	
<code>-q</code>	—	be more quiet on stdout (e.g. when using statistics)	Suppress per-packet output, for use with <code>-z</code> statistics.
<code>-Q</code>	—	only log true errors to stderr (quieter than -q)	
<code>-g</code>	—	enable group read access on the output file(s)	
<code>-W</code>	n	Save extra information in the file, if supported. n = write network address resolution information	
<code>-U</code>	tap_name	PDU export mode, see the man page for details	
<code>-z</code>	statistics	various statistics, see the man page for details	Run a statistics tap (conversations, endpoints, protocol tree).
<code>--export-tls-session-keys</code>	keyfile	export TLS Session Keys to a file named "keyfile"	
<code>--color</code>	—	color output text similarly to the Wireshark GUI, requires a terminal with 24-bit color support Also supplies color attributes to pdml and psml formats (Note that attributes are nonstandard)	
<code>--no-duplicate-keys</code>	—	If -T json is specified, merge duplicate keys in an object into a single key with as value a json array containing all values	
<code>--temp-dir</code>	directory	write temporary files to this directory (default: /tmp)	
<code>--log-level</code>	level	sets the active log level ("critical", "warning", etc.)	

Flag	Argument	What it does	When you would use it
<code>--log-fatal</code>	level	sets level to abort the program ("critical" or "warning")	
<code>--log-domains</code>	[!]list	comma separated list of the active log domains	
<code>--log-debug</code>	[!]list	comma separated list of domains with "debug" level	
<code>--log-noisy</code>	[!]list	comma separated list of domains with "noisy" level	
<code>--log-file</code>	path	file to output messages to (in addition to stderr)	
<code>-h</code>	—	display this help and exit	
<code>--help</code>	—	display this help and exit	
<code>-v</code>	—	display version info and exit	
<code>--version</code>	—	display version info and exit	
<code>-K</code>	keytab	keytab file to use for kerberos decryption	
<code>-G</code>	report	dump one of several available reports and exit default report="fields" use "-G help" for more help	

Gotchas

- `-f` and `-Y` are different languages. `-f` is BPF and applies at capture time; `-Y` is the Wireshark display filter and applies to a file. Passing display syntax to `-f` fails, sometimes silently.
- Name resolution is on by default and will emit DNS queries from the analysis host. Use `-n` when touching evidence.
- `-T fields` prints nothing useful without at least one `-e`.

See also

`dumpcap`, `capinfos`, `ngrep`, `tcpflow`

vshadowinfo

Kit: SIFT Workstation (libyal) **Capability:** Inspect or mount a forensic image container **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: vshadowinfo

Synopsis

```
vshadowinfo [ -o offset ] [ -ahvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
vshadowinfo -V
# from cyberlab 06-windows-artifact-libs
vshadowinfo exercise/volume.raw
```

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	shows allocation information	
<code>-h</code>	—	shows this help	
<code>-o</code>	—	specify the volume offset in bytes	
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	

Gotchas

TODO: operational traps.

See also

ewfinfo , ewfmount , ewfverify , ewfexport , affinfo , affcat , img_stat , ntfs-3g

log2timeline.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - log2timeline version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract timestamped events from evidence into a Plaso storage file, the first half of a super-timeline.

Synopsis

```
log2timeline.py [-h] [--troubles] [-V] [--artifact_definitions PATH]
[--custom_artifact_definitions PATH] [--data PATH]
[--archives TYPES]
[--artifact_filters ARTIFACT_FILTERS]
[--artifact_filters_file PATH]
[--extract_winreg_binary] [--preferred_year YEAR]
```

Common invocations

```
# from cyberlab 03-timeline-analysis
log2timeline.py --version
# from cyberlab 03-timeline-analysis
log2timeline.py --status_view none --storage-file /tmp/case.plaso exercise/sample_fs/
# from cyberlab 49-intrusion-timeline-case
log2timeline.py --storage-file timeline.plaso "$IMAGE"
```

Options

All 100 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Show this help message and exit.	
--help	—	Show this help message and exit.	
--troubles	—	Show troubleshooting information.	
-V	—	Show the version information.	
--version	—	Show the version information.	

Flag	Argument	What it does	When you would use it
<code>--artifact_definitions</code>	PATH	Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows	
<code>--artifact-definitions</code>	PATH	Path to a directory or file containing artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or Windows	
<code>--custom_artifact_definitions</code>	PATH	Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or	
<code>--custom-artifact-definitions</code>	PATH	Path to a directory or file containing custom artifact definitions, which are .yaml files. Artifact definitions can be used to describe and quickly collect data of interest, such as specific files or	
<code>--data</code>	PATH	Path to a directory containing the data files.	
<code>--archives</code>	TYPES	Define a list of archive and storage media image types for which to process embedded file entries, such as TAR (archive.tar) or ZIP (archive.zip). This is a comma separated list where each entry is th	
<code>--artifact_filters</code>	ARTIFACT_FILTERS	Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj	
<code>--artifact-filters</code>	ARTIFACT_FILTERS	Names of forensic artifact definitions, provided on the command command line (comma separated). Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions proj	
<code>--artifact_filters_file</code>	PATH	Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project.	

Flag	Argument	What it does	When you would use it
<code>--artifact-filters_file</code>	PATH	Names of forensic artifact definitions, provided in a file with one artifact name per line. Forensic artifacts are stored in .yaml files that are directly pulled from the artifact definitions project.	
<code>--extract_winreg_binary</code>	—	Extract binary Windows Registry values. WARNING: This can make processing significantly slower.	
<code>--extract-winreg-binary</code>	—	Extract binary Windows Registry values. WARNING: This can make processing significantly slower.	
<code>--preferred_year</code>	YEAR	When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.	
<code>--preferred-year</code>	YEAR	When a format's timestamp does not include a year, e.g. syslog, use this as the initial year instead of attempting auto-detection.	
<code>--skip_compressed_streams</code>	—	Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.	
<code>--skip-compressed-streams</code>	—	Skip processing file content within compressed streams, such as syslog.gz and syslog.bz2.	
<code>-f</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	Use a file filter to limit what gets processed.
<code>--filter-file</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	
<code>--filter_file</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	
<code>--file-filter</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	

Flag	Argument	What it does	When you would use it
<code>--file_filter</code>	FILE_FILTER	List of files to include for targeted collection of files to parse, one line per file path, setup is /path file - where each element can contain either a variable set in the preprocessing stage or a	
<code>--hasher_file_size_limit</code>	SIZE	Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.	
<code>--hasher-file-size-limit</code>	SIZE	Define the maximum file size in bytes that hashers should process. Any larger file will be skipped. A size of 0 represents no limit.	
<code>--hashers</code>	HASHER_LIST	Define a list of hashers to use by the tool. This is a comma separated list where each entry is the name of a hasher, such as "md5,sha256". "all" indicates that all hashers should be enabled. "none" d	Compute hashes during extraction, saving a second pass.
<code>--parsers</code>	PARSER_FILTER_EXPRESSION	Define which presets, parsers and/or plugins to use, or show possible values. The expression is a comma separated string where each element is a preset, parser or plugin name. Each element can be prep	Restrict to specific parsers. A targeted run is minutes instead of hours.
<code>--yara_rules</code>	PATH	Path to a file containing Yara rules definitions.	
<code>--yara-rules</code>	PATH	Path to a file containing Yara rules definitions.	
<code>--partitions</code>	PARTITIONS	Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb	
<code>--partition</code>	PARTITIONS	Define partitions to be processed. A range of partitions can be defined as: "3..5". Multiple partitions can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be comb	
<code>--volumes</code>	VOLUMES	Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:	
<code>--volume</code>	VOLUMES	Define volumes to be processed. A range of volumes can be defined as: "3..5". Multiple volumes can be defined as: "1,3,5" (a list of comma separated values). Ranges and lists can also be combined as:	

Flag	Argument	What it does	When you would use it
<code>--codepage</code>	CODEPAGE	The preferred codepage, which is used for decoding single-byte or multi-byte character extracted strings.	
<code>--language</code>	LANGUAGE_TAG	The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " --language list" to see a list of supported language tags. The en-US (LCID 0x0409) language	
<code>--no_extract_winevt_resources</code>	—	Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.	
<code>--no-extract-winevt-resources</code>	—	Do not extract Windows EventLog resources such as event message template strings. By default Windows EventLog resources will be extracted when a Windows EventLog parser is enabled.	
<code>-z</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	Time zone of the source machine.
<code>--zone</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	
<code>--timezone</code>	TIME_ZONE	preferred time zone of extracted date and time values that are stored without a time zone indicator. The time zone is determined based on the source data where possible otherwise it will default to UT	
<code>--no_vss</code>	—	Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.	
<code>--no-vss</code>	—	Do not scan for Volume Shadow Snapshots (VSS). This means that Volume Shadow Snapshots (VSS) are not processed. WARNING: this option is deprecated use --vss_stores=none instead.	

Flag	Argument	What it does	When you would use it
<code>--vss_only</code>	—	Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.	
<code>--vss-only</code>	—	Do not process the current volume if Volume Shadow Snapshots (VSS) have been selected.	
<code>--vss_stores</code>	VSS_STORES	Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val	Also process Volume Shadow Copies — often where the pre-attack state survives.
<code>--vss-stores</code>	VSS_STORES	Define Volume Shadow Snapshots (VSS) (or stores) that need to be processed. A range of snapshots can be defined as: "3..5". Multiple snapshots can be defined as: "1,3,5" (a list of comma separated val	
<code>-d</code>	—	Enable debug output.	
<code>--debug</code>	—	Enable debug output.	
<code>-q</code>	—	Disable informational output.	
<code>--quiet</code>	—	Disable informational output.	
<code>-u</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--unattended</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--info</code>	—	Print out information about supported plugins and parsers.	
<code>--use_markdown</code>	—	Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"	
<code>--use-markdown</code>	—	Output lists in Markdown format use in combination with "--hashers list", "--parsers list" or "--timezone list"	
<code>--no_dependencies_check</code>	—	Disable the dependencies check.	
<code>--no-dependencies-check</code>	—	Disable the dependencies check.	
<code>--logfile</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	

Flag	Argument	What it does	When you would use it
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "log2timeline-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--status_view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	Change progress display; <code>none</code> for clean logs.
<code>--status-view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	
<code>--status_view_file</code>	PATH	The name of the status view file.	
<code>--status-view-file</code>	PATH	The name of the status view file.	
<code>--status_view_interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--status-view-interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--buffer_size</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--buffer-size</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--bs</code>	BUFFER_SIZE	The buffer size for the output (defaults to 196MiB).	
<code>--queue_size</code>	QUEUE_SIZE	The maximum number of queued items per worker (defaults to 125000)	
<code>--queue-size</code>	QUEUE_SIZE	The maximum number of queued items per worker (defaults to 125000)	
<code>--single_process</code>	—	Indicate that the tool should run in a single process.	
<code>--single-process</code>	—	Indicate that the tool should run in a single process.	

Flag	Argument	What it does	When you would use it
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--temporary_directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--temporary-directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--vfs_back_end</code>	TYPE	The preferred dfVFS back-end: "auto", "fsext", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".	
<code>--vfs-back-end</code>	TYPE	The preferred dfVFS back-end: "auto", "fsext", "fsfat", "fshfs", "fsntfs", "tsk" or "vsgpt".	
<code>--worker_memory_limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker-memory-limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker_timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	

Flag	Argument	What it does	When you would use it
<code>--worker-timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--workers</code>	WORKERS	Number of worker processes. The default is the number of available system CPUs minus one, for the main (foreman) process.	Number of worker processes; tune to the host's cores.
<code>--sigsegv_handler</code>	—	Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari	
<code>--sigsegv-handler</code>	—	Enables the SIGSEGV handler. WARNING this functionality is experimental and will a deadlock worker process if a real segfault is caught, but not signal SIGSEGV. This functionality is therefore primari	
<code>--profilers</code>	PROFILERS_LIST	List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.	
<code>--profiling_directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling-directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling_sample_rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--profiling-sample-rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--storage_file</code>	PATH	The path of the storage file. If not specified, one will be made in the form -.plaso	Where the .plaso output goes.
<code>--storage-file</code>	PATH	The path of the storage file. If not specified, one will be made in the form -.plaso	
<code>--storage_format</code>	FORMAT	Format of the storage file, the default is: sqlite. Supported options: sqlite	

Flag	Argument	What it does	When you would use it
<code>--storage-format</code>	FORMAT	Format of the storage file, the default is: sqlite. Supported options: sqlite	
<code>--task_storage_format</code>	FORMAT	Format for task storage, the default is: sqlite. Supported options: redis, sqlite	
<code>--task-storage-format</code>	FORMAT	Format for task storage, the default is: sqlite. Supported options: redis, sqlite	

Gotchas

- This produces a .plaso database, **not** a timeline you can read. `psort.py` is the second half; running only this looks like nothing happened.
- A full run on a disk image can take hours and tens of GB. Scope with `--parsers` unless you genuinely need everything.

See also

`psort.py` , `pinfo.py`

mactime

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Turn a TSK body file into a human-readable chronological timeline.

Common invocations

```
# from cyberlab 01-disk-forensics
mactime -b /tmp/bodyfile.txt -d > /tmp/timeline.csv
# from cyberlab 03-timeline-analysis
mactime -V
# from cyberlab 03-timeline-analysis
mactime -b /tmp/body.txt -d > /tmp/mactime.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d > exercise/timeline.csv
# from cyberlab 22-sleuthkit-mastery
mactime -b exercise/bodyfile.txt -d | tail
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt > timeline.csv
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' ' $3 ~ /b/' | sort
# from cyberlab 23-plaso-supertimeline
mactime -d -b exercise/bodyfile.txt | awk -F',' ' $3 ~ /b/' | sort | head -n 1
```

Options

All 9 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-b</code>	—	Specifies the body file location, else STDIN is used	Read the body file produced by <code>fls -m</code> — the normal input.
<code>-d</code>	—	Output in comma delimited format	Emit CSV rather than the default text, for spreadsheets or further tooling.
<code>-h</code>	—	Display a header with session information	Produce HTML output for a report.

Flag	Argument	What it does	When you would use it
<code>-y</code>	—	Dates are displayed in ISO 8601 format	Print dates ISO-style (year first), which sorts correctly.
<code>-m</code>	—	Dates have month as number instead of word (does not work with <code>-y</code>)	Print month numerically instead of by name.
<code>-z</code>	—	Specify the timezone the data came from (in the local system format) (does not work with <code>-y</code>)	Time zone of the evidence machine. Getting this wrong shifts the whole timeline.
<code>-g</code>	—	Specifies the group file location, else GIDs are used	Map group IDs to names using a supplied group file.
<code>-p</code>	—	Specifies the password file location, else UIDs are used	Map user IDs to names using a supplied passwd file.
<code>-V</code>	—	Prints the version to STDOUT	

Gotchas

- `mactime` reports in the time zone you give it, not the one embedded in the evidence. An unstated `-z` silently produces a plausible, wrong timeline — state it explicitly every run.
- A date range is passed as a trailing argument (`mactime -b body.txt 2026-01-01..2026-02-01`), not as a flag.

See also

`fls`, `tsk_gettimes`

pininfo.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - pininfo version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Shows information about a Plaso storage file, for example how it was collected, what information was extracted from a source, etc.

Synopsis

```
pininfo.py [-h] [--troubles] [-V] [--logfile FILENAME]
[--process_memory_limit SIZE] [--compare STORAGE_FILE]
[--output_format FORMAT] [--hash TYPE] [--report TYPE]
[--sections SECTIONS_LIST] [-v] [-w OUTPUTFILE]
[PATH]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	Show this help message and exit.	
--help	—	Show this help message and exit.	
--troubles	—	Show troubleshooting information.	
-V	—	Show the version information.	
--version	—	Show the version information.	
--logfile	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	

Flag	Argument	What it does	When you would use it
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "pinfo- YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--compare</code>	STORAGE_FILE	The path of the storage file to compare against.	
<code>--output_format</code>	FORMAT	Format of the output, the default is: text. Supported options: json, markdown, text.	
<code>--output-format</code>	FORMAT	Format of the output, the default is: text. Supported options: json, markdown, text.	
<code>--hash</code>	TYPE	Type of hash to output in file_hashes report. Supported options: md5, sha1, sha256	
<code>--report</code>	TYPE	Report on specific information. Supported options: browser_search, chrome_extension, environment_variables, file_hashes, list, none, windows_services, winevt_providers	
<code>--sections</code>	SECTIONS_LIST	List of sections to output. This is a comma separated list where each entry is the name of a section. Use " --sections list" to list the available sections and " --sections all" to show all available	
<code>-v</code>	—	Print verbose output.	
<code>--verbose</code>	—	Print verbose output.	
<code>-w</code>	OUTPUTFILE	Output filename.	
<code>--write</code>	OUTPUTFILE	Output filename.	

Gotchas

TODO: operational traps.

See also

[log2timeline.py](#), [psort.py](#)

psort.py

Kit: Kali Linux · SIFT Workstation **Capability:** Build a super-timeline from many artifact sources **Version:** plaso - psort version 20260512 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Filter, sort and output the events in a Plaso storage file.

Synopsis

```
psort.py [-h] [--troubles] [-V] [--analysis PLUGIN_LIST]
[--process_memory_limit SIZE]
[--temporary_directory DIRECTORY] [--worker_memory_limit SIZE]
[--worker_timeout MINUTES] [--logfile FILENAME] [-d] [-q] [-u]
[--status_view TYPE] [--status_view_file PATH]
[--status_view_interval SECONDS] [--slice DATE_TIME]
```

Common invocations

```
# from cyberlab 03-timeline-analysis
psort.py --version
# from cyberlab 03-timeline-analysis
psort.py -o l2tcsv -w /tmp/timeline.csv /tmp/case.plaso
# from cyberlab 49-intrusion-timeline-case
psort.py -o l2tcsv -w super_timeline.csv timeline.plaso "date > '2024-01-10 00:00:00' AND date < '2024-01-10 00:00:00'"
```

Options

All 59 options parsed from the captured help text; 4 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Show this help message and exit.	
--help	—	Show this help message and exit.	
--troubles	—	Show troubleshooting information.	
-V	—	Show the version information.	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Show the version information.	
<code>--analysis</code>	PLUGIN_LIST	A comma separated list of analysis plugin names to be loaded or "--analysis list" to see a list of available plugins.	Run analysis plugins (tagging, sessionizing) over events.
<code>--process_memory_limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--process-memory-limit</code>	SIZE	Maximum amount of memory (data segment) a process is allowed to allocate in bytes, where 0 represents no limit. The default limit is 4294967296 (4 GiB). This applies to both the main (foreman) process	
<code>--temporary_directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--temporary-directory</code>	DIRECTORY	Path to the directory that should be used to store temporary files created during processing.	
<code>--worker_memory_limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker-memory-limit</code>	SIZE	Maximum amount of memory (data segment and shared memory) a worker process is allowed to consume in bytes, where 0 represents no limit. The default limit is 2147483648 (2 GiB). If a worker process exc	
<code>--worker_timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--worker-timeout</code>	MINUTES	Number of minutes before a worker process that is not providing status updates is considered inactive. The default timeout is 15.0 minutes. If a worker process exceeds this timeout it is killed by the	
<code>--logfile</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	

Flag	Argument	What it does	When you would use it
<code>--log_file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>--log-file</code>	FILENAME	Path of the file in which to store log messages, by default this file will be named: "psort-YYYYMMDDThhmmss.log.gz". Note that the file will be gzip compressed if the extension is ".gz".	
<code>-d</code>	—	Enable debug output.	
<code>--debug</code>	—	Enable debug output.	
<code>-q</code>	—	Disable informational output.	Quiet, for scripted runs.
<code>--quiet</code>	—	Disable informational output.	
<code>-u</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--unattended</code>	—	Enable unattended mode and do not ask the user for additional input when needed, but terminate with an error instead.	
<code>--status_view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	
<code>--status-view</code>	TYPE	The processing status view mode: "file", "linear", "none" or "window".	
<code>--status_view_file</code>	PATH	The name of the status view file.	
<code>--status-view-file</code>	PATH	The name of the status view file.	
<code>--status_view_interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--status-view-interval</code>	SECONDS	Number of seconds to update the status view.	
<code>--slice</code>	DATE_TIME	Date and time to create a time slice around. This parameter, if defined, will display all events that happened X minutes before and after the defined date, where X is controlled by the --slice_size op	
<code>--slice_size</code>	SLICE_SIZE	Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the --slicer it determines the number of events before and after a	

Flag	Argument	What it does	When you would use it
<code>--slice-size</code>	SLICE_SIZE	Defines the slice size. In the case of a regular time slice it defines the number of minutes the slice size should be. In the case of the <code>--slicer</code> it determines the number of events before and after a	
<code>--slicer</code>	—	Create a time slice around every filter match. This parameter, if defined will save all X events before and after a filter match has been made. X is defined by the <code>--slice_size</code> parameter.	
<code>--data</code>	PATH	Path to a directory containing the data files.	
<code>-a</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--include_all</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--include-all</code>	—	By default the psort removes duplicate entries from the output. This parameter changes that behavior so all events are included.	
<code>--language</code>	LANGUAGE_TAG	The preferred language, which is used for extracting and formatting Windows EventLog message strings. Use " <code>--language list</code> " to see a list of supported language tags. The en-US (LCID 0x0409) language	
<code>--additional_fields</code>	ADDITIONAL_FIELDS	Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit	
<code>--additional-fields</code>	ADDITIONAL_FIELDS	Defines additional fields to be included in the output besides the default fields. Multiple additional field names can be defined as a list of comma separated values. Output formats that support addit	
<code>--custom_fields</code>	CUSTOM_FIELDS	Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.	
<code>--custom-fields</code>	CUSTOM_FIELDS	Defines custom fields to be included in the output besides the default fields. A custom field is defined as "name:value". Multiple custom field names can be defined as list of comma separated values.	

Flag	Argument	What it does	When you would use it
<code>--custom_formatter_definitions</code>	PATH	Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte	
<code>--custom-formatter-definitions</code>	PATH	Path to a file containing custom event formatter definitions, which is a .yaml file. Custom event formatter definitions can be used to customize event messages and override the built-in event formatte	
<code>--dynamic_time</code>	—	Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic	
<code>--dynamic-time</code>	—	Indicate that the output should use dynamic time. Output formats that support dynamic time are: dynamic	
<code>--output_time_zone</code>	TIME_ZONE	time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam	
<code>--output-time-zone</code>	TIME_ZONE	time zone of date and time values written to the output, if supported by the output format. Use "list" to see a list of available time zones. Output formats that support an output time zone are: dynam	
<code>-o</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	Output format — <code>l2tcsv</code> , <code>dynamic</code> , <code>json</code> , or a timeline tool.
<code>--output_format</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	
<code>--output-format</code>	FORMAT	The output format. Use "-o list" to see a list of available output formats.	
<code>-w</code>	OUTPUT_FILE	Output filename.	Write output to a file rather than stdout.
<code>--write</code>	OUTPUT_FILE	Output filename.	
<code>--fields</code>	FIELDS	Defines which fields should be included in the output.	
<code>--profilers</code>	PROFILERS_LIST	List of profilers to use by the tool. This is a comma separated list where each entry is the name of a profiler. Use "--profilers list" to list the available profilers.	

Flag	Argument	What it does	When you would use it
<code>--profiling_directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling-directory</code>	DIRECTORY	Path to the directory that should be used to store the profiling sample files. By default the sample files are stored in the current working directory.	
<code>--profiling_sample_rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	
<code>--profiling-sample-rate</code>	SAMPLE_RATE	Profiling sample rate (defaults to a sample every 1000 files).	

Gotchas

- The date filter is a positional argument, not a flag. Filtering to the incident window is what makes a multi-million-event timeline usable.
- Output ordering follows the storage file, so always sort or filter explicitly rather than assuming chronology.

See also

`log2timeline.py`, `pinfo.py`

tsk_gettimes

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Build a filesystem MAC-time timeline **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.sleuthkit.org/sleuthkit>

[← Capability index](#) · [Kit tool list](#)

Purpose

Analyze disk images and recover files from them.

Synopsis

```
tsk_gettimes [-vVm] [-i imgtype] [-b dev_sector_size] [-z zone] [-s seconds] image [image]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-i	imgtype	The format of the image file (use '-i list' for supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-m	—	Calculate MD5 hash in output (slow)	
-v	—	verbose output to stderr	
-V	—	Print version	
-z	—	Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)	
-s	seconds	Time skew of original machine (in seconds) (only useful with -l & -m)	

Gotchas

TODO: operational traps.

See also

[fls](#), [mactime](#)

cyberchef

Kit: REMnux · FLARE-VM · Security Onion **Capability:** Decode, decrypt or transform encoded data **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://github.com/gchq/CyberChef/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Decode and otherwise analyze data using this browser app.

Common invocations

```
# from cyberlab 09-deobfuscation
cyberchef &
# from cyberlab 25-cyberchef-recipes
cyberchef --help | head -n 5
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt 2>/dev/null | cyberchef --recipe "XOR"
# from cyberlab 25-cyberchef-recipes
cyberchef --recipe "From Base64" --input encoded_payload.ps1.txt | cyberchef --recipe "XOR({'option': '"
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[base64dump.py](#), [rax2](#), [xxd](#), [openssl](#), [numbers-to-string.py](#)

hashcat

Kit: Kali Linux **Capability:** Crack passwords and hashes **Version:** v6.2.6 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

hashcat (v6.2.6) starting in help mode

Synopsis

```
hashcat [options]... hash|hashfile|hccapxfile [dictionary|mask|directory]...
```

Common invocations

```
# from cyberlab 11-offensive-kali
hashcat --version
# from cyberlab 11-offensive-kali
hashcat -b -m 0 2>&1 | head -n 8
# from cyberlab 11-offensive-kali
hashcat -m 0 -a 0 exercise/lab_hash.txt /usr/share/wordlists/rockyou.txt
# from cyberlab 11-offensive-kali
hashcat -m 0 --show exercise/lab_hash.txt
# from cyberlab 40-password-cracking
hashcat prints a version string like "v6.2.6".
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 exercise/hash.txt exercise/wordlist.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 --show exercise/hash.txt --potfile-path exercise/hc.pot
# from cyberlab 40-password-cracking
hashcat -m 0 -a 0 -r /usr/share/hashcat/rules/best64.rule exercise/hash.txt exercise/wordlist.txt --po
```

Options

All 143 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-m</code>	—	Num Hash-type, references below (otherwise autodetect) -m 1000	
<code>--hash-type</code>	—	Num Hash-type, references below (otherwise autodetect) -m 1000	
<code>-a</code>	—	Num Attack-mode, see references below -a 3	
<code>--attack-mode</code>	—	Num Attack-mode, see references below -a 3	
<code>-V</code>	—	Print version	
<code>--version</code>	—	Print version	
<code>-h</code>	—	Print help	
<code>--help</code>	—	Print help	
<code>--quiet</code>	—	Suppress output	
<code>--hex-charset</code>	—	Assume charset is given in hex	
<code>--hex-salt</code>	—	Assume salt is given in hex	
<code>--hex-wordlist</code>	—	Assume words in wordlist are given in hex	
<code>--force</code>	—	Ignore warnings	
<code>--status</code>	—	Enable automatic update of the status screen	
<code>--status-json</code>	—	Enable JSON format for status output	
<code>--status-timer</code>	—	Num Sets seconds between status screen updates to X --status-timer=1	
<code>--stdin-timeout-abort</code>	—	Num Abort if there is no input from stdin for X seconds --stdin-timeout-abort=300	
<code>--machine-readable</code>	—	Display the status view in a machine-readable format	
<code>--keep-guessing</code>	—	Keep guessing the hash after it has been cracked	
<code>--self-test-disable</code>	—	Disable self-test functionality on startup	
<code>--loopback</code>	—	Add new plains to induct directory	
<code>--markov-hcstat2</code>	—	File Specify hcstat2 file to use --markov-hcstat2=my.hcstat2	
<code>--markov-disable</code>	—	Disables markov-chains, emulates classic brute-force	
<code>--markov-classic</code>	—	Enables classic markov-chains, no per-position	
<code>--markov-inverse</code>	—	Enables inverse markov-chains, no per-position	
<code>-t</code>	—	Num Threshold X when to stop accepting new markov-chains -t 50	
<code>--markov-threshold</code>	—	Num Threshold X when to stop accepting new markov-chains -t 50	
<code>--runtime</code>	—	Num Abort session after X seconds of runtime --runtime=10	
<code>--session</code>	—	Str Define specific session name --session=mysession	
<code>--restore</code>	—	Restore session from --session	
<code>--restore-disable</code>	—	Do not write restore file	

Flag	Argument	What it does	When you would use it
<code>--restore-file-path</code>	—	File Specific path to restore file <code>--restore-file-path=x.restore</code>	
<code>-o</code>	—	File Define outfile for recovered hash <code>-o outfile.txt</code>	
<code>--outfile</code>	—	File Define outfile for recovered hash <code>-o outfile.txt</code>	
<code>--outfile-format</code>	—	Str Outfile format to use, separated with commas <code>--outfile-format=1,3</code>	
<code>--outfile-autohex-disable</code>	—	Disable the use of \$HEX[] in output plains	
<code>--outfile-check-timer</code>	—	Num Sets seconds between outfile checks to X <code>--outfile-check-timer=30</code>	
<code>-p</code>	—	Char Separator char for hashlists and outfile <code>-p :</code>	
<code>--separator</code>	—	Char Separator char for hashlists and outfile <code>-p :</code>	
<code>--stdout</code>	—	Do not crack a hash, instead print candidates only	
<code>--show</code>	—	Compare hashlist with potfile; show cracked hashes	
<code>--left</code>	—	Compare hashlist with potfile; show uncracked hashes	
<code>--username</code>	—	Enable ignoring of usernames in hashfile	
<code>--remove</code>	—	Enable removal of hashes once they are cracked	
<code>--remove-timer</code>	—	Num Update input hash file each X seconds <code>--remove-timer=30</code>	
<code>--potfile-disable</code>	—	Do not write potfile	
<code>--potfile-path</code>	—	File Specific path to potfile <code>--potfile-path=my.pot</code>	
<code>--encoding-from</code>	—	Code Force internal wordlist encoding from X <code>--encoding-from=iso-8859-15</code>	
<code>--encoding-to</code>	—	Code Force internal wordlist encoding to X <code>--encoding-to=utf-32le</code>	
<code>--debug-mode</code>	—	Num Defines the debug mode (hybrid only by using rules) <code>--debug-mode=4</code>	
<code>--debug-file</code>	—	File Output file for debugging rules <code>--debug-file=good.log</code>	
<code>--induction-dir</code>	—	Dir Specify the induction directory to use for loopback <code>--induction=inducts</code>	
<code>--outfile-check-dir</code>	—	Dir Specify the outfile directory to monitor for plains <code>--outfile-check-dir=x</code>	
<code>--logfile-disable</code>	—	Disable the logfile	
<code>--hccapx-message-pair</code>	—	Num Load only message pairs from hccapx matching X <code>--hccapx-message-pair=2</code>	
<code>--nonce-error-corrections</code>	—	Num The BF size range to replace AP's nonce last bytes <code>--nonce-error-corrections=16</code>	
<code>--keyboard-layout-mapping</code>	—	File Keyboard layout mapping table for special hash-modes <code>--keyb=german.hckmap</code>	

Flag	Argument	What it does	When you would use it
<code>--truecrypt-keyfiles</code>	—	File Keyfiles to use, separated with commas <code>--truecrypt-keyf=x.png</code>	
<code>--veracrypt-keyfiles</code>	—	File Keyfiles to use, separated with commas <code>--veracrypt-keyf=x.txt</code>	
<code>--veracrypt-pim-start</code>	—	Num VeraCrypt personal iterations multiplier start <code>--veracrypt-pim-start=450</code>	
<code>--veracrypt-pim-stop</code>	—	Num VeraCrypt personal iterations multiplier stop <code>--veracrypt-pim-stop=500</code>	
<code>-b</code>	—	Run benchmark of selected hash-modes	
<code>--benchmark</code>	—	Run benchmark of selected hash-modes	
<code>--benchmark-all</code>	—	Run benchmark of all hash-modes (requires -b)	
<code>--speed-only</code>	—	Return expected speed of the attack, then quit	
<code>--progress-only</code>	—	Return ideal progress step size and time to process	
<code>-c</code>	—	Num Sets size in MB to cache from the wordfile to X <code>-c 32</code>	
<code>--segment-size</code>	—	Num Sets size in MB to cache from the wordfile to X <code>-c 32</code>	
<code>--bitmap-min</code>	—	Num Sets minimum bits allowed for bitmaps to X <code>--bitmap-min=24</code>	
<code>--bitmap-max</code>	—	Num Sets maximum bits allowed for bitmaps to X <code>--bitmap-max=24</code>	
<code>--cpu-affinity</code>	—	Str Locks to CPU devices, separated with commas <code>--cpu-affinity=1,2,3</code>	
<code>--hook-threads</code>	—	Num Sets number of threads for a hook (per compute unit) <code>--hook-threads=8</code>	
<code>--hash-info</code>	—	Show information for each hash-mode	
<code>--example-hashes</code>	—	Alias of <code>--hash-info</code>	
<code>--backend-ignore-cuda</code>	—	Do not try to open CUDA interface on startup	
<code>--backend-ignore-hip</code>	—	Do not try to open HIP interface on startup	
<code>--backend-ignore-metal</code>	—	Do not try to open Metal interface on startup	
<code>--backend-ignore-opengl</code>	—	Do not try to open OpenGL interface on startup	
<code>-I</code>	—	Show system/environment/backend API info <code>-I</code> or <code>-II</code>	
<code>--backend-info</code>	—	Show system/environment/backend API info <code>-I</code> or <code>-II</code>	
<code>-d</code>	—	Str Backend devices to use, separated with commas <code>-d 1</code>	
<code>--backend-devices</code>	—	Str Backend devices to use, separated with commas <code>-d 1</code>	
<code>-D</code>	—	Str OpenCL device-types to use, separated with commas <code>-D 1</code>	

Flag	Argument	What it does	When you would use it
<code>--opencl-device-types</code>	—	Str OpenCL device-types to use, separated with commas -D 1	
<code>-O</code>	—	Enable optimized kernels (limits password length)	
<code>--optimized-kernel-enable</code>	—	Enable optimized kernels (limits password length)	
<code>-M</code>	—	Disable multiply kernel-accel with processor count	
<code>--multiply-accel-disable</code>	—	Disable multiply kernel-accel with processor count	
<code>-w</code>	—	Num Enable a specific workload profile, see pool below -w 3	
<code>--workload-profile</code>	—	Num Enable a specific workload profile, see pool below -w 3	
<code>-n</code>	—	Num Manual workload tuning, set outerloop step size to X -n 64	
<code>--kernel-accel</code>	—	Num Manual workload tuning, set outerloop step size to X -n 64	
<code>-u</code>	—	Num Manual workload tuning, set innerloop step size to X -u 256	
<code>--kernel-loops</code>	—	Num Manual workload tuning, set innerloop step size to X -u 256	
<code>-T</code>	—	Num Manual workload tuning, set thread count to X -T 64	
<code>--kernel-threads</code>	—	Num Manual workload tuning, set thread count to X -T 64	
<code>--backend-vector-width</code>	—	Num Manually override backend vector-width to X --backend-vector=4	
<code>--spin-damp</code>	—	Num Use CPU for device synchronization, in percent --spin-damp=10	
<code>--hwmon-disable</code>	—	Disable temperature and fanspeed reads and triggers	
<code>--hwmon-temp-abort</code>	—	Num Abort if temperature reaches X degrees Celsius --hwmon-temp-abort=100	
<code>--scrypt-tmto</code>	—	Num Manually override TMTO value for scrypt to X --scrypt-tmto=3	
<code>-s</code>	—	Num Skip X words from the start -s 1000000	
<code>--skip</code>	—	Num Skip X words from the start -s 1000000	
<code>-l</code>	—	Num Limit X words from the start + skipped words -l 1000000	
<code>--limit</code>	—	Num Limit X words from the start + skipped words -l 1000000	
<code>--keyspace</code>	—	Show keyspace base:mod values and quit	
<code>-j</code>	—	Rule Single rule applied to each word from left wordlist -j 'c'	
<code>--rule-left</code>	—	Rule Single rule applied to each word from left wordlist -j 'c'	
<code>-k</code>	—	Rule Single rule applied to each word from right wordlist -k '^_'	
<code>--rule-right</code>	—	Rule Single rule applied to each word from right wordlist -k '^_'	
<code>-r</code>	—	File Multiple rules applied to each word from wordlists -r rules/best64.rule	

Flag	Argument	What it does	When you would use it
<code>--rules-file</code>	—	File Multiple rules applied to each word from wordlists -r rules/best64.rule	
<code>-g</code>	—	Num Generate X random rules -g 10000	
<code>--generate-rules</code>	—	Num Generate X random rules -g 10000	
<code>--generate-rules-func-min</code>	—	Num Force min X functions per rule	
<code>--generate-rules-func-max</code>	—	Num Force max X functions per rule	
<code>--generate-rules-func-sel</code>	—	Str Pool of rule operators valid for random rule engine --generate-rules-func-sel=ioTlc	
<code>--generate-rules-seed</code>	—	Num Force RNG seed set to X	
<code>-1</code>	—	CS User-defined charset ?1 -1 ?!d?u	
<code>--custom-charset1</code>	—	CS User-defined charset ?1 -1 ?!d?u	
<code>-2</code>	—	CS User-defined charset ?2 -2 ?!d?s	
<code>--custom-charset2</code>	—	CS User-defined charset ?2 -2 ?!d?s	
<code>-3</code>	—	CS User-defined charset ?3	
<code>--custom-charset3</code>	—	CS User-defined charset ?3	
<code>-4</code>	—	CS User-defined charset ?4	
<code>--custom-charset4</code>	—	CS User-defined charset ?4	
<code>--identify</code>	—	Shows all supported algorithms for input hashes --identify my.hash	
<code>-i</code>	—	Enable mask increment mode	
<code>--increment</code>	—	Enable mask increment mode	
<code>--increment-min</code>	—	Num Start mask incrementing at X --increment-min=4	
<code>--increment-max</code>	—	Num Stop mask incrementing at X --increment-max=8	
<code>-S</code>	—	Enable slower (but advanced) candidate generators	
<code>--slow-candidates</code>	—	Enable slower (but advanced) candidate generators	
<code>--brain-server</code>	—	Enable brain server	
<code>--brain-server-timer</code>	—	Num Update the brain server dump each X seconds (min:60) --brain-server-timer=300	
<code>-z</code>	—	Enable brain client, activates -S	
<code>--brain-client</code>	—	Enable brain client, activates -S	
<code>--brain-client-features</code>	—	Num Define brain client features, see below --brain-client-features=3	
<code>--brain-host</code>	—	Str Brain server host (IP or domain) --brain-host=127.0.0.1	

Flag	Argument	What it does	When you would use it
<code>--brain-port</code>	—	Port Brain server port <code>--brain-port=13743</code>	
<code>--brain-password</code>	—	Str Brain server authentication password <code>--brain-password=bZfhCvGUSjRq</code>	
<code>--brain-session</code>	—	Hex Overrides automatically calculated brain session <code>--brain-session=0x2ae611db</code>	
<code>--brain-session-whitelist</code>	—	Hex Allow given sessions only, separated with commas <code>--brain-session-whitelist=0x2ae611db</code>	

Gotchas

TODO: operational traps.

See also

`john`, `hydra`

hydra

Kit: Kali Linux · SIFT Workstation **Capability:** Crack passwords and hashes **Version:** Hydra v9.4 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TODO: one-line imperative purpose.

Common invocations

```
# from cyberlab 11-offensive-kali
hydra -h 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
hydra -U ssh 2>&1 | head -n 5
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[hashcat](#) , [john](#)

john

Kit: Kali Linux **Capability:** Crack passwords and hashes **Version:** stat: version: No such file or directory **Captured:** cyberlab-aio via (no args) on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

John the Ripper password cracker, version 1.9.0

Synopsis

```
john [OPTIONS] [PASSWORD-FILES]
```

Common invocations

```
# from cyberlab 11-offensive-kali
john --list=build-info 2>&1 | head -n 1
# from cyberlab 11-offensive-kali
john --format=Raw-MD5 --wordlist=/usr/share/wordlists/rockyou.txt exercise/lab_hash.txt
# from cyberlab 11-offensive-kali
john --show --format=Raw-MD5 exercise/lab_hash.txt
# from cyberlab 40-password-cracking
john --version 2>&1 | head -n 1
# from cyberlab 40-password-cracking
john --list=formats | tr ',' '\n' | grep -i -m 5 md5
# from cyberlab 40-password-cracking
john --format=raw-md5 --wordlist=exercise/wordlist.txt exercise/hash.txt
# from cyberlab 40-password-cracking
john --format=raw-md5 --show exercise/hash.txt
```

Options

All 21 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--single</code>	—	"single crack" mode	
<code>--wordlist</code>	FILE	wordlist mode, read words from FILE or stdin	
<code>--stdin</code>	—	wordlist mode, read words from FILE or stdin	

Flag	Argument	What it does	When you would use it
<code>--rules</code>	—	enable word mangling rules for wordlist mode	
<code>--incremental</code>	MODE (optional)	"incremental" mode [using section MODE]	
<code>--external</code>	MODE	external mode or word filter	
<code>--stdout</code>	LENGTH (optional)	just output candidate passwords [cut at LENGTH]	
<code>--restore</code>	NAME (optional)	restore an interrupted session [called NAME]	
<code>--session</code>	NAME	give a new session the NAME	
<code>--status</code>	NAME (optional)	print status of a session [called NAME]	
<code>--make-charset</code>	FILE	make a charset, FILE will be overwritten	
<code>--show</code>	—	show cracked passwords	
<code>--test</code>	TIME (optional)	run tests and benchmarks for TIME seconds each	
<code>--users</code>	[-]LOGIN UID[,...]	[do not] load this (these) user(s) only	
<code>--groups</code>	[-]GID[,...]	load users [not] of this (these) group(s) only	
<code>--shells</code>	[-]SHELL[,...]	load users with[out] this (these) shell(s) only	
<code>--salts</code>	[-]N	load salts with[out] at least N passwords only	
<code>--save-memory</code>	LEVEL	enable memory saving, at LEVEL 1..3	
<code>--node</code>	MIN[-MAX]	this node's number range out of TOTAL count	
<code>--fork</code>	N	fork N processes	
<code>--format</code>	NAME	force hash type NAME: descrypt/bsdicrypt/md5crypt/bcrypt/LM/AFS/tripcode/dummy/crypt	

Gotchas

TODO: operational traps.

See also

[hashcat](#) , [hydra](#)

openssl

Kit: Base OS — present on every Linux image **Capability:** Decode, decrypt or transform encoded data **Version:** Invalid command '--version'; type "help" for a list. **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Standard commands

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[cyberchef](#), [base64dump.py](#), [rax2](#), [xxd](#), [numbers-to-string.py](#)

rax2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Decode, decrypt or transform encoded data **Version:** rax2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rax2 [-h|...] [- | expr ...] # convert between numeric bases
int      ->  hex          ; rax2 10
hex      ->  int          ; rax2 0xa
-int     ->  hex          ; rax2 -77
-hex     ->  int          ; rax2 0xffffffffb3
int      ->  bin          ; rax2 b30
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 30 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	show ascii table ; rax2 -a	
-b	base	output in ; rax2 -b 10 0x46	
-c	—	output in C string ; rax2 -c 0x1234 # \x34\x12\x00\x00	
-C	—	dump as C byte array ; rax2 -C < bytes	
-d	—	force integer ; rax2 -d 3 -> 3 instead of 0x3	
-e	—	swap endianness ; rax2 -e 0x33	
-D	—	base64 decode ; rax2 -D "aGVsbG8="	
-E	—	base64 encode ; rax2 -E "hello"	
-f	—	floating point ; rax2 -f 6.3+2.1	

Flag	Argument	What it does	When you would use it
<code>-F</code>	—	stdin slurp code hex ; rax2 -F < shellcode.[c/py/js]	
<code>-h</code>	—	help ; rax2 -h	
<code>-H</code>	—	hash string ; rax2 -H linux osx	
<code>-i</code>	—	IP address <-> LONG ; rax2 -i 3530468537	
<code>-j</code>	—	json format output ; rax2 -j 0x1234 # same as r2 -c '?j 0x1234'	
<code>-k</code>	—	keep base ; rax2 -k 33+3 -> 36	
<code>-K</code>	—	randomart ; rax2 -K 0x34 1020304050	
<code>-n</code>	—	newline ; append newline to output (for -E/-D/-r/..)	
<code>-o</code>	—	octalstr -> raw ; rax2 -o \162 \62 # r2	
<code>-q</code>	—	quiet mode ; rax2 -qC < /etc/hosts # be quiet	
<code>-r</code>	—	r2 style output ; rax2 -r 0x1234 # same as r2 -c '? 0x1234'	
<code>-s</code>	—	hexstr -> raw ; rax2 -s 43 4a 50	
<code>-S</code>	—	raw -> hexstr ; rax2 -S < /bin/lS > lS.hex	
<code>-t</code>	—	tstamp -> str ; rax2 -t 1234567890	
<code>-u</code>	—	units ; rax2 -u 389289238 # 317.0M	
<code>-v</code>	—	version ; rax2 -v	
<code>-w</code>	—	signed word ; rax2 -w 0xffff 0xffff_ffff '0xff&0xffff'	
<code>-x</code>	—	output in hexpairs ; rax2 -x 0x1234 # 34120000	
<code>-X</code>	—	bin -> hex(bignum) ; rax2 -X 111111111 # 0x1ff	
<code>-z</code>	—	str -> bin ; rax2 -z hello	
<code>-Z</code>	—	bin -> str ; rax2 -Z 01000101 01110110	

Gotchas

TODO: operational traps.

See also

[cyberchef](#) , [base64dump.py](#) , [xxd](#) , [openssl](#) , [numbers-to-string.py](#)

binwalk

Kit: REMnux · Kali Linux **Capability:** Carve files out of unstructured data; Detect and reverse packing; Find hidden data **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/ReFirmLabs/binwalk>

← [Capability index](#) · [Kit tool list](#)

Purpose

Find and extract embedded files and filesystems inside a binary blob or firmware image.

Synopsis

```
binwalk [OPTIONS] [FILE1] [FILE2] [FILE3] ...
```

Common invocations

```
# from cyberlab 33-binwalk-firmware
binwalk --help | head -n 3
# from cyberlab 33-binwalk-firmware
binwalk firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -e firmware.bin
# from cyberlab 33-binwalk-firmware
binwalk -E firmware.bin
```

Options

All 102 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-B	—	Scan target file(s) for common file signatures	
--signature	—	Scan target file(s) for common file signatures	
-R	str	Scan target file(s) for the specified sequence of bytes	
--raw	str	Scan target file(s) for the specified sequence of bytes	
-A	—	Scan target file(s) for common executable opcode signatures	Scan for executable opcodes to identify architecture.

Flag	Argument	What it does	When you would use it
<code>--opcodes</code>	—	Scan target file(s) for common executable opcode signatures	
<code>-m</code>	file	Specify a custom magic file to use	
<code>--magic</code>	file	Specify a custom magic file to use	
<code>-b</code>	—	Disable smart signature keywords	
<code>--dumb</code>	—	Disable smart signature keywords	
<code>-I</code>	—	Show results marked as invalid	
<code>--invalid</code>	—	Show results marked as invalid	
<code>-x</code>	str	Exclude results that match	Exclude signatures matching this string, to cut false hits.
<code>--exclude</code>	str	Exclude results that match	
<code>-y</code>	str	Only show results that match	Only report signatures matching this string.
<code>--include</code>	str	Only show results that match	
<code>-e</code>	—	Automatically extract known file types	Extract what is found, rather than only listing it.
<code>--extract</code>	—	Automatically extract known file types	
<code>-D</code>	type[:ext[:cmd]]	Extract signatures (regular expression), give the files an extension of , and execute	
<code>--dd</code>	type[:ext[:cmd]]	Extract signatures (regular expression), give the files an extension of , and execute	
<code>-M</code>	—	Recursively scan extracted files	Recurse into extracted files (matryoshka) — for nested firmware.
<code>--matryoshka</code>	—	Recursively scan extracted files	
<code>-d</code>	int	Limit matryoshka recursion depth (default: 8 levels deep)	Limit recursion depth; unbounded <code>-M</code> can explode.
<code>--depth</code>	int	Limit matryoshka recursion depth (default: 8 levels deep)	
<code>-C</code>	str	Extract files/folders to a custom directory (default: current working directory)	Choose the output directory for extractions.
<code>--directory</code>	str	Extract files/folders to a custom directory (default: current working directory)	
<code>-j</code>	int	Limit the size of each extracted file	
<code>--size</code>	int	Limit the size of each extracted file	
<code>-n</code>	int	Limit the number of extracted files	
<code>--count</code>	int	Limit the number of extracted files	
<code>-0</code>	str	Execute external extraction utilities with the specified user's privileges	

Flag	Argument	What it does	When you would use it
<code>--run-as</code>	str	Execute external extraction utilities with the specified user's privileges	
<code>-1</code>	—	Do not sanitize extracted symlinks that point outside the extraction directory (dangerous)	
<code>--preserve-symlinks</code>	—	Do not sanitize extracted symlinks that point outside the extraction directory (dangerous)	
<code>-r</code>	—	Delete carved files after extraction	
<code>--rm</code>	—	Delete carved files after extraction	
<code>-z</code>	—	Carve data from files, but don't execute extraction utilities	
<code>--carve</code>	—	Carve data from files, but don't execute extraction utilities	
<code>-V</code>	—	Extract into sub-directories named by the offset	
<code>--subdirs</code>	—	Extract into sub-directories named by the offset	
<code>-E</code>	—	Calculate file entropy	Entropy analysis — the fast way to spot encryption or compression.
<code>--entropy</code>	—	Calculate file entropy	
<code>-F</code>	—	Use faster, but less detailed, entropy analysis	
<code>--fast</code>	—	Use faster, but less detailed, entropy analysis	
<code>-J</code>	—	Save plot as a PNG	
<code>--save</code>	—	Save plot as a PNG	
<code>-Q</code>	—	Omit the legend from the entropy plot graph	
<code>--nlegend</code>	—	Omit the legend from the entropy plot graph	
<code>-N</code>	—	Do not generate an entropy plot graph	
<code>--nplot</code>	—	Do not generate an entropy plot graph	
<code>-H</code>	float	Set the rising edge entropy trigger threshold (default: 0.95)	
<code>--high</code>	float	Set the rising edge entropy trigger threshold (default: 0.95)	
<code>-L</code>	float	Set the falling edge entropy trigger threshold (default: 0.85)	
<code>--low</code>	float	Set the falling edge entropy trigger threshold (default: 0.85)	
<code>-W</code>	—	Perform a hexdump / diff of a file or files	
<code>--hexdump</code>	—	Perform a hexdump / diff of a file or files	
<code>-G</code>	—	Only show lines containing bytes that are the same among all files	

Flag	Argument	What it does	When you would use it
<code>--green</code>	—	Only show lines containing bytes that are the same among all files	
<code>-i</code>	—	Only show lines containing bytes that are different among all files	
<code>--red</code>	—	Only show lines containing bytes that are different among all files	
<code>-U</code>	—	Only show lines containing bytes that are different among some files	
<code>--blue</code>	—	Only show lines containing bytes that are different among some files	
<code>-u</code>	—	Only display lines that are the same between all files	
<code>--similar</code>	—	Only display lines that are the same between all files	
<code>-w</code>	—	Diff all files, but only display a hex dump of the first file	
<code>--terse</code>	—	Diff all files, but only display a hex dump of the first file	
<code>-X</code>	—	Scan for raw deflate compression streams	
<code>--deflate</code>	—	Scan for raw deflate compression streams	
<code>-Z</code>	—	Scan for raw LZMA compression streams	
<code>--lzma</code>	—	Scan for raw LZMA compression streams	
<code>-P</code>	—	Perform a superficial, but faster, scan	
<code>--partial</code>	—	Perform a superficial, but faster, scan	
<code>-S</code>	—	Stop after the first result	
<code>--stop</code>	—	Stop after the first result	
<code>-l</code>	int	Number of bytes to scan	
<code>--length</code>	int	Number of bytes to scan	
<code>-o</code>	int	Start scan at this file offset	
<code>--offset</code>	int	Start scan at this file offset	
<code>-O</code>	int	Add a base address to all printed offsets	
<code>--base</code>	int	Add a base address to all printed offsets	
<code>-K</code>	int	Set file block size	
<code>--block</code>	int	Set file block size	
<code>-g</code>	int	Reverse every n bytes before scanning	
<code>--swap</code>	int	Reverse every n bytes before scanning	
<code>-f</code>	file	Log results to file	

Flag	Argument	What it does	When you would use it
<code>--log</code>	file	Log results to file	
<code>-c</code>	—	Log results to file in CSV format	
<code>--csv</code>	—	Log results to file in CSV format	
<code>-t</code>	—	Format output to fit the terminal window	
<code>--term</code>	—	Format output to fit the terminal window	
<code>-q</code>	—	Suppress output to stdout	
<code>--quiet</code>	—	Suppress output to stdout	
<code>-v</code>	—	Enable verbose output	
<code>--verbose</code>	—	Enable verbose output	
<code>-h</code>	—	Show help output	
<code>--help</code>	—	Show help output	
<code>-a</code>	str	Only scan files whose names match this regex	
<code>--finclude</code>	str	Only scan files whose names match this regex	
<code>-p</code>	str	Do not scan files whose names match this regex	
<code>--fexclude</code>	str	Do not scan files whose names match this regex	
<code>-s</code>	int	Enable the status server on the specified port	
<code>--status</code>	int	Enable the status server on the specified port	

Gotchas

- **Always timeout-guard binwalk in a harness.** Signature scanning on a large or synthetic image can run effectively forever; this project has been bitten by exactly that.
- `-e -M` on an unknown blob can produce an enormous directory tree. Set `-d` and carve into a scratch filesystem, not your case directory.
- A signature hit is a guess based on magic bytes. Confirm with `file` or entropy before reporting an embedded filesystem as fact.

See also

`foremost`, `scalpel`, `bulk_extractor`, `tcpextract`, `upx`, `die`, `diec`, `7za`

blkls

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Extract filesystem blocks — by default the unallocated ones, which is the input a carver wants.

Synopsis

```
blkls [-aAelvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vo
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 13 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-e	—	every block (including file system metadata blocks)	Every block, including filesystem metadata.
-l	—	print details in time machine list format	List block details rather than emitting their contents.
-a	—	Display allocated blocks	Allocated blocks only — the inverse, when isolating live data.
-A	—	Display unallocated blocks	Unallocated blocks. The default and the usual intent: pipe this into <code>foremost</code> or <code>scalpel</code> so the carver reads only free space instead of the whole image.
-f	fstype	File system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
-i	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
-b	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.

Flag	Argument	What it does	When you would use it
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mm1s</code> .
<code>-P</code>	pooltype	Pool container type (use '-P list' for supported types)	Pool type, for APFS or LVM containers.
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-s</code>	—	print slack space only (other flags are ignored)	Slack space only: the tail of the last block of each file, where fragments of previous contents survive. A distinct hunt from carving free space, and it ignores the other flags.
<code>-v</code>	—	verbose to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	print version	

Gotchas

- Output goes to stdout and is the size of the free space — redirect it to a file on a volume that can hold it, not into a pager.
- Block offsets in the extracted stream do **not** match offsets in the original image, because only unallocated blocks were written. Use `-l` if you need to map a hit back to its real location.

See also

`tsk_recover`, `icat`, `photorec`, `testdisk`

bulk_extractor

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Search raw data for a pattern; Recover encryption keys from memory **Version:** bulk_extractor 2.2.0 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** https://github.com/simsong/bulk_extractor/

← [Capability index](#) · [Kit tool list](#)

Purpose

Scan an image for features — email addresses, URLs, credit card numbers, EXIF, keys — without parsing the filesystem at all, so deleted and unallocated content is included.

Synopsis

```
bulk_extractor [OPTION...] image_name
```

Common invocations

```
# from cyberlab 02-memory-forensics
bulk_extractor -V
# from cyberlab 02-memory-forensics
bulk_extractor -o be_out sample.mem
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/bulk_out exercise/sample.dd
# from cyberlab 05-file-carving
bulk_extractor -o /tmp/ak_bulk exercise/sample.dd
# from cyberlab 20-volatility-deep
bulk_extractor -o be_out exercise/memdump.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -E zip -o be_out memory.raw
# from cyberlab 47-ransomware-memory-case
bulk_extractor -E aes -o exercise/be_out exercise/memory.raw
# from cyberlab 51-linux-triage-workflow
bulk_extractor -o be_out disk.raw
```

Options

All 69 options parsed from the captured help text; 58 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-A</code>	arg	Offset added (in bytes) to feature locations (default: 0)	Add an offset to reported feature locations — use it when scanning a carved fragment so offsets still refer to the original image.
<code>--offset_add</code>	arg	Offset added (in bytes) to feature locations (default: 0)	Add an offset to reported feature locations.
<code>-b</code>	arg	Path of file whose contents are prepended to top of all feature files	Prepend a banner file to every feature file, e.g. a case header.
<code>--banner_file</code>	arg	Path of file whose contents are prepended to top of all feature files	Prepend a banner file to every feature file.
<code>-C</code>	arg	Size of context window reported in bytes (default: 16)	Bytes of context stored around each hit. Raise it when a bare match is not enough to judge relevance.
<code>--context_window</code>	arg	Size of context window reported in bytes (default: 16)	Bytes of context stored around each hit.
<code>-d</code>	—	enable debug-level diagnostic logging	Debug-level diagnostic logging.
<code>--debug</code>	—	enable debug-level diagnostic logging	
<code>-E</code>	arg	disable all scanners except the one specified. Same as -x all -E scanner.	Run exactly one scanner and disable the rest. The fastest way to answer a single question instead of a full sweep.
<code>--enable_exclusive</code>	arg	disable all scanners except the one specified. Same as -x all -E scanner.	Run exactly one scanner and disable the rest.
<code>-e</code>	arg	enable a scanner (can be repeated)	Enable a scanner that is off by default, repeatable.
<code>--enable</code>	arg	enable a scanner (can be repeated)	Enable a scanner that is off by default, repeatable.
<code>-x</code>	arg	disable a scanner (can be repeated)	Disable a scanner, repeatable. Turning off the noisy ones is usually a bigger speed win than adding threads.
<code>--disable</code>	arg	disable a scanner (can be repeated)	Disable a scanner, repeatable.
<code>-f</code>	arg	search for a pattern (can be repeated)	Search for a regex pattern, repeatable.
<code>--find</code>	arg	search for a pattern (can be repeated)	Search for a regex pattern, repeatable.
<code>-F</code>	arg	read patterns to search from a file (can be repeated)	Read search patterns from a file — the practical form when hunting a list of indicators.
<code>--find_file</code>	arg	read patterns to search from a file (can be repeated)	Read search patterns from a file.
<code>--find-case-sensitive</code>	—	make -f and -F patterns case-sensitive	Make <code>-f</code> / <code>-F</code> case-sensitive; they are not by default.

Flag	Argument	What it does	When you would use it
<code>-G</code>	arg	page size in bytes (default: 16777216)	Page size read per pass.
<code>--pagesize</code>	arg	page size in bytes (default: 16777216)	Page size read per pass.
<code>-g</code>	arg	margin size in bytes (default: 4194304)	Margin carried between pages, so a feature spanning a page boundary is still found.
<code>--marginsize</code>	arg	margin size in bytes (default: 4194304)	Margin carried between pages, so features spanning a page boundary are still found.
<code>-j</code>	arg	number of threads (default: 12)	Thread count. Defaults to the core count; lower it when the scan is competing with other work.
<code>--threads</code>	arg	number of threads (default: 12)	Thread count.
<code>-J</code>	—	read and process data in the primary thread	Single-threaded. Slow, but the first thing to try when a scan crashes or results look non-deterministic.
<code>--no_threads</code>	—	read and process data in the primary thread	Single-threaded — use when debugging a crash.
<code>-M</code>	arg	max recursion depth (default: 12)	Maximum recursion depth into nested containers. Lower it if a zip bomb or deeply nested archive stalls the scan.
<code>--max_depth</code>	arg	max recursion depth (default: 12)	Maximum recursion depth into nested containers.
<code>--max_bad_alloc_errors</code>	arg	max bad allocation errors (default: 3)	Allocation failures tolerated before aborting.
<code>--max_minute_wait</code>	arg	maximum number of minutes to wait until all data are read (default: 60)	How long to wait for all data to be read before giving up — raise it for slow or failing media.
<code>--log-level</code>	arg	diagnostic log level: trace, debug, info, warning, error, critical, or off	Diagnostic log level.
<code>--log-file</code>	arg	diagnostic log file (default: /bulk_extractor.log)	Diagnostic log file; defaults inside the output directory.
<code>--notify_main_thread</code>	—	Display notifications in the main thread after phase1 completes. Useful for running with ThreadSanitizer	
<code>--notify_async</code>	—	Display notificaitons asynchronously (default)	
<code>-o</code>	arg	output directory [REQUIRED]	Output directory. Required, and it must not already exist unless you also pass <code>-Z</code> .
<code>--outdir</code>	arg	output directory [REQUIRED]	Output directory. Required.

Flag	Argument	What it does	When you would use it
<code>-P</code>	arg	directories for scanner shared libraries (can be repeated). Default directories include /usr/local/lib/bulk_extractor, /usr/lib/bulk_extractor and any directories specified in the BE_PATH environment	Additional directories to load scanner plugins from.
<code>--scanner_dir</code>	arg	directories for scanner shared libraries (can be repeated). Default directories include /usr/local/lib/bulk_extractor, /usr/lib/bulk_extractor and any directories specified in the BE_PATH environment	Additional directories to load scanner plugins from.
<code>-p</code>	arg	print the value of [:length][h][r] with optional length, hex output, or raw output.	Print the value at a path, with optional length and hex or raw output — inspection rather than scanning.
<code>--path</code>	arg	print the value of [:length][h][r] with optional length, hex output, or raw output.	Print the value at a path, for inspection rather than scanning.
<code>-q</code>	—	no status or performance output	Suppress status and performance output.
<code>--quit</code>	—	no status or performance output	Suppress status and performance output.
<code>-r</code>	arg	file to read alert list from	Alert list: features to flag prominently — the inverse of a stop list, for known-bad indicators.
<code>--alert_list</code>	arg	file to read alert list from	Alert list of features to flag prominently.
<code>-R</code>	—	treat image file as a directory to recursively explore	Treat the input as a directory and recurse it, rather than as a disk image.
<code>--recurse</code>	—	treat image file as a directory to recursively explore	Treat the input as a directory and recurse it.
<code>-S</code>	arg	set a name=value option (can be repeated)	Set a scanner option as <code>name=value</code> , repeatable.
<code>--set</code>	arg	set a name=value option (can be repeated)	Set a scanner option as <code>name=value</code> , repeatable.
<code>-s</code>	arg	random sampling parameter frac[:passes]	Random sampling as <code>frac[:passes]</code> . Scans a fraction of a huge image to judge whether a full run is worth the hours.
<code>--sampling</code>	arg	random sampling parameter frac[:passes]	Random sampling as <code>frac[:passes]</code> .
<code>-V</code>	—	Display PACKAGE_VERSION (currently) 2.2.0-DEVELOP	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Display PACKAGE_VERSION (currently) 2.2.0-DEVELOP	
<code>-w</code>	arg	file to read stop list from	Stop list: features to suppress. This is how you cut the known-good noise that otherwise buries the findings.
<code>--stop_list</code>	arg	file to read stop list from	Stop list of features to suppress.
<code>-Y</code>	arg	specify [-end] of area on disk to scan	Restrict the scan to a byte range, when <code>mmls</code> already told you which region matters.
<code>--scan</code>	arg	specify [-end] of area on disk to scan	Restrict the scan to a <code><start>[-end]</code> byte range.
<code>-z</code>	arg	specify a starting page number	Start at a given page number, to resume a long scan.
<code>--page_start</code>	arg	specify a starting page number	Start at a given page number.
<code>-Z</code>	—	wipe the output directory (recursively) before starting	Wipe the output directory first. Convenient for reruns and destructive by definition — never point it at a directory holding results you still need.
<code>--zap</code>	—	wipe the output directory (recursively) before starting	Wipe the output directory first — destructive.
<code>-0</code>	—	disable real-time notification	
<code>--no_notify</code>	—	disable real-time notification	
<code>-1</code>	—	version 1.0 notification (console-output)	
<code>--version1</code>	—	version 1.0 notification (console-output)	
<code>-H</code>	—	report information about each scanner	Report what each scanner does. Worth reading once; the scanner set is the tool.
<code>--info_scanners</code>	—	report information about each scanner	Report what each scanner does.
<code>-h</code>	—	print help screen	
<code>--help</code>	—	print help screen	

Gotchas

- It ignores the filesystem completely. That is the point — it finds content in unallocated space and slack that a filesystem-aware tool cannot reach — but it also means a hit carries no filename and no timestamp. Map the offset back with `ffind` / `istat` before naming a file in a report.
- Feature files are raw pattern matches, not verified findings. The credit-card scanner in particular flags anything passing a Luhn check, which includes plenty of ordinary numbers.

- Output is large and the scan is long. Sample with `-s` on a multi-terabyte image before committing to a full pass.

See also

`foremost` , `scalpel` , `binwalk` , `tcpextract` , `rafind2` , `strings` , `grep` , `xxd`

ffind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#) <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Find the file name that points at a given inode — the reverse of a directory lookup.

Synopsis

```
ffind [-aduvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vol]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 11 options parsed from the captured help text; 10 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Find all occurrences	Show every name for the inode. Hard-linked files have more than one, and stopping at the first hides that.
<code>-d</code>	—	Find deleted entries ONLY	Deleted names only. The direct answer to "what was this inode called before it was removed?"
<code>-u</code>	—	Find undeleted entries ONLY	Undeleted names only, when a recycled inode is returning stale hits.
<code>-f</code>	fstype	Image file system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mmls</code> .

Flag	Argument	What it does	When you would use it
<code>-P</code>	pooltype	Pool container type (use '-p list' for supported types)	Pool type, for APFS or LVM containers.
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-v</code>	—	Verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	Print version	

Gotchas

- Inodes are reused. A name returned for a deleted inode may belong to whatever claimed it next, not to the file you are chasing — corroborate with `istat` timestamps before naming it in a report.
- This is the tool for the question `icat` provokes: you carved data out by inode and now need to say what it was called.

See also

`fls`, `ils`, `tsk_recover`

file

Kit: REMnux · FLARE-VM · SIFT Workstation **Capability:** Inspect metadata for one file or inode; Identify what a file actually is **Version:** file-5.44 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](https://github.com/file/file) <https://github.com/file/file>

← [Capability index](#) · [Kit tool list](#)

Purpose

Identify file type using "magic" numbers.

Synopsis

```
file [OPTION...] [FILE...]
```

Common invocations

```
# from cyberlab 33-binwalk-firmware
file foremost_out/jpg/*.jpg 2>/dev/null || file foremost_out/**/*.jpg
# from cyberlab 35-radare2-intro
file exercise/hello
# from cyberlab 38-network-emulation
file /tmp/beacon_reply.bin # -> data / HTML (INetSim default object)
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	display this help and exit	
<code>-v</code>	—	output version information and exit	
<code>--version</code>	—	output version information and exit	
<code>-m</code>	LIST	use LIST as a colon-separated list of magic number files	
<code>--magic-file</code>	LIST	use LIST as a colon-separated list of magic number files	
<code>-z</code>	—	try to look inside compressed files	
<code>--uncompress</code>	—	try to look inside compressed files	

Flag	Argument	What it does	When you would use it
<code>-Z</code>	—	only print the contents of compressed files	
<code>--uncompress-noreport</code>	—	only print the contents of compressed files	
<code>-b</code>	—	do not prepend filenames to output lines	
<code>--brief</code>	—	do not prepend filenames to output lines	
<code>-c</code>	—	print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it	
<code>--checking-printout</code>	—	print the parsed form of the magic file, use in conjunction with <code>-m</code> to debug a new magic file before installing it	
<code>-e</code>	TEST	exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens	
<code>--exclude</code>	TEST	exclude TEST from the list of test to be performed for file. Valid tests are: apptype, ascii, cdf, compress, csv, elf, encoding, soft, tar, json, text, tokens	
<code>--exclude-quiet</code>	TEST	like exclude, but ignore unknown tests	
<code>-f</code>	FILE	read the filenames to be examined from FILE	
<code>--files-from</code>	FILE	read the filenames to be examined from FILE	
<code>-F</code>	STRING	use string as separator instead of `:'	
<code>--separator</code>	STRING	use string as separator instead of `:'	
<code>-i</code>	—	output MIME type strings (<code>--mime-type</code> and <code>--mime-encoding</code>)	
<code>--mime</code>	—	output MIME type strings (<code>--mime-type</code> and <code>--mime-encoding</code>)	
<code>--apple</code>	—	output the Apple CREATOR/TYPE	
<code>--extension</code>	—	output a slash-separated list of extensions	
<code>--mime-type</code>	—	output the MIME type	
<code>--mime-encoding</code>	—	output the MIME encoding	
<code>-k</code>	—	don't stop at the first match	
<code>--keep-going</code>	—	don't stop at the first match	
<code>-l</code>	—	list magic strength	
<code>--list</code>	—	list magic strength	
<code>-L</code>	—	follow symlinks (default if POSIXLY_CORRECT is set)	
<code>--dereference</code>	—	follow symlinks (default if POSIXLY_CORRECT is set)	
<code>-h</code>	—	don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)	
<code>--no-dereference</code>	—	don't follow symlinks (default if POSIXLY_CORRECT is not set) (default)	
<code>-n</code>	—	do not buffer output	

Flag	Argument	What it does	When you would use it
<code>--no-buffer</code>	—	do not buffer output	
<code>-N</code>	—	do not pad output	
<code>--no-pad</code>	—	do not pad output	
<code>-0</code>	—	terminate filenames with ASCII NUL	
<code>--print0</code>	—	terminate filenames with ASCII NUL	
<code>-p</code>	—	preserve access times on files	
<code>--preserve-date</code>	—	preserve access times on files	
<code>-P</code>	—	set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce	
<code>--parameter</code>	—	set file engine parameter limits bytes 7340032 max bytes to look inside file elf_notes 256 max ELF notes processed elf_phnum 2048 max ELF prog sections processed elf_shnum 32768 max ELF sections proce	
<code>-r</code>	—	don't translate unprintable chars to \ooo	
<code>--raw</code>	—	don't translate unprintable chars to \ooo	
<code>-s</code>	—	treat special (block/char devices) files as ordinary ones	
<code>--special-files</code>	—	treat special (block/char devices) files as ordinary ones	
<code>-S</code>	—	disable system call sandboxing	
<code>--no-sandbox</code>	—	disable system call sandboxing	
<code>-C</code>	—	compile file specified by -m	
<code>--compile</code>	—	compile file specified by -m	
<code>-d</code>	—	print debugging messages	
<code>--debug</code>	—	print debugging messages	

Gotchas

TODO: operational traps.

See also

`istat`, `ils`, `stat`, `die`, `diec`

fls

Kit: SIFT Workstation · REMnux · Kali (`sleuthkit`) **Category:** Filesystem analysis **Version:** The Sleuth Kit 4.11.1
Docs: <https://sleuthkit.org/sleuthkit/man/fls.html> **Verified:** 2026-07-29 from `cyberlab-aio:v1` — raw output in `capture/cyberlab-aio/help/fls.help.txt`

Worked example of the format. Every option below was read off the real binary, not written from memory.
See [docs/FORMAT.md](#).

Purpose

List file and directory names in a disk image, including deleted entries.

Synopsis

```
fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/]  
    [-o imgoffset] [-z ZONE] [-s seconds] image [images] [inode]
```

If `[inode]` is not given, the root directory is used.

Note: the usage line above advertises only `[-adDFlhpruvV]`, but the tool also accepts `-P`, `-B`, `-S` and `-k`. They are documented below.

Common invocations

```
# List the root directory of an image
fls {{path/to/image.dd}}

# List a partition that starts at a sector offset (the usual case for a full disk)
fls -o {{2048}} {{path/to/image.dd}}

# Show deleted entries only – the fast answer to "what was removed?"
fls -d -o {{2048}} {{path/to/image.dd}}

# Recurse the whole filesystem showing full paths, deleted files included
fls -rp -o {{2048}} {{path/to/image.dd}}

# Recurse showing ONLY deleted files with full paths
fls -rpd -o {{2048}} {{path/to/image.dd}}

# Long listing (sizes, MAC times) like ls -l, in a named time zone
fls -l -z {{EST5EDT}} -o {{2048}} {{path/to/image.dd}}

# Emit a body file for timelining, then build the timeline with mactime
fls -r -m {{/}} -o {{2048}} {{path/to/image.dd}} > {{path/to/body.txt}}

# List the contents of one directory by inode number
fls -o {{2048}} {{path/to/image.dd}} {{inode}}
```

Options — complete

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Display <code>.</code> and <code>..</code> entries	Rarely; they are hidden by default as noise
<code>-d</code>	—	Display deleted entries only	Recovery hunts — isolates what was removed
<code>-D</code>	—	Display only directories	Mapping structure without file noise
<code>-F</code>	—	Display only files	Excluding directories from a listing
<code>-l</code>	—	Display long version (like <code>ls -l</code>)	When you need sizes, UID/GID and MAC times
<code>-p</code>	—	Display full path for each file	Almost always with <code>-r</code> ; otherwise output is ambiguous
<code>-r</code>	—	Recurse on directory entries	Walking a whole filesystem rather than one directory
<code>-u</code>	—	Display undeleted entries only	The inverse of <code>-d</code> ; filtering live files

Flag	Argument	What it does	When you would use it
<code>-m</code>	<code>dir/</code>	Output in mactime input format, with <code>dir/</code> as the mount point	Building a super-timeline; feeds <code>mactime</code>
<code>-h</code>	—	Include MD5 checksum hash in mactime output	Timeline work where you also need file hashes
<code>-f</code>	<code>fstype</code>	Filesystem type (<code>-f list</code> for supported types)	When auto-detection fails or is wrong
<code>-i</code>	<code>imgtype</code>	Image format (<code>-i list</code> for supported types)	Non-raw images: E01, AFF, etc.
<code>-b</code>	<code>dev_sector_size</code>	Device sector size in bytes	4Kn drives, where the 512-byte assumption breaks
<code>-o</code>	<code>imgoffset</code>	Offset into the image, in sectors	The most common flag — targeting a partition in a full-disk image
<code>-P</code>	<code>pooltype</code>	Pool container type (<code>-P list</code> for supported types)	APFS / logical volume containers
<code>-B</code>	<code>pool_volume_block</code>	Starting block, pool volumes only	Selecting a volume inside a pool
<code>-S</code>	<code>snap_id</code>	Snapshot ID (APFS only)	Examining a specific APFS snapshot
<code>-k</code>	<code>password</code>	Decryption password for encrypted volumes	BitLocker/FileVault-style encrypted volumes
<code>-z</code>	<code>ZONE</code>	Time zone of the original machine (e.g. <code>EST5EDT</code> , <code>GMT</code>)	Only meaningful with <code>-l</code> ; wrong TZ silently skews every timestamp
<code>-s</code>	<code>seconds</code>	Time skew of the original machine, in seconds	Only with <code>-l</code> / <code>-m</code> ; correcting a known-bad system clock
<code>-v</code>	—	Verbose output to stderr	Debugging why an image will not parse
<code>-V</code>	—	Print version	Recording tool provenance in your notes

Gotchas

- `-o` **is in sectors, not bytes**. Get it from `mmfs` first. A wrong offset usually yields "Cannot determine file system type", not an obvious error.
- `-z` **only affects `-l` output**. Setting it without `-l` changes nothing, which makes a wrong-timezone timeline easy to produce and hard to notice.
- **Deleted entries may have unrecoverable content**. `fls -d` listing a name does not mean `icat` will return its data — the blocks may be reallocated.
- `-r` **on a large image is slow and noisy**. Pipe to a file, then grep it, rather than scrolling.

See also

- `mmls` — partition table; where you get the `-o` value
- `icat` — extract the content of an inode `fls` found
- `istat` — detailed metadata for one inode
- `mactime` — consumes the `-m` body file to build the timeline
- `tsk_recover` — bulk export of allocated/deleted files

foremost

Kit: Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** 1.5.7 **Captured:** cyberlab-aio via -h on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Carve files out of an image or raw data by header and footer signatures.

Common invocations

```
# from cyberlab 05-file-carving
foremost -V
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/foremost_out
# from cyberlab 05-file-carving
foremost -t jpg,pdf -i exercise/sample.dd -o /tmp/ak_foremost
# from cyberlab 33-binwalk-firmware
foremost -i firmware.bin -o foremost_out
```

Options

All 11 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-V	—	- display copyright information and exit	
-t	—	- specify file type. (-t jpeg,pdf ...)	Restrict carving to given types (jpg , pdf , all). Narrowing this is the difference between a usable result and 40,000 files.
-d	—	- turn on indirect block detection (for UNIX file-systems)	
-i	—	- specify input file (default is stdin)	Input file or device to carve from.
-a	—	- Write all headers, perform no error detection (corrupted files)	Write all headers found, even without a valid footer.
-w	—	- Only write the audit file, do not write any detected files to the disk	Write the audit file only, carving nothing — a cheap dry run.
-o	—	- set output directory (defaults to output)	Output directory — must be empty or foremost refuses to run.

Flag	Argument	What it does	When you would use it
<code>-c</code>	—	- set configuration file to use (defaults to foremost.conf)	Use a custom configuration file to add signatures.
<code>-q</code>	—	- enables quick mode. Search are performed on 512 byte boundaries.	Quick mode: scan only sector boundaries. Much faster, misses embedded files.
<code>-Q</code>	—	- enables quiet mode. Suppress output messages.	
<code>-v</code>	—	- verbose mode. Logs all messages to screen	Verbose output.

Gotchas

- Carving recovers content but **not filenames or timestamps** — those live in filesystem metadata that carving bypasses. Use `tsk_recover` when the filesystem is intact and carve only what it cannot reach.
- The output directory must be empty; foremost aborts otherwise. This trips scripted reruns constantly.

See also

`scalpel`, `binwalk`, `bulk_extractor`, `tcpextract`, `tcpflow`

fsstat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Report a filesystem's layout and parameters: type, block size, inode range, and the geometry every other TSK tool needs.

Synopsis

```
fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
```

Common invocations

```
# from cyberlab 01-disk-forensics
fsstat -V
# from cyberlab 01-disk-forensics
fsstat -o 2048 exercise/sample.dd
# from cyberlab 01-disk-forensics
fsstat -o 0 exercise/sample.dd | grep -Ei "file system type|sector size|cluster size"
# from cyberlab 22-sleuthkit-mastery
fsstat -o 2048 exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
fsstat -o 2048 disk.raw
```

Options

All 10 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-t</code>	—	display type only	Print only the filesystem type. The scriptable form when all you need is a yes/no on what this volume is.
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Set the image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Device sector size. Needed on 4Kn drives, where the 512-byte default silently computes every offset wrong.

Flag	Argument	What it does	When you would use it
<code>-f</code>	<code>fstype</code>	File system type (use '-f list' for supported types)	Force the filesystem type when detection is wrong or the superblock is damaged (<code>-f list</code> shows the options).
<code>-o</code>	<code>imgoffset</code>	The offset of the file system in the image (in sectors)	Offset, in sectors , of the filesystem inside the image. Take it from <code>mmls</code> ; this is the flag that ties the two tools together.
<code>-P</code>	<code>pooltype</code>	Pool container type (use '-P list' for supported types)	Pool type, for APFS or LVM containers that hold the volume.
<code>-B</code>	<code>pool_volume_block</code>	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr, useful when detection fails.
<code>-V</code>	—	Print version	
<code>-k</code>	<code>password</code>	Decryption password for encrypted volumes	Password for an encrypted volume.

Gotchas

- Run this first. Block size and inode range from `fsstat` are what make `blkls` , `icat` and `ils` output interpretable — starting anywhere else means guessing at the numbers they print.
- If it reports the wrong type or refuses the volume, the `-o` offset is wrong far more often than the image is corrupt.

See also

`mmls` , `img_stat` , `testdisk`

grep

Kit: Base OS — present on every Linux image **Capability:** Search raw data for a pattern **Version:** grep (GNU grep) 3.8 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Search for PATTERNS in each FILE.

Synopsis

```
grep [OPTION]... PATTERNS [FILE]...
```

Common invocations

```
# from cyberlab 02-memory-forensics
grep -i "benign.lab.local" be_out/url.txt
# from cyberlab 02-memory-forensics
grep -i "analyst@lab.local" be_out/email.txt
# from cyberlab 03-timeline-analysis
grep filestat /tmp/timeline.csv | sort -t',' -k1,2 | head -n 1
# from cyberlab 04-registry-analysis
grep -i "ComputerName" /tmp/system_dump.txt | head
# from cyberlab 05-file-carving
grep -v '^#' /etc/scalpel/scalpel.conf | grep -qi jpg || printf '\njpg y 200000000 \\xff\\xd8\\xff \\xf
# from cyberlab 05-file-carving
grep -E 'jpg:|pdf:' /tmp/ak_foremost/audit.txt
# from cyberlab 06-windows-artifact-libs
grep -c "Record number" /tmp/security_events.txt
# from cyberlab 06-windows-artifact-libs
grep "Event Identifier" /tmp/security_events.txt | awk -F: '{print $2}' | sort | uniq -c | sort -rn |
```

Options

All 82 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-E</code>	—	PATTERNS are extended regular expressions	
<code>--extended-regexp</code>	—	PATTERNS are extended regular expressions	
<code>-F</code>	—	PATTERNS are strings	
<code>--fixed-strings</code>	—	PATTERNS are strings	
<code>-G</code>	—	PATTERNS are basic regular expressions	
<code>--basic-regexp</code>	—	PATTERNS are basic regular expressions	
<code>-P</code>	—	PATTERNS are Perl regular expressions	
<code>--perl-regexp</code>	—	PATTERNS are Perl regular expressions	
<code>-e</code>	PATTERNS	use PATTERNS for matching	
<code>--regexp</code>	PATTERNS	use PATTERNS for matching	
<code>-f</code>	FILE	take PATTERNS from FILE	
<code>--file</code>	FILE	take PATTERNS from FILE	
<code>-i</code>	—	ignore case distinctions in patterns and data	
<code>--ignore-case</code>	—	ignore case distinctions in patterns and data	
<code>--no-ignore-case</code>	—	do not ignore case distinctions (default)	
<code>-w</code>	—	match only whole words	
<code>--word-regexp</code>	—	match only whole words	
<code>-x</code>	—	match only whole lines	
<code>--line-regexp</code>	—	match only whole lines	
<code>-z</code>	—	a data line ends in 0 byte, not newline	
<code>--null-data</code>	—	a data line ends in 0 byte, not newline	
<code>-s</code>	—	suppress error messages	
<code>--no-messages</code>	—	suppress error messages	
<code>-v</code>	—	select non-matching lines	
<code>--invert-match</code>	—	select non-matching lines	
<code>-V</code>	—	display version information and exit	
<code>--version</code>	—	display version information and exit	
<code>--help</code>	—	display this help text and exit	
<code>-m</code>	NUM	stop after NUM selected lines	
<code>--max-count</code>	NUM	stop after NUM selected lines	
<code>-b</code>	—	print the byte offset with output lines	
<code>--byte-offset</code>	—	print the byte offset with output lines	
<code>-n</code>	—	print line number with output lines	
<code>--line-number</code>	—	print line number with output lines	

Flag	Argument	What it does	When you would use it
<code>--line-buffered</code>	—	flush output on every line	
<code>-H</code>	—	print file name with output lines	
<code>--with-filename</code>	—	print file name with output lines	
<code>-h</code>	—	suppress the file name prefix on output	
<code>--no-filename</code>	—	suppress the file name prefix on output	
<code>--label</code>	LABEL	use LABEL as the standard input file name prefix	
<code>-o</code>	—	show only nonempty parts of lines that match	
<code>--only-matching</code>	—	show only nonempty parts of lines that match	
<code>-q</code>	—	suppress all normal output	
<code>--quiet</code>	—	suppress all normal output	
<code>--silent</code>	—	suppress all normal output	
<code>--binary-files</code>	TYPE	assume that binary files are TYPE; TYPE is 'binary', 'text', or 'without-match'	
<code>-a</code>	—	equivalent to <code>--binary-files=text</code>	
<code>--text</code>	—	equivalent to <code>--binary-files=text</code>	
<code>-I</code>	—	equivalent to <code>--binary-files=without-match</code>	
<code>-d</code>	ACTION	how to handle directories; ACTION is 'read', 'recurse', or 'skip'	
<code>--directories</code>	ACTION	how to handle directories; ACTION is 'read', 'recurse', or 'skip'	
<code>-D</code>	ACTION	how to handle devices, FIFOs and sockets; ACTION is 'read' or 'skip'	
<code>--devices</code>	ACTION	how to handle devices, FIFOs and sockets; ACTION is 'read' or 'skip'	
<code>-r</code>	—	like <code>--directories=recurse</code>	
<code>--recursive</code>	—	like <code>--directories=recurse</code>	
<code>-R</code>	—	likewise, but follow all symlinks	
<code>--dereference-recursive</code>	—	likewise, but follow all symlinks	
<code>--include</code>	GLOB	search only files that match GLOB (a file pattern)	
<code>--exclude</code>	GLOB	skip files that match GLOB	
<code>--exclude-from</code>	FILE	skip files that match any file pattern from FILE	
<code>--exclude-dir</code>	GLOB	skip directories that match GLOB	
<code>-L</code>	—	print only names of FILEs with no selected lines	
<code>--files-without-match</code>	—	print only names of FILEs with no selected lines	
<code>-l</code>	—	print only names of FILEs with selected lines	

Flag	Argument	What it does	When you would use it
<code>--files-with-matches</code>	—	print only names of FILES with selected lines	
<code>-c</code>	—	print only a count of selected lines per FILE	
<code>--count</code>	—	print only a count of selected lines per FILE	
<code>-T</code>	—	make tabs line up (if needed)	
<code>--initial-tab</code>	—	make tabs line up (if needed)	
<code>-Z</code>	—	print 0 byte after FILE name	
<code>--null</code>	—	print 0 byte after FILE name	
<code>-B</code>	NUM	print NUM lines of leading context	
<code>--before-context</code>	NUM	print NUM lines of leading context	
<code>-A</code>	NUM	print NUM lines of trailing context	
<code>--after-context</code>	NUM	print NUM lines of trailing context	
<code>-C</code>	NUM	print NUM lines of output context	
<code>--context</code>	NUM	print NUM lines of output context	
<code>--group-separator</code>	SEP	print SEP on line between matches with context	
<code>--no-group-separator</code>	—	do not print separator for matches with context	
<code>--colour</code>	WHEN (optional)	use markers to highlight the matching strings; WHEN is 'always', 'never', or 'auto'	
<code>-U</code>	—	do not strip CR characters at EOL (MSDOS/Windows)	
<code>--binary</code>	—	do not strip CR characters at EOL (MSDOS/Windows)	

Gotchas

TODO: operational traps.

See also

`rafind2`, `strings`, `xxd`, `bulk_extractor`

icat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Extract the contents of a file by inode, including deleted files.

Synopsis

```
icat [-hrRsvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image [images] inum[-typ[-
```

Common invocations

```
# from cyberlab 01-disk-forensics
icat -o 2048 exercise/sample.dd 5 > /tmp/recovered_file.bin
# from cyberlab 01-disk-forensics
icat -o 0 exercise/sample.dd 5 > /tmp/recovered.txt
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 4 | head
# from cyberlab 22-sleuthkit-mastery
icat -o 2048 exercise/practice.dd 6
# from cyberlab 51-linux-triage-workflow
icat -o 2048 disk.raw 5 > recovered_file.bin
# from cyberlab 51-linux-triage-workflow
icat exercise/triage_sample.raw $(find -p exercise/triage_sample.raw | awk '/eicar.com/{gsub(/:/,"",$2)
```

Options

All 14 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-h	—	Do not display holes in sparse files	Skip holes in sparse files so the output is not padded with zeroes.
-r	—	Recover deleted file	Attempt recovery of a deleted file — the reason you usually reach for icat.

Flag	Argument	What it does	When you would use it
<code>-R</code>	—	Recover deleted file and suppress recovery errors	Recover with slack space included, when you want everything the blocks still hold.
<code>-S</code>	—	Display slack space at end of file	Include slack space in the output.
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	
<code>-f</code>	fstype	File system type (use '-f list' for supported types)	Force the filesystem type when detection is wrong.
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Partition offset in sectors, from <code>mmLS</code> .
<code>-P</code>	pooltype	Pool container type (use '-P list' for supported types)	
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	
<code>-S</code>	snap_id	Snapshot ID (for APFS only)	
<code>-v</code>	—	verbose to stderr	
<code>-V</code>	—	Print version	
<code>-k</code>	password	Decryption password for encrypted volumes	Supply a decryption password for an encrypted volume.

Gotchas

- Always redirect to a file (`icat ... > out.bin`). Binary content dumped to a terminal will corrupt your session.
- A deleted inode listed by `fls -d` may return nothing or garbage: the metadata survived but the blocks were reallocated. Empty output is evidence, not a tool failure.

See also

`tsk_recover`, `photorec`, `testdisk`, `blkls`

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

List inode metadata, including inodes that no longer have a name pointing at them.

Synopsis

```
ils [-emOpvV] [-aAlLzZ] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 19 options parsed from the captured help text; 18 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-e	—	Display all inodes	Every inode, allocated or not.
-m	—	Display output in the mactime format	mactime format — the form <code>mactime</code> consumes to build a timeline. This is how deleted-file metadata reaches the timeline at all.
-O	—	Display inodes that are unallocated, but were sill open (UFS/ExtX only)	Unallocated inodes that were still open at the time of imaging (UFS/ExtX). The same trick, caught mid-deletion.
-p	—	Display orphan inodes (unallocated with no file name)	Orphan inodes — allocated content with no directory entry. Files that were unlinked while still open, and a standard hiding place worth checking explicitly.
-s	seconds	Time skew of original machine (in seconds)	Correct for a known clock skew on the source machine, in seconds, so times line up with other evidence.
-a	—	Allocated inodes	Allocated inodes only.

Flag	Argument	What it does	When you would use it
<code>-A</code>	—	Unallocated inodes	Unallocated inodes only — deleted file metadata that often survives after the name is gone.
<code>-l</code>	—	Linked inodes	Linked inodes (a name still points at them).
<code>-L</code>	—	Unlinked inodes	Unlinked inodes (nothing does).
<code>-z</code>	—	Unused inodes	Unused inodes.
<code>-Z</code>	—	Used inodes	Used inodes.
<code>-i</code>	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Device sector size; required on 4Kn drives.
<code>-f</code>	fstype	File system type (use '-f list' for supported types)	Force the filesystem type (<code>-f list</code> shows the options).
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Offset, in sectors, from <code>mmls</code> .
<code>-p</code>	pooltype	Pool container type (use '-p list' for supported types)	Pool type, for APFS or LVM containers.
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	Starting block within a pool volume.
<code>-v</code>	—	verbose output to stderr	Verbose diagnostics to stderr.
<code>-V</code>	—	Display version number	

Gotchas

- `ils` finds metadata with no name; `ffind` turns an inode back into a name; `icat` extracts its content. Deleted-file work is usually all three in sequence.
- An inode surviving does not mean its data did. The blocks it points at may already be reallocated, so `icat` can return another file's contents entirely.

See also

`fls`, `ffind`, `tsk_recover`, `istat`, `file`, `stat`

istat

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect metadata for one file or inode **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Show the full metadata for one inode: times, size, and the blocks it occupies.

Synopsis

```
istat [-N num] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] [-P pooltype] [-B pool_vol]
```

Common invocations

```
# from cyberlab 22-sleuthkit-mastery
istat -o 2048 exercise/practice.dd 4
```

Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-N	num	force the display of NUM address of block pointers	Limit how many block addresses are printed for a large file.
-r	—	display run list instead of list of block addresses	Include recovery information for a deleted inode.
-z	zone	time zone of original machine (i.e. EST5EDT or GMT)	Set the time zone for the displayed timestamps.
-s	seconds	Time skew of original machine (in seconds)	Apply a clock skew in seconds for a known-bad system clock.
-i	imgtype	The format of the image file (use '-i list' for supported types)	
-b	dev_sector_size	The size (in bytes) of the device sectors	
-f	fstype	File system type (use '-f list' for supported types)	

Flag	Argument	What it does	When you would use it
<code>-o</code>	imgoffset	The offset of the file system in the image (in sectors)	Partition offset in sectors, from <code>mm1s</code> .
<code>-P</code>	pooltype	Pool container type (use '-p list' for supported types)	
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	
<code>-S</code>	snap_id	Snapshot ID (for APFS only)	
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	
<code>-k</code>	password	Decryption password for encrypted volumes	

Gotchas

- The block list is what lets you prove whether a deleted file is still recoverable — cross-check it before promising a recovery.

See also

`ils`, `file`, `stat`

mmls

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** See the partition and volume layout **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Display the partition layout of a disk image, including unallocated gaps.

Common invocations

```
# from cyberlab 01-disk-forensics
mmls -V
# from cyberlab 01-disk-forensics
mmls exercise/sample.dd
# from cyberlab 22-sleuthkit-mastery
mmls exercise/practice.dd
# from cyberlab 51-linux-triage-workflow
mmls disk.raw
# from cyberlab 57-forensic-acquisition
sudo mmls /mnt/ewf/ewf1
```

Options

All 12 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-t</code>	vstype	The type of volume system (use '-t list' for list of supported types)	Force the volume-system type when auto-detection guesses wrong (<code>-t list</code> shows the options).
<code>-i</code>	imgtype	The format of the image file (use '-i list' for list supported types)	Set the image format for non-raw evidence such as E01 or AFF.
<code>-b</code>	dev_sector_size	The size (in bytes) of the device sectors	Set the device sector size; required on 4Kn drives where the 512-byte default is wrong.
<code>-o</code>	imgoffset	Offset to the start of the volume that contains the partition system (in sectors)	Read a volume system nested at an offset — rare, but needed for nested containers.
<code>-B</code>	—	print the rounded length in bytes	Print volume sizes in bytes rather than sectors, when reporting.
<code>-r</code>	—	recurse and look for other partition tables in partitions (DOS Only)	Recurse into nested volume systems (e.g. an extended partition).

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	verbose output	Verbose diagnostics to stderr when an image will not parse.
<code>-V</code>	—	print the version	
<code>-a</code>	—	Show allocated volumes	Show allocated volumes only, when the gap entries are noise.
<code>-A</code>	—	Show unallocated volumes	Show unallocated space only — where a hidden or deleted partition would show up.
<code>-m</code>	—	Show metadata volumes	
<code>-M</code>	—	Hide metadata volumes	

Gotchas

- The **Start** column is in sectors. That value is what every other TSK tool wants for `-o`. Multiplying by the sector size here is the single most common mistake in a TSK workflow.
- If `mmfs` reports no partition table, the image may be a single volume rather than a whole disk — try `fsstat` on it directly at offset 0 before assuming the image is corrupt.

See also

`fsstat`, `img_stat`, `testdisk`

photorec

Kit: SIFT Workstation **Capability:** Recover deleted or lost files **Version:** PhotoRec 7.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

PhotoRec 7.1, Data Recovery Utility, July 2019

Synopsis

```
photorec [/log] [/debug] [/d recup_dir] [file.dd|file.e01|device]
photorec /version
```

Common invocations

```
# from cyberlab 01-disk-forensics
photorec /version
# from cyberlab 01-disk-forensics
photorec /log /d /tmp/carved /cmd exercise/sample.dd partition_none,options,mode_ext2,fileopt,everythi
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
/log	—	create a photorec.log file	
/debug	—	add debug information	

Gotchas

TODO: operational traps.

See also

[tsk_recover](#), [icat](#), [testdisk](#), [blkls](#)

rafind2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Search raw data for a pattern **Version:** rafind2 6.1.9
Captured: `cyberlab-aio` via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rafind2 [-mBXnzZhqv] [-a align] [-b sz] [-f/t from/to] [-[e|s|S] str] [-x hex] [-R str] [-I str] [-g]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`strings`, `grep`, `xxd`, `bulk_extractor`

scalpel

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Carve files out of unstructured data **Version:** Scalpel version 1.60 **Captured:** cyberlab-aio via -h on 2026-08-02 **raw Docs:** <https://github.com/sleuthkit/scalpel>

← [Capability index](#) · [Kit tool list](#)

Purpose

Carve contents out of binary files, such as partitions.

Synopsis

```
scalpel [-b] [-c <config file>] [-d] [-h|V] [-i <file>]
[-m blocksize] [-n] [-o <outputdir>] [-O num] [-q clustersize]
[-r] [-s num] [-t <blockmap file>] [-u] [-v]
<imgfile> [<imgfile>] ...
```

Common invocations

```
# from cyberlab 05-file-carving
scalpel -V
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/scalpel_out exercise/sample.dd
# from cyberlab 05-file-carving
scalpel -c /etc/scalpel/scalpel.conf -o /tmp/ak_scalpel exercise/sample.dd
```

Options

All 17 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-b	—	Carve files even if defined footers aren't discovered within maximum carve size for file type [foremost 0.69 compat mode].	
-c	—	Choose configuration file.	
-d	—	Generate header/footer database; will bypass certain optimizations and discover all footers, so performance suffers. Doesn't affect the set of files carved. EXPERIMENTAL	
-h	—	Print this help message and exit.	

Flag	Argument	What it does	When you would use it
<code>-i</code>	—	Read names of disk images from specified file.	
<code>-m</code>	—	Generate/update carve coverage blockmap file. The first 32bit unsigned int in the file identifies the block size. Thereafter each 32bit unsigned int entry in the blockmap file corresponds to one block	
<code>-n</code>	—	Don't add extensions to extracted files.	
<code>-o</code>	—	Set output directory for carved files.	
<code>-O</code>	—	Don't organize carved files by type. Default is to organize carved files into subdirectories.	
<code>-p</code>	—	Perform image file preview; audit log indicates which files would have been carved, but no files are actually carved.	
<code>-q</code>	—	Carve only when header is cluster-aligned.	
<code>-r</code>	—	Find only first of overlapping headers/footers [foremost 0.69 compat mode].	
<code>-s</code>	—	Skip n bytes in each disk image before carving.	
<code>-t</code>	—	Set directory for coverage blockmap. EXPERIMENTAL	
<code>-u</code>	—	Use carve coverage blockmap when carving. Carve only sections of the image whose entries in the blockmap are 0. These areas are treated as contiguous regions. EXPERIMENTAL	
<code>-V</code>	—	Print copyright information and exit.	
<code>-v</code>	—	Verbose mode.	

Gotchas

TODO: operational traps.

See also

`foremost`, `binwalk`, `bulk_extractor`, `tcpxtract`

stat

Kit: Base OS — present on every Linux image **Capability:** Inspect metadata for one file or inode **Version:** stat (GNU coreutils) 9.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Display file or file system status.

Synopsis

```
stat [OPTION]... FILE...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-L	—	follow links	
--dereference	—	follow links	
-f	—	display file system status instead of file status	
--file-system	—	display file system status instead of file status	
--cached	MODE	specify how to use cached attributes; useful on remote file systems. See MODE below	
-c	—	use the specified FORMAT instead of the default; output a newline after each use of FORMAT	
--format	FORMAT	use the specified FORMAT instead of the default; output a newline after each use of FORMAT	
--printf	FORMAT	like --format, but interpret backslash escapes, and do not output a mandatory trailing newline; if you want a newline, include \n in FORMAT	
-t	—	print the information in terse form	

Flag	Argument	What it does	When you would use it
<code>--terse</code>	—	print the information in terse form	
<code>--help</code>	—	display this help and exit	
<code>--version</code>	—	output version information and exit	

Gotchas

TODO: operational traps.

See also

`istat`, `ils`, `file`

strings

Kit: Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Extract strings, including obfuscated ones **Version:** GNU strings (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Print sequences of printable characters found in a binary file.

Synopsis

```
strings [option(s)] [file(s)]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 14 options parsed from the captured help text; 3 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-d	—	Only scan the data sections in the file	
--data	—	Only scan the data sections in the file	
-f	—	Print the name of the file before each string	
--print-file-name	—	Print the name of the file before each string	
-n	number	Locate & print any sequence of at least	Minimum length. Default 4 is noisy; 8–10 cuts most false hits.
--bytes	number	displayable characters. (The default is 4).	
-t	o,d,x	Print the location of the string in base 8, 10 or 16	
--radix	o,d,x	Print the location of the string in base 8, 10 or 16	
-o	—	An alias for --radix=o	Print the byte offset of each string — lets you seek back to it.

Flag	Argument	What it does	When you would use it
<code>-T</code>	—	Specify the binary file format	
<code>--target</code>	BFDNAME	Specify the binary file format	
<code>-U</code>	d s i x e h	Specify how to treat UTF-8 encoded unicode characters	Control how unicode is handled; needed for UTF-16 Windows strings.
<code>-h</code>	—	Display this information	
<code>--help</code>	—	Display this information	

Gotchas

- GNU `strings` reads only initialised, loaded sections by default. Use `-a` to scan the whole file — malware routinely hides outside them.
- It finds ASCII by default and will miss UTF-16LE strings that Windows binaries are full of. When a PE looks empty, that is usually why — reach for `floss` instead.

See also

`rafind2`, `grep`, `xxd`, `bulk_extractor`, `floss`, `base64dump.py`, `numbers-to-string.py`

tcpxtract

Kit: REMnux · SIFT Workstation **Capability:** Carve files out of unstructured data; Extract files and payloads from traffic **Version:** tcpxtract v1.0.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <http://tcpxtract.sourceforge.net/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract files from network traffic.

Synopsis

```
tcpxtract [OPTIONS] [[-d <DEVICE>] [-f <FILE>]]
```

Common invocations

```
# from cyberlab 05-file-carving
tcpxtract --version 2>&1 | head -n 1
# from cyberlab 05-file-carving
tcpxtract -f exercise/sample.pcap -o /tmp/tcpxtract_out
# from cyberlab 05-file-carving
tcpxtract -f exercise/sample.pcap -o /tmp/ak_tcp
```

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--file</code>	FILE	to specify an input capture file instead of a device	
<code>-f</code>	FILE	to specify an input capture file instead of a device	
<code>--device</code>	DEVICE	to specify an input device (i.e. eth0)	
<code>-d</code>	DEVICE	to specify an input device (i.e. eth0)	
<code>--config</code>	FILE	use FILE as the config file	
<code>-c</code>	FILE	use FILE as the config file	
<code>--output</code>	DIRECTORY	dump files to DIRECTORY instead of current directory	
<code>-o</code>	DIRECTORY	dump files to DIRECTORY instead of current directory	
<code>--version</code>	—	display the version number of this program	

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	display the version number of this program	
<code>--help</code>	—	display this lovely screen	
<code>-h</code>	—	display this lovely screen	

Gotchas

TODO: operational traps.

See also

`foremost`, `scalpel`, `binwalk`, `bulk_extractor`, `tcpflow`

testdisk

Kit: SIFT Workstation **Capability:** See the partition and volume layout; Recover deleted or lost files **Version:** TestDisk 7.1 **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TestDisk 7.1, Data Recovery Utility, July 2019

Synopsis

```
testdisk [/log] [/debug] [file.dd|file.e01|device]
testdisk /list  [/log]   [file.dd|file.e01|device]
testdisk /version
```

Common invocations

```
# from cyberlab 01-disk-forensics
testdisk /version
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>/log</code>	—	create a testdisk.log file	
<code>/debug</code>	—	add debug information	
<code>/list</code>	—	display current partitions	

Gotchas

TODO: operational traps.

See also

`mmls`, `fsstat`, `img_stat`, `tsk_recover`, `icat`, `photorec`, `blkls`

tsk_recover

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** List files and directories, including deleted ones; Recover deleted or lost files **Version:** The Sleuth Kit ver 4.11.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.sleuthkit.org/sleuthkit>

← [Capability index](#) · [Kit tool list](#)

Purpose

Bulk-export files from an image to a directory.

Synopsis

```
tsk_recover [-vVae] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o sector_offset] [-P pooltype] [-B
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 11 options parsed from the captured help text; 6 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-i	imgtype	The format of the image file (use '-i list' for supported types)	Image format for non-raw evidence.
-b	dev_sector_size	The size (in bytes) of the device sectors	
-f	fstype	The file system type (use '-f list' for supported types)	Force the filesystem type.
-v	—	verbose output to stderr	
-V	—	Print version	
-a	—	Recover allocated files only	Recover allocated (live) files only.
-e	—	Recover all files (allocated and unallocated)	Recover every file, allocated and deleted — the usual choice.

Flag	Argument	What it does	When you would use it
<code>-o</code>	sector_offset	sector offset for a volume to recover (recovers only that volume)	Partition offset in sectors, from <code>mmfs</code> .
<code>-P</code>	pooltype	Pool container type (use '-P list' for supported types)	
<code>-B</code>	pool_volume_block	Starting block (for pool volumes only)	
<code>-d</code>	dir_inum	Directory inum to recover from (must also specify a specific partition using -o or there must not be a volume system)	Recover from a specified directory inode rather than the root.

Gotchas

- Default behaviour recovers only *deleted* files, which surprises people expecting a full export. Use `-e` for everything.
- This preserves paths and names, unlike carving. Prefer it whenever the filesystem metadata is intact, and carve only what it cannot reach.

See also

`fls`, `ffind`, `ils`, `icat`, `photorec`, `testdisk`, `blkls`

xxd

Kit: Base OS — present on every Linux image **Capability:** Search raw data for a pattern; Decode, decrypt or transform encoded data; Inspect files by hand **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

xxd [options] [infile [outfile]]

Synopsis

```
xxd [options] [infile [outfile]]
or
xxd -r [-s [-]offset] [-c cols] [-ps] [infile [outfile]]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 16 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	toggle autoskip: A single '*' replaces nul-lines. Default off.	
-b	—	binary digit dump (incompatible with -ps,-i,-r). Default hex.	
-C	—	capitalize variable names in C include file style (-i).	
-c	cols	format octets per line. Default 16 (-i: 12, -ps: 30).	
-E	—	show characters in EBCDIC. Default ASCII.	
-e	—	little-endian dump (incompatible with -ps,-i,-r).	
-g	bytes	number of octets per group in normal output. Default 2 (-e: 4).	
-h	—	print this summary.	
-i	—	output in C include file style.	
-l	len	stop after octets.	
-n	name	set the variable name used in C include output (-i).	
-o	off	add to the displayed file position.	

Flag	Argument	What it does	When you would use it
<code>-r</code>	—	reverse operation: convert (or patch) hexdump into binary.	
<code>-d</code>	—	show offset in decimal instead of hex.	
<code>-u</code>	—	use upper case hex letters.	
<code>-v</code>	—	show version: "xxd 2022-01-14 by Juergen Weigert et al."	

Gotchas

TODO: operational traps.

See also

`rafind2`, `strings`, `grep`, `bulk_extractor`, `cyberchef`, `base64dump.py`, `rax2`, `openssl`

mraptor

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** MacroRaptor 0.56.2 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
mraptor [options] <filename> [filename2 ...]
```

Common invocations

```
# from cyberlab 36-oletools-deep
mraptor --version
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc
# from cyberlab 36-oletools-deep
mraptor sample_macro.doc # automated verdict
```

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

Flag	Argument	What it does	When you would use it
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>-l</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
<code>--loglevel</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
<code>-m</code>	—	Show matched strings.	
<code>--matches</code>	—	Show matched strings.	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

msodde

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

← [Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
msodde [-h] [-j] [--nounquote] [-l LOGLEVEL] [-p PASSWORD] [-d] [-f]
[-a]
FILE
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 15 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-j	—	Output in json format. Do not use with -ldebug	
--json	—	Output in json format. Do not use with -ldebug	
--nounquote	—	don't unquote values	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
-p	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	
--password	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	
-d	—	Return only DDE and DDEAUTO fields	
--dde-only	—	Return only DDE and DDEAUTO fields	

Flag	Argument	What it does	When you would use it
<code>-f</code>	—	Return all fields except harmless ones	
<code>--filter</code>	—	Return all fields except harmless ones	
<code>-a</code>	—	Return all fields, irrespective of their contents	
<code>--all-fields</code>	—	Return all fields, irrespective of their contents	

Gotchas

TODO: operational traps.

See also

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`

msoffcrypto-tool

Kit: REMnux **Capability:** Analyse other container formats **Captured:** cyberlab-aio via --help on 2026-08-02
raw Docs: <https://github.com/nolze/msoffcrypto-tool>

← [Capability index](#) · [Kit tool list](#)

Purpose

Decrypt a Microsoft Office file with password, intermediate key, or private key which generated its escrow key.

Synopsis

```
msoffcrypto-tool [-h] (-p [PASSWORD] | -t) [-e] [-v] [infile] [outfile]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-p	PASSWORD	password text	
--password	PASSWORD	password text	
-t	—	test if the file is encrypted	
--test	—	test if the file is encrypted	
-e	—	encryption mode (default is false)	
-v	—	print verbose information	

Gotchas

TODO: operational traps.

See also

7za, unzip

olebrowse

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olefile`

oledir

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oledir 0.54 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oledir [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [olebrowse](#), [olefile](#)

oledump.py

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oledump.py 0.0.85 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oledump.py [options] [file]
```

Common invocations

```
# from cyberlab 10-malicious-documents
oledump.py -h | head -n 1
# from cyberlab 10-malicious-documents
oledump.py exercise/sample.doc
# from cyberlab 10-malicious-documents
oledump.py -s 3 -v exercise/sample.doc
# from cyberlab 36-oletools-deep
oledump.py --help | head -n 3
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc
# from cyberlab 36-oletools-deep
oledump.py sample_macro.doc # locate the 'M' stream (3)
# from cyberlab 36-oletools-deep
oledump.py -s 3 -v sample_macro.doc # dump VBA source
```

Options

All 64 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	show program's version number and exit	

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-m</code>	—	Print manual	
<code>--man</code>	—	Print manual	
<code>-s</code>	SELECT	select item nr for dumping (a for all)	
<code>--select</code>	SELECT	select item nr for dumping (a for all)	
<code>-d</code>	—	perform dump	
<code>--dump</code>	—	perform dump	
<code>-x</code>	—	perform hex dump	
<code>--hexdump</code>	—	perform hex dump	
<code>-a</code>	—	perform ascii dump	
<code>--asciidump</code>	—	perform ascii dump	
<code>-A</code>	—	perform ascii dump with RLE	
<code>--asciidumprle</code>	—	perform ascii dump with RLE	
<code>-S</code>	—	perform strings dump	
<code>--strings</code>	—	perform strings dump	
<code>-T</code>	—	do head & tail	
<code>--headtail</code>	—	do head & tail	
<code>-v</code>	—	VBA decompression	
<code>--vbadecompress</code>	—	VBA decompression	
<code>--vbadecompressskipattributes</code>	—	VBA decompression, skipping initial attributes	
<code>--vbadecompresscorrupt</code>	—	VBA decompression, display beginning if corrupted	
<code>-r</code>	—	read raw file (use with options -v or -p	
<code>--raw</code>	—	read raw file (use with options -v or -p	
<code>-t</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	
<code>--translate</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	
<code>-e</code>	—	extract OLE embedded file	
<code>--extract</code>	—	extract OLE embedded file	
<code>-i</code>	—	print extra info for selected item	
<code>--info</code>	—	print extra info for selected item	
<code>-p</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>--plugins</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	

Flag	Argument	What it does	When you would use it
<code>--pluginoptions</code>	PLUGINOPTIONS	options for the plugin	
<code>--plugindir</code>	PLUGINDIR	directory for the plugin	
<code>-q</code>	—	only print output from plugins	
<code>--quiet</code>	—	only print output from plugins	
<code>-y</code>	YARA	YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option)	
<code>--yara</code>	YARA	YARA rule-file, @file, directory or #rule to check streams (YARA search doesn't work with -s option)	
<code>-D</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>--decoderdir</code>	DECODERDIR	directory for the decoder	
<code>--yarastrings</code>	—	Print YARA strings	
<code>-M</code>	—	Print metadata	
<code>--metadata</code>	—	Print metadata	
<code>-c</code>	—	Add extra calculated data to output, like hashes	
<code>--calc</code>	—	Add extra calculated data to output, like hashes	
<code>--decompress</code>	—	Search for compressed data in the stream and decompress it	
<code>-V</code>	—	verbose output with decoder errors and YARA rules	
<code>--verbose</code>	—	verbose output with decoder errors and YARA rules	
<code>-C</code>	CUT	cut data	
<code>--cut</code>	CUT	cut data	
<code>-E</code>	EXTRA	add extra info (environment variable: OLEDUMP_EXTRA)	
<code>--extra</code>	EXTRA	add extra info (environment variable: OLEDUMP_EXTRA)	
<code>--storages</code>	—	Include storages in report	
<code>-f</code>	FIND	Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)	
<code>--find</code>	FIND	Find D0CF11E0 MAGIC sequence (use l for listing, number for selecting)	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	
<code>-u</code>	—	Include unused data after end of stream	

Flag	Argument	What it does	When you would use it
<code>--unuseddata</code>	—	Include unused data after end of stream	
<code>--password</code>	PASSWORD	The ZIP password to be used (default infected)	
<code>--trimnull</code>	—	Trim starting at first null byte	

Gotchas

TODO: operational traps.

See also

`olevba`, `oleid`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

olefile

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://github.com/decalage2/olefile>

[← Capability index](#) · [Kit tool list](#)

Purpose

Python package to parse, read and write MS OLE2 files.

Synopsis

```
olefile [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	—	check all streams (for debugging purposes)	
-p	—	extract all user-defined properties	
-d	—	debug mode, shortcut for -l debug (displays a lot of debug information, for developers only)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

oleid

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oleid 0.60.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oleid [-h] [FILE ...]
```

Common invocations

```
# from cyberlab 10-malicious-documents
oleid exercise/sample.doc
# from cyberlab 36-oletools-deep
oleid -h | head -n 3
# from cyberlab 36-oletools-deep
oleid gives a quick risk indicator table (VBA macros: Yes/No, encryption, etc.)
# from cyberlab 36-oletools-deep
oleid sample_macro.doc
# from cyberlab 48-phishing-doc-case
oleid --version
# from cyberlab 48-phishing-doc-case
oleid exercise/invoice_sample.doc
```

Options

All 2 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

olemap

Kit: REMnux **Capability:** Analyse Office documents and macros **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [oledir](#), [olebrowse](#), [olefile](#)

olemeta

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** olemeta 0.54 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
olemeta [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

oleobj

Kit: REMnux **Capability:** Analyse Office documents and macros; Analyse RTF documents and embedded objects

Version: oleobj 0.60.1 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:**

<https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
usage: oleobj [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 14 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-r	—	find files recursively in subdirectories.	
-d	OUTPUT_DIR	use specified directory to output files.	
-z	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
--zip	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
-f	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
--zipfname	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
-l	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
--loglevel	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	

Flag	Argument	What it does	When you would use it
<code>-i</code>	FILE	Additional file to parse (same as positional arguments)	
<code>--more-input</code>	FILE	Additional file to parse (same as positional arguments)	
<code>-v</code>	—	verbose mode, set logging to DEBUG (overwrites -l)	
<code>--verbose</code>	—	verbose mode, set logging to DEBUG (overwrites -l)	

Gotchas

TODO: operational traps.

See also

`olevba`, `oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`

oletimes

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** oletimes 0.54 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://www.decorage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
oletimes [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password (requires Python 2.6+)	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [olemap](#), [oledir](#), [olebrowse](#), [olefile](#)

olevba

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** olevba 0.60.2 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract and analyse VBA macros from Office documents.

Synopsis

```
usage: olevba [options] <filename> [filename2 ...]
```

Common invocations

```
# from cyberlab 10-malicious-documents
olevba --version
# from cyberlab 10-malicious-documents
olevba --decode exercise/sample.doc
# from cyberlab 10-malicious-documents
olevba exercise/sample.doc | grep -E "AutoExec|AutoOpen|Suspicious"
# from cyberlab 36-oletools-deep
olevba sample_macro.doc
# from cyberlab 36-oletools-deep
olevba sample_macro.doc # keyword analysis table
# from cyberlab 48-phishing-doc-case
olevba --decode exercise/invoice_sample.doc
```

Options

All 29 options parsed from the captured help text; 10 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password.	Read documents from inside a zip archive.

Flag	Argument	What it does	When you would use it
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open all files from it, using the provided password.	
<code>-p</code>	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	Supply a password for an encrypted document.
<code>--password</code>	PASSWORD	if encrypted office files are encountered, try decryption with this password. May be repeated.	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *)	
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default: *)	
<code>-a</code>	—	display only analysis results, not the macro source code	Show only the analysis results, skipping the macro source.
<code>--analysis</code>	—	display only analysis results, not the macro source code	
<code>-c</code>	—	display only VBA source code, do not analyze it	Show only the macro source code.
<code>--code</code>	—	display only VBA source code, do not analyze it	
<code>--decode</code>	—	display all the obfuscated strings with their decoded content (Hex, Base64, StrReverse, Dridex, VBA).	Display all the decoded strings the analyser recovered.
<code>--attr</code>	—	display the attribute lines at the beginning of VBA source code	
<code>--reveal</code>	—	display the macro source code after replacing all the obfuscated strings by their decoded content.	Substitute deobfuscated values back into the source, which is the most readable view of a hostile macro.
<code>-l</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	Set log level when diagnosing a parse failure.
<code>--loglevel</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
<code>--deobf</code>	—	Attempt to deobfuscate VBA expressions (slow)	
<code>--relaxed</code>	—	Do not raise errors if opening of substream fails (this option is now deprecated, enabled by default)	
<code>--show-pcode</code>	—	Show disassembled P-code (using pcodedmp)	Disassemble the stored p-code — catches macros where the VBA source and compiled p-code disagree.
<code>--no-pcode</code>	—	Disable extraction and analysis of pcode	
<code>--no-xlm</code>	—	Do not extract XLM Excel macros. This may speed up analysis of large files.	
<code>-t</code>	—	triage mode, display results as a summary table (default for multiple files)	Triage mode — one compact line per file, ideal for a batch.
<code>--triage</code>	—	triage mode, display results as a summary table (default for multiple files)	

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	detailed mode, display full results (default for single file)	Show the deobfuscated macro source.
<code>--detailed</code>	—	detailed mode, display full results (default for single file)	
<code>-j</code>	—	json mode, detailed in json format (never default)	
<code>--json</code>	—	json mode, detailed in json format (never default)	

Gotchas

- **VBA source and p-code can differ.** Malware abuses this so the readable source looks benign while the p-code executes. If a document is suspicious but the source is clean, check `--show-pcode`.
- Run this on a copy, in an isolated VM. Nothing here executes the macro, but the sample itself is still live malware.

See also

`oleid`, `oledump.py`, `olemeta`, `oletimes`, `olemap`, `oledir`, `olebrowse`, `olefile`

pcodedmp

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** pcodedmp version 1.2.6 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://github.com/bontchev/pcodedmp>

[← Capability index](#) · [Kit tool list](#)

Purpose

Disassemble VBA p-code.

Synopsis

```
pcodedmp [-h] [-v] [-n] [-d] [-b] [-o OUTPUTFILE]
fileOrDir [fileOrDir ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-v</code>	—	show program's version number and exit	
<code>--version</code>	—	show program's version number and exit	
<code>-n</code>	—	Don't recurse into directories	
<code>--norecurse</code>	—	Don't recurse into directories	
<code>-d</code>	—	Only disassemble, no stream dumps	
<code>--disasmonly</code>	—	Only disassemble, no stream dumps	
<code>-b</code>	—	Dump the stream contents	
<code>--verbose</code>	—	Dump the stream contents	
<code>-o</code>	OUTPUTFILE	Output file name	
<code>--output</code>	OUTPUTFILE	Output file name	

Gotchas

TODO: operational traps.

See also

[olevba](#) , [oleid](#) , [oledump.py](#) , [olemeta](#) , [oletimes](#) , [olemap](#) , [oledir](#) , [olebrowse](#)

pdf-parser

Kit: Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

pdf-parser, use it to parse a PDF document

Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

Common invocations

```
# from cyberlab 37-pdf-analysis
pdf-parser --version
# from cyberlab 37-pdf-analysis
pdf-parser --search JavaScript exercise/sample.pdf
# from cyberlab 37-pdf-analysis
pdf-parser --object 4 --filter --raw exercise/sample.pdf
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-m	—	Print manual	
--man	—	Print manual	
-s	SEARCH	string to search in indirect objects (except streams)	
--search	SEARCH	string to search in indirect objects (except streams)	

Flag	Argument	What it does	When you would use it
<code>-f</code>	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
<code>--filter</code>	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
<code>-o</code>	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
<code>--object</code>	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
<code>-r</code>	REFERENCE	id of indirect object being referenced (version independent)	
<code>--reference</code>	REFERENCE	id of indirect object being referenced (version independent)	
<code>-e</code>	ELEMENTS	type of elements to select (ctxsi)	
<code>--elements</code>	ELEMENTS	type of elements to select (ctxsi)	
<code>-w</code>	—	raw output for data and filters	
<code>--raw</code>	—	raw output for data and filters	
<code>-a</code>	—	display stats for pdf document	
<code>--stats</code>	—	display stats for pdf document	
<code>-t</code>	TYPE	type of indirect object to select	
<code>--type</code>	TYPE	type of indirect object to select	
<code>-0</code>	—	parse stream of /ObjStm objects	
<code>--objstm</code>	—	parse stream of /ObjStm objects	
<code>-v</code>	—	display malformed PDF elements	
<code>--verbose</code>	—	display malformed PDF elements	
<code>-x</code>	EXTRACT	filename to extract malformed content to	
<code>--extract</code>	EXTRACT	filename to extract malformed content to	
<code>-H</code>	—	display hash of objects	
<code>--hash</code>	—	display hash of objects	
<code>-n</code>	—	do not canonicalize the output	
<code>--nocalonicalizedoutput</code>	—	do not canonicalize the output	
<code>-d</code>	DUMP	filename to dump stream content to	
<code>--dump</code>	DUMP	filename to dump stream content to	
<code>-D</code>	—	display debug info	
<code>--debug</code>	—	display debug info	

Flag	Argument	What it does	When you would use it
<code>-c</code>	—	display the content for objects without streams or with streams without filters	
<code>--content</code>	—	display the content for objects without streams or with streams without filters	
<code>--searchstream</code>	SEARCHSTREAM	string to search in streams	
<code>--unfiltered</code>	—	search in unfiltered streams	
<code>--casesensitive</code>	—	case sensitive search in streams	
<code>--regex</code>	—	use regex to search in streams	
<code>--overridingfilters</code>	OVERRIDINGFILTERS	override filters with given filters (use raw for the raw stream content)	
<code>-g</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generate</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generateembedded</code>	GENERATEEMBEDDED	generate a Python program that embeds the selected indirect object as a file	
<code>-y</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yara</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yarastrings</code>	—	Print YARA strings	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>-k</code>	KEY	key to search in dictionaries	
<code>--key</code>	KEY	key to search in dictionaries	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	

Gotchas

TODO: operational traps.

See also

`pdfid` , `pdfid.py` , `pdf-parser.py`

pdf-parser.py

Kit: REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdf-parser.py 0.7.14 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://blog.didierstevens.com/programs/pdf-tools/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine elements of the PDF file.

Synopsis

```
pdf-parser.py [options] pdf-file|zip-file|url
```

Common invocations

```
# from cyberlab 10-malicious-documents
pdf-parser.py --version
# from cyberlab 10-malicious-documents
pdf-parser.py --search JavaScript exercise/sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --search OpenAction exercise/resume_sample.pdf
# from cyberlab 48-phishing-doc-case
pdf-parser.py --type /JavaScript exercise/resume_sample.pdf
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-m	—	Print manual	
--man	—	Print manual	
-s	SEARCH	string to search in indirect objects (except streams)	
--search	SEARCH	string to search in indirect objects (except streams)	

Flag	Argument	What it does	When you would use it
<code>-f</code>	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
<code>--filter</code>	—	pass stream object through filters (FlateDecode, ASCIIHexDecode, ASCII85Decode, LZWDecode and RunLengthDecode only)	
<code>-o</code>	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
<code>--object</code>	OBJECT	id(s) of indirect object(s) to select, use comma (,) to separate ids (version independent)	
<code>-r</code>	REFERENCE	id of indirect object being referenced (version independent)	
<code>--reference</code>	REFERENCE	id of indirect object being referenced (version independent)	
<code>-e</code>	ELEMENTS	type of elements to select (ctxsi)	
<code>--elements</code>	ELEMENTS	type of elements to select (ctxsi)	
<code>-w</code>	—	raw output for data and filters	
<code>--raw</code>	—	raw output for data and filters	
<code>-a</code>	—	display stats for pdf document	
<code>--stats</code>	—	display stats for pdf document	
<code>-t</code>	TYPE	type of indirect object to select	
<code>--type</code>	TYPE	type of indirect object to select	
<code>-0</code>	—	parse stream of /ObjStm objects	
<code>--objstm</code>	—	parse stream of /ObjStm objects	
<code>-v</code>	—	display malformed PDF elements	
<code>--verbose</code>	—	display malformed PDF elements	
<code>-x</code>	EXTRACT	filename to extract malformed content to	
<code>--extract</code>	EXTRACT	filename to extract malformed content to	
<code>-H</code>	—	display hash of objects	
<code>--hash</code>	—	display hash of objects	
<code>-n</code>	—	do not canonicalize the output	
<code>--nocanonicaledoutput</code>	—	do not canonicalize the output	
<code>-d</code>	DUMP	filename to dump stream content to	
<code>--dump</code>	DUMP	filename to dump stream content to	
<code>-D</code>	—	display debug info	
<code>--debug</code>	—	display debug info	

Flag	Argument	What it does	When you would use it
<code>-c</code>	—	display the content for objects without streams or with streams without filters	
<code>--content</code>	—	display the content for objects without streams or with streams without filters	
<code>--searchstream</code>	SEARCHSTREAM	string to search in streams	
<code>--unfiltered</code>	—	search in unfiltered streams	
<code>--casesensitive</code>	—	case sensitive search in streams	
<code>--regex</code>	—	use regex to search in streams	
<code>--overridingfilters</code>	OVERRIDINGFILTERS	override filters with given filters (use raw for the raw stream content)	
<code>-g</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generate</code>	—	generate a Python program that creates the parsed PDF file	
<code>--generateembedded</code>	GENERATEEMBEDDED	generate a Python program that embeds the selected indirect object as a file	
<code>-y</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yara</code>	YARA	YARA rule (or directory or @file) to check streams (can be used with option --unfiltered)	
<code>--yarastrings</code>	—	Print YARA strings	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>-k</code>	KEY	key to search in dictionaries	
<code>--key</code>	KEY	key to search in dictionaries	
<code>-j</code>	—	produce json output	
<code>--jsonoutput</code>	—	produce json output	

Gotchas

TODO: operational traps.

See also

`pdfid` , `pdfid.py` , `pdf-parser`

pdfid

Kit: Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Triage a PDF by counting the structural keywords that indicate active content.

Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url|@file] ...
```

Common invocations

```
# from cyberlab 37-pdf-analysis
pdfid --version
# from cyberlab 37-pdf-analysis
pdfid exercise/sample.pdf
```

Options

All 31 options parsed from the captured help text; 7 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-s	—	scan the given directory	Scan a directory of PDFs rather than a single file.
--scan	—	scan the given directory	
-a	—	display all the names	Display all keyword counts, including zero entries.
--all	—	display all the names	
-e	—	display extra data, like dates	Include extra keywords beyond the default set.
--extra	—	display extra data, like dates	
-f	—	force the scan of the file, even without proper %PDF header	

Flag	Argument	What it does	When you would use it
<code>--force</code>	—	force the scan of the file, even without proper %PDF header	
<code>-d</code>	—	disable JavaScript and auto launch	Disarm the PDF — neutralise /JS and /JavaScript. Produces a safe-to-open copy; never overwrite the original.
<code>--disarm</code>	—	disable JavaScript and auto launch	
<code>-p</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>--plugins</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>-c</code>	—	output csv data when using plugins	CSV output for batch triage.
<code>--csv</code>	—	output csv data when using plugins	
<code>-m</code>	MINIMUMSCORE	minimum score for plugin results output	Only report files scoring above a minimum, for a large sweep.
<code>--minimumscore</code>	MINIMUMSCORE	minimum score for plugin results output	
<code>-v</code>	—	verbose (will also raise caught exceptions)	
<code>--verbose</code>	—	verbose (will also raise caught exceptions)	
<code>-S</code>	SELECT	selection expression	
<code>--select</code>	SELECT	selection expression	
<code>-n</code>	—	supress output for counts equal to zero	Hide zero-count keywords to shorten the output.
<code>--nozero</code>	—	supress output for counts equal to zero	
<code>-o</code>	OUTPUT	output to log file	
<code>--output</code>	OUTPUT	output to log file	
<code>--pluginoptions</code>	PLUGINOPTIONS	options for the plugin	
<code>-l</code>	—	take filenames literally, no wildcard matching	
<code>--literalfilenames</code>	—	take filenames literally, no wildcard matching	
<code>--recursemdir</code>	—	Recurse directories (wildcards and here files (@...) allowed)	

Gotchas

- `pdfid` **counts keywords; it does not parse**. A high `/JS` count is a reason to look, not proof of malice — and a crafted PDF can hide content from it. Follow up with `pdf-parser`.
- `/OpenAction` plus `/JS` is the classic auto-execute pair worth prioritising in triage.

See also

`pdfid.py` , `pdf-parser` , `pdf-parser.py`

pdfid.py

Kit: REMnux · Kali Linux **Capability:** Analyse PDFs **Version:** pdfid.py 0.2.10 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://blog.didierstevens.com/programs/pdf-tools/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Identify suspicious elements of the PDF file.

Synopsis

```
pdfid.py [options] [pdf-file|zip-file|url|@file] ...
```

Common invocations

```
# from cyberlab 10-malicious-documents
pdfid.py --version
# from cyberlab 10-malicious-documents
pdfid.py exercise/sample.pdf
```

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-s	—	scan the given directory	
--scan	—	scan the given directory	
-a	—	display all the names	
--all	—	display all the names	
-e	—	display extra data, like dates	
--extra	—	display extra data, like dates	
-f	—	force the scan of the file, even without proper %PDF header	

Flag	Argument	What it does	When you would use it
<code>--force</code>	—	force the scan of the file, even without proper %PDF header	
<code>-d</code>	—	disable JavaScript and auto launch	
<code>--disarm</code>	—	disable JavaScript and auto launch	
<code>-p</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>--plugins</code>	PLUGINS	plugins to load (separate plugins with a comma , ; @file supported)	
<code>-c</code>	—	output csv data when using plugins	
<code>--csv</code>	—	output csv data when using plugins	
<code>-m</code>	MINIMUMSCORE	minimum score for plugin results output	
<code>--minimumscore</code>	MINIMUMSCORE	minimum score for plugin results output	
<code>-v</code>	—	verbose (will also raise caught exceptions)	
<code>--verbose</code>	—	verbose (will also raise caught exceptions)	
<code>-S</code>	SELECT	selection expression	
<code>--select</code>	SELECT	selection expression	
<code>-n</code>	—	supress output for counts equal to zero	
<code>--nozero</code>	—	supress output for counts equal to zero	
<code>-o</code>	OUTPUT	output to log file	
<code>--output</code>	OUTPUT	output to log file	
<code>--pluginoptions</code>	PLUGINOPTIONS	options for the plugin	
<code>-l</code>	—	take filenames literally, no wildcard matching	
<code>--literalfilenames</code>	—	take filenames literally, no wildcard matching	
<code>--recursedir</code>	—	Recurse directories (wildcards and here files (@...) allowed)	

Gotchas

TODO: operational traps.

See also

`pdfid`, `pdf-parser`, `pdf-parser.py`

pyxswf

Kit: REMnux **Capability:** Analyse Office documents and macros **Version:** pyxswf 0.54 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
pyxswf.py
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-x</code>	—	Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed	
<code>--extract</code>	—	Extracts the embedded SWF(s), names it MD5HASH.swf & saves it in the working dir. No addition args needed	
<code>-y</code>	—	Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed	
<code>--yara</code>	—	Scans the SWF(s) with yara. If the SWF(s) is compressed it will be deflated. No addition args needed	
<code>-s</code>	—	Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed	
<code>--md5scan</code>	—	Scans the SWF(s) for MD5 signatures. Please see func checkMD5 to define hashes. No addition args needed	
<code>-H</code>	—	Displays the SWFs file header. No addition args needed	

Flag	Argument	What it does	When you would use it
<code>--header</code>	—	Displays the SWFs file header. No addition args needed	
<code>-d</code>	—	Deflates compressed SWFS(s)	
<code>--decompress</code>	—	Deflates compressed SWFS(s)	
<code>-r</code>	PATH	Will recursively scan a directory for files that contain SWFs. Must provide path in quotes	
<code>--recdir</code>	PATH	Will recursively scan a directory for files that contain SWFs. Must provide path in quotes	
<code>-c</code>	—	Compresses the SWF using Zlib	
<code>--compress</code>	—	Compresses the SWF using Zlib	
<code>-o</code>	—	Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream	
<code>--ole</code>	—	Parse an OLE file (e.g. Word, Excel) to look for SWF in each stream	
<code>-f</code>	—	Parse an RTF file to look for SWF in each embedded object	
<code>--rtf</code>	—	Parse an RTF file to look for SWF in each embedded object	

Gotchas

TODO: operational traps.

See also

[olevba](#) , [oleid](#) , [oledump.py](#) , [olemeta](#) , [oletimes](#) , [olemap](#) , [oledir](#) , [olebrowse](#)

rtfobj

Kit: REMnux **Capability:** Analyse RTF documents and embedded objects **Version:** rtfobj 0.60.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Synopsis

```
rtfobj [options] <filename> [filename2 ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-r</code>	—	find files recursively in subdirectories.	
<code>-z</code>	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
<code>--zip</code>	ZIP_PASSWORD	if the file is a zip archive, open first file from it, using the provided password (requires Python 2.6+)	
<code>-f</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>--zipfname</code>	ZIP_FNAME	if the file is a zip archive, file(s) to be opened within the zip. Wildcards * and ? are supported. (default:*)	
<code>-l</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	
<code>--loglevel</code>	LOGLEVEL	logging level debug/info/warning/error/critical (default=warning)	

Flag	Argument	What it does	When you would use it
<code>-s</code>	SAVE_OBJECT	Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once.	
<code>--save</code>	SAVE_OBJECT	Save the object corresponding to the provided number to a file, for example "-s 2". Use "-s all" to save all objects at once.	
<code>-d</code>	OUTPUT_DIR	use specified directory to save output files.	

Gotchas

TODO: operational traps.

See also

`oleobj`

xlmdeobfuscator

Kit: REMnux **Capability:** Analyse Office documents and macros; Break simple obfuscation **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://github.com/DissectMalware/XLMMacroDeobfuscator>

[← Capability index](#) · [Kit tool list](#)

Purpose

Deobfuscate XLM macros (also known as Excel 4.0 macros) from Microsoft Office files.

Synopsis

```
xlmdeobfuscator [-h] [-c FILE_PATH] [-f FILE_PATH] [-n] [-x]
[--sort-formulas] [--defined-names] [-2]
[--with-ms-excel] [-s] [-d DAY]
[--output-formula-format OUTPUT_FORMULA_FORMAT]
[--extract-formula-format EXTRACT_FORMULA_FORMAT]
[--no-indent] [--silent] [--export-json FILE_PATH]
```

Common invocations

```
# from cyberlab 10-malicious-documents
xlmdeobfuscator --help | head -n 1
# from cyberlab 10-malicious-documents
xlmdeobfuscator --file exercise/sample.xls
```

Options

All 30 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-c</code>	FILE_PATH	Specify a config file (must be a valid JSON file)	
<code>--config-file</code>	FILE_PATH	Specify a config file (must be a valid JSON file)	
<code>-f</code>	FILE_PATH	The path of a XLSM file	
<code>--file</code>	FILE_PATH	The path of a XLSM file	

Flag	Argument	What it does	When you would use it
<code>-n</code>	—	Disable interactive shell	
<code>--noninteractive</code>	—	Disable interactive shell	
<code>-x</code>	—	Only extract cells without any emulation	
<code>--extract-only</code>	—	Only extract cells without any emulation	
<code>--sort-formulas</code>	—	Sort extracted formulas based on their cell address (requires -x)	
<code>--defined-names</code>	—	Extract all defined names	
<code>-2</code>	—	[Deprecated] Do not use MS Excel to process XLS files	
<code>--no-ms-excel</code>	—	[Deprecated] Do not use MS Excel to process XLS files	
<code>--with-ms-excel</code>	—	Use MS Excel to process XLS files	
<code>-s</code>	—	Open an XLM shell before interpreting the macros in the input	
<code>--start-with-shell</code>	—	Open an XLM shell before interpreting the macros in the input	
<code>-d</code>	DAY	Specify the day of month	
<code>--day</code>	DAY	Specify the day of month	
<code>--output-formula-format</code>	OUTPUT_FORMULA_FORMAT	Specify the format for output formulas ([[CELL-ADDR]], [[INT-FORMULA]], and [[STATUS]])	
<code>--extract-formula-format</code>	EXTRACT_FORMULA_FORMAT	Specify the format for extracted formulas ([[CELL-ADDR]], [[CELL-FORMULA]], and [[CELL-VALUE]])	
<code>--no-indent</code>	—	Do not show indent before formulas	
<code>--silent</code>	—	Do not print output	
<code>--export-json</code>	FILE_PATH	Export the output to JSON	
<code>--start-point</code>	CELL_ADDR	Start interpretation from a specific cell address	
<code>-p</code>	PASSWORD	Password to decrypt the protected document	
<code>--password</code>	PASSWORD	Password to decrypt the protected document	
<code>-o</code>	OUTPUT_LEVEL	Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands).	
<code>--output-level</code>	OUTPUT_LEVEL	Set the level of details to be shown (0:all commands, 1: commands no jump 2:important commands 3:strings in important commands).	
<code>--timeout</code>	N	stop emulation after N seconds (0: not interruption N>0: stop emulation after N seconds)	

Gotchas

TODO: operational traps.

See also

[olevba](#), [oleid](#), [oledump.py](#), [olemeta](#), [oletimes](#), [olemap](#), [oledir](#), [olebrowse](#)

7za

Kit: REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Captured:** cyberlab-aio
via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.7-zip.org>

[← Capability index](#) · [Kit tool list](#)

Purpose

Compress and decompress files using a variety of algorithms.

Synopsis

```
7za <command> [<switches>...] <archive_name> [<file_names>...] [@listfile]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 1 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-y</code>	—	assume Yes on all queries	

Gotchas

TODO: operational traps.

See also

`upx`, `die`, `diec`, `binwalk`, `unzip`, `msoffcrypto-tool`

base64dump.py

Kit: REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data

Version: base64dump.py 0.0.30 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#)

<https://blog.didierstevens.com/2020/07/03/update-base64dump-py-version-0-0-12/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Locate and decode strings encoded in Base64 and other common encodings.

Synopsis

```
base64dump.py [options] [file]
```

Common invocations

```
# from cyberlab 09-deobfuscation
base64dump.py --version 2>&1 | head -n 1
# from cyberlab 09-deobfuscation
base64dump.py exercise/encoded_payload.bin
# from cyberlab 09-deobfuscation
base64dump.py -s 1 -d exercise/encoded_payload.bin | head -c 200
# from cyberlab 25-cyberchef-recipes
base64dump.py --version
# from cyberlab 25-cyberchef-recipes
base64dump.py encoded_payload.ps1.txt
# from cyberlab 25-cyberchef-recipes
base64dump.py -s 1 -d encoded_payload.ps1.txt | xxd | head -n 2
```

Options

All 54 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--version	—	show program's version number and exit	
-h	—	show this help message and exit	
--help	—	show this help message and exit	

Flag	Argument	What it does	When you would use it
<code>-m</code>	—	Print manual	
<code>--man</code>	—	Print manual	
<code>-e</code>	ENCODING	select encoding to use, use "all" to try all encodings (default base64)	
<code>--encoding</code>	ENCODING	select encoding to use, use "all" to try all encodings (default base64)	
<code>-s</code>	SELECT	select item nr for dumping (a for all, L for largest)	
<code>--select</code>	SELECT	select item nr for dumping (a for all, L for largest)	
<code>-d</code>	—	perform dump	
<code>--dump</code>	—	perform dump	
<code>-x</code>	—	perform hex dump	
<code>--hexdump</code>	—	perform hex dump	
<code>-a</code>	—	perform ascii dump	
<code>--asciidump</code>	—	perform ascii dump	
<code>-A</code>	—	perform ascii dump with RLE	
<code>--asciidumprle</code>	—	perform ascii dump with RLE	
<code>-S</code>	—	perform strings dump	
<code>--strings</code>	—	perform strings dump	
<code>-t</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	
<code>--translate</code>	TRANSLATE	string translation, like utf16 or .decode("utf8")	
<code>-n</code>	NUMBER	minimum number of bytes in decoded data	
<code>--number</code>	NUMBER	minimum number of bytes in decoded data	
<code>-c</code>	CUT	cut data	
<code>--cut</code>	CUT	cut data	
<code>-w</code>	—	ignore whitespace	
<code>--ignorewhitespace</code>	—	ignore whitespace	
<code>-u</code>	—	do not repeat identical decoded data	
<code>--unique</code>	—	do not repeat identical decoded data	
<code>-z</code>	—	ignore null (zero) bytes	
<code>--ignorenullbytes</code>	—	ignore null (zero) bytes	
<code>-i</code>	IGNORE	characters to ignore	
<code>--ignore</code>	IGNORE	characters to ignore	
<code>-I</code>	IGNOREHEX	characters to ignore (hexadecimal)	

Flag	Argument	What it does	When you would use it
<code>--ignorehex</code>	IGNOREHEX	characters to ignore (hexadecimal)	
<code>-y</code>	YARA	YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option)	
<code>--yara</code>	YARA	YARA rule-file, @file, directory or #rule to check data (YARA search doesn't work with -s option)	
<code>-D</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoders</code>	DECODERS	decoders to load (separate decoders with a comma , ; @file supported)	
<code>--decoderoptions</code>	DECODEROPTIONS	options for the decoder	
<code>--decoderdir</code>	DECODERDIR	directory for the decoder	
<code>--yarastrings</code>	—	Print YARA strings	
<code>-V</code>	—	verbose output with decoder errors and YARA rules	
<code>--verbose</code>	—	verbose output with decoder errors and YARA rules	
<code>--jsonoutput</code>	—	produce json output	
<code>--jsoninput</code>	—	consume json input	
<code>-T</code>	—	do head & tail	
<code>--headtail</code>	—	do head & tail	
<code>-p</code>	PREPROCESS	Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)	
<code>--preprocess</code>	PREPROCESS	Python function to process encodings prior to decoding (like L4, lambda bytes: bytes[:-1], ...)	
<code>-P</code>	POSTPROCESS	Python function to post-process decodings	
<code>--postprocess</code>	POSTPROCESS	Python function to post-process decodings	
<code>--sort</code>	SORT	Sort output	
<code>--stats</code>	—	provide statistics for encoded data	

Gotchas

TODO: operational traps.

See also

[floss](#), [strings](#), [numbers-to-string.py](#), [cyberchef](#), [rax2](#), [xxd](#), [openssl](#)

capa

Kit: REMnux · FLARE-VM **Capability:** Identify capabilities in a binary **Version:** capa 9.4.0 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://github.com/mandiant/capa>

← [Capability index](#) · [Kit tool list](#)

Purpose

Identify the capabilities of a binary by matching rules against its disassembly, mapped to MITRE ATT&CK.

Synopsis

```
capa [-h] [--version] [-v] [-vv] [-d] [-q]
[--color {auto,always,never}]
[-f {auto,pe,dotnet,elf,sc32,sc64,cape,drakvuf,vmray,freeze,binexport2,binja_database}]
[-b {auto,vivisect,ida,pefile,binja,dotnet,binexport2,ghidra,freeze,cape,drakvuf,vmray}]
[--restrict-to-functions RESTRICT_TO_FUNCTIONS]
[--restrict-to-processes RESTRICT_TO_PROCESSES]
```

Common invocations

```
# from cyberlab 08-malware-static-triage
capa --version
# from cyberlab 08-malware-static-triage
capa exercise/sample.bin | head -n 40
# from cyberlab 21-yara-authoring
capa -q /bin/ls | head -n 30
# from cyberlab 12-static-re
capa exercise\sample_static.exe
# from cyberlab 12-static-re
capa -v exercise\sample_static.exe
# from cyberlab 42-floss-strings
capa .\exercise\sample.exe
# from cyberlab 42-floss-strings
capa -v .\exercise\sample.exe | Select-String -Pattern "XOR|encode|T1027"
# from cyberlab 46-cutter-windows
capa -v .\exercise\sample.exe
```

Options

All 21 options parsed from the captured help text; 8 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>--version</code>	—	show program's version number and exit	
<code>-v</code>	—	enable verbose result document (no effect with <code>--json</code>)	Verbose: show which rules matched and where.
<code>--verbose</code>	—	enable verbose result document (no effect with <code>--json</code>)	
<code>-d</code>	—	enable debugging output on STDERR	
<code>--debug</code>	—	enable debugging output on STDERR	
<code>-q</code>	—	disable all output but errors	Quiet output for scripted use.
<code>--quiet</code>	—	disable all output but errors	
<code>--color</code>	auto,always,never	enable ANSI color codes in results, default: only during interactive session	
<code>--restrict-to-functions</code>	RESTRICT_TO_FUNCTIONS	provide a list of comma-separated function virtual addresses to analyze (static analysis).	Analyse only the named functions — useful on a large binary you have already triaged.
<code>--restrict-to-processes</code>	RESTRICT_TO_PROCESSES	provide a list of comma-separated process IDs to analyze (dynamic analysis).	
<code>--os</code>	auto,linux,macos,windows	select sample OS: auto (detect OS automatically - default), linux, macos, windows	Force the target OS when auto-detection is wrong.
<code>-r</code>	RULES	path to rule file or directory, use embedded rules by default	Use an alternate rules directory.
<code>--rules</code>	RULES	path to rule file or directory, use embedded rules by default	
<code>-s</code>	SIGNATURES	path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default	Point at the library-identification signatures.
<code>--signatures</code>	SIGNATURES	path to .sig/.pat file or directory used to identify library functions, use embedded signatures by default	
<code>-t</code>	TAG	filter on rule meta field values	Only run rules carrying a given tag.
<code>--tag</code>	TAG	filter on rule meta field values	
<code>-j</code>	—	emit JSON instead of text	JSON output, for feeding a triage pipeline.
<code>--json</code>	—	emit JSON instead of text	

- capa reports *capabilities*, not verdicts. 'Can encrypt data' describes a backup tool as readily as ransomware — it narrows where to look.
- Packed samples yield almost nothing useful. Unpack first, or capa will just describe the packer stub.

See also

floss

clamscan

Kit: REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.clamav.net>

[← Capability index](#) · [Kit tool list](#)

Purpose

Scan files and directories with the ClamAV signature engine.

Common invocations

```
# from cyberlab 34-clamav-scanning
clamscan --version
# from cyberlab 34-clamav-scanning
clamscan -r -i /tmp/samples
# from cyberlab 34-clamav-scanning
clamscan /tmp/eicar.com
# from cyberlab 34-clamav-scanning
clamscan exercise/sample.txt
# from cyberlab 34-clamav-scanning
clamscan --database=exercise/lab34.yar exercise/sample.txt
# from cyberlab 51-linux-triage-workflow
clamscan -r --infected --stdout be_out recovered_file.bin
# from cyberlab 51-linux-triage-workflow
clamscan --infected --stdout exercise/extract/eicar.com
```

Options

All 53 options parsed from the captured help text; 11 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Show this help	
<code>-h</code>	—	Show this help	
<code>--version</code>	—	Print version number	
<code>-V</code>	—	Print version number	
<code>--verbose</code>	—	Be verbose	
<code>-v</code>	—	Be verbose	
<code>--archive-verbose</code>	—	Show filenames inside scanned archives	

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Show filenames inside scanned archives	
<code>--debug</code>	—	Enable libclamav's debug messages	
<code>--quiet</code>	—	Only output error messages	Only report errors, for scripted runs.
<code>--stdout</code>	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
<code>--no-summary</code>	—	Disable summary at end of scanning	Suppress the trailing summary block when parsing output.
<code>--infected</code>	—	Only print infected files	
<code>-i</code>	—	Only print infected files	Print infected files only — the flag that makes output usable.
<code>--bell</code>	—	Sound bell on virus detection	Audible alert on detection; useful during a long live scan.
<code>--tempdir</code>	DIRECTORY	Create temporary files in DIRECTORY	Redirect temp extraction; important when scanning large archives on a small root filesystem.
<code>--database</code>	FILE	-d FILE/DIR Load virus database from FILE or load all supported db files from DIR	
<code>--fail-if-cvd-older-than</code>	days	Return with a nonzero error code if virus database outdated.	
<code>--log</code>	FILE	-l FILE Save scan report to FILE	Write results to a log file for the case record.
<code>--file-list</code>	FILE	-f FILE Scan files from FILE	Scan exactly the paths listed in a file.
<code>--move</code>	DIRECTORY	Move infected files into DIRECTORY	Quarantine detections into a directory. Never point this at evidence you must preserve.
<code>--copy</code>	DIRECTORY	Copy infected files into DIRECTORY	Copy rather than move detections, preserving the original.
<code>--exclude</code>	REGEX	Don't scan file names matching REGEX	Skip paths matching a regex — noisy or irrelevant trees.
<code>--exclude-dir</code>	REGEX	Don't scan directories matching REGEX	
<code>--include</code>	REGEX	Only scan file names matching REGEX	Restrict the scan to paths matching a regex.
<code>--include-dir</code>	REGEX	Only scan directories matching REGEX	
<code>--bytecode-timeout</code>	N	Set bytecode timeout (in milliseconds)	
<code>--exclude-pua</code>	CAT	Skip PUA sigs of category CAT	
<code>--include-pua</code>	CAT	Load PUA sigs of category CAT	
<code>--structured-ssn-format</code>	X	SSN format (0=normal,1=stripped,2=both)	

Flag	Argument	What it does	When you would use it
<code>--structured-ssn-count</code>	N	Min SSN count to generate a detect	
<code>--structured-cc-count</code>	N	Min CC count to generate a detect	
<code>--structured-cc-mode</code>	X	CC mode (0=credit debit and private label, 1=credit cards only)	
<code>--nocerts</code>	—	Disable authenticode certificate chain verification in PE files	
<code>--dumpcerts</code>	—	Dump authenticode certificate chain in PE files	
<code>--max-scantime</code>	#n	Scan time longer than this will be skipped and assumed clean (milliseconds)	
<code>--max-filesize</code>	#n	Files larger than this will be skipped and assumed clean	
<code>--max-scansize</code>	#n	The maximum amount of data to scan for each container file (**)	
<code>--max-files</code>	#n	The maximum number of files to scan for each container file (**)	
<code>--max-recursion</code>	#n	Maximum archive recursion level for container file (**)	
<code>--max-dir-recursion</code>	#n	Maximum directory recursion level	
<code>--max-embeddedpe</code>	#n	Maximum size file to check for embedded PE	
<code>--max-htmlnormalize</code>	#n	Maximum size of HTML file to normalize	
<code>--max-htmlnotags</code>	#n	Maximum size of normalized HTML file to scan	
<code>--max-scriptnormalize</code>	#n	Maximum size of script file to normalize	
<code>--max-ziptypercg</code>	#n	Maximum size zip to type reanalyze	
<code>--max-partitions</code>	#n	Maximum number of partitions in disk image to be scanned	
<code>--max-iconspe</code>	#n	Maximum number of icons in PE file to be scanned	
<code>--max-rechwp3</code>	#n	Maximum recursive calls to HWP3 parsing function	
<code>--pcre-match-limit</code>	#n	Maximum calls to the PCRE match function.	
<code>--pcre-recmatch-limit</code>	#n	Maximum recursive calls to the PCRE match function.	
<code>--pcre-max-filesize</code>	#n	Maximum size file to perform PCRE subsig matching.	

Flag	Argument	What it does	When you would use it
<code>--disable-cache</code>	—	Disable caching and cache checks for hash sums of scanned files.	

Gotchas

- Without `-i` the output lists every clean file too, which buries the hits in a large tree.
- `--move` and `--remove` MUTATE evidence. On an investigation, scan a copy or use `--copy` — a quarantine action on original evidence is not defensible.
- Signatures must be current for the result to mean anything; a stale database yields a comfortable and worthless clean result.

See also

`yara`, `yarac`, `freshclam`, `sigtool`

die

Kit: FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** Detect It Easy v3.10
Captured: `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

qt.qpa.xcb: could not connect to display

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`file`, `diec`, `upx`, `binwalk`, `7za`, `unzip`

Detect It Easy (GUI)

Kit: REMnux · FLARE-VM **Capability:** Identify a file's format, compiler and packer **Version:** Detect It Easy v3.10 (x86_64) **CLI counterpart:** `diec` **Captured:** `C:\Tools\die\die.exe` on FLARE-VM, 2026-08-02 — control tree in `capture/gui/die/die.tree.txt`

Worked example of the GUI format. Every control below was read from the application's accessibility tree, not from a screenshot or from memory. See [docs/GUI-FORMAT.md](#).

Purpose

Identify what a binary actually is — its format, architecture, compiler, linker and any packer or protector — before committing to a deeper analysis path.

Use the GUI or the CLI?

`diec` does the same detection and is the better choice for scripting, batch work and anything reproducible. The GUI earns its place for three things the CLI cannot do:

- browsing the **Signatures** that produced a verdict, when you need to know *why* something was flagged
- switching detection engines and re-scanning interactively to compare results
- the **Advanced** panes — entropy, section and hex views — which are exploratory by nature

Anything you would put in a report or a pipeline belongs in `diec`.

Window map

 Detect It Easy main window

One window, `GuiMainWindow`, in three regions:

Region	Automation container	What it is for
File name	<code>groupBoxFileName</code>	Choosing the target file
Formats	<code>widgetFormats</code>	The detection result, and the scan controls
Extra	<code>groupBoxExtra</code>	Advanced views, options and about

Controls

All 45 nodes come from the capture; the leaf controls are listed here.

Control	Type	AutomationId	What it does
File name	Edit	lineEditFileName	Path of the file being examined
...	Button	toolButtonOpenFile	Open the file chooser
>	Button	toolButtonRecentFiles	Recently opened files
File type	ComboBox	comboBoxFileType	Detected container format; selectable when a file has more than one interpretation
File size	Edit	lineEditFileSize	Size of the target, read-only
Scan	ComboBox	comboBoxScanEngine	Which detection engine runs — see below
Endianness	Edit	lineEditEndianness	Byte order, read-only
Mode	Edit	lineEditMode	32- or 64-bit, read-only
Architecture	Edit	lineEditArch	Target CPU, read-only
Type	Edit	lineEditType	Executable, DLL, driver, read-only
Signatures	Button	pushButtonDieSignatures	Open the signature browser — the "why was this flagged" view
Directory	Button	pushButtonDieScanDirectory	Scan a whole directory instead of one file
Log	Button	pushButtonDieLog	Scan log
Scan	Button	pushButtonDieScanStart	Run the scan
0 msec	Button	toolButtonElapsedTime	Elapsed time of the last scan
Advanced	CheckBox	checkBoxAdvanced	Reveal entropy, sections and hex views
Shortcuts	Button	toolButtonShortcuts	Keyboard shortcut list
Options	Button	toolButtonOptions	Settings
About	Button	toolButtonAbout	Version and credits
Exit	Button	toolButtonExit	Close

Two combo boxes carry no label in the tree and sit beside **Signatures**: `comboBoxFlags` and `comboBoxDatabases`, which select the signature flags and the signature database in use.

Scan engine values

Enumerated from `comboBoxScanEngine`, so this list is the actual set of choices in this build:

Value	When to use it
Automatic	The default; picks an engine from the detected format
Detect It Easy (DiE)	DiE's own signature set — the packer and compiler detection it is known for
Nauz File Detector (NFD)	A second opinion from a different signature set, when DiE returns nothing
Yara rules	Run YARA rules instead of signatures, when you have rules of your own

Identify an unknown binary

1. Select the file with ..., or drag it onto **File name**.

2. Leave **Scan** on **Automatic** for a first pass.
3. Press **Scan**.
4. Read **Architecture**, **Mode** and **Type** — these are structural facts, not guesses.
5. Press **Signatures** to see which signature produced the verdict.
6. If nothing is detected, switch **Scan** to **Nauz File Detector (NFD)** and scan again.
7. Tick **Advanced** and check entropy before concluding a file is unpacked.

Gotchas

- A packer verdict is a **signature match, not proof**. Packers are routinely misidentified, and a custom packer will not match anything at all. High entropy with no signature is the interesting case, not the boring one.
- "Nothing detected" is a statement about the signature set, not about the file. Try the other engine before concluding anything.
- The read-only fields are parsed from headers, which malware edits freely. **Type** saying `DLL` means the header says DLL.
- The GUI scans one file at a time unless you use **Directory**. For a corpus, use `diec` — this window is for looking at one sample closely.

diec

Kit: REMnux · FLARE-VM **Capability:** Identify what a file actually is; Detect and reverse packing **Version:** die 3.10
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://github.com/horsicq/Detect-It-Easy>

← [Capability index](#) · [Kit tool list](#)

Purpose

Determine types of files and examine file properties.

Synopsis

```
diec [options] target
```

Common invocations

```
# from cyberlab 08-malware-static-triage
diec --version
# from cyberlab 08-malware-static-triage
diec -j exercise/sample.bin | head -n 40
# from cyberlab 32-remnux-static-triage
diec sample.exe
# from cyberlab 32-remnux-static-triage
diec exercise/sample.exe | grep -iE "compiler|packer|linker"
```

Options

All 47 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	Displays help on commandline options.	
--help	—	Displays help on commandline options.	
--help-all	—	Displays help including Qt specific options.	
-v	—	Displays version information.	
--version	—	Displays version information.	
-r	—	Recursive scan.	
--recursivescan	—	Recursive scan.	
-d	—	Deep scan.	

Flag	Argument	What it does	When you would use it
<code>--deepscan</code>	—	Deep scan.	
<code>-u</code>	—	Heuristic scan.	
<code>--heuristicscan</code>	—	Heuristic scan.	
<code>-b</code>	—	Verbose.	
<code>--verbose</code>	—	Verbose.	
<code>-g</code>	—	Aggressive scan.	
<code>--aggressivescan</code>	—	Aggressive scan.	
<code>-a</code>	—	Scan all types.	
<code>--alltypes</code>	—	Scan all types.	
<code>-l</code>	—	Profiling signatures.	
<code>--profiling</code>	—	Profiling signatures.	
<code>-U</code>	—	Hide unknown.	
<code>--hideunknown</code>	—	Hide unknown.	
<code>-e</code>	—	Show entropy.	
<code>--entropy</code>	—	Show entropy.	
<code>-i</code>	—	Show file info.	
<code>--info</code>	—	Show file info.	
<code>-S</code>	method	Special file info for . For example -S "Hash" or -S "Hash#MD5".	
<code>--special</code>	method	Special file info for . For example -S "Hash" or -S "Hash#MD5".	
<code>-x</code>	—	Result as XML.	
<code>--xml</code>	—	Result as XML.	
<code>-j</code>	—	Result as JSON.	
<code>--json</code>	—	Result as JSON.	
<code>-c</code>	—	Result as CSV.	
<code>--csv</code>	—	Result as CSV.	
<code>-t</code>	—	Result as TSV.	
<code>--tsv</code>	—	Result as TSV.	
<code>-p</code>	—	Result as Plain Text.	
<code>--plaintext</code>	—	Result as Plain Text.	
<code>-D</code>	path	Set database.	
<code>--database</code>	path	Set database.	
<code>-E</code>	path	Set extra database.	
<code>--extradatabase</code>	path	Set extra database.	
<code>-C</code>	path	Set custom database.	

Flag	Argument	What it does	When you would use it
<code>--customdatabase</code>	path	Set custom database.	
<code>-s</code>	—	Show database.	
<code>--showdatabase</code>	—	Show database.	
<code>-m</code>	—	Show all special methods for the file.	
<code>--showmethods</code>	—	Show all special methods for the file.	

Gotchas

TODO: operational traps.

See also

`file` , `die` , `upx` , `binwalk` , `7za` , `unzip`

floss

Kit: REMnux · FLARE-VM **Capability:** Identify capabilities in a binary; Extract strings, including obfuscated ones; Break simple obfuscation **Version:** floss 3.1.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs: https://github.com/mandiant/flare-floss](https://github.com/mandiant/flare-floss)

← [Capability index](#) · [Kit tool list](#)

Purpose

Extract strings from a binary, including obfuscated strings that `strings` cannot see.

Synopsis

```
floss [-h] [-H] [-n MIN_LENGTH]
[--no {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[--only {static,stack,tight,decoded} [{static,stack,tight,decoded} ...]]
[-j] [-v] [-d] [-q] [--color {auto,always,never}]
sample
```

Common invocations

```
# from cyberlab 08-malware-static-triage
floss --version

# from cyberlab 08-malware-static-triage
floss --no-color exercise/sample.bin | head -n 60

# from cyberlab 12-static-re
floss --no-static exercise\sample_static.exe

# from cyberlab 12-static-re
floss exercise\sample_static.exe

# from cyberlab 42-floss-strings
floss --help

# from cyberlab 42-floss-strings
floss .\exercise\sample.exe

# from cyberlab 42-floss-strings
floss --json --only decoded stack .\exercise\sample.exe > .\exercise\floss.json

# from cyberlab 42-floss-strings
floss --only static .\exercise\sample.exe | Select-String "HELLO"
```

Options

All 14 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-H</code>	—	show advanced options and exit	
<code>-n</code>	MIN_LENGTH	minimum string length	Minimum string length; raise it to cut noise on large binaries.
<code>--minimum-length</code>	MIN_LENGTH	minimum string length	
<code>-j</code>	—	emit JSON instead of text	Emit JSON, for feeding results into other tooling.
<code>--json</code>	—	emit JSON instead of text	
<code>-v</code>	—	enable verbose results, e.g. including function offsets (does not affect JSON output)	Verbose progress; FLOSS emulation can be slow and silent otherwise.
<code>--verbose</code>	—	enable verbose results, e.g. including function offsets (does not affect JSON output)	
<code>-d</code>	—	enable debugging output on STDERR, specify multiple times to increase verbosity	
<code>--debug</code>	—	enable debugging output on STDERR, specify multiple times to increase verbosity	
<code>-q</code>	—	disable all status output on STDOUT except fatal errors	Quiet mode — just the strings, for piping.
<code>--quiet</code>	—	disable all status output on STDOUT except fatal errors	
<code>--color</code>	auto,always,never	enable ANSI color codes in results, default: only during interactive session	Control colour output when redirecting to a file.

Gotchas

- FLOSS emulates code to recover decoded strings, so it is far slower than `strings` and can take minutes on a large sample. Budget for it.
- It is built for Windows PE files. Pointing it at an ELF or a script gives you little more than plain `strings` would.

See also

`capa`, `strings`, `base64dump.py`, `numbers-to-string.py`, `xortool`, `xlmdeobfuscator`

freshclam

Kit: REMnux · SIFT Workstation **Capability:** Scan with signatures for known-bad **Version:** ClamAV 1.4.3 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.clamav.net>

[← Capability index](#) · [Kit tool list](#)

Purpose

Scan files for malware signatures.

Common invocations

```
# from cyberlab 34-clamav-scanning
freshclam --version
# from cyberlab 34-clamav-scanning
sudo freshclam
```

Options

All 27 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Show this help	
<code>-h</code>	—	Show this help	
<code>--version</code>	—	Print version number and exit	
<code>-V</code>	—	Print version number and exit	
<code>--verbose</code>	—	Be verbose	
<code>-v</code>	—	Be verbose	
<code>--debug</code>	—	Enable debug messages	
<code>--quiet</code>	—	Only output error messages	
<code>--no-warnings</code>	—	Don't print and log warnings	
<code>--stdout</code>	—	Write to stdout instead of stderr. Does not affect 'debug' messages.	
<code>--show-progress</code>	—	Show download progress percentage	
<code>--config-file</code>	FILE	Read configuration from FILE.	

Flag	Argument	What it does	When you would use it
<code>--log</code>	FILE	-l FILE Log into FILE	
<code>--daemon</code>	—	Run in daemon mode	
<code>-d</code>	—	Run in daemon mode	
<code>--pid</code>	FILE	-p FILE Write the daemon's pid to FILE	
<code>--foreground</code>	—	Don't fork into background (for use in daemon mode).	
<code>-F</code>	—	Don't fork into background (for use in daemon mode).	
<code>--user</code>	USER	-u USER Run as USER	
<code>--no-dns</code>	—	Force old non-DNS verification method	
<code>--checks</code>	#n	-c #n Number of checks per day, 1 <= n <= 50	
<code>--datadir</code>	DIRECTORY	Download new databases into DIRECTORY NOTE: DIRECTORY must already exist, be an absolute path, and be writeable by freshclam and readable by clamd/clamscan. --daemon-notify[=/path/clamd.conf] Send REL	
<code>--local-address</code>	IP	-a IP Bind to IP for HTTP downloads	
<code>--on-update-execute</code>	COMMAND	Execute COMMAND after successful update. Use EXIT_1 to return 1 after successful database update.	
<code>--on-error-execute</code>	COMMAND	Execute COMMAND if errors occurred	
<code>--on-outdated-execute</code>	COMMAND	Execute COMMAND when software is outdated	
<code>--update-db</code>	DBNAME	Only update database DBNAME	

Gotchas

TODO: operational traps.

See also

`yara`, `yarac`, `clamscan`, `sigtool`

numbers-to-string.py

Kit: REMnux **Capability:** Extract strings, including obfuscated ones; Decode, decrypt or transform encoded data

Version: numbers-to-string.py 0.0.11 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs:](#)

<https://blog.didierstevens.com/2020/12/12/update-numbers-to-string-py-version-0-0-11/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Translate number sequences into ASCII characters.

Synopsis

```
numbers-to-string.py [options] [expression [[@]file ...]]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	show program's version number and exit	
<code>-h</code>	—	show this help message and exit	
<code>--help</code>	—	show this help message and exit	
<code>-m</code>	—	Print manual	
<code>--man</code>	—	Print manual	
<code>-o</code>	OUTPUT	Output to file	
<code>--output</code>	OUTPUT	Output to file	
<code>-e</code>	—	Generate error when error occurs in Python expression	
<code>--error</code>	—	Generate error when error occurs in Python expression	
<code>-i</code>	—	Ignore numbers greater than 255	
<code>--ignore</code>	—	Ignore numbers greater than 255	
<code>-n</code>	NUMBER	Minimum number of numbers (3 by default)	
<code>--number</code>	NUMBER	Minimum number of numbers (3 by default)	

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	Verbose	
<code>--verbose</code>	—	Verbose	
<code>-j</code>	—	Join output	
<code>--join</code>	—	Join output	
<code>-S</code>	—	Generate statistics	
<code>--statistics</code>	—	Generate statistics	
<code>-t</code>	TABLE	Translation table	
<code>--table</code>	TABLE	Translation table	
<code>-T</code>	BEGINTABLE	Begin translation table	
<code>--begintable</code>	BEGINTABLE	Begin translation table	
<code>-b</code>	—	Produce binary output	
<code>--binary</code>	—	Produce binary output	
<code>-l</code>	LINE	Select line with given number	
<code>--line</code>	LINE	Select line with given number	
<code>--grep</code>	GREP	Grep expression	
<code>--grepoptions</code>	GREOPTIONS	Grep options	
<code>--begin</code>	BEGIN	Begin substring	
<code>--end</code>	END	End substring	

Gotchas

TODO: operational traps.

See also

`floss`, `strings`, `base64dump.py`, `cyberchef`, `rax2`, `xxd`, `openssl`

objdump

Kit: REMnux **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary **Version:** GNU objdump (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs:](#) <https://en.wikipedia.org/wiki/Objdump>

← [Capability index](#) · [Kit tool list](#)

Purpose

Disassemble binary files.

Synopsis

```
objdump <option(s)> <file(s)>
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 78 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	Display archive header information	
<code>--archive-headers</code>	—	Display archive header information	
<code>-f</code>	—	Display the contents of the overall file header	
<code>--file-headers</code>	—	Display the contents of the overall file header	
<code>-p</code>	—	Display object format specific file header contents	
<code>--private-headers</code>	—	Display object format specific file header contents	
<code>-x</code>	—	Display the contents of all headers	
<code>--all-headers</code>	—	Display the contents of all headers	
<code>-d</code>	—	Display assembler contents of executable sections	
<code>--disassemble</code>	—	Display assembler contents of executable sections	
<code>-D</code>	—	Display assembler contents of all sections	

Flag	Argument	What it does	When you would use it
<code>--disassemble-all</code>	—	Display assembler contents of all sections	
<code>-S</code>	—	Intermix source code with disassembly <code>--source-comment[=]</code> Prefix lines of source code with	
<code>--source</code>	—	Intermix source code with disassembly <code>--source-comment[=]</code> Prefix lines of source code with	
<code>-s</code>	—	Display the full contents of all sections requested	
<code>--full-contents</code>	—	Display the full contents of all sections requested	
<code>-g</code>	—	Display debug information in object file	
<code>--debugging</code>	—	Display debug information in object file	
<code>-e</code>	—	Display debug information using ctags style	
<code>--debugging-tags</code>	—	Display debug information using ctags style	
<code>-G</code>	—	Display (in raw form) any STABS info in the file	
<code>--stabs</code>	—	Display (in raw form) any STABS info in the file	
<code>-L</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies <code>-WK</code>)	
<code>--process-links</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies <code>-WK</code>)	
<code>--ctf</code>	SECTION (optional)	Display CTF info from SECTION, (default <code>`.ctf'</code>)	
<code>--sframe</code>	SECTION (optional)	Display SFrame info from SECTION, (default <code>`.sframe'</code>)	
<code>-t</code>	—	Display the contents of the symbol table(s)	
<code>--syms</code>	—	Display the contents of the symbol table(s)	
<code>-T</code>	—	Display the contents of the dynamic symbol table	
<code>--dynamic-syms</code>	—	Display the contents of the dynamic symbol table	
<code>-r</code>	—	Display the relocation entries in the file	
<code>--reloc</code>	—	Display the relocation entries in the file	
<code>-R</code>	—	Display the dynamic relocation entries in the file	
<code>--dynamic-reloc</code>	—	Display the dynamic relocation entries in the file	
<code>-v</code>	—	Display this program's version number	
<code>--version</code>	—	Display this program's version number	
<code>-i</code>	—	List object formats and architectures supported	
<code>--info</code>	—	List object formats and architectures supported	
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	

Flag	Argument	What it does	When you would use it
<code>-b</code>	BFDNAME	Specify the target object format as BFDNAME	
<code>--target</code>	BFDNAME	Specify the target object format as BFDNAME	
<code>-m</code>	MACHINE	Specify the target architecture as MACHINE	
<code>--architecture</code>	MACHINE	Specify the target architecture as MACHINE	
<code>-j</code>	NAME	Only display information for section NAME	
<code>--section</code>	NAME	Only display information for section NAME	
<code>--file-start-context</code>	—	Include context from start of file (with -S)	
<code>-I</code>	DIR	Add DIR to search list for source files	
<code>--include</code>	DIR	Add DIR to search list for source files	
<code>-l</code>	—	Include line numbers and filenames in output	
<code>--line-numbers</code>	—	Include line numbers and filenames in output	
<code>-F</code>	—	Include file offsets when displaying information	
<code>--file-offsets</code>	—	Include file offsets when displaying information	
<code>-C</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	
<code>--demangle</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	
<code>--recurse-limit</code>	—	Enable a limit on recursion whilst demangling (default)	
<code>--no-recurse-limit</code>	—	Disable a limit on recursion whilst demangling	
<code>-w</code>	—	Format output for more than 80 columns	
<code>--wide</code>	—	Format output for more than 80 columns	
<code>-z</code>	—	Do not skip blocks of zeroes when disassembling	
<code>--disassemble-zeroes</code>	—	Do not skip blocks of zeroes when disassembling	
<code>--start-address</code>	ADDR	Only process data whose address is >= ADDR	
<code>--stop-address</code>	ADDR	Only process data whose address is < ADDR	
<code>--no-addresses</code>	—	Do not print address alongside disassembly	
<code>--prefix-addresses</code>	—	Print complete address alongside disassembly	
<code>--insn-width</code>	WIDTH	Display WIDTH bytes on a single line for -d	
<code>--adjust-vma</code>	OFFSET	Add OFFSET to all displayed section addresses	
<code>--show-all-symbols</code>	—	When disassembling, display all symbols at a given address	
<code>--special-syms</code>	—	Include special symbols in symbol dumps	

Flag	Argument	What it does	When you would use it
<code>--inlines</code>	—	Print all inlines for source line (with -l)	
<code>--prefix</code>	PREFIX	Add PREFIX to absolute paths for -S	
<code>--prefix-strip</code>	LEVEL	Strip initial directory names for -S	
<code>--dwarf-depth</code>	N	Do not display DIEs at depth N or greater	
<code>--dwarf-start</code>	N	Display DIEs starting at offset N	
<code>--dwarf-check</code>	—	Make additional dwarf consistency checks.	
<code>--ctf-parent</code>	NAME	Use CTF archive member NAME as the CTF parent	
<code>--visualize-jumps</code>	—	Visualize jumps by drawing ASCII art lines	
<code>--disassembler-color</code>	off	Disable disassembler color output. (default)	

Gotchas

TODO: operational traps.

See also

`rabin2`, `readelf`, `readelf.py`, `r2`, `rasm2`, `vivbin`, `vdbbin`

rabin2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Inspect PE / ELF structure; Disassemble and explore a binary
Version: rabin2 6.1.9 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw Docs:](#)
<https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rabin2 [-AcdeEghHiIjJlLMqrRsSuvVxzZ] [-@ at] [-a arch] [-b bits] [-B addr]
[-C F:C:D] [-f str] [-m addr] [-n str] [-N m:M] [-P[-P] pdb]
[-o str] [-O help] [-k query] [-D lang mangledsymbol] file
```

Common invocations

```
# from cyberlab 35-radare2-intro
rabin2 (shipped with radare2) lists strings from data sections; r2 disassembles main
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello
# from cyberlab 35-radare2-intro
rabin2 -z exercise/hello | grep r2rules
```

Options

All 47 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-@	addr	show section, symbol or import at addr	
-A	—	list sub-binaries and their arch-bits pairs	
-a	arch	set arch (x86, arm, .. or _)	
-b	bits	set bits (32, 64 ...)	
-B	addr	override base address (pie bins)	
-c	—	list classes	

Flag	Argument	What it does	When you would use it
-C	fmt:C:D	create [elf,mach0,pe] with Code and Data hexpairs (see -a)	
-d	—	show debug/dwarf information	
-e	—	program entrypoint	
-E	—	globally exportable symbols	
-f	str	select sub-bin named str	
-F	binfmt	force to use that bin plugin (ignore header check)	
-g	—	same as -SMZIHVResizcld -SS -SSS -ee (show all info)	
-G	addr	load address . address to header	
-h	—	this help message	
-H	—	header fields	
-i	—	imports (symbols imported from libraries)	
-I	—	binary info	
-j	—	output in json	
-k	sdb-query	run sdb query. for example: '*'	
-K	algo	calculate checksums (md5, sha1, ..)	
-l	—	linked libraries	
-L	plugin	list supported bin plugins or plugin details	
-m	addr	show source line at addr	
-M	—	main (show address of main symbol)	
-n	str	show section, symbol or import named str	
-N	min:max	force min:max number of chars per string (see -z and -zz)	
-o	str	output file or extraction directory for write operations	
-O	str	write/extract operations (-O help)	
-p	—	show always physical addresses	
-P	—	show debug/pdb information	
-q	—	be quiet, just show fewer data	
-Q	—	show load address used by dlopen (non-aslr libs)	
-r	—	radare output	
-R	—	relocations	
-s	—	symbols	
-S	—	sections	
-t	—	display file hashes	
-T	—	display file signature	
-u	—	resources	

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	display version and quit	
<code>-V</code>	—	show binary version information	
<code>-w</code>	—	display try/catch blocks	
<code>-x</code>	—	extract sub-binaries (-xS sections, -xSS segments, -xu resources)	
<code>-y</code>	—	show types (structs, enums, function signatures)	
<code>-z</code>	—	strings (from data section)	
<code>-Z</code>	—	guess size of binary program	

Gotchas

TODO: operational traps.

See also

[readelf](#), [readelf.py](#), [objdump](#), [r2](#), [rasm2](#), [vivbin](#), [vdbbin](#)

radiff2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Compare or cluster samples; Diff two binaries **Version:** radiff2 6.1.9 **Captured:** cyberlab-aio via -h on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

← [Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
radiff2 [-options] [-A[A]] [-B #] [-g sym] [-m graph_mode][-t %] [file] [file]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 29 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	arch	specify architecture plugin to use (x86, arm, ..)	
-A	-A	run aaa or aaaa after loading each binary (see -C)	
-b	bits	specify register size for arch (16 (thumb), 32, 64, ..)	
-c	cmd	run given command on every RCore instance	
-C	—	graphdiff code (columns: off-A, match-ratio, off-B) (see -A)	
-d	—	use delta diffing	
-D	—	show disasm instead of hexpairs	
-e	k=v	set eval config var value for all RCore instances	
-f	help	select output format (see '-f help' for details)	
-g	arg	graph diff of [sym] or functions in [off1,off2]	
-i	help	compare bin information (symbols, strings, classes, ..)	
-j	—	output in json format (see -f json)	
-l	addr	specify final offset (length) for diffing, same format as -o	

Flag	Argument	What it does	When you would use it
<code>-m</code>	mode	choose the graph output mode (aditsjJ)	
<code>-n</code>	—	count of changes	
<code>-o</code>	addr	specify initial offset for diffing, can use A:B format	
<code>-O</code>	—	code diffing with opcode bytes only	
<code>-p</code>	—	use physical addressing (io.va=false) (only for radiff2 -AC)	
<code>-q</code>	—	quiet mode (disable colors, reduce output)	
<code>-r</code>	—	output in radare commands	
<code>-s</code>	—	compute edit distance (no substitution, Eugene W. Myers O(ND) diff algorithm)	
<code>-S</code>	name	sort code diff (name, namelen, addr, size, type, dist) (only for -C or -g)	
<code>-T</code>	—	analyze files in threads (EXPERIMENTAL, 30% faster and crashy)	
<code>-x</code>	—	show two column hexdump diffing	
<code>-X</code>	—	use xpatch format for the diffing output	
<code>-u</code>	—	unified output (---+++)	
<code>-U</code>	—	unified output using system 'diff'	
<code>-v</code>	—	show version information	
<code>-V</code>	—	be verbose (current only for -s)	

Gotchas

TODO: operational traps.

See also

[ssdeep](#)

readelf

Kit: Base OS — present on every Linux image **Capability:** Inspect PE / ELF structure **Version:** GNU readelf (GNU Binutils for Debian) 2.40 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Display information about the contents of ELF format files

Synopsis

```
readelf <option(s)> elf-file(s)
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 68 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	Equivalent to: -h -l -S -s -r -d -V -A -l	
--all	—	Equivalent to: -h -l -S -s -r -d -V -A -l	
-h	—	Display the ELF file header	
--file-header	—	Display the ELF file header	
-l	—	Display the program headers	
--program-headers	—	Display the program headers	
--segments	—	An alias for --program-headers	
-S	—	Display the sections' header	
--section-headers	—	Display the sections' header	
--sections	—	An alias for --section-headers	
-g	—	Display the section groups	

Flag	Argument	What it does	When you would use it
<code>--section-groups</code>	—	Display the section groups	
<code>-t</code>	—	Display the section details	
<code>--section-details</code>	—	Display the section details	
<code>-e</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>--headers</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>-s</code>	—	Display the symbol table	
<code>--syms</code>	—	Display the symbol table	
<code>--symbols</code>	—	An alias for <code>--syms</code>	
<code>--dyn-syms</code>	—	Display the dynamic symbol table	
<code>--lto-syms</code>	—	Display LTO symbol tables	
<code>--sym-base</code>	0 8 10 16	Force base for symbol sizes. The options are mixed (the default), octal, decimal, hexadecimal.	
<code>-C</code>	—	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	
<code>--demangle</code>	STYLE (optional)	Decode mangled/processed symbol names STYLE can be "none", "auto", "gnu-v3", "java", "gnat", "dlang", "rust"	
<code>--no-demangle</code>	—	Do not demangle low-level symbol names. (default)	
<code>--recurse-limit</code>	—	Enable a demangling recursion limit. (default)	
<code>--no-recurse-limit</code>	—	Disable a demangling recursion limit	
<code>-n</code>	—	Display the core notes (if present)	
<code>--notes</code>	—	Display the core notes (if present)	
<code>-r</code>	—	Display the relocations (if present)	
<code>--relocs</code>	—	Display the relocations (if present)	
<code>-u</code>	—	Display the unwind info (if present)	
<code>--unwind</code>	—	Display the unwind info (if present)	
<code>-d</code>	—	Display the dynamic section (if present)	
<code>--dynamic</code>	—	Display the dynamic section (if present)	
<code>-V</code>	—	Display the version sections (if present)	
<code>--version-info</code>	—	Display the version sections (if present)	
<code>-A</code>	—	Display architecture specific information (if any)	
<code>--arch-specific</code>	—	Display architecture specific information (if any)	

Flag	Argument	What it does	When you would use it
<code>-c</code>	—	Display the symbol/file index in an archive	
<code>--archive-index</code>	—	Display the symbol/file index in an archive	
<code>-D</code>	—	Use the dynamic section info when displaying symbols	
<code>--use-dynamic</code>	—	Use the dynamic section info when displaying symbols	
<code>-x</code>	—	Dump the contents of section as bytes	
<code>--hex-dump</code>	number name	Dump the contents of section as bytes	
<code>-p</code>	—	Dump the contents of section as strings	
<code>--string-dump</code>	number name	Dump the contents of section as strings	
<code>-R</code>	—	Dump the relocated contents of section	
<code>--relocated-dump</code>	number name	Dump the relocated contents of section	
<code>-z</code>	—	Decompress section before dumping it	
<code>--decompress</code>	—	Decompress section before dumping it	
<code>-P</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -wK)	
<code>--process-links</code>	—	Display the contents of non-debug sections in separate debuginfo files. (Implies -wK)	
<code>--dwarf-depth</code>	N	Do not display DIEs at depth N or greater	
<code>--dwarf-start</code>	N	Display DIEs starting at offset N	
<code>--ctf</code>	number name	Display CTF info from section	
<code>--ctf-parent</code>	name	Use CTF archive member as the CTF parent	
<code>--ctf-symbols</code>	number name	Use section as the CTF external symtab	
<code>--ctf-strings</code>	number name	Use section as the CTF external strtab	
<code>--sframe</code>	NAME (optional)	Display SFrame info from section NAME, (default '.sframe')	
<code>-I</code>	—	Display histogram of bucket list lengths	
<code>--histogram</code>	—	Display histogram of bucket list lengths	
<code>-W</code>	—	Allow output width to exceed 80 characters	
<code>--wide</code>	—	Allow output width to exceed 80 characters	
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	
<code>-v</code>	—	Display the version number of readelf	
<code>--version</code>	—	Display the version number of readelf	

Gotchas

TODO: operational traps.

See also

[rabin2](#), [readelf.py](#), [objdump](#)

readelf.py

Kit: REMnux **Capability:** Inspect PE / ELF structure **Version:** based on pyelftools 0.33 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://github.com/eliben/pyelftools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Python library for parsing and analyzing ELF files and DWARF debugging information.

Synopsis

```
usage: readelf.py [options] <elf-file>
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	show program's version number and exit	
<code>--version</code>	—	show program's version number and exit	
<code>-d</code>	—	Display the dynamic section	
<code>--dynamic</code>	—	Display the dynamic section	
<code>-H</code>	—	Display this information	
<code>--help</code>	—	Display this information	
<code>-h</code>	—	Display the ELF file header	
<code>--file-header</code>	—	Display the ELF file header	
<code>-l</code>	—	Display the program headers	
<code>--program-headers</code>	—	Display the program headers	
<code>--segments</code>	—	Display the program headers	
<code>-S</code>	—	Display the sections' headers	

Flag	Argument	What it does	When you would use it
<code>--section-headers</code>	—	Display the sections' headers	
<code>--sections</code>	—	Display the sections' headers	
<code>-e</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>--headers</code>	—	Equivalent to: <code>-h -l -S</code>	
<code>-s</code>	—	Display the symbol table	
<code>--symbols</code>	—	Display the symbol table	
<code>--syms</code>	—	Display the symbol table	
<code>-n</code>	—	Display the core notes (if present)	
<code>--notes</code>	—	Display the core notes (if present)	
<code>-r</code>	—	Display the relocations (if present)	
<code>--relocs</code>	—	Display the relocations (if present)	
<code>-x</code>	number name	Dump the contents of section as bytes	
<code>--hex-dump</code>	number name	Dump the contents of section as bytes	
<code>-p</code>	number name	Dump the contents of section as strings	
<code>--string-dump</code>	number name	Dump the contents of section as strings	
<code>-V</code>	—	Display the version sections (if present)	
<code>--version-info</code>	—	Display the version sections (if present)	
<code>-A</code>	—	Display the architecture-specific information (if present)	
<code>--arch-specific</code>	—	Display the architecture-specific information (if present)	
<code>--debug-dump</code>	what	Display the contents of DWARF debug sections. can one of {info,decodedline,frames,frames- interp,aranges,pubtypes,pubnames,loc,Ranges}	
<code>--traceback</code>	—	Dump the Python traceback on ELFError exceptions from elftools	

Gotchas

TODO: operational traps.

See also

`rabin2`, `readelf`, `objdump`

unzip

Kit: REMnux **Capability:** Detect and reverse packing; Analyse other container formats **Version:** UnZip 6.00
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <http://infozip.sourceforge.net>

[← Capability index](#) · [Kit tool list](#)

Purpose

Compress and decompress files using the zip algorithm.

Synopsis

```
unzip [-Z] [-opts[modifiers]] file[.zip] [list] [-x xlist] [-d exdir]
Default action is to extract files in list, except those in xlist, to exdir;
file[.zip] may be a wildcard. -Z => ZipInfo mode ("unzip -Z" for usage).
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 12 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-p	—	extract files to pipe, no messages -l list files (short format)	
-f	—	freshen existing files, create none -t test compressed archive data	
-u	—	update files, create if necessary -z display archive comment only	
-v	—	list verbosely/show version info -T timestamp archive to latest	
-x	—	exclude files that follow (in xlist) -d extract files into exdir	
-n	—	never overwrite existing files -q quiet mode (-qq => quieter)	
-o	—	overwrite files WITHOUT prompting -a auto-convert any text files	
-j	—	junk paths (do not make directories) -aa treat ALL files as text	
-U	—	use escapes for all non-ASCII Unicode -UU ignore any Unicode fields	
-C	—	match filenames case-insensitively -L make (some) names lowercase	
-X	—	restore UID/GID info -V retain VMS version numbers	
-K	—	keep setuid/setgid/tacky permissions -M pipe through "more" pager	

Gotchas

TODO: operational traps.

See also

[upx](#), [die](#), [diec](#), [binwalk](#), [7za](#), [msoffcrypto-tool](#)

upx

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Detect and reverse packing **Version:** upx 4.2.4
Captured: cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://upx.github.io>

← [Capability index](#) · [Kit tool list](#)

Purpose

Pack and unpack executables with the UPX compressor.

Synopsis

```
upx [-123456789dlthVL] [-qvfk] [-o file] file..
```

Common invocations

```
# from cyberlab 28-x64dbg-workflow
upx --best .\exercise\packed_hello.exe
# from cyberlab 28-x64dbg-workflow
upx -d .\exercise\packed_hello.exe -o .\exercise\unpacked_ref.exe
# from cyberlab 43-pe-bear-deep
upx --best -o C:\work\pe-lab\exercise\packed_hello.exe C:\work\pe-lab\exercise\hello.exe
# from cyberlab 52-unpacking-case
upx --best .\exercise\sample_packed.exe
```

Options

All 37 options parsed from the captured help text; 5 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-1	—	compress faster -9 compress better	
-d	—	decompress -l list compressed file	Decompress — the only flag you normally want in analysis.
-t	—	test compressed file -V display version number	Test the file's integrity without writing anything.
-h	—	give this help -L display software license	
-q	—	be quiet -v be verbose	Quiet output.
-f	—	force compression of suspicious files	Force the operation on a file UPX considers questionable.

Flag	Argument	What it does	When you would use it
<code>--no-color</code>	—	change look	
<code>--mono</code>	—	change look	
<code>--color</code>	—	change look	
<code>--no-progress</code>	—	change look	
<code>--lzma</code>	—	try LZMA [slower but tighter than NRV]	
<code>--brute</code>	—	try all available compression methods & filters [slow]	
<code>--ultra-brute</code>	—	try even more compression variants [very slow]	
<code>-k</code>	—	keep backup files	Keep a backup of the original file.
<code>--backup</code>	—	keep backup files	
<code>--no-backup</code>	—	no backup files [default]	
<code>--overlay</code>	copy	copy any extra data attached to the file [default]	
<code>--force-overwrite</code>	—	force overwrite of output files	
<code>--link</code>	—	preserve hard links (Unix only) [USE WITH CARE]	
<code>--no-link</code>	—	do not preserve hard links but rename files [default]	
<code>--no-mode</code>	—	do not preserve file mode (aka permissions)	
<code>--no-owner</code>	—	do not preserve file ownership	
<code>--no-time</code>	—	do not preserve file timestamp	
<code>--coff</code>	—	produce COFF output [default: EXE]	
<code>--8086</code>	—	make compressed com work on any 8086	
<code>--no-reloc</code>	—	put no relocations in to the exe header	
<code>--8-bit</code>	—	uses 8 bit size compression [default: 32 bit]	
<code>--8mib-ram</code>	—	8 megabyte memory limit [default: 2 MiB]	
<code>--boot-only</code>	—	disables client/host transfer compatibility	
<code>--no-align</code>	—	don't align to 2048 bytes [enables: --console-run]	
<code>--le</code>	—	produce LE output [default: EXE]	
<code>--compress-exports</code>	0	do not compress the export section	
<code>--compress-icons</code>	0	do not compress any icons	
<code>--compress-resources</code>	0	do not compress any resources at all	

Flag	Argument	What it does	When you would use it
<code>--keep-resource</code>	list	do not compress resources specified by list	
<code>--strip-relocs</code>	0	do not strip relocations	
<code>--preserve-build-id</code>	—	copy .gnu.note.build-id to compressed output	

Gotchas

- **Work on a copy.** `upx -d` rewrites the file in place, mutating your sample and invalidating its hash.
- Malware routinely uses a modified UPX header so stock `upx -d` refuses. Failure to unpack is itself an indicator, not a dead end.

See also

`die`, `diec`, `binwalk`, `7za`, `unzip`

yara

Kit: REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://virustotal.github.io/yara/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Scan files, directories or live processes against YARA rules.

Synopsis

```
yara [OPTION]... [NAMESPACE:]RULES_FILE... FILE | DIR | PID
```

Common invocations

```
# from cyberlab 08-malware-static-triage
yara --version
# from cyberlab 08-malware-static-triage
yara -r exercise/triage.yar exercise/sample.bin
# from cyberlab 21-yara-authoring
yara --help | head -n 20
# from cyberlab 21-yara-authoring
yara -w /tmp/hunt_eicar.yar /tmp/hunt_eicar.yar
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar /tmp
# from cyberlab 21-yara-authoring
yara -r -s /tmp/hunt_eicar.yar exercise/
# from cyberlab 21-yara-authoring
yara your_rule.yar exercise/eicar_sample.txt
# from cyberlab 34-clamav-scanning
yara -s exercise/lab34.yar exercise/sample.txt
```

Options

All 55 options parsed from the captured help text; 12 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>--atom-quality-table</code>	FILE	path to a file with the atom quality table	

Flag	Argument	What it does	When you would use it
<code>-C</code>	—	load compiled rules	Load a pre-compiled ruleset (from <code>yarac</code>) instead of source.
<code>--compiled-rules</code>	—	load compiled rules	
<code>-c</code>	—	print only number of matches	Print only the count of matches per file, for a quick sweep.
<code>--count</code>	—	print only number of matches	
<code>-d</code>	VAR=VALUE	define external variable	Define an external variable a rule expects, e.g. <code>-d filename=x</code> .
<code>--define</code>	VAR=VALUE	define external variable	
<code>--fail-on-warnings</code>	—	fail on warnings	
<code>-f</code>	—	fast matching mode	Fast matching mode; faster but can miss some overlapping matches.
<code>--fast-scan</code>	—	fast matching mode	
<code>-h</code>	—	show this help and exit	
<code>--help</code>	—	show this help and exit	
<code>-i</code>	IDENTIFIER	print only rules named IDENTIFIER	Scan only the rule with the given identifier.
<code>--identifier</code>	IDENTIFIER	print only rules named IDENTIFIER	
<code>--max-process-memory-chunk</code>	NUMBER	set maximum chunk size while reading process memory (default=1073741824)	
<code>-l</code>	NUMBER	abort scanning after matching a NUMBER of rules	
<code>--max-rules</code>	NUMBER	abort scanning after matching a NUMBER of rules	
<code>--max-strings-per-rule</code>	NUMBER	set maximum number of strings per rule (default=10000)	
<code>-x</code>	MODULE=FILE	pass FILE's content as extra data to MODULE	
<code>--module-data</code>	MODULE=FILE	pass FILE's content as extra data to MODULE	
<code>-n</code>	—	print only not satisfied rules (negate)	Print files that do NOT match — useful for finding the odd one out.
<code>--negate</code>	—	print only not satisfied rules (negate)	
<code>-N</code>	—	do not follow symlinks when scanning	
<code>--no-follow-symlinks</code>	—	do not follow symlinks when scanning	
<code>-w</code>	—	disable warnings	Suppress warnings, which are noisy on large third-party rulesets.

Flag	Argument	What it does	When you would use it
<code>--no-warnings</code>	—	disable warnings	
<code>-m</code>	—	print metadata	Print rule metadata, which usually carries author and reference.
<code>--print-meta</code>	—	print metadata	
<code>-D</code>	—	print module data	
<code>--print-module-data</code>	—	print module data	
<code>-e</code>	—	print rules' namespace	
<code>--print-namespace</code>	—	print rules' namespace	
<code>-S</code>	—	print rules' statistics	
<code>--print-stats</code>	—	print rules' statistics	
<code>-s</code>	—	print matching strings	Print the matching strings and their offsets. Essential for verifying a hit is real rather than trusting the rule name.
<code>--print-strings</code>	—	print matching strings	
<code>-L</code>	—	print length of matched strings	
<code>--print-string-length</code>	—	print length of matched strings	
<code>-g</code>	—	print tags	
<code>--print-tags</code>	—	print tags	
<code>-r</code>	—	recursively search directories	Recurse into directories — the usual mode when triaging a tree.
<code>--recursive</code>	—	recursively search directories	
<code>--scan-list</code>	—	scan files listed in FILE, one per line	
<code>-z</code>	NUMBER	skip files larger than the given size when scanning a directory	
<code>--skip-larger</code>	NUMBER	skip files larger than the given size when scanning a directory	
<code>-k</code>	SLOTS	set maximum stack size (default=16384)	
<code>--stack-size</code>	SLOTS	set maximum stack size (default=16384)	
<code>-t</code>	TAG	print only rules tagged as TAG	Scan only rules carrying a given tag.
<code>--tag</code>	TAG	print only rules tagged as TAG	
<code>-p</code>	NUMBER	use the specified NUMBER of threads to scan a directory	Use N threads to speed up a large scan.
<code>--threads</code>	NUMBER	use the specified NUMBER of threads to scan a directory	

Flag	Argument	What it does	When you would use it
<code>-a</code>	SECONDS	abort scanning after the given number of SECONDS	
<code>--timeout</code>	SECONDS	abort scanning after the given number of SECONDS	
<code>-v</code>	—	show version information	
<code>--version</code>	—	show version information	

Gotchas

- A rule that references an external variable fails to compile unless you supply it with `-d`. This is the most common 'the rule is broken' false alarm.
- Scanning a live process needs a PID rather than a path, and root.
- Compile large rulesets once with `yarac` and scan with `-C`; recompiling thousands of rules per scan dominates runtime.

See also

`yarac`, `clamscan`, `freshclam`, `sigtool`

yarac

Kit: REMnux · Kali Linux · FLARE-VM **Capability:** Scan with signatures for known-bad **Version:** 4.2.3 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://virustotal.github.io/yara/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Identify and classify malware samples using Yara rules.

Synopsis

```
yarac [OPTION]... [NAMESPACE:]SOURCE_FILE... OUTPUT_FILE
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--atom-quality-table</code>	FILE	path to a file with the atom quality table	
<code>-d</code>	VAR=VALUE	define external variable	
<code>--define</code>	VAR=VALUE	define external variable	
<code>--fail-on-warnings</code>	—	fail on warnings	
<code>-h</code>	—	show this help and exit	
<code>--help</code>	—	show this help and exit	
<code>--max-strings-per-rule</code>	NUMBER	set maximum number of strings per rule (default=10000)	
<code>-w</code>	—	disable warnings	
<code>--no-warnings</code>	—	disable warnings	
<code>-v</code>	—	show version information	
<code>--version</code>	—	show version information	

Gotchas

TODO: operational traps.

See also

`yara`, `clamscan`, `freshclam`, `sigtool`

aeskeyfind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://citp.princeton.edu/our-work/memory/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Find 128-bit and 256-bit AES keys in a memory image.

Synopsis

```
aeskeyfind [OPTION]... MEMORY-IMAGE
```

Common invocations

```
# from cyberlab 02-memory-forensics
aeskeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
aeskeyfind sample.mem
```

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	verbose output -- prints the extended keys and the constraints on the rows of the key schedule	
<code>-q</code>	—	don't display a progress bar	
<code>-h</code>	—	displays this help message	

Gotchas

TODO: operational traps.

See also

```
rsa_keyfind, bulk_extractor
```

rsakeyfind

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Recover encryption keys from memory **Captured:** cyberlab-aio via (no args) on 2026-08-02 **raw Docs:** <https://citp.princeton.edu/our-work/memory/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Find BER-encoded RSA private keys in a memory image.

Synopsis

```
rsakeyfind MEMORY-IMAGE [MODULUS-FILE]
```

Common invocations

```
# from cyberlab 02-memory-forensics
rsakeyfind 2>&1 | head -n 1
# from cyberlab 02-memory-forensics
rsakeyfind sample.mem
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[aeskeyfind](#) , [bulk_extractor](#)

vol

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Extract and analyse artifacts from a memory image using Volatility 3 plugins.

Synopsis

```
vol [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [--filters FILTERS]
[--hide-columns [HIDE_COLUMNS ...]] [-r RENDERER]
```

Common invocations

```
# from cyberlab 02-memory-forensics
vol --help | head -n 3
# from cyberlab 02-memory-forensics
vol -h | grep -i -E "pslist|netscan|windows.info" | head -n 10
# from cyberlab 02-memory-forensics
vol -f $IMAGE windows.pslist | head -n 20
# from cyberlab 20-volatility-deep
vol --version
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.info
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pslist
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.pstree
# from cyberlab 20-volatility-deep
vol -f exercise/memdump.raw windows.psscan
```

Options

All 35 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	Show this help message and exit, for specific plugin options use 'vol --help'	
<code>--help</code>	—	Show this help message and exit, for specific plugin options use 'vol --help'	
<code>-c</code>	CONFIG	Load the configuration from a json file	Load saved configuration from a JSON file.
<code>--config</code>	CONFIG	Load the configuration from a json file	
<code>--parallelism</code>	{processes,threads,off}	Enables parallelism (defaults to off if no argument given)	
<code>-e</code>	EXTEND	Extend the configuration with a new (or changed) setting	
<code>--extend</code>	EXTEND	Extend the configuration with a new (or changed) setting	
<code>-p</code>	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Add a directory of custom plugins.
<code>--plugin-dirs</code>	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	
<code>-s</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Point at a local symbol-table directory — the fix for an air-gapped host that cannot download symbols.
<code>--symbol-dirs</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	
<code>-v</code>	—	Increase output verbosity	Increase verbosity while diagnosing a symbol-table failure.
<code>--verbosity</code>	—	Increase output verbosity	
<code>-l</code>	LOG	Log output to a file as well as the console	
<code>--log</code>	LOG	Log output to a file as well as the console	
<code>-o</code>	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin dumps (processes, DLLs, files).
<code>--output-dir</code>	OUTPUT_DIR	Directory in which to output any generated files	
<code>-q</code>	—	Remove progress feedback	Quiet — suppress the progress and INFO banner when scripting.
<code>--quiet</code>	—	Remove progress feedback	
<code>-f</code>	FILE	Shorthand for --single-location=file:// if single- location is not defined	The memory image to analyse — required for almost every plugin.
<code>--file</code>	FILE	Shorthand for --single-location=file:// if single- location is not defined	

Flag	Argument	What it does	When you would use it
<code>--write-config</code>	—	Write configuration JSON file out to config.json	
<code>--save-config</code>	SAVE_CONFIG	Save configuration JSON file to a file	
<code>--clear-cache</code>	—	Clears out all short-term cached items	
<code>--cache-path</code>	CACHE_PATH	Change the default path (/root/.cache/volatility3) used to store the cache	
<code>--offline</code>	—	Do not search online for additional JSON files	Never attempt a network fetch for symbols. Use this on evidence networks so a scan cannot stall on a download.
<code>-u</code>	URL	Search online for ISF json files	
<code>--remote-isf-url</code>	URL	Search online for ISF json files	
<code>--filters</code>	FILTERS	List of filters to apply to the output (in the form of [+ -]columnname,pattern[!])	
<code>--hide-columns</code>	HIDE_COLUMNS ...	Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided	
<code>-r</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Choose the renderer: <code>csv</code> / <code>json</code> when feeding another tool.
<code>--renderer</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	
<code>--stackers</code>	STACKERS ...	List of stackers	
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	

Gotchas

- **Timeout-guard** `vol` **in any harness.** On a synthetic or truncated memory image it has pinned a core at 99% indefinitely on this project. Never run it unbounded in an automated pass.
- Volatility 3 takes plugin names, not v2 syntax: `windows.pslist`, not `--profile=... pslist`. Most 'plugin not found' errors are pasted v2 commands.
- The first run against an unfamiliar image downloads symbol tables. On an isolated network that hangs — pre-stage symbols and use `--offline`.

See also

`volatility3`, `volshell`

volatility3

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Analyse a memory image. These are the framework-wide options; each plugin adds its own, shown by `volatility3 <plugin> --help`.

Synopsis

```
volatility3 [-h] [-c CONFIG] [--parallelism [{processes,threads,off}]]
[-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS] [-v] [-l LOG]
[-o OUTPUT_DIR] [-q] [-f FILE] [--write-config]
[--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL]
[--filters FILTERS] [--hide-columns [HIDE_COLUMNS ...]]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 35 options parsed from the captured help text; 33 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	Show this help message and exit, for specific plugin options use 'volatility3 --help'	
<code>--help</code>	—	Show this help message and exit, for specific plugin options use 'volatility3 --help'	
<code>-c</code>	CONFIG	Load the configuration from a json file	Load options from a JSON config.
<code>--config</code>	CONFIG	Load the configuration from a json file	Load options from a JSON config.

Flag	Argument	What it does	When you would use it
<code>--parallelism</code>	{processes,threads,off}	Enables parallelism (defaults to off if no argument given)	Enable process or thread parallelism. Off by default, and worth turning on for a long scan.
<code>-e</code>	EXTEND	Extend the configuration with a new (or changed) setting	Override a single configuration setting.
<code>--extend</code>	EXTEND	Extend the configuration with a new (or changed) setting	Override a single configuration setting.
<code>-p</code>	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Additional plugin directories, for plugins outside the tree.
<code>--plugin-dirs</code>	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	Additional plugin directories.
<code>-s</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables. The usual fix when a Linux or macOS image will not resolve: the ISF for that exact kernel has to be reachable.
<code>--symbol-dirs</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	Where to find symbol tables — the usual fix when a Linux or macOS image will not resolve.
<code>-v</code>	—	Increase output verbosity	More verbose output; repeat for more.
<code>--verbosity</code>	—	Increase output verbosity	More verbose output; repeat for more.
<code>-l</code>	LOG	Log output to a file as well as the console	Also write output to a log file.
<code>--log</code>	LOG	Log output to a file as well as the console	Also write output to a log file.
<code>-o</code>	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin writes — dumped processes, extracted DLLs. Required before any <code>--dump</code> plugin option does anything useful.
<code>--output-dir</code>	OUTPUT_DIR	Directory in which to output any generated files	Directory for files the plugin writes. Required before any <code>--dump</code> plugin option is useful.
<code>-q</code>	—	Remove progress feedback	Suppress progress feedback, for scripted runs.
<code>--quiet</code>	—	Remove progress feedback	Suppress progress feedback, for scripted runs.
<code>-f</code>	FILE	Shorthand for <code>--single-location=file://</code> if single- location is not defined	The memory image. Almost every invocation starts here.
<code>--file</code>	FILE	Shorthand for <code>--single-location=file://</code> if single- location is not defined	The memory image. Almost every invocation starts here.

Flag	Argument	What it does	When you would use it
<code>--write-config</code>	—	Write configuration JSON file out to config.json	Write the resolved configuration to config.json — useful for making a complex run reproducible.
<code>--save-config</code>	SAVE_CONFIG	Save configuration JSON file to a file	Write the resolved configuration to a named file.
<code>--clear-cache</code>	—	Clears out all short-term cached items	Drop cached items. First thing to try when results look stale or an image was replaced in place.
<code>--cache-path</code>	CACHE_PATH	Change the default path (/root/.cache/volatility3) used to store the cache	Move the cache somewhere with room; it grows.
<code>--offline</code>	—	Do not search online for additional JSON files	Never fetch symbols online. Use it on an analysis host that must not touch the network, and expect failures unless the ISFs are already local.
<code>-u</code>	URL	Search online for ISF json files	Point at an alternative ISF repository.
<code>--remote-isf-url</code>	URL	Search online for ISF json files	Point at an alternative ISF repository.
<code>--filters</code>	FILTERS	List of filters to apply to the output (in the form of [+]-columnname,pattern(!))	Filter rows as <code>[+]-column,pattern</code> , so a plugin that returns thousands of rows can be narrowed without a second pass.
<code>--hide-columns</code>	HIDE_COLUMNS ...	Case-insensitive space separated list of prefixes to determine which columns to hide in the output if provided	Drop columns from the output to keep a wide table readable.
<code>-r</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Output renderer. <code>csv</code> or <code>jsonl</code> when the result feeds another tool; <code>pretty</code> is for reading, not for parsing.
<code>--renderer</code>	RENDERER	Determines how to render the output (quick, none, csv, pretty, json, jsonl, arrow, parquet)	Output renderer. <code>csv</code> or <code>jsonl</code> when the result feeds another tool; <code>pretty</code> is for reading.
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	The image URI, when it is not a plain local file (<code>-f</code> is shorthand for this).
<code>--stackers</code>	STACKERS ...	List of stackers	Control the layer stackers used to interpret the image.
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	Supply swap files alongside the image, so paged-out memory can be resolved.

Gotchas

- Symbols are the usual failure, not the image. Windows profiles are generated automatically, but Linux and macOS need an ISF matching the exact kernel build — same version is not enough.

- Volatility 3 dropped the Volatility 2 profile system entirely, so `--profile` from older notes and blog posts does not exist here.
- It has pinned CPU indefinitely on a malformed image before. Run long analyses under a timeout rather than assuming progress.

See also

`vol`, `volshell`

volshell

Kit: SIFT Workstation (Volatility 3) **Capability:** Analyse a memory image **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Volshell (Volatility 3 Framework) 2.28.0

Synopsis

```
volshell [-h] [-c CONFIG] [-e EXTEND] [-p PLUGIN_DIRS] [-s SYMBOL_DIRS]
[-v] [-o OUTPUT_DIR] [-q] [--log LOG] [-f FILE]
[--write-config] [--save-config SAVE_CONFIG] [--clear-cache]
[--cache-path CACHE_PATH] [--offline | -u URL] [-w | -l | -m]
[--single-location SINGLE_LOCATION]
[--stackers [STACKERS ...]]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 38 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	CONFIG	Load the configuration from a json file	
--config	CONFIG	Load the configuration from a json file	
-e	EXTEND	Extend the configuration with a new (or changed) setting	
--extend	EXTEND	Extend the configuration with a new (or changed) setting	
-p	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	
--plugin-dirs	PLUGIN_DIRS	Semi-colon separated list of paths to find plugins	

Flag	Argument	What it does	When you would use it
<code>-s</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	
<code>--symbol-dirs</code>	SYMBOL_DIRS	Semi-colon separated list of paths to find symbols	
<code>-v</code>	—	Increase output verbosity	
<code>--verbosity</code>	—	Increase output verbosity	
<code>-o</code>	OUTPUT_DIR	Directory in which to output any generated files	
<code>--output-dir</code>	OUTPUT_DIR	Directory in which to output any generated files	
<code>-q</code>	—	Remove progress feedback	
<code>--quiet</code>	—	Remove progress feedback	
<code>--log</code>	LOG	Log output to a file as well as the console	
<code>-f</code>	FILE	Shorthand for <code>--single-location=file://</code> if single-location is not defined	
<code>--file</code>	FILE	Shorthand for <code>--single-location=file://</code> if single-location is not defined	
<code>--write-config</code>	—	Write configuration JSON file out to config.json	
<code>--save-config</code>	SAVE_CONFIG	Save configuration JSON file to a file	
<code>--clear-cache</code>	—	Clears out all short-term cached items	
<code>--cache-path</code>	CACHE_PATH	Change the default path (<code>/root/.cache/volatility3</code>) used to store the cache	
<code>--offline</code>	—	Do not search online for additional JSON files	
<code>-u</code>	URL	Search online for ISF json files	
<code>--remote-isf-url</code>	URL	Search online for ISF json files	
<code>-w</code>	—	Run a Windows volshell	
<code>--windows</code>	—	Run a Windows volshell	
<code>-l</code>	—	Run a Linux volshell	
<code>--linux</code>	—	Run a Linux volshell	
<code>-m</code>	—	Run a Mac volshell	
<code>--mac</code>	—	Run a Mac volshell	
<code>--single-location</code>	SINGLE_LOCATION	Specifies a base location on which to stack	
<code>--stackers</code>	STACKERS ...	List of stackers	
<code>--single-swap-locations</code>	SINGLE_SWAP_LOCATIONS ...	Specifies a list of swap layer URIs for use with single-location	
<code>--script</code>	SCRIPT	File to load and execute at start	
<code>--script-only</code>	—	Exit volshell after the script specified in <code>--script</code> completes	
<code>--pid</code>	PID	Process ID	

Gotchas

TODO: operational traps.

See also

`vol`, `volatility3`

arp-scan

Kit: SIFT Workstation **Capability:** Probe or scan hosts and services **Version:** arp-scan 1.10.0 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Target hosts must be specified on the command line unless the `--file` or

Synopsis

```
arp-scan [options] [hosts...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

[nmap](#), [nping](#)

capinfos

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Read and filter packet captures **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
capinfos [options] <infile> ...
```

Common invocations

```
# from cyberlab 24-wireshark-deep
capinfos exercise/sample.pcap
```

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	display this help and exit	
--help	—	display this help and exit	
-v	—	display version info and exit	
--version	—	display version info and exit	

Gotchas

TODO: operational traps.

See also

[tshark](#), [ngrep](#), [tcpflow](#)

editcap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Cut, split, deduplicate and convert capture files — the tool that makes an unmanageable pcap workable before analysis starts.

Synopsis

```
editcap [options] ... <infile> <outfile> [ <packet#>[-<packet#>] ... ]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 25 options parsed from the captured help text; 21 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-r</code>	—	keep the selected packets; default is to delete them.	Invert the selection: keep the specified packets instead of deleting them. Easy to forget, and it reverses the meaning of the whole command.
<code>-A</code>	start time	only read packets whose timestamp is after (or equal to) the given time.	Keep only packets at or after this timestamp.
<code>-B</code>	stop time	only read packets whose timestamp is before the given time. Time format for -A/-B options is YYYY-MM-DDThh:mm:ss[.nnnnnnnnn][Z -hh:mm] Unix epoch timestamps are also supported.	Keep only packets before this timestamp. With <code>-A</code> , this is how a multi-gigabyte capture becomes the incident window.
<code>--novlan</code>	—	remove vlan info from packets before checking for duplicates.	Ignore VLAN tags when comparing for duplicates, so the same frame seen on two VLANs collapses to one.
<code>-d</code>	—	remove packet if duplicate (window == 5).	Drop duplicate packets using the default 5-packet window. Captures taken from a SPAN port routinely see each packet twice, which distorts every count downstream.

Flag	Argument	What it does	When you would use it
<code>-D</code>	dup window	remove packet if duplicate; configurable . Valid values are 0 to 1000000. NOTE: A of 0 with -V (verbose option) is useful to print MD5 hashes.	Drop duplicates with an explicit window, when <code>-d</code> 's default is too narrow.
<code>-W</code>	dup time window	remove packet if duplicate packet is found EQUAL TO OR LESS THAN prior to current packet. A is specified in relative seconds (e.g. 0.000001). NOTE: The use of the '	Drop duplicates within a time window rather than a packet count.
<code>-s</code>	snaplen	truncate each packet to max. bytes of data.	Truncate each packet to N bytes. Strips payload while keeping headers — the usual way to share a capture without its contents.
<code>-L</code>	—	adjust the frame (i.e. reported) length when chopping and/or snapping.	Adjust the recorded frame length to match after truncating, so the file is not self-inconsistent.
<code>-t</code>	time adjustment	adjust the timestamp of each packet. is in relative seconds (e.g. -0.5).	Shift every timestamp by a relative amount. This is how a capture from a host with a skewed clock is aligned to the rest of the timeline.
<code>-o</code>	change offset	When used in conjunction with -E, skip some bytes from the beginning of the packet. This allows one to preserve some bytes, in order to have some headers untouched.	With <code>-E</code> , skip bytes before introducing errors.
<code>--seed</code>	seed	When used in conjunction with -E, set the seed to use for the pseudo-random number generator. This allows one to repeat a particular sequence of errors.	With <code>-E</code> , fix the random seed so a corrupted-capture test is reproducible.
<code>-I</code>	bytes to ignore	ignore the specified number of bytes at the beginning of the frame during MD5 hash calculation, unless the frame is too short, then the full frame is used. Useful to remove duplicated packets taken on	Ignore N leading bytes when comparing for duplicates.
<code>-c</code>	packets per file	split the packet output to different files based on uniform packet counts with a maximum of each.	Split into files of N packets each. The standard fix for a capture too large for Wireshark to open.
<code>-i</code>	seconds per file	split the packet output to different files based on uniform time intervals with a maximum of each.	Split into files covering N seconds each — the same fix, when time is the natural unit.
<code>-F</code>	capture type	set the output file type; default is pcapng. An empty "-F" option will list the file types.	Output file format; pcapng by default. An empty <code>-F</code> lists the choices.
<code>-T</code>	encap type	set the output file encapsulation type; default is the same as the input file. An empty "-T" option will list the encapsulation types.	Output encapsulation type, when the link type must change.
<code>--discard-all-secrets</code>	—	Discard all decryption secrets from the input file when writing the output file. Does not discard secrets added by "--inject-secrets" in the same command line.	Strip embedded decryption secrets before handing the file to someone else.
<code>--capture-comment</code>	comment	Add a capture file comment, if supported.	Attach a comment to the file — a place to record provenance that travels with the capture.

Flag	Argument	What it does	When you would use it
<code>--discard-capture-comment</code>	—	Discard capture file comments from the input file when writing the output file. Does not discard comments added by "--capture-comment" in the same command line.	Remove existing comments on output.
<code>-h</code>	—	display this help and exit.	
<code>--help</code>	—	display this help and exit.	
<code>-V</code>	—	verbose output. If -V is used with any of the 'Duplicate Packet Removal' options (-d, -D or -w) then Packet lengths and MD5 hashes are printed to standard-error.	Verbose; with the duplicate options it reports what was removed rather than silently dropping packets.
<code>-v</code>	—	print version information and exit.	
<code>--version</code>	—	print version information and exit.	

Gotchas

- `-r` inverts the selection. Without it the named packets are **deleted**, which is the opposite of what most people intend the first time.
- Deduplication is a heuristic over a window, not a proof. A genuine retransmission looks like a duplicate, and dropping it destroys the evidence that a retransmission occurred.
- Splitting rennumbers packets per output file. Frame numbers cited from a split file do not refer to the original capture.

See also

`mergecap`, `reordercap`

inetsim

Kit: REMnux · Kali Linux **Capability:** Simulate network services for detonation **Version:** INetSim 1.3.2 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.inetsim.org/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Emulate common network services and interact with malware.

Synopsis

```
/usr/bin/inetsim [options]
```

Common invocations

```
# from cyberlab 38-network-emulation
inetsim --version 2>&1 | head -n 3
# from cyberlab 38-network-emulation
sudo inetsim --config ~/lab38/inetsim.conf --data-dir /var/lib/inetsim --log-dir ~/lab38/log &
```

Options

All 14 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Print this help message.	
<code>--version</code>	—	Show version information.	
<code>--config</code>	filename	Configuration file to use.	
<code>--log-dir</code>	directory	Directory logfiles are written to.	
<code>--data-dir</code>	directory	Directory containing service data.	
<code>--report-dir</code>	directory	Directory reports are written to.	
<code>--bind-address</code>	IP address	Default IP address to bind services to. Overrides configuration option 'default_bind_address'.	
<code>--max-childs</code>	num	Default maximum number of child processes per service. Overrides configuration option 'default_max_childs'.	

Flag	Argument	What it does	When you would use it
<code>--user</code>	username	Default user to run services. Overrides configuration option 'default_run_as_user'.	
<code>--faketime-init-delta</code>	secs	Initial faketime delta (seconds). Overrides configuration option 'faketime_init_delta'.	
<code>--faketime-auto-delay</code>	secs	Delay for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_delay'.	
<code>--faketime-auto-incr</code>	secs	Delta for auto incrementing faketime (seconds). Overrides configuration option 'faketime_auto_increment'.	
<code>--session</code>	id	Session id to use. Defaults to main process id.	
<code>--pidfile</code>	filename	Pid file to use. Defaults to '/var/run/inetsim.pid'.	

Gotchas

TODO: operational traps.

mergecap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 **raw Docs:** <https://www.wireshark.org>

← [Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
mergecap [options] -w <outfile>|- <infile> [<infile> ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 7 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-a	—	concatenate rather than merge files. default is to merge based on frame timestamps.	
-s	snaplen	truncate packets to bytes of data.	
-h	—	display this help and exit.	
--help	—	display this help and exit.	
-V	—	verbose output.	
-v	—	print version information and exit.	
--version	—	print version information and exit.	

Gotchas

TODO: operational traps.

See also

`editcap`, `reordercap`

ngrep

Kit: REMnux · SIFT Workstation **Capability:** Read and filter packet captures **Version:** V1.47.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://github.com/jpr5/ngrep/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Grep packet payloads, live or from a capture, with BPF filtering — pattern matching on the wire.

Synopsis

```
ngrep <-hNXViqpevxLDtTRM> <-IO pcap_dump> <-n num> <-d dev> <-A num>
<-s snaplen> <-S limitlen> <-W normal|byline|single|none> <-c cols>
<-P char> <-F file> <-K count>
<match expression> <bpf filter>
```

Common invocations

```
# from cyberlab 07-network-pcap
ngrep -V 2>&1 | head -n 1
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent'
# from cyberlab 07-network-pcap
ngrep -I exercise/sample.pcap -q -W byline 'User-Agent' | grep -i 'User-Agent'
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent"
# from cyberlab 24-wireshark-deep
ngrep -I exercise/sample.pcap -q -W byline "User-Agent" | grep -i "User-Agent"
```

Options

All 29 options parsed from the captured help text; 27 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-h</code>	—	is help/usage	
<code>-V</code>	—	is version information	

Flag	Argument	What it does	When you would use it
<code>-q</code>	—	is be quiet (don't print packet reception hash marks)	Suppress the reception hash marks.
<code>-e</code>	—	is show empty packets	Show empty packets, which are otherwise hidden.
<code>-i</code>	—	is ignore case	Case-insensitive match — usually what you want for protocol keywords and hostnames.
<code>-v</code>	—	is invert match	Invert the match, to see everything that is <i>not</i> the known traffic.
<code>-R</code>	—	is don't do privilege revocation logic	Skip privilege revocation.
<code>-x</code>	—	is print in alternate hexdump format	Print payloads as a hexdump — necessary when the protocol is not text.
<code>-X</code>	—	is interpret match expression as hexadecimal	Treat the pattern as hexadecimal, for matching binary signatures rather than text.
<code>-w</code>	—	is word-regex (expression must match as a word)	Match the pattern as a whole word, to stop a short string matching inside longer ones.
<code>-p</code>	—	is don't go into promiscuous mode	Do not enter promiscuous mode — capture only traffic addressed to this host.
<code>-l</code>	—	is make stdout line buffered	Line-buffer stdout, so output appears when piped.
<code>-D</code>	—	is replay pcap_dumps with their recorded time intervals	Replay a pcap at its recorded timing rather than as fast as possible.
<code>-t</code>	—	is print timestamp every time a packet is matched	Print a timestamp on every match.
<code>-T</code>	—	is print delta timestamp every time a packet is matched specify twice for delta from first match	Print the delta since the previous match; twice for delta from the first. Useful for spotting beaconing intervals.
<code>-M</code>	—	is don't do multi-line match (do single-line match instead)	Single-line matching instead of multi-line.
<code>-I</code>	—	is read packet stream from pcap format file pcap_dump	Read from a pcap file instead of an interface. The safe mode: no capture privileges and no risk of touching live traffic.
<code>-O</code>	—	is dump matched packets in pcap format to pcap_dump	Write matched packets to a pcap, turning a search into a smaller capture that Wireshark can open.
<code>-n</code>	—	is look at only num packets	Stop after N packets.
<code>-A</code>	—	is dump num packets after a match	Also print N packets following each match, for the response to the request that matched.
<code>-s</code>	—	is set the bpf caplen	Set the BPF capture length.
<code>-S</code>	—	is set the limitlen on matched packets	Limit how much of a matched packet is shown.

Flag	Argument	What it does	When you would use it
<code>-W</code>	—	is set the dump format (normal, byline, single, none)	Output format: <code>byline</code> is far more readable for text protocols than the default.
<code>-c</code>	—	is force the column width to the specified size	Force the column width.
<code>-P</code>	—	is set the non-printable display char to what is specified	Set the character shown for non-printable bytes.
<code>-F</code>	—	is read the bpf filter from the specified file	Read the BPF filter from a file, when it is too long to be comfortable on a command line.
<code>-N</code>	—	is show sub protocol number	Show sub-protocol numbers.
<code>-d</code>	—	is use specified device instead of the pcap default	Choose the capture interface rather than the pcap default.
<code>-K</code>	—	is send N packets to kill observed connections	Send packets to kill matched connections. This writes to the network — it is not an analysis option, and it does not belong anywhere near evidence handling.

Gotchas

- It matches within individual packets. A string split across TCP segments will not be found — reassembly is `tshark`'s job, not this one's.
- `-K` is the one flag here that changes the world instead of observing it. Everything else reads; that one transmits.
- Matching payload on a live interface needs capture privileges, and on a busy link `ngrep` drops packets silently. Capture first, search the file afterwards, when the answer has to be complete.

See also

`tshark`, `capinfos`, `tcpflow`

nmap

Kit: Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Version:** Nmap version 7.93 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Nmap 7.93 (<https://nmap.org>)

Synopsis

```
nmap [Scan Type(s)] [Options] {target specification}
```

Common invocations

```
# from cyberlab 11-offensive-kali
nmap --version
# from cyberlab 11-offensive-kali
nmap -sT -Pn -p 1-1024 127.0.0.1
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -p 1-1000 -oN scan.txt "$TARGET"
# from cyberlab 26-metasploit-workflow
nmap -sV -Pn -oX scan.xml "$TARGET"
# from cyberlab 41-web-app-testing
nmap -sV -p 80,443,8080 "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 80,443 --script http-title,http-headers,http-server-header "$TARGET"
# from cyberlab 41-web-app-testing
nmap -p 8000 --script http-title 127.0.0.1
```

Options

All 66 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--exclude</code>	host1[,host2] [,host3],...	Exclude hosts/networks	
<code>--excludefile</code>	exclude_file	Exclude list from file	

Flag	Argument	What it does	When you would use it
<code>--dns-servers</code>	serv1[,serv2],...	Specify custom DNS servers	
<code>--system-dns</code>	—	Use OS's DNS resolver	
<code>--traceroute</code>	—	Trace hop path to each host	
<code>--scanflags</code>	flags	Customize TCP scan flags	
<code>-b</code>	FTP relay host	FTP bounce scan	
<code>-p</code>	port ranges	Only scan specified ports Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9	
<code>--exclude-ports</code>	port ranges	Exclude the specified ports from scanning	
<code>-F</code>	—	Fast mode - Scan fewer ports than the default scan	
<code>-r</code>	—	Scan ports sequentially - don't randomize	
<code>--top-ports</code>	number	Scan most common ports	
<code>--port-ratio</code>	ratio	Scan ports more common than	
<code>--version-intensity</code>	level	Set from 0 (light) to 9 (try all probes)	
<code>--version-light</code>	—	Limit to most likely probes (intensity 2)	
<code>--version-all</code>	—	Try every single probe (intensity 9)	
<code>--version-trace</code>	—	Show detailed version scan activity (for debugging)	
<code>--script</code>	Lua scripts	is a comma separated list of directories, script-files or script-categories	
<code>--script-args</code>	n1=v1,[n2=v2,...]	provide arguments to scripts	
<code>--script-args-file</code>	filename	provide NSE script args in a file	
<code>--script-trace</code>	—	Show all data sent and received	
<code>--script-updatedb</code>	—	Update the script database.	
<code>--script-help</code>	Lua scripts	Show help about scripts. is a comma-separated list of script-files or script-categories.	
<code>-O</code>	—	Enable OS detection	
<code>--osscan-limit</code>	—	Limit OS detection to promising targets	
<code>--osscan-guess</code>	—	Guess OS more aggressively	
<code>-T</code>	0-5	Set timing template (higher is faster)	
<code>--min-hostgroup</code>	—	Parallel host scan group sizes	
<code>--min-parallelism</code>	—	Probe parallelization	
<code>--min-rtt-timeout</code>	—	Specifies probe round trip time.	

Flag	Argument	What it does	When you would use it
<code>--max-retries</code>	tries	Caps number of port scan probe retransmissions.	
<code>--host-timeout</code>	time	Give up on target after this long	
<code>--min-rate</code>	number	Send packets no slower than per second	
<code>--max-rate</code>	number	Send packets no faster than per second	
<code>-f</code>	—	fragment packets (optionally w/given MTU)	
<code>--mtu</code>	val	fragment packets (optionally w/given MTU)	
<code>-D</code>	decoy1,decoy2[,ME],...	Cloak a scan with decoys	
<code>-S</code>	IP_Address	Spoof source address	
<code>-e</code>	iface	Use specified interface	
<code>--proxies</code>	url1,[url2],...	Relay connections through HTTP/SOCKS4 proxies	
<code>--data</code>	hex string	Append a custom payload to sent packets	
<code>--data-string</code>	string	Append a custom ASCII string to sent packets	
<code>--data-length</code>	num	Append random data to sent packets	
<code>--ip-options</code>	options	Send packets with specified ip options	
<code>--ttl</code>	val	Set IP time-to-live field	
<code>--spoof-mac</code>	mac address	Spoof your MAC address	
<code>--badsum</code>	—	Send packets with a bogus TCP/UDP/SCTP checksum	
<code>-v</code>	—	Increase verbosity level (use -vv or more for greater effect)	
<code>-d</code>	—	Increase debugging level (use -dd or more for greater effect)	
<code>--reason</code>	—	Display the reason a port is in a particular state	
<code>--open</code>	—	Only show open (or possibly open) ports	
<code>--packet-trace</code>	—	Show all packets sent and received	
<code>--iflist</code>	—	Print host interfaces and routes (for debugging)	
<code>--append-output</code>	—	Append to rather than clobber specified output files	
<code>--resume</code>	filename	Resume an aborted scan	
<code>--noninteractive</code>	—	Disable runtime interactions via keyboard	
<code>--stylesheet</code>	path	XSL stylesheet to transform XML output to HTML	
<code>--webxml</code>	—	Reference stylesheet from Nmap.Org for more portable XML	
<code>--no-stylesheet</code>	—	Prevent associating of XSL stylesheet w/XML output	
<code>-6</code>	—	Enable IPv6 scanning	
<code>-A</code>	—	Enable OS detection, version detection, script scanning, and traceroute	
<code>--datadir</code>	dirname	Specify custom Nmap data file location	
<code>--privileged</code>	—	Assume that the user is fully privileged	

Flag	Argument	What it does	When you would use it
<code>--unprivileged</code>	—	Assume the user lacks raw socket privileges	
<code>-V</code>	—	Print version number	
<code>-h</code>	—	Print this help summary page.	

Gotchas

TODO: operational traps.

See also

`nping`, `arp-scan`

nping

Kit: Kali Linux · FLARE-VM **Capability:** Probe or scan hosts and services **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Nping 0.7.93 (<https://nmap.org/nping>)

Synopsis

```
nping [Probe mode] [Options] {target specification}
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 85 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>--tcp-connect</code>	—	Unprivileged TCP connect probe mode.	
<code>--tcp</code>	—	TCP probe mode.	
<code>--udp</code>	—	UDP probe mode.	
<code>--icmp</code>	—	ICMP probe mode.	
<code>--arp</code>	—	ARP/RARP probe mode.	
<code>--tr</code>	—	Traceroute mode (can only be used with TCP/UDP/ICMP modes).	
<code>--traceroute</code>	—	Traceroute mode (can only be used with TCP/UDP/ICMP modes).	
<code>-p</code>	port spec	Set destination port(s).	
<code>--dest-port</code>	port spec	Set destination port(s).	
<code>-g</code>	portnumber	Try to use a custom source port.	
<code>--source-port</code>	portnumber	Try to use a custom source port.	
<code>--seq</code>	seqnumber	Set sequence number.	

Flag	Argument	What it does	When you would use it
<code>--flags</code>	flag list	Set TCP flags (ACK,PSH,RST,SYN,FIN...)	
<code>--ack</code>	acknumber	Set ACK number.	
<code>--win</code>	size	Set window size.	
<code>--badsum</code>	—	Use a random invalid checksum.	
<code>--icmp-type</code>	type	ICMP type.	
<code>--icmp-code</code>	code	ICMP code.	
<code>--icmp-id</code>	id	Set identifier.	
<code>--icmp-seq</code>	n	Set sequence number.	
<code>--icmp-redirect-addr</code>	addr	Set redirect address.	
<code>--icmp-param-pointer</code>	pnt	Set parameter problem pointer.	
<code>--icmp-advert-lifetime</code>	time	Set router advertisement lifetime.	
<code>--icmp-advert-entry</code>	IP,pref	Add router advertisement entry.	
<code>--icmp-orig-time</code>	—	: Set originate timestamp.	
<code>--icmp-recv-time</code>	—	: Set receive timestamp.	
<code>--icmp-trans-time</code>	timestamp	Set transmit timestamp.	
<code>--arp-type</code>	type	Type: ARP, ARP-reply, RARP, RARP-reply.	
<code>--arp-sender-mac</code>	mac	Set sender MAC address.	
<code>--arp-sender-ip</code>	—	: Set sender IP address.	
<code>--arp-target-mac</code>	mac	Set target MAC address.	
<code>--arp-target-ip</code>	—	: Set target IP address.	
<code>-S</code>	—	Set source IP address.	
<code>--source-ip</code>	—	Set source IP address.	
<code>--dest-ip</code>	addr	Set destination IP address (used as an alternative to {target specification}).	
<code>--tos</code>	tos	Set type of service field (8bits).	
<code>--id</code>	—	: Set identification field (16 bits).	
<code>--df</code>	—	Set Don't Fragment flag.	
<code>--mf</code>	—	Set More Fragments flag.	
<code>--evil</code>	—	Set Reserved / Evil flag.	
<code>--ttl</code>	hops	Set time to live [0-255].	
<code>--badsum-ip</code>	—	Use a random invalid checksum.	

Flag	Argument	What it does	When you would use it
<code>--ip-options</code>	S R [route]]L [route]]T U ...	Set IP options	
<code>--mtu</code>	size	Set MTU. Packets get fragmented if MTU is small enough.	
<code>-6</code>	—	Use IP version 6.	
<code>--IPv6</code>	—	Use IP version 6.	
<code>--hop-limit</code>	—	Set hop limit (same as IPv4 TTL).	
<code>--traffic-class</code>	class	: Set traffic class.	
<code>--flow</code>	label	Set flow label.	
<code>--dest-mac</code>	mac	Set destination mac address. (Disables ARP resolution)	
<code>--source-mac</code>	mac	Set source MAC address.	
<code>--ether-type</code>	type	Set EtherType value.	
<code>--data</code>	hex string	Include a custom payload.	
<code>--data-string</code>	text	Include a custom ASCII text.	
<code>--data-length</code>	len	Include len random bytes as payload.	
<code>--echo-client</code>	passphrase	Run Nping in client mode.	
<code>--echo-server</code>	passphrase	Run Nping in server mode.	
<code>--echo-port</code>	port	Use custom to listen or connect.	
<code>--no-crypto</code>	—	Disable encryption and authentication.	
<code>--once</code>	—	Stop the server after one connection.	
<code>--safe-payloads</code>	—	Erase application data in echoed packets.	
<code>--delay</code>	time	Adjust delay between probes.	
<code>--rate</code>	—	: Send num packets per second.	
<code>-h</code>	—	Display help information.	
<code>--help</code>	—	Display help information.	
<code>-V</code>	—	Display current version number.	
<code>--version</code>	—	Display current version number.	
<code>-c</code>	n	Stop after rounds.	
<code>--count</code>	n	Stop after rounds.	
<code>-e</code>	name	Use supplied network interface.	
<code>--interface</code>	name	Use supplied network interface.	
<code>-H</code>	—	Do not display sent packets.	
<code>--hide-sent</code>	—	Do not display sent packets.	
<code>-N</code>	—	Do not try to capture replies.	
<code>--no-capture</code>	—	Do not try to capture replies.	

Flag	Argument	What it does	When you would use it
<code>--privileged</code>	—	Assume user is fully privileged.	
<code>--unprivileged</code>	—	Assume user lacks raw socket privileges.	
<code>--send-eth</code>	—	Send packets at the raw Ethernet layer.	
<code>--send-ip</code>	—	Send packets using raw IP sockets.	
<code>--bpf-filter</code>	filter spec	Specify custom BPF filter.	
<code>-v</code>	—	Increment verbosity level by one.	
<code>-d</code>	—	Increment debugging level by one.	
<code>-q</code>	—	Decrease verbosity level by one.	
<code>--quiet</code>	—	Set verbosity and debug level to minimum.	
<code>--debug</code>	—	Set verbosity and debug to the max level.	

Gotchas

TODO: operational traps.

See also

`nmap`, `arp-scan`

reordercap

Kit: REMnux · Kali Linux · FLARE-VM · SIFT Workstation **Capability:** Split, merge or repair capture files **Version:** Git v4.0.17 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://www.wireshark.org>

[← Capability index](#) · [Kit tool list](#)

Purpose

Capture and analyze network traffic with this sniffer.

Synopsis

```
reordercap [options] <infile> <outfile>
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-n	—	don't write to output file if the input file is ordered.	
-h	—	display this help and exit.	
-v	—	print version information and exit.	

Gotchas

TODO: operational traps.

See also

[editcap](#), [mergecap](#)

tcpflow

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Read and filter packet captures; Extract files and payloads from traffic **Version:** TCPFLOW 1.6.1 **Captured:** cyberlab-aio via `--help` on 2026-08-02 [raw Docs:](#) <https://downloads.digitalcorpora.org/downloads/tcpflow/>

← [Capability index](#) · [Kit tool list](#)

Purpose

Analyze the flow of network traffic.

Synopsis

```
tcpflow [-aBcCDhIpsvVZ] [-b max_bytes] [-d debug_level]
[-[eE] scanner] [-f max_fds] [-F[ctTXMkmg]] [-h|--help] [-i iface]
[-l files...] [-L semlock] [-m min_bytes] [-o outdir] [-r file] [-R file]
[-S name=value] [-T template] [-U|--relinquish-privileges user] [-v|--verbose]
[-w file] [-x scanner] [-X xmlfile] [-z|--chroot dir] [expression]
```

Common invocations

```
# from cyberlab 07-network-pcap
tcpflow --version | head -n 1
```

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-a</code>	—	do ALL post-processing.	
<code>-b</code>	max_bytes	max number of bytes per flow to save	
<code>-d</code>	debug_level	debug level; default is 1	
<code>-f</code>	—	maximum number of file descriptors to use	
<code>-h</code>	—	print this help message (-hh for more help)	
<code>-H</code>	—	print detailed information about each scanner	
<code>-i</code>	—	network interface on which to listen	
<code>-I</code>	—	write for each flow another file *.findx to provide byte-indexed timestamps	
<code>-g</code>	—	output each flow in alternating colors (note change!)	

Flag	Argument	What it does	When you would use it
<code>-l</code>	—	treat non-flag arguments as input files rather than a pcap expression	
<code>-L</code>	—	semlock - specifies that writes are locked using a named semaphore	
<code>-p</code>	—	don't use promiscuous mode	
<code>-q</code>	—	quiet mode - do not print warnings	
<code>-r</code>	file	read packets from tcpdump pcap file (may be repeated)	
<code>-R</code>	file	read packets from tcpdump pcap file TO FINISH CONNECTIONS	
<code>-v</code>	—	verbose operation equivalent to -d 10	
<code>-V</code>	—	print version number and exit	
<code>-w</code>	—	file : write packets not processed to file	
<code>-o</code>	—	outdir : specify output directory (default '.')	
<code>-X</code>	—	filename : DFXML output to filename	
<code>-m</code>	—	bytes : specifies skip that starts a new stream (default 16777216).	
<code>-Z</code>	—	do not decompress gzip-compressed HTTP transactions	
<code>-K</code>	—	output keep pcap flow structure.	
<code>-U</code>	user	relinquish privileges and become user (if running as root)	
<code>-z</code>	dir	chroot to dir (requires that -U be used).	
<code>-E</code>	scanner	- turn off all scanners except scanner	
<code>-B</code>	—	binary output, even with -c or -C (normally -c or -C turn it off)	
<code>-c</code>	—	console print only (don't create files)	
<code>-C</code>	—	console print only, but without the display of source/dest header	
<code>-0</code>	—	don't print newlines after packets when printing to console	
<code>-s</code>	—	strip non-printable characters (change to '.')	
<code>-J</code>	—	output json format.	
<code>-D</code>	—	output in hex (useful to combine with -c or -C)	

Gotchas

TODO: operational traps.

See also

`tshark`, `capinfos`, `ngrep`, `tcpxtract`, `foremost`

curl

Kit: REMnux **Capability:** Fetch and verify external references **Version:** curl 7.88.1 **Captured:** cyberlab-ai0 via --help on 2026-08-02 **raw Docs:** <https://curl.se>

[← Capability index](#) · [Kit tool list](#)

Purpose

Interact with servers via supported protocols, including HTTP, HTTPS, FTP, IMAP, etc. using this command-line tool.

Synopsis

```
curl [options...] <url>
```

Common invocations

```
# from cyberlab 38-network-emulation
curl -s http://127.0.0.1/malware.bin -o /tmp/served.bin && file /tmp/served.bin
# from cyberlab 38-network-emulation
curl -s "http://$TARGET_DOMAIN/gate.php?id=203.0.113.10" -o /tmp/beacon_reply.bin
```

Options

All 22 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-d	data	HTTP POST data	
--data	data	HTTP POST data	
-f	—	Fail fast with no output on HTTP errors	
--fail	—	Fail fast with no output on HTTP errors	
-h	category	Get help for commands	
--help	category	Get help for commands	
-i	—	Include protocol response headers in the output	
--include	—	Include protocol response headers in the output	
-o	file	Write to file instead of stdout	
--output	file	Write to file instead of stdout	
-O	—	Write output to a file named as the remote file	

Flag	Argument	What it does	When you would use it
<code>--remote-name</code>	—	Write output to a file named as the remote file	
<code>-s</code>	—	Silent mode	
<code>--silent</code>	—	Silent mode	
<code>-T</code>	file	Transfer local FILE to destination	
<code>--upload-file</code>	file	Transfer local FILE to destination	
<code>-A</code>	name	Send User-Agent to server	
<code>--user-agent</code>	name	Send User-Agent to server	
<code>-v</code>	—	Make the operation more talkative	
<code>--verbose</code>	—	Make the operation more talkative	
<code>-V</code>	—	Show version number and quit	
<code>--version</code>	—	Show version number and quit	

Gotchas

TODO: operational traps.

See also

`wget`

ezhexviewer

Kit: REMnux **Capability:** Inspect files by hand **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#) **Docs:** <https://www.decalage.info/python/oletools>

[← Capability index](#) · [Kit tool list](#)

Purpose

Microsoft Office OLE2 compound documents.

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`xxd`, `less`

less

Kit: Base OS — present on every Linux image **Capability:** Inspect files by hand **Version:** version: No such file or directory **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

← Capability index · Kit tool list

Purpose

S S U U M M M M A A R R Y Y O O F F L L E E S S S S C C O O M M M M A A N N D D S S

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 62 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-?	—	 --help Display help (from command line).	
-a	—	 --search-skip-screen Search skips current screen.	
-A	—	 --SEARCH-SKIP-SCREEN Search starts just after target line.	
-b	_ON	 --buffers=[_ON] Number of buffers.	
-B	—	 --auto-buffers Don't automatically allocate buffers for pipes.	
-c	—	 --clear-screen Repaint by clearing rather than scrolling.	
-d	—	 --dumb Dumb terminal.	
-e	—	 --quit-at-eof --QUIT-AT-EOF Quit at end of file.	
-E	—	 --quit-at-eof --QUIT-AT-EOF Quit at end of file.	
-f	—	 --force Force open non-regular files.	
-F	—	 --quit-if-one-screen Quit if entire file fits on first screen.	
-g	—	 --hilite-search Highlight only last match for searches.	
-G	—	 --HILITE-SEARCH Don't highlight any matches for searches.	
-h	_ON	 --max-back-scroll=[_ON] Backward scroll limit.	
-i	—	 --ignore-case Ignore case in searches that do not contain uppercase.	

Flag	Argument	What it does	When you would use it
<code>-I</code>	—	 --IGNORE-CASE Ignore case in all searches.	
<code>-j</code>	<code>_ON</code>	 --jump-target=[<code>_ON</code>] Screen position of target lines.	
<code>-J</code>	—	 --status-column Display a status column at left edge of screen.	
<code>-k</code>	<code>□/□i_□l_□e</code>	. --lesskey-file=[<code>□/□i_□l_□e</code>] Use a lesskey file.	
<code>-K</code>	—	 --quit-on-intr Exit less in response to ctrl-C.	
<code>-L</code>	—	 --no-lessopen Ignore the LESSOPEN environment variable.	
<code>-m</code>	—	 --long-prompt --LONG-PROMPT Set prompt style.	
<code>-M</code>	—	 --long-prompt --LONG-PROMPT Set prompt style.	
<code>-n</code>	—	 --line-numbers --LINE-NUMBERS Don't use line numbers.	
<code>-N</code>	—	 --line-numbers --LINE-NUMBERS Don't use line numbers.	
<code>-o</code>	<code>□/□i_□l_□e</code>	. --log-file=[<code>□/□i_□l_□e</code>] Copy to log file (standard input only).	
<code>-O</code>	<code>□/□i_□l_□e</code>	. --LOG-FILE=[<code>□/□i_□l_□e</code>] Copy to log file (unconditionally overwrite).	
<code>-p</code>	<code>□p□a_□t_□t_□e_□r_□n</code>	Start at pattern (from command line).	
<code>--pattern</code>	<code>□p□a_□t_□t_□e_□r_□n</code>	Start at pattern (from command line).	
<code>-P</code>	<code>□p□r_□o_□m_□p_□t</code>	Define new prompt.	
<code>--prompt</code>	<code>□p□r_□o_□m_□p_□t</code>	Define new prompt.	
<code>-q</code>	—	 --quiet --QUIET --silent --SILENT Quiet the terminal bell.	
<code>-Q</code>	—	 --quiet --QUIET --silent --SILENT Quiet the terminal bell.	
<code>-r</code>	—	 --raw-control-chars --RAW-CONTROL-CHARS Output "raw" control characters.	
<code>-R</code>	—	 --raw-control-chars --RAW-CONTROL-CHARS Output "raw" control characters.	
<code>-s</code>	—	 --squeeze-blank-lines Squeeze multiple blank lines.	
<code>-S</code>	—	 --chop-long-lines Chop (truncate) long lines rather than wrapping.	
<code>-t</code>	<code>□t□a_□g</code>	.. --tag=[<code>□t□a_□g</code>] Find a tag.	
<code>-T</code>	<code>□t□a_□g_□s_□f_□i_□l_□e</code>	Use an alternate tags file.	
<code>--tag-file</code>	<code>□t□a_□g_□s_□f_□i_□l_□e</code>	Use an alternate tags file.	
<code>-u</code>	—	 --underline-special --UNDERLINE-SPECIAL Change handling of backspaces.	
<code>-U</code>	—	 --underline-special --UNDERLINE-SPECIAL Change handling of backspaces.	
<code>-V</code>	—	 --version Display the version number of "less".	

Flag	Argument	What it does	When you would use it
<code>-w</code>	—	 --hilite-unread Highlight first new line after forward-screen.	
<code>-W</code>	—	 --HILITE-UNREAD Highlight first new line after any forward movement.	
<code>-X</code>	—	 --no-init Don't use termcap init/deinit strings.	
<code>-y</code>	<code>_ON</code>	... --max-forw-scroll=[<code>_ON</code>] Forward scroll limit.	
<code>-z</code>	<code>_ON</code>	... --window=[<code>_ON</code>] Set size of window.	
<code>-#</code>	<code>_ON</code>	... --shift=[<code>_ON</code>] Set horizontal scroll amount (0 = one half screen width).	
<code>--file-size</code>	—	Automatically determine the size of the input file.	
<code>--follow-name</code>	—	The F command changes files if the input file is renamed.	
<code>--incsearch</code>	—	Search file as each pattern character is typed in.	
<code>--line-num-width</code>	<code>N</code>	Set the width of the -N line number field to N characters.	
<code>--mouse</code>	—	Enable mouse input.	
<code>--no-keypad</code>	—	Don't send termcap keypad init/deinit strings.	
<code>--no-histdups</code>	—	Remove duplicates from command history.	
<code>--rscroll</code>	<code>C</code>	Set the character used to mark truncated lines.	
<code>--save-marks</code>	—	Retain marks across invocations of less.	
<code>--status-col-width</code>	<code>N</code>	Set the width of the -J status column to N characters.	
<code>--use-backslash</code>	—	Subsequent options use backslash as escape char.	
<code>--use-color</code>	—	Enables colored text.	
<code>--wheel-lines</code>	<code>N</code>	Each click of the mouse wheel moves N lines.	

Gotchas

TODO: operational traps.

See also

`xxd`, `ezhexviewer`

wget

Kit: REMnux **Capability:** Fetch and verify external references **Version:** GNU Wget 1.21.3 **Captured:** cyberlab-aio
via `--help` on 2026-08-02 **raw Docs:** <https://www.gnu.org/software/wget/>

[← Capability index](#) · [Kit tool list](#)

Purpose

Interact with servers via HTTP, HTTPS, FTP, and FTPS using this command-line tool.

Synopsis

```
wget [OPTION]... [URL]...
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 185 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-V</code>	—	display the version of Wget and exit	
<code>--version</code>	—	display the version of Wget and exit	
<code>-h</code>	—	print this help	
<code>--help</code>	—	print this help	
<code>-b</code>	—	go to background after startup	
<code>--background</code>	—	go to background after startup	
<code>-e</code>	COMMAND	execute a '.wgetrc'-style command	
<code>--execute</code>	COMMAND	execute a '.wgetrc'-style command	
<code>-o</code>	FILE	log messages to FILE	
<code>--output-file</code>	FILE	log messages to FILE	
<code>-a</code>	FILE	append messages to FILE	
<code>--append-output</code>	FILE	append messages to FILE	
<code>-d</code>	—	print lots of debugging information	

Flag	Argument	What it does	When you would use it
<code>--debug</code>	—	print lots of debugging information	
<code>-q</code>	—	quiet (no output)	
<code>--quiet</code>	—	quiet (no output)	
<code>-v</code>	—	be verbose (this is the default)	
<code>--verbose</code>	—	be verbose (this is the default)	
<code>--report-speed</code>	TYPE	output bandwidth as TYPE. TYPE can be bits	
<code>-i</code>	FILE	download URLs found in local or external FILE	
<code>--input-file</code>	FILE	download URLs found in local or external FILE	
<code>-F</code>	—	treat input file as HTML	
<code>--force-html</code>	—	treat input file as HTML	
<code>-B</code>	URL	resolves HTML input-file links (-i -F) relative to URL	
<code>--base</code>	URL	resolves HTML input-file links (-i -F) relative to URL	
<code>--config</code>	FILE	specify config file to use	
<code>--no-config</code>	—	do not read any config file	
<code>--rejected-log</code>	FILE	log reasons for URL rejection to FILE	
<code>-t</code>	NUMBER	set number of retries to NUMBER (0 unlimits)	
<code>--tries</code>	NUMBER	set number of retries to NUMBER (0 unlimits)	
<code>--retry-connrefused</code>	—	retry even if connection is refused	
<code>--retry-on-http-error</code>	ERRORS	comma-separated list of HTTP errors to retry	
<code>-O</code>	FILE	write documents to FILE	
<code>--output-document</code>	FILE	write documents to FILE	
<code>--no-netrc</code>	—	don't try to obtain credentials from .netrc	
<code>-c</code>	—	resume getting a partially-downloaded file	
<code>--continue</code>	—	resume getting a partially-downloaded file	
<code>--start-pos</code>	OFFSET	start downloading from zero-based position OFFSET	
<code>--progress</code>	TYPE	select progress gauge type	
<code>--show-progress</code>	—	display the progress bar in any verbosity mode	
<code>-N</code>	—	don't re-retrieve files unless newer than local	
<code>--timestamping</code>	—	don't re-retrieve files unless newer than local	
<code>--no-if-modified-since</code>	—	don't use conditional if-modified-since get requests in timestamping mode	

Flag	Argument	What it does	When you would use it
<code>--no-use-server-timestamps</code>	—	don't set the local file's timestamp by the one on the server	
<code>-S</code>	—	print server response	
<code>--server-response</code>	—	print server response	
<code>--spider</code>	—	don't download anything	
<code>-T</code>	SECONDS	set all timeout values to SECONDS	
<code>--timeout</code>	SECONDS	set all timeout values to SECONDS	
<code>--dns-timeout</code>	SECS	set the DNS lookup timeout to SECS	
<code>--connect-timeout</code>	SECS	set the connect timeout to SECS	
<code>--read-timeout</code>	SECS	set the read timeout to SECS	
<code>-w</code>	SECONDS	wait SECONDS between retrievals (applies if more than 1 URL is to be retrieved)	
<code>--wait</code>	SECONDS	wait SECONDS between retrievals (applies if more than 1 URL is to be retrieved)	
<code>--waitretry</code>	SECONDS	wait 1..SECONDS between retries of a retrieval (applies if more than 1 URL is to be retrieved)	
<code>--random-wait</code>	—	wait from 0.5WAIT...1.5WAIT secs between retrievals (applies if more than 1 URL is to be retrieved)	
<code>--no-proxy</code>	—	explicitly turn off proxy	
<code>-Q</code>	NUMBER	set retrieval quota to NUMBER	
<code>--quota</code>	NUMBER	set retrieval quota to NUMBER	
<code>--bind-address</code>	ADDRESS	bind to ADDRESS (hostname or IP) on local host	
<code>--limit-rate</code>	RATE	limit download rate to RATE	
<code>--no-dns-cache</code>	—	disable caching DNS lookups	
<code>--restrict-file-names</code>	OS	restrict chars in file names to ones OS allows	
<code>--ignore-case</code>	—	ignore case when matching files/directories	
<code>-4</code>	—	connect only to IPv4 addresses	
<code>--inet4-only</code>	—	connect only to IPv4 addresses	
<code>-6</code>	—	connect only to IPv6 addresses	
<code>--inet6-only</code>	—	connect only to IPv6 addresses	
<code>--prefer-family</code>	FAMILY	connect first to addresses of specified family, one of IPv6, IPv4, or none	
<code>--user</code>	USER	set both ftp and http user to USER	
<code>--password</code>	PASS	set both ftp and http password to PASS	

Flag	Argument	What it does	When you would use it
<code>--ask-password</code>	—	prompt for passwords	
<code>--use-askpass</code>	COMMAND	specify credential handler for requesting username and password. If no COMMAND is specified the WGET_ASKPASS or the SSH_ASKPASS environment variable is used.	
<code>--no-iri</code>	—	turn off IRI support	
<code>--local-encoding</code>	ENC	use ENC as the local encoding for IRIs	
<code>--remote-encoding</code>	ENC	use ENC as the default remote encoding	
<code>--unlink</code>	—	remove file before clobber	
<code>--xattr</code>	—	turn on storage of metadata in extended file attributes	
<code>-x</code>	—	force creation of directories	
<code>--force-directories</code>	—	force creation of directories	
<code>--protocol-directories</code>	—	use protocol name in directories	
<code>-P</code>	PREFIX	save files to PREFIX/..	
<code>--directory-prefix</code>	PREFIX	save files to PREFIX/..	
<code>--cut-dirs</code>	NUMBER	ignore NUMBER remote directory components	
<code>--http-user</code>	USER	set http user to USER	
<code>--http-password</code>	PASS	set http password to PASS	
<code>--no-cache</code>	—	disallow server-cached data	
<code>--default-page</code>	NAME	change the default page name (normally this is 'index.html').	
<code>-E</code>	—	save HTML/CSS documents with proper extensions	
<code>--adjust-extension</code>	—	save HTML/CSS documents with proper extensions	
<code>--ignore-length</code>	—	ignore 'Content-Length' header field	
<code>--header</code>	STRING	insert STRING among the headers	
<code>--compression</code>	TYPE	choose compression, one of auto, gzip and none. (default: none)	
<code>--max-redirect</code>	—	maximum redirections allowed per page	
<code>--proxy-user</code>	USER	set USER as proxy username	
<code>--proxy-password</code>	PASS	set PASS as proxy password	
<code>--referer</code>	URL	include 'Referer: URL' header in HTTP request	
<code>--save-headers</code>	—	save the HTTP headers to file	
<code>-U</code>	AGENT	identify as AGENT instead of Wget/VERSION	

Flag	Argument	What it does	When you would use it
<code>--user-agent</code>	AGENT	identify as AGENT instead of Wget/VERSION	
<code>--no-http-keep-alive</code>	—	disable HTTP keep-alive (persistent connections)	
<code>--no-cookies</code>	—	don't use cookies	
<code>--load-cookies</code>	FILE	load cookies from FILE before session	
<code>--save-cookies</code>	FILE	save cookies to FILE after session	
<code>--keep-session-cookies</code>	—	load and save session (non-permanent) cookies	
<code>--post-data</code>	STRING	use the POST method; send STRING as the data	
<code>--post-file</code>	FILE	use the POST method; send contents of FILE	
<code>--method</code>	HTTPMethod	use method "HTTPMethod" in the request	
<code>--body-data</code>	STRING	send STRING as data. --method MUST be set	
<code>--body-file</code>	FILE	send contents of FILE. --method MUST be set	
<code>--content-disposition</code>	—	honor the Content-Disposition header when choosing local file names (EXPERIMENTAL)	
<code>--content-on-error</code>	—	output the received content on server errors	
<code>--auth-no-challenge</code>	—	send Basic HTTP authentication information without first waiting for the server's challenge	
<code>--secure-protocol</code>	PR	choose secure protocol, one of auto, SSLv2, SSLv3, TLSv1, TLSv1_1, TLSv1_2, TLSv1_3 and PFS	
<code>--https-only</code>	—	only follow secure HTTPS links	
<code>--no-check-certificate</code>	—	don't validate the server's certificate	
<code>--certificate</code>	FILE	client certificate file	
<code>--certificate-type</code>	TYPE	client certificate type, PEM or DER	
<code>--private-key</code>	FILE	private key file	
<code>--private-key-type</code>	TYPE	private key type, PEM or DER	
<code>--ca-certificate</code>	FILE	file with the bundle of CAs	
<code>--ca-directory</code>	DIR	directory where hash list of CAs is stored	
<code>--crl-file</code>	FILE	file with bundle of CRLs	
<code>--pinnedpubkey</code>	FILE	Public key (PEM/DER) file, or any number of base64 encoded sha256 hashes preceded by 'sha256/' and separated by ';', to verify peer against	

Flag	Argument	What it does	When you would use it
<code>--ciphers</code>	STR	Set the priority string (GnuTLS) or cipher list string (OpenSSL) directly. Use with care. This option overrides <code>--secure-protocol</code> . The format and syntax of this string depend on the specific SSL/TLS e	
<code>--no-hsts</code>	—	disable HSTS	
<code>--hsts-file</code>	—	path of HSTS database (will override default)	
<code>--ftp-user</code>	USER	set ftp user to USER	
<code>--ftp-password</code>	PASS	set ftp password to PASS	
<code>--no-remove-listing</code>	—	don't remove '.listing' files	
<code>--no-glob</code>	—	turn off FTP file name globbing	
<code>--no-passive-ftp</code>	—	disable the "passive" transfer mode	
<code>--preserve-permissions</code>	—	preserve remote file permissions	
<code>--retr-symlinks</code>	—	when recursing, get linked-to files (not dir)	
<code>--ftps-implicit</code>	—	use implicit FTPS (default port is 990)	
<code>--ftps-resume-ssl</code>	—	resume the SSL/TLS session started in the control connection when opening a data connection	
<code>--ftps-clear-data-connection</code>	—	cipher the control channel only; all the data will be in plaintext	
<code>--ftps-fallback-to-ftp</code>	—	fall back to FTP if FTPS is not supported in the target server	
<code>--warc-file</code>	FILENAME	save request/response data to a .warc.gz file	
<code>--warc-header</code>	STRING	insert STRING into the warcinfo record	
<code>--warc-max-size</code>	NUMBER	set maximum size of WARC files to NUMBER	
<code>--warc-cdx</code>	—	write CDX index files	
<code>--warc-dedup</code>	FILENAME	do not store records listed in this CDX file	
<code>--no-warc-compression</code>	—	do not compress WARC files with GZIP	
<code>--no-warc-digests</code>	—	do not calculate SHA1 digests	
<code>--no-warc-keep-log</code>	—	do not store the log file in a WARC record	
<code>--warc-tempdir</code>	DIRECTORY	location for temporary files created by the WARC writer	
<code>-r</code>	—	specify recursive download	
<code>--recursive</code>	—	specify recursive download	
<code>-l</code>	NUMBER	maximum recursion depth (inf or 0 for infinite)	

Flag	Argument	What it does	When you would use it
<code>--level</code>	NUMBER	maximum recursion depth (inf or 0 for infinite)	
<code>--delete-after</code>	—	delete files locally after downloading them	
<code>-k</code>	—	make links in downloaded HTML or CSS point to local files	
<code>--convert-links</code>	—	make links in downloaded HTML or CSS point to local files	
<code>--convert-file-only</code>	—	convert the file part of the URLs only (usually known as the basename)	
<code>--backups</code>	N	before writing file X, rotate up to N backup files	
<code>-K</code>	—	before converting file X, back up as X.orig	
<code>--backup-converted</code>	—	before converting file X, back up as X.orig	
<code>-m</code>	—	shortcut for <code>-N -r -l inf --no-remove-listing</code>	
<code>--mirror</code>	—	shortcut for <code>-N -r -l inf --no-remove-listing</code>	
<code>-p</code>	—	get all images, etc. needed to display HTML page	
<code>--page-requisites</code>	—	get all images, etc. needed to display HTML page	
<code>--strict-comments</code>	—	turn on strict (SGML) handling of HTML comments	
<code>-A</code>	LIST	comma-separated list of accepted extensions	
<code>--accept</code>	LIST	comma-separated list of accepted extensions	
<code>-R</code>	LIST	comma-separated list of rejected extensions	
<code>--reject</code>	LIST	comma-separated list of rejected extensions	
<code>--accept-regex</code>	REGEX	regex matching accepted URLs	
<code>--reject-regex</code>	REGEX	regex matching rejected URLs	
<code>--regex-type</code>	TYPE	regex type (posix pcre)	
<code>-D</code>	LIST	comma-separated list of accepted domains	
<code>--domains</code>	LIST	comma-separated list of accepted domains	
<code>--exclude-domains</code>	LIST	comma-separated list of rejected domains	
<code>--follow-ftp</code>	—	follow FTP links from HTML documents	
<code>--follow-tags</code>	LIST	comma-separated list of followed HTML tags	
<code>--ignore-tags</code>	LIST	comma-separated list of ignored HTML tags	
<code>-H</code>	—	go to foreign hosts when recursive	
<code>--span-hosts</code>	—	go to foreign hosts when recursive	
<code>-L</code>	—	follow relative links only	
<code>--relative</code>	—	follow relative links only	

Flag	Argument	What it does	When you would use it
<code>-I</code>	LIST	list of allowed directories	
<code>--include-directories</code>	LIST	list of allowed directories	
<code>--trust-server-names</code>	—	use the name specified by the redirection URL's last component	
<code>-X</code>	LIST	list of excluded directories	
<code>--exclude-directories</code>	LIST	list of excluded directories	

Gotchas

TODO: operational traps.

See also

[curl](#)

frida

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://frida.re>

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida [options] target
```

Common invocations

```
# from cyberlab 39-frida-tracing
frida --version
# from cyberlab 39-frida-tracing
frida -f ./target -l /dev/stdin --no-pause <<'EOF'
```

Options

All 66 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	

Flag	Argument	What it does	When you would use it
<code>--certificate</code>	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
<code>--origin</code>	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
<code>--token</code>	TOKEN	authenticate with HOST using TOKEN	
<code>--keepalive-interval</code>	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
<code>--device-option</code>	option	override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
<code>--p2p</code>	—	establish a peer-to-peer connection with target	
<code>--stun-server</code>	ADDRESS	set STUN server ADDRESS to use with --p2p	
<code>-f</code>	TARGET	spawn FILE	
<code>--file</code>	TARGET	spawn FILE	
<code>-F</code>	—	attach to frontmost application	
<code>--attach-frontmost</code>	—	attach to frontmost application	
<code>-n</code>	NAME	attach to NAME	
<code>--attach-name</code>	NAME	attach to NAME	
<code>-N</code>	IDENTIFIER	attach to IDENTIFIER	
<code>--attach-identifier</code>	IDENTIFIER	attach to IDENTIFIER	
<code>-p</code>	PID	attach to PID	
<code>--attach-pid</code>	PID	attach to PID	
<code>-W</code>	PATTERN	await spawn matching PATTERN	
<code>--await</code>	PATTERN	await spawn matching PATTERN	
<code>--stdio</code>	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
<code>--aux</code>	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
<code>--realm</code>	native,emulated	realm to attach in	
<code>--exceptor</code>	full,handler-only,off	configure the exception handling mode	
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	

Flag	Argument	What it does	When you would use it
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	
<code>-l</code>	SCRIPT	load SCRIPT	
<code>--load</code>	SCRIPT	load SCRIPT	
<code>-P</code>	PARAMETERS_JSON	parameters as JSON, same as Gadget	
<code>--parameters</code>	PARAMETERS_JSON	parameters as JSON, same as Gadget	
<code>-C</code>	USER_CMODULE	load CMODULE	
<code>--cmodule</code>	USER_CMODULE	load CMODULE	
<code>--toolchain</code>	any,internal,external	CModule toolchain to use when compiling from source code	
<code>-c</code>	CODESHARE_URI	load CODESHARE_URI	
<code>--codeshare</code>	CODESHARE_URI	load CODESHARE_URI	
<code>-e</code>	CODE	evaluate CODE	
<code>--eval</code>	CODE	evaluate CODE	
<code>-q</code>	—	quiet mode (no prompt) and quit after -l and -e	
<code>-t</code>	TIMEOUT	seconds to wait before terminating in quiet mode (or 'inf' to run forever)	
<code>--timeout</code>	TIMEOUT	seconds to wait before terminating in quiet mode (or 'inf' to run forever)	
<code>--pause</code>	—	leave main thread paused after spawning program	
<code>-o</code>	LOGFILE	output to log file	
<code>--output</code>	LOGFILE	output to log file	
<code>--eternalize</code>	—	eternalize the script before exit	
<code>--exit-on-error</code>	—	exit with code 1 after encountering any exception in the SCRIPT	
<code>--kill-on-exit</code>	—	kill the spawned program when Frida exits	
<code>--auto-perform</code>	—	wrap entered code with Java.perform	
<code>--auto-reload</code>	—	Enable auto reload of provided scripts and c module (on by default, will be required in the future)	
<code>--no-auto-reload</code>	—	Disable auto reload of provided scripts and c module	

Gotchas

TODO: operational traps.

See also

`frida-trace`, `frida-ps`, `frida-discover`, `frida-kill`, `frida-ls-devices`

frida-discover

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://frida.re>

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-discover [options] target
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 43 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	

Flag	Argument	What it does	When you would use it
<code>--keepalive-interval</code>	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
<code>--device-option</code>	option	override a backend-specific option, such as "control-endpoint=(string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
<code>--p2p</code>	—	establish a peer-to-peer connection with target	
<code>--stun-server</code>	ADDRESS	set STUN server ADDRESS to use with --p2p	
<code>-f</code>	TARGET	spawn FILE	
<code>--file</code>	TARGET	spawn FILE	
<code>-F</code>	—	attach to frontmost application	
<code>--attach-frontmost</code>	—	attach to frontmost application	
<code>-n</code>	NAME	attach to NAME	
<code>--attach-name</code>	NAME	attach to NAME	
<code>-N</code>	IDENTIFIER	attach to IDENTIFIER	
<code>--attach-identifier</code>	IDENTIFIER	attach to IDENTIFIER	
<code>-p</code>	PID	attach to PID	
<code>--attach-pid</code>	PID	attach to PID	
<code>-W</code>	PATTERN	await spawn matching PATTERN	
<code>--await</code>	PATTERN	await spawn matching PATTERN	
<code>--stdio</code>	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
<code>--aux</code>	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
<code>--realm</code>	native,emulated	realm to attach in	
<code>--exceptor</code>	full,handler-only,off	configure the exception handling mode	
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	

Flag	Argument	What it does	When you would use it
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	

Gotchas

TODO: operational traps.

See also

`frida`, `frida-trace`, `frida-ps`, `frida-kill`, `frida-ls-devices`

frida-kill

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-kill [options] process
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 20 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with “Origin” header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	

Flag	Argument	What it does	When you would use it
<code>--keepalive-interval</code>	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
<code>--device-option</code>	option	override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
<code>--p2p</code>	—	establish a peer-to-peer connection with target	
<code>--stun-server</code>	ADDRESS	set STUN server ADDRESS to use with --p2p	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	

Gotchas

TODO: operational traps.

See also

`frida`, `frida-trace`, `frida-ps`, `frida-discover`, `frida-ls-devices`

frida-ls-devices

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-ls-devices [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-O	FILE	text file containing additional command line options	
--options-file	FILE	text file containing additional command line options	
--version	—	show program's version number and exit	

Gotchas

TODO: operational traps.

See also

[frida](#), [frida-trace](#), [frida-ps](#), [frida-discover](#), [frida-kill](#)

frida-ps

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://frida.re>

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-ps [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 28 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	
--certificate	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
--origin	ORIGIN	connect to remote server with “Origin” header set to ORIGIN	
--token	TOKEN	authenticate with HOST using TOKEN	

Flag	Argument	What it does	When you would use it
<code>--keepalive-interval</code>	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
<code>--device-option</code>	option	override a backend-specific option, such as "control-endpoint=(string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
<code>--p2p</code>	—	establish a peer-to-peer connection with target	
<code>--stun-server</code>	ADDRESS	set STUN server ADDRESS to use with --p2p	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	
<code>-a</code>	—	list only applications	
<code>--applications</code>	—	list only applications	
<code>-i</code>	—	include all installed applications	
<code>--installed</code>	—	include all installed applications	
<code>-j</code>	—	output results as JSON	
<code>--json</code>	—	output results as JSON	
<code>-e</code>	—	exclude icons in output	
<code>--exclude-icons</code>	—	exclude icons in output	

Gotchas

TODO: operational traps.

See also

`frida`, `frida-trace`, `frida-discover`, `frida-kill`, `frida-ls-devices`

frida-trace

Kit: REMnux **Capability:** Emulate or instrument execution **Version:** 17.16.3 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#) **Docs:** <https://frida.re>

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Purpose

Trace the execution of a process to analyze its behavior.

Synopsis

```
frida-trace [options] target
```

Common invocations

```
# from cyberlab 39-frida-tracing
frida-trace --version
# from cyberlab 39-frida-tracing
timeout 8 frida-trace -f ./target -i getenv
```

Options

All 82 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-D	ID	connect to device with the given ID	
--device	ID	connect to device with the given ID	
-U	—	connect to USB device	
--usb	—	connect to USB device	
-R	—	connect to remote frida-server	
--remote	—	connect to remote frida-server	
-H	HOST	connect to remote frida-server on HOST	
--host	HOST	connect to remote frida-server on HOST	

Flag	Argument	What it does	When you would use it
<code>--certificate</code>	CERTIFICATE	speak TLS with HOST, expecting CERTIFICATE	
<code>--origin</code>	ORIGIN	connect to remote server with "Origin" header set to ORIGIN	
<code>--token</code>	TOKEN	authenticate with HOST using TOKEN	
<code>--keepalive-interval</code>	INTERVAL	set keepalive interval in seconds, or 0 to disable (defaults to -1 to auto-select based on transport)	
<code>--device-option</code>	option	override a backend-specific option, such as "control- endpoint= (string)localabstract:/my-frida-server" (supported types are: string, bool, int)	
<code>--p2p</code>	—	establish a peer-to-peer connection with target	
<code>--stun-server</code>	ADDRESS	set STUN server ADDRESS to use with --p2p	
<code>-f</code>	TARGET	spawn FILE	
<code>--file</code>	TARGET	spawn FILE	
<code>-F</code>	—	attach to frontmost application	
<code>--attach-frontmost</code>	—	attach to frontmost application	
<code>-n</code>	NAME	attach to NAME	
<code>--attach-name</code>	NAME	attach to NAME	
<code>-N</code>	IDENTIFIER	attach to IDENTIFIER	
<code>--attach-identifier</code>	IDENTIFIER	attach to IDENTIFIER	
<code>-p</code>	PID	attach to PID	
<code>--attach-pid</code>	PID	attach to PID	
<code>-W</code>	PATTERN	await spawn matching PATTERN	
<code>--await</code>	PATTERN	await spawn matching PATTERN	
<code>--stdio</code>	inherit,pipe	stdio behavior when spawning (defaults to "inherit")	
<code>--aux</code>	option	set aux option when spawning, such as "uid=(int)42" (supported types are: string, bool, int)	
<code>--realm</code>	native,emulated	realm to attach in	
<code>--exceptor</code>	full,handler-only,off	configure the exception handling mode	
<code>--disable-unwind-broker</code>	—	disable the unwind broker	
<code>--disable-exit-monitor</code>	—	disable the exit monitor	
<code>--disable-thread-suspend-monitor</code>	—	disable the thread suspend monitor	
<code>--linker-notifier-offset</code>	OFFSET	add a linker notifier OFFSET (may be specified multiple times)	
<code>--runtime</code>	qjs,v8	script runtime to use	

Flag	Argument	What it does	When you would use it
<code>--debug</code>	—	enable the Node.js compatible script debugger	
<code>--squelch-crash</code>	—	if enabled, will not dump crash report to console	
<code>-O</code>	FILE	text file containing additional command line options	
<code>--options-file</code>	FILE	text file containing additional command line options	
<code>--version</code>	—	show program's version number and exit	
<code>-I</code>	MODULE	include MODULE	
<code>--include-module</code>	MODULE	include MODULE	
<code>-X</code>	MODULE	exclude MODULE	
<code>--exclude-module</code>	MODULE	exclude MODULE	
<code>-i</code>	FUNCTION	include [MODULE!]FUNCTION	
<code>--include</code>	FUNCTION	include [MODULE!]FUNCTION	
<code>-x</code>	FUNCTION	exclude [MODULE!]FUNCTION	
<code>--exclude</code>	FUNCTION	exclude [MODULE!]FUNCTION	
<code>-T</code>	INCLUDE_IMPORTS	include program's imports	
<code>--include-imports</code>	INCLUDE_IMPORTS	include program's imports	
<code>-t</code>	MODULE	include MODULE imports	
<code>--include-module-imports</code>	MODULE	include MODULE imports	
<code>-m</code>	OBJC_METHOD	include OBJC_METHOD	
<code>--include-objc-method</code>	OBJC_METHOD	include OBJC_METHOD	
<code>-M</code>	OBJC_METHOD	exclude OBJC_METHOD	
<code>--exclude-objc-method</code>	OBJC_METHOD	exclude OBJC_METHOD	
<code>-y</code>	SWIFT_FUNC	include SWIFT_FUNC	
<code>--include-swift-func</code>	SWIFT_FUNC	include SWIFT_FUNC	
<code>-Y</code>	SWIFT_FUNC	exclude SWIFT_FUNC	
<code>--exclude-swift-func</code>	SWIFT_FUNC	exclude SWIFT_FUNC	
<code>-j</code>	JAVA_METHOD	include JAVA_METHOD	
<code>--include-java-method</code>	JAVA_METHOD	include JAVA_METHOD	
<code>-J</code>	JAVA_METHOD	exclude JAVA_METHOD	
<code>--exclude-java-method</code>	JAVA_METHOD	exclude JAVA_METHOD	

Flag	Argument	What it does	When you would use it
<code>-s</code>	DEBUG_SYMBOL	include DEBUG_SYMBOL	
<code>--include-debug-symbol</code>	DEBUG_SYMBOL	include DEBUG_SYMBOL	
<code>-q</code>	—	do not format output messages	
<code>--quiet</code>	—	do not format output messages	
<code>-d</code>	—	add module name to generated onEnter log statement	
<code>--decorate</code>	—	add module name to generated onEnter log statement	
<code>-S</code>	PATH	path to JavaScript file used to initialize the session	
<code>--init-session</code>	PATH	path to JavaScript file used to initialize the session	
<code>-P</code>	PARAMETERS_JSON	parameters as JSON, exposed as a global named 'parameters'	
<code>--parameters</code>	PARAMETERS_JSON	parameters as JSON, exposed as a global named 'parameters'	
<code>-o</code>	OUTPUT	dump messages to file	
<code>--output</code>	OUTPUT	dump messages to file	
<code>--ui-host</code>	UI_HOST	the host to serve the UI on (default localhost)	
<code>--ui-port</code>	UI_PORT	the TCP port to serve the UI on	
<code>--ui-allow-origin</code>	ORIGIN	allow browser requests from ORIGIN; may be specified multiple times	

Gotchas

TODO: operational traps.

See also

`frida`, `frida-ps`, `frida-discover`, `frida-kill`, `frida-ls-devices`

r2

Kit: REMnux · Kali Linux · SIFT Workstation **Capability:** Disassemble and explore a binary **Version:** 6.1.9-124 r2
Captured: cyberlab-aio via -h on 2026-08-02 [raw Docs:](#) <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
r2 [-ACdfjLMnNqStuvwzX] [-P patch] [-p prj] [-a arch] [-b bits] [-c cmd]
[-s addr] [-B baddr] [-m maddr] [-i script] [-e k=v] file|pid|-|--|=
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 31 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-0	—	print \x00 after init and every command	
-1	—	redirect stderr to stdout	
-2	—	close stderr file descriptor (silent warning messages)	
-a	arch	set asm.arch	
-A	—	run 'aaa' command to analyze all referenced code	
-b	bits	set asm.bits	
-B	baddr	set base address for PIE binaries	
-C	—	file is host:port (alias for -c+=http://%s/cmd/)	
-d	—	debug the executable 'file' or running process 'pid'	
-E	—	open cfg.editor with the user rc file	
-f	—	block size = file size	
-i	file	run rlang program, r2script file or load plugin	

Flag	Argument	What it does	When you would use it
<code>-I</code>	file	run script file before the file is opened	
<code>-j</code>	—	use json for -v, -L and maybe others	
<code>-m</code>	addr	map file at given address (loadaddr)	
<code>-M</code>	—	do not demangle symbol names	
<code>-N</code>	—	do not load user settings and scripts	
<code>-q</code>	—	quiet mode (no prompt) and quit after -i	
<code>-Q</code>	—	quiet mode (no prompt) and quit faster (quickLeak=true)	
<code>-p</code>	prj	use project, list if no arg, load if no file	
<code>-P</code>	file	apply rapatch file and quit	
<code>-r</code>	rarun2	specify rarun2 profile to load (same as -e dbg.profile=X)	
<code>-s</code>	addr	initial seek	
<code>-S</code>	—	start r2 in sandbox mode	
<code>-t</code>	—	load rabin2 info in thread	
<code>-u</code>	—	set bin.filter=false to get raw sym/sec/cls names	
<code>-v</code>	—	show radare2 version (-V show lib versions)	
<code>-V</code>	—	show radare2 version (-V show lib versions)	
<code>-w</code>	—	open file in write mode	
<code>-x</code>	—	open without exec-flag (asm.emu will not work), See io.exec	
<code>-X</code>	—	same as -e bin.usextr=false (useful for dyldcache)	

Gotchas

TODO: operational traps.

See also

`rabin2`, `rasm2`, `objdump`, `vivbin`, `vdbbin`

rasm2

Kit: REMnux **Capability:** Disassemble and explore a binary; Analyse shellcode **Version:** rasm2 6.1.9 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://www.radare.org/n/radare2.html>

[← Capability index](#) · [Kit tool list](#)

Purpose

Examine binary files, including disassembling and debugging. Includes r2ai and decai plugins for LLM-powered analysis (API key or local Ollama required), plus the r2ghidra plugin for Ghidra decompilation via the pdg command.

Synopsis

```
rasm2 [-ACdDehHLBvw] [-a arch] [-b bits] [-s addr] [-S syntax]
[-f file] [-o file] [-F fil:ter] [-i skip] [-l len] 'code'|hex|0101b|-
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 25 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-a</code>	arch	set architecture to assemble/disassemble (see -L)	
<code>-A</code>	—	show Analysis information from given hexpairs	
<code>-b</code>	bits	set cpu register size (8, 16, 32, 64) (RASM2_BITS)	
<code>-B</code>	—	binary input/output (-l is mandatory for binary input)	
<code>-c</code>	cpu	select specific CPU (depends on arch)	
<code>-C</code>	—	output in C format	
<code>-d</code>	—	disassemble from hexpair bytes (-D show hexpairs)	
<code>-D</code>	—	disassemble from hexpair bytes (-D show hexpairs)	
<code>-e</code>	—	use big endian instead of little endian	
<code>-E</code>	—	display ESIL expression (same input as in -d)	
<code>-f</code>	file	read data from file	
<code>-F</code>	parser	specify which parse filter use (see -LL)	

Flag	Argument	What it does	When you would use it
<code>-i</code>	len	ignore/skip N bytes of the input buffer	
<code>-j</code>	—	output in json format	
<code>-k</code>	kernel	select operating system (linux, windows, darwin, android, ios, ..)	
<code>-l</code>	len	input/Output length	
<code>-N</code>	—	same as r2 -N (or R2_NOPLUGINS) (not load any plugin)	
<code>-o</code>	file	output file name (rasm2 -Bf a.asm -o a)	
<code>-p</code>	—	run SPP over input for assembly	
<code>-q</code>	—	quiet mode	
<code>-r</code>	—	output in radare commands	
<code>-S</code>	syntax	select syntax (intel, att)	
<code>-v</code>	—	show version information	
<code>-x</code>	—	use hex dwords instead of hex pairs when assembling.	
<code>-w</code>	—	what's this instruction for? describe opcode	

Gotchas

TODO: operational traps.

See also

`r2` , `rabin2` , `objdump` , `vivbin` , `vdbbin` , `xortool`

vdbbin

Kit: REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-02
raw Docs: <https://github.com/vivisect/vivisect>

[← Capability index](#) · [Kit tool list](#)

Purpose

Statically examine and emulate binary files.

Synopsis

```
vdbbin [options] [platformopt=foo, ...]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 27 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-c	COMMAND	Debug a fired command	
--cmd	COMMAND	Debug a fired command	
-p	PROCESS	Attach to process by name or pid	
--process	PROCESS	Attach to process by name or pid	
-Q	—	Run the QT gui	
--qt	—	Run the QT gui	
-R	REMOTEHOST	Attach to remote VDB server	
--remote	REMOTEHOST	Attach to remote VDB server	
-r	—	Do not stop on attach	
--run	—	Do not stop on attach	
-s	SNAPSHOT	Load a vtrace snapshot file	
--snapshot	SNAPSHOT	Load a vtrace snapshot file	

Flag	Argument	What it does	When you would use it
<code>-S</code>	—	—	
<code>--server</code>	—	—	
<code>-v</code>	—	—	
<code>--verbose</code>	—	—	
<code>-t</code>	TARGET	Activate special vdb target (-t ? for list)	
<code>--target</code>	TARGET	Activate special vdb target (-t ? for list)	
<code>--android</code>	—	Debug Android with ADB!	
<code>-e</code>	EVENTID	Used for Windows JIT	
<code>--eventid</code>	EVENTID	Used for Windows JIT	
<code>-w</code>	WAITFOR	Wait for process with name	
<code>--waitfor</code>	WAITFOR	Wait for process with name	
<code>--LI</code>	—	Breakpoint on Library initialization	
<code>--LL</code>	—	Breakpoint on Library load time	

Gotchas

TODO: operational traps.

See also

[r2](#) , [rabin2](#) , [rasm2](#) , [objdump](#) , [vivbin](#)

Kit: REMnux **Capability:** Disassemble and explore a binary **Captured:** cyberlab-aio via --help on 2026-08-02
raw Docs: <https://github.com/vivisect/vivisect>

← [Capability index](#) · [Kit tool list](#)

Purpose

Statically examine and emulate binary files.

Synopsis

```
vivbin [options] <workspace|binaries...>
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 34 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	show this help message and exit	
--help	—	show this help message and exit	
-M	MODNAME	run the file listed as an analysis module in non-gui mode and exit	
--module	MODNAME	run the file listed as an analysis module in non-gui mode and exit	
-A	—	Do <i>not</i> do an initial auto-analysis pass	
--skip-analysis	—	Do <i>not</i> do an initial auto-analysis pass	
-B	—	Do <i>not</i> start the gui, just load, analyze and save	
--bulk	—	Do <i>not</i> start the gui, just load, analyze and save	
-C	—	Output vivisect performace profiling (cProfile) info	
--cprofile	—	Output vivisect performace profiling (cProfile) info	
-E	ENTRYPOINTS	Add Entry Point for bulk analysis (can have multiple "-E " args	
--entrypoint	ENTRYPOINTS	Add Entry Point for bulk analysis (can have multiple "-E " args	
-O	OPTION	.= (optval must be json syntax)	

Flag	Argument	What it does	When you would use it
<code>--option</code>	OPTION	.= (optval must be json syntax)	
<code>-p</code>	PARSEMOD	Manually specify the parser module (pe/elf/blob/...)	
<code>--parser</code>	PARSEMOD	Manually specify the parser module (pe/elf/blob/...)	
<code>-s</code>	STORAGE_NAME	Specify a storage module by name	
<code>--storage</code>	STORAGE_NAME	Specify a storage module by name	
<code>-S</code>	—	Run Vivisect Server. Last argument should be the VivWorkspaces directory.	
<code>--server</code>	—	Run Vivisect Server. Last argument should be the VivWorkspaces directory.	
<code>-P</code>	SERVER_PORT	Port to run Vivisect Server on (only meaningful with --server)	
<code>--port</code>	SERVER_PORT	Port to run Vivisect Server on (only meaningful with --server)	
<code>-v</code>	—	Enable verbose mode (multiples matter: -vvvv)	
<code>--verbose</code>	—	Enable verbose mode (multiples matter: -vvvv)	
<code>-V</code>	VERSION	Add file version (if available) to save file name	
<code>--version</code>	VERSION	Add file version (if available) to save file name	
<code>-c</code>	CONFIG	Path to a directory to use for config data	
<code>--config</code>	CONFIG	Path to a directory to use for config data	
<code>-a</code>	—	Autosave configuration data	
<code>--autosave</code>	—	Autosave configuration data	
<code>-o</code>	OUTFILE	Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace)	
<code>--outfile</code>	OUTFILE	Name of VivWorkspace file to create (useful for loading multiple binaries into one workspace)	
<code>-m</code>	—	List Architectures and their version/maturity	
<code>--archmaturity</code>	—	List Architectures and their version/maturity	

Gotchas

TODO: operational traps.

See also

`r2`, `rabin2`, `rasm2`, `objdump`, `vdbbin`

xortool

Kit: REMnux **Capability:** Analyse shellcode; Break simple obfuscation **Version:** 1.1.0 **Captured:** cyberlab-aio via `--help` on 2026-08-02 **raw Docs:** <https://github.com/hellman/xortool>

[← Capability index](#) · [Kit tool list](#)

Purpose

Recover the key length and key of an XOR-encrypted file by frequency analysis.

Synopsis

```
xortool [-x] [-m MAX-LEN] [-f] [-t CHARSET] [FILE]
xortool [-x] [-l LEN] [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-x] [-m MAX-LEN| -l LEN] [-c CHAR | -b | -o] [-f] [-t CHARSET] [-p PLAIN] [-r PERCENT] [FILE]
xortool [-h | --help]
xortool --version
```

Common invocations

```
# from cyberlab 09-deobfuscation
xortool --version
# from cyberlab 09-deobfuscation
xortool -c 20 exercise/encoded_payload.bin
```

Options

All 20 options parsed from the captured help text; 4 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-x</code>	—	input is hex-encoded str	
<code>--hex</code>	—	input is hex-encoded str	
<code>-l</code>	LEN	length of the key	Fix the key length when you already know it.
<code>--key-length</code>	LEN	length of the key	
<code>-m</code>	MAX-LEN	maximum key length to probe [default: 65]	Maximum key length to consider.
<code>--max-keylen</code>	MAX-LEN	maximum key length to probe [default: 65]	

Flag	Argument	What it does	When you would use it
<code>-c</code>	CHAR	most frequent char (one char or hex code)	Give the most frequent character of the plaintext — usually <code>20</code> (space) for text, <code>00</code> for binaries. This is the flag that makes or breaks the attack.
<code>--char</code>	CHAR	most frequent char (one char or hex code)	
<code>-b</code>	—	brute force all possible most frequent chars	Brute-force the most frequent character rather than guessing.
<code>--brute-chars</code>	—	brute force all possible most frequent chars	
<code>-o</code>	—	same as -b but will only check printable chars	
<code>--brute-printable</code>	—	same as -b but will only check printable chars	
<code>-f</code>	—	filter outputs based on the charset	
<code>--filter-output</code>	—	filter outputs based on the charset	
<code>-p</code>	PLAIN	use known plaintext for decoding	
<code>--known-plaintext</code>	PLAIN	use known plaintext for decoding	
<code>-r</code>	PERCENT	threshold validity percentage [default: 95]	
<code>--threshold</code>	PERCENT	threshold validity percentage [default: 95]	
<code>-h</code>	—	show this help	
<code>--help</code>	—	show this help	

Gotchas

- Output lands in an `xortool_out/` directory, not stdout. People routinely think it did nothing.
- `-c 00` is right far more often than the default for packed binaries, because null padding dominates.

See also

`rasm2`, `floss`, `xlmedeobfuscator`

AmcacheParser

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases; Parse execution and persistence artifacts **Version:** 2026.5.0+76dc8354aa98ce1e1c6f942abcfb09f583f411dd **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse Amcache.hve — the record of programs present on a host, with SHA-1 hashes, including binaries that have since been deleted.

Synopsis

```
AmcacheParser [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 15 options parsed from the captured help text; 9 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-f	f	Amcache.hve file to parse	The Amcache.hve to parse.
-i	—	Include file entries for Programs entries	Include file entries associated with Programs entries. More complete, and noisier.
-w	w	Path to file containing SHA-1 hashes to <i>exclude</i> from the results. Blacklisting overrides whitelisting	Blacklist of SHA-1 hashes to exclude. Blacklisting overrides whitelisting, so a hash in both is dropped — the safe default when suppressing known-good noise.
-b	b	Path to file containing SHA-1 hashes to <i>include</i> from the results. Blacklisting overrides whitelisting	Whitelist of SHA-1 hashes to include.
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. The usual output.
--csvf	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.

Flag	Argument	What it does	When you would use it
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options. Default is: yyyy-MM-dd HH:mm:ss [default: yyyy-MM-dd HH:mm:ss]	Custom timestamp format for the output.
<code>--mp</code>	—	When true, display higher precision for timestamps	Higher-precision timestamps.
<code>--nl</code>	—	When true, ignore transaction log files for dirty hives. Default is FALSE	Ignore transaction logs for a dirty hive. Leave this off unless you know why you want it: skipping the logs means parsing a hive that is missing its most recent changes.
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

- Amcache records that a binary was **present**, not that it ran. It is evidence of existence; Prefetch and event logs are evidence of execution. Conflating the two is the standard error with this artifact.
- It carries SHA-1 for entries, which makes it the fastest way to tie a deleted binary to threat intelligence long after the file is gone.
- The hive is usually dirty when collected from a live host. Let the transaction logs replay — the entries only in the logs are the most recent, which is normally the part you care about.

See also

`esedbexport`, `esedbinfo`, `SrumECmd`, `PECmd`, `AppCompatCacheParser`, `MFTECmd`

AppCompatCacheParser

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+0cf059f40c2f7b31acdccb142461945402217398 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
AppCompatCacheParser [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 13 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	Full path to SYSTEM hive to process. If this option is not specified, the live Registry will be used	
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	
<code>-c</code>	c	The ControlSet to parse. Default is to extract all control sets [default: -1]	
<code>-t</code>	—	Sorts last modified timestamps in descending order	
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	
<code>--nl</code>	—	When true, ignore transaction log files for dirty hives	
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

TODO: operational traps.

See also

[PECmd](#), [MFTECmd](#), [AmcacheParser](#)

chainsaw

Kit: SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** chainsaw 2.16.0 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Rapidly work with Forensic Artefacts

Synopsis

```
chainsaw [OPTIONS] <COMMAND>
```

Common invocations

```
# from cyberlab 58-eventlog-hunting
chainsaw hunt Security_sample.evtx -s sigma/ --mapping mapping.yml
```

Options

All 6 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
--no-banner	—	Hide Chainsaw's banner	
--num-threads	NUM_THREADS	Limit the thread number (default: num of CPUs)	
-h	—	Print help	
--help	—	Print help	
-V	—	Print version	
--version	—	Print version	

Gotchas

TODO: operational traps.

See also

[evtxexport](#) , [evtxinfo](#) , [EvtxECmd](#) , [hayabusa](#)

esedbexport

Kit: SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: esedbexport

Synopsis

```
esedbexport [ -c codepage ] [ -l logfile ] [ -m mode ] [ -t target ]
[ -T table_name ] [ -hvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
esedbexport -V
# from cyberlab 06-windows-artifact-libs
esedbexport -t /tmp/edb_out exercise/Current.edb
```

Options

All 8 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254 windows-1255, windows-1256, windows-125	
-h	—	shows this help	
-l	—	logs information about the exported items	
-m	—	export mode, option: all, tables (default) 'all' exports all the tables or a single specified table with indexes, 'tables' exports all the tables or a single specified table	
-t	—	specify the basename of the target directory to export to (default is the source filename) esedbexport will add the suffix .export to the basename	

Flag	Argument	What it does	When you would use it
<code>-T</code>	—	exports only a specific table	
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	

Gotchas

TODO: operational traps.

See also

[esedbinfo](#) , [SrumECmd](#) , [AmcacheParser](#)

esedbinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse ESE / SRUM / Amcache databases **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: esedbinfo

Synopsis

```
esedbinfo [ -hvV ] source
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-h	—	shows this help	
-v	—	verbose output to stderr	
-V	—	print version	

Gotchas

TODO: operational traps.

See also

[esedbexport](#) , [SrumECmd](#) , [AmcacheParser](#)

EvtxECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse Windows event logs **Version:** 2026.5.0+bfc7f47ccbf65ffc9a3777cde5498db2fdd94664 **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Parse Windows event logs into a normalised, filterable CSV, mapping the useful fields out of the XML payload.

Synopsis

```
EvtxECmd [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 26 options parsed from the captured help text; 20 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
-f	f	File to process. Either this or -d is required	A single log, when you already know which one matters.
-d	d	Directory to recursively process. Either this or -f is required	Recurse a directory of .evtx files — the usual mode when working from a collected winevt\Logs .
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. The normal output, and the form the rest of a timeline workflow expects.
--csvf	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
--json	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	JSON output, for feeding another tool.
--jsonf	jsonf	File name to save JSON formatted results to. When present, overrides default name	Override the generated JSON filename.
--xml	xml	Directory to save XML formatted results to	XML output.

Flag	Argument	What it does	When you would use it
<code>--xmlf</code>	xmlf	File name to save XML formatted results to. When present, overrides default name	Override the generated XML filename.
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	Custom timestamp format for the output.
<code>--inc</code>	inc	List of Event IDs to process. All others are ignored. Overrides <code>--exc</code> Format is 4624,4625,5410,5500-5600	Process only these Event IDs. The fastest way to cut a multi-gigabyte log down to the question being asked — ranges are allowed (4624,4625,5410-5500).
<code>--exc</code>	exc	List of Event IDs to IGNORE. All others are included. Format is 4624,4625,5410,5500-5600	Process everything except these Event IDs. <code>--inc</code> wins if both are given.
<code>--sd</code>	sd	Start date for including events (UTC). Anything OLDER than this is dropped. Format should match <code>--dt</code>	Drop events older than this date (UTC).
<code>--ed</code>	ed	End date for including events (UTC). Anything NEWER than this is dropped. Format should match <code>--dt</code>	Drop events newer than this date (UTC). With <code>--sd</code> , scopes the parse to the incident window instead of all history.
<code>--fj</code>	—	When true, export all available data when using <code>--json</code>	Export all available data in JSON rather than the mapped subset.
<code>--tdt</code>	tdt	The number of seconds to use for time discrepancy detection. Default is 1 [default: 1]	Seconds of tolerance for time-discrepancy detection — flags records whose timestamps disagree, a clock-tampering signal.
<code>--met</code>	—	When true, show metrics about processed event log. Default is true	Show per-log metrics about what was processed.
<code>--maps</code>	maps	The path where event maps are located. Defaults to 'Maps' folder where program was executed [default: /opt/ezttools/EvtxECmd/Maps]	Where the event maps live. The maps are what turn raw XML into named columns; without the right ones, useful fields stay buried in the payload.
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by <code>-f</code> or <code>-d</code>	Also parse every Volume Shadow Copy on the drive, which is where cleared or rotated logs may survive.
<code>--dedupe</code>	—	Deduplicate <code>-f</code> or <code>-d</code> & VSCs based on SHA-1. First file found wins	Drop duplicates by SHA-1 across the source and shadow copies. Use it whenever <code>--vss</code> is on.
<code>--sync</code>	—	If true, the latest maps from https://github.com/EricZimmerman/evtx/tree/master/evtx/Maps are downloaded and local maps updated	Pull the latest maps from upstream. Worth doing before a big parse — map coverage improves continuously.

Flag	Argument	What it does	When you would use it
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

- Without a map for an Event ID, the interesting values stay inside the XML payload rather than becoming columns. If an expected field is missing, the map is usually the reason, not the log.
- A cleared log is itself the finding: Security 1102 and System 104 record the clearing. Include them explicitly when hunting anti-forensics.
- Event log timestamps are recorded in UTC but `--sd` / `--ed` are only as good as your assumption about the host's clock. Corroborate before building a timeline on them.

See also

`evtxexport`, `evtxinfo`, `chainsaw`, `hayabusa`

evtxexport

Kit: SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: evtxexport

Synopsis

```
evtxexport [ -c codepage ] [ -f format ] [ -l log_file ]
[ -m mode ] [ -p resource_files_path ]
[ -r registry_files_path ] [ -s system_file ]
[ -S software_file ] [ -t event_log_type ]
[ -hTvV ] source
```

Common invocations

```
# from cyberlab 06-windows-artifact-libs
evtxexport -V
# from cyberlab 06-windows-artifact-libs
evtxexport -h
# from cyberlab 06-windows-artifact-libs
evtxexport -f text exercise/Security.evtx > /tmp/security_events.txt
```

Options

All 13 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-f	—	output format, options: xml, text (default)	
-h	—	shows this help	

Flag	Argument	What it does	When you would use it
<code>-l</code>	—	logs information about the exported items	
<code>-m</code>	—	export mode, option: all, items (default), recovered 'all' exports the (allocated) items and recovered items, 'items' exports the (allocated) items and 'recovered' exports the recovered items	
<code>-p</code>	—	search PATH for the resource files	
<code>-r</code>	—	name of the directory containing the SOFTWARE and SYSTEM (Windows) Registry file	
<code>-s</code>	—	filename of the SYSTEM (Windows) Registry file. This option overrides the path provided by -r	
<code>-S</code>	—	filename of the SOFTWARE (Windows) Registry file. This option overrides the path provided by -r	
<code>-t</code>	—	event log type, options: application, security, system if not specified the event log type is determined based on the filename.	
<code>-T</code>	—	use event template definitions to parse the event record data	
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	

Gotchas

TODO: operational traps.

See also

[evtxinfo](#), [EvtxECmd](#), [chainsaw](#), [hayabusa](#)

evtxinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse Windows event logs **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: evtxinfo

Synopsis

```
evtxinfo [ -c codepage ] [ -hvV ] source
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-v	—	verbose output to stderr	
-V	—	print version	

Gotchas

TODO: operational traps.

See also

evtxexport , EvtxECmd , chainsaw , hayabusa

hayabusa

Kit: SIFT / Security Onion (Sigma-based log hunting) **Capability:** Parse Windows event logs **Version:** error: unexpected argument '--version' found **Captured:** `cyberlab-aio` via `help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Hayabusa v3.9.0 - Showa Day Release

Synopsis

```
hayabusa.exe <COMMAND> [OPTIONS]
hayabusa.exe help <COMMAND> or hayabusa.exe <COMMAND> -h
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`evtxexport`, `evtxinfo`, `EvtxECmd`, `chainsaw`

hivexsh

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

hivexsh: failed to open hive file: help: No such file or directory

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`rip.pl`, `regripper`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

MFTECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+4fd94a6bd12237e8501baff5fc9e5b1b01c53862 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

[← Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
MFTECmd [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 32 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	File to process (\$MFT \$J \$Boot \$SDS \$I30). Required	
<code>-m</code>	m	\$MFT file to use when -f points to a \$J file (Use this to resolve parent path in \$J CSV output)	
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes. This or --json required unless --de or --body is specified	
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	
<code>--json</code>	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes. This or --csv required unless --de or --body is specified	
<code>--jsonf</code>	jsonf	File name to save JSON formatted results to. When present, overrides default name	
<code>--body</code>	body	Directory to save bodyfile formatted results to. --bdl is also required when using this option	
<code>--bodyf</code>	bodyf	File name to save body formatted results to. When present, overrides default name	
<code>--bdl</code>	bdl	Drive letter (C, D, etc.) to use with bodyfile. Only the drive letter itself should be provided	

Flag	Argument	What it does	When you would use it
<code>--blf</code>	—	When true, use LF vs CRLF for newlines	
<code>--dd</code>	dd	Directory to save exported \$MFT FILE record. --do is also required when using this option	
<code>--do</code>	do	Offset of the \$MFT FILE record to dump as decimal or hex. Ex: 5120 or 0x1400 Use --de or --debug to see offsets	
<code>--de</code>	de	Dump full details for \$MFT entry/sequence #. Format is 'Entry' or 'Entry-Seq' as decimal or hex. Example: 5, 624-5 or 0x270-0x5.	
<code>--dr</code>	—	When true, dump \$MFT resident files to dir specified by --csv or --json, in 'Resident' subdirectory. Files will be named '--_bin'	
<code>--fls</code>	—	When true, displays contents of directory from \$MFT specified by --de. Ignored when --de points to a file	
<code>--ds</code>	ds	Dump full details for Security Id from \$SDS as decimal or hex. Example: 624 or 0x270	
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	
<code>--sn</code>	—	Include DOS file name types in \$MFT output	
<code>--fl</code>	—	Generate condensed file listing of parsed \$MFT contents. Requires --csv	
<code>--at</code>	—	When true, include all timestamps from 0x30 attribute vs only when they differ from 0x10 in the \$MFT	
<code>--rs</code>	—	When true, recover slack space from FILE records when processing \$MFT files. This option has no effect for \$I30 files	
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f	
<code>--dedupe</code>	—	Deduplicate -f & VSCs based on SHA-1. First file found wins	
<code>--ir</code>	—	Include resident data in JSON/CSV output	
<code>--re</code>	re	Comma-separated list of extensions to include for resident data (e.g., '.txt,.ps1,.bat'). If omitted, includes all	
<code>--rm</code>	rm	Maximum size in bytes for resident data to include (max: 1024000) [default: 1024]	
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

TODO: operational traps.

See also

[PECmd](#) , [AppCompatCacheParser](#) , [AmcacheParser](#)

PECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse execution and persistence artifacts **Version:** 2026.5.0+bde430c69ba4d97fea8b71fdddb6df7849419c10 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Parse Windows Prefetch files into evidence of what executed, when, how often, and which files each run touched.

Synopsis

```
PECmd [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 20 options parsed from the captured help text; 14 reviewed with usage guidance.

Flag	Argument	What it does	When you would use it
<code>-f</code>	f	File to process. Either this or -d is required	A single .pf file, when chasing one binary.
<code>-d</code>	d	Directory to recursively process. Either this or -f is required	Recurse a directory — the normal mode, since Prefetch is only meaningful as a set. Point it at <code>C:\Windows\Prefetch</code> .
<code>-k</code>	k	Comma separated list of keywords to highlight in output. By default, 'temp' and 'tmp' are highlighted. Any additional keywords will be added to these	Highlight extra keywords in the output. <code>temp</code> and <code>tmp</code> are highlighted by default; add the names you are hunting.
<code>-o</code>	o	When specified, save prefetch file bytes to the given path. Useful to look at decompressed Win10 files	Save the decompressed Prefetch bytes. Win10+ Prefetch is MAM-compressed, so this is how you get something another tool or a hex editor can read.
<code>-q</code>	—	Do not dump full details about each file processed. Speeds up processing when using --json or --csv	Suppress the per-file detail. Worth it on a large directory when the CSV is the real output.
<code>--csv</code>	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	Write CSV to a directory. This is the output that matters: Prefetch is a timeline source, and Timeline Explorer or a spreadsheet is where the pattern shows up.

Flag	Argument	What it does	When you would use it
<code>--csvf</code>	csvf	File name to save CSV formatted results to. When present, overrides default name	Override the generated CSV filename.
<code>--json</code>	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	JSON output, when the results feed another tool.
<code>--jsonf</code>	jsonf	File name to save JSON formatted results to. When present, overrides default name	Override the generated JSON filename.
<code>--html</code>	html	Directory to save xhtml formatted results to. Be sure to include the full path in double quotes	XHTML report, for handing to someone who will not open a CSV.
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	Custom timestamp format for the output.
<code>--mp</code>	—	When true, display higher precision for timestamps	Higher-precision timestamps, when ordering events within the same second matters.
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f or -d	Also parse every Volume Shadow Copy on the drive. Prefetch rolls over at 1024 entries on Win10+, so shadow copies are often the only place an older execution still exists.
<code>--dedupe</code>	—	Deduplicate -f or -d & VSCs based on SHA-1. First file found wins	Drop duplicates by SHA-1 across the source and the shadow copies. Effectively mandatory with <code>--vss</code> , which otherwise returns the same file many times over.
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

- Prefetch proves a program **ran**; it does not prove who ran it or what it did. Pair it with event logs or `AmcacheParser` before attributing anything.
- Absence is not evidence of absence. Prefetch can be disabled, is commonly off on SSD-era server builds, and rolls over — a missing entry means nothing on its own.
- The last-run timestamps are the eight most recent executions only. Older runs are gone from the file even though the run count keeps counting them.

See also

`AppCompatCacheParser` , `MFTECmd` , `AmcacheParser`

RECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse registry hives **Version:** 2026.5.0+bcd0ac33ed98de61ea6de551eef96052bddbbd49 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
RECmd [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 33 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	f	Hive to search. -f or -d is required	
-d	d	Directory to look for hives (recursively). -f or -d is required	
--kn	kn	Display details for key name. Includes subkeys and values	
--vn	vn	Value name. Only this value will be dumped	
--bn	bn	Use settings from supplied file to find keys/values. See included sample file for examples	
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
--csvf	csvf	File name to save CSV formatted results to. When present, overrides default name	
--saveTo	saveTo	Saves --vn value data in binary form to file. Expects path to a FILE	
--json	json	Directory to save JSON formatted results to. Be sure to include the full path in double quotes	
--jsonf	jsonf	File name to save JSON formatted results to. When present, overrides default name	

Flag	Argument	What it does	When you would use it
<code>--details</code>	—	Show more details when displaying results	
<code>--base64</code>	base64	Find Base64 encoded values with size >= Base64 (specified in bytes)	
<code>--minSize</code>	minSize	Find values with data size >= MinSize (specified in bytes)	
<code>--sa</code>	sa	Search for in keys, values, data, and slack	
<code>--sk</code>	sk	Search for in value record's key names	
<code>--sv</code>	sv	Search for in value record's value names	
<code>--sd</code>	sd	Search for in value record's value data	
<code>--ss</code>	ss	Search for in value record's value slack	
<code>--literal</code>	—	If true, --sd and --ss search value will not be interpreted as ASCII or Unicode byte strings	
<code>--nd</code>	—	If true, do not show data when using --sd or --ss	
<code>--regex</code>	—	If present, treat in --sk, --sv, --sd, and --ss as a regular expression	
<code>--dt</code>	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss.ffffff]	
<code>--nl</code>	—	When true, allow transaction log files to not exist for dirty hives	
<code>--recover</code>	—	If true, recover deleted keys/values. Default is true	
<code>--vss</code>	—	Process all Volume Shadow Copies that exist on drive specified by -f or -d	
<code>--dedupe</code>	—	Deduplicate -f or -d & VSCs based on SHA-1. First file found wins	
<code>--sync</code>	—	If true, the latest batch files from https://github.com/EricZimmerman/RECmd/tree/master/BatchExamples are downloaded and local files updated	
<code>--debug</code>	—	Show debug information during processing	
<code>--trace</code>	—	Show trace information during processing	
<code>-?</code>	—	Show help and usage information	
<code>-h</code>	—	Show help and usage information	
<code>--help</code>	—	Show help and usage information	
<code>--version</code>	—	Show version information	

Gotchas

TODO: operational traps.

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

regfexport

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfexport

Synopsis

```
regfexport [ -c codepage ] [ -K key_path ] [ -l logfile ]
[ -hvV ] source
```

Common invocations

```
# from cyberlab 04-registry-analysis
regfexport -V
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive > /tmp/system_dump.txt
# from cyberlab 04-registry-analysis
regfexport exercise/SYSTEM_sample.hive | grep -i "ComputerName"
```

Options

All 6 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-K	—	show information about a specific key path.	
-l	—	logs information about the exported items	

Flag	Argument	What it does	When you would use it
<code>-v</code>	—	verbose output to stderr	
<code>-V</code>	—	print version	

Gotchas

TODO: operational traps.

See also

`rip.pl`, `regripper`, `hivexsh`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

regfinfo

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfinfo

Synopsis

```
regfinfo [ -B bodyfile ] [ -c codepage ] [ -hHvV ] source
```

Common invocations

```
# from cyberlab 04-registry-analysis
regfinfo -V
# from cyberlab 04-registry-analysis
regfinfo exercise/SYSTEM_sample.hive
# from cyberlab 04-registry-analysis
regfinfo /tmp/SYSTEM_recovered.hive
```

Options

All 6 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-B	—	output key and value hierarchy as a bodyfile	
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-H	—	shows the key and value hierarchy	
-v	—	verbose output to stderr	

Flag	Argument	What it does	When you would use it
<code>-V</code>	<code>—</code>	print version	

Gotchas

TODO: operational traps.

See also

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

regfmount

Kit: SIFT Workstation (libyal) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Invalid argument: regfmount

Synopsis

```
regfmount [ -c codepage ] [ -X extended_options ] [ -hvV ] file
mount_point
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 5 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-c	—	codepage of ASCII strings, options: ascii, windows-874, windows-932, windows-936, windows-949, windows-950, windows-1250, windows-1251, windows-1252 (default), windows-1253, windows-1254, windows-1255	
-h	—	shows this help	
-v	—	verbose output to stderr, while regfmount will remain running in the foreground	
-V	—	print version	
-X	—	extended options to pass to sub system	

Gotchas

TODO: operational traps.

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regipy-dump](#), [regipy-parse-header](#), [regipy-plugins-run](#)

regipy-diff

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TODO: one-line imperative purpose.

Synopsis

```
regipy-diff [OPTIONS] FIRST_HIVE_PATH SECOND_HIVE_PATH
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 4 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	—	
-v	—	Verbosity	
--verbose	—	Verbosity	
--help	—	Show this message and exit.	

Gotchas

TODO: operational traps.

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-parse-header](#)

regipy-dump

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

identified for some reason

Synopsis

```
regipy-dump [OPTIONS] HIVE_PATH
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 18 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	—	
-p	TEXT	A registry path to start iterating from	
--registry-path	TEXT	A registry path to start iterating from	
-t	—	Create a CSV timeline instead	
--timeline	—	Create a CSV timeline instead	
-l	TEXT	Specify a hive type, if it could not be identified for some reason	
--hive-type	TEXT	Specify a hive type, if it could not be identified for some reason	
-r	TEXT	The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
--partial_hive_path	TEXT	The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
-v	—	Verbosity	
--verbose	—	Verbosity	

Flag	Argument	What it does	When you would use it
<code>-d</code>	—	Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned	
<code>--do-not-fetch-values</code>	—	Not fetching the values for each subkey makes the iteration way faster. Values count will still be returned	
<code>-s</code>	TEXT	If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat	
<code>--start-date</code>	TEXT	If "-s" was specified, fetch only values for subkeys starting this timestamp in isoformat	
<code>-e</code>	TEXT	If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat	
<code>--end-date</code>	TEXT	If "-e" was specified, fetch only values for subkeys until this timestamp in isoformat	
<code>--help</code>	—	Show this message and exit.	

Gotchas

TODO: operational traps.

See also

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-parse-header`, `regipy-plugins-run`

regipy-parse-header

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

TODO: one-line imperative purpose.

Synopsis

```
regipy-parse-header [OPTIONS] HIVE_PATH
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 3 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-v	—	Verbosity	
--verbose	—	Verbosity	
--help	—	Show this message and exit.	

Gotchas

TODO: operational traps.

See also

[rip.pl](#), [regripper](#), [hivexsh](#), [regfexport](#), [regfinfo](#), [regfmount](#), [regipy-dump](#), [regipy-plugins-run](#)

regipy-plugins-run

Kit: SIFT Workstation (regipy) **Capability:** Parse registry hives **Captured:** cyberlab-aio via --help on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

identified for some reason

Synopsis

```
regipy-plugins-run [OPTIONS] HIVE_PATH
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-o	FILE	Output path for plugins result [required]	
-p	TEXT	A plugin or list of plugins to execute command separated	
--plugins	TEXT	A plugin or list of plugins to execute command separated	
-t	TEXT	Specify a hive type, if it could not be identified for some reason	
--hive-type	TEXT	Specify a hive type, if it could not be identified for some reason	
-r	TEXT	The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
--partial_hive_path	TEXT	The path from which the partial hive actually starts, for example: -t ntuser - r "/Software" would mean this is actually a HKCU hive, starting from HKCU/Software	
-v	—	Verbosity	
--verbose	—	Verbosity	
--include-unvalidated	—	Include plugins that don't have validation test cases. These plugins may return incomplete or inaccurate data. Use at your own risk.	

Flag	Argument	What it does	When you would use it
<code>--help</code>	—	Show this message and exit.	

Gotchas

TODO: operational traps.

See also

`rip.pl`, `regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`

regripper

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Rip v.3.0 - CLI RegRipper tool

Common invocations

TODO: up to 8 task-titled invocations.

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`rip.pl`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

rip.pl

Kit: Kali Linux **Capability:** Parse registry hives **Captured:** `cyberlab-aio` via `--help` on 2026-08-02 [raw](#)

[← Capability index](#) · [Kit tool list](#)

Purpose

Rip v.3.0 - CLI RegRipper tool

Common invocations

```
# from cyberlab 04-registry-analysis
rip.pl -h
# from cyberlab 04-registry-analysis
rip.pl -r exercise/SYSTEM_sample.hive -p compname
# from cyberlab 49-intrusion-timeline-case
rip.pl -h | head -n 3
# from cyberlab 49-intrusion-timeline-case
rip.pl -r NTUSER.DAT -p run
# from cyberlab 49-intrusion-timeline-case
rip.pl -r SYSTEM -p usbstor
```

Options

No option definitions could be parsed from this tool's help output. It may be subcommand-driven or have no flags; needs manual review.

Gotchas

TODO: operational traps.

See also

`regripper`, `hivexsh`, `regfexport`, `regfinfo`, `regfmount`, `regipy-dump`, `regipy-parse-header`, `regipy-plugins-run`

SrumECmd

Kit: FLARE-VM / SIFT (Eric Zimmerman tools) **Capability:** Parse ESE / SRUM / Amcache databases **Version:** 2026.5.0+880ad26bcb011976a8fc521eea63fc5e6e65ba02 **Captured:** cyberlab-aio via --help on 2026-08-02 raw

← [Capability index](#) · [Kit tool list](#)

Purpose

Description:

Synopsis

```
SrumECmd [options]
```

Common invocations

TODO: up to 8 task-titled invocations.

Options

All 11 options parsed from the captured help text. The final column is filled in by review.

Flag	Argument	What it does	When you would use it
-f	f	SRUDB.dat file to parse	
-r	r	SOFTWARE hive to process. This is optional, but recommended	
-d	d	Directory to recursively process, looking for SRUDB.dat and SOFTWARE hive. This mode is primarily used with KAPE so both SRUDB.dat and SOFTWARE hive can be located	
--csv	csv	Directory to save CSV formatted results to. Be sure to include the full path in double quotes	
--dt	dt	The custom date/time format to use when displaying time stamps. See https://goo.gl/CNVq0k for options [default: yyyy-MM-dd HH:mm:ss]	
--debug	—	Show debug information during processing	
--trace	—	Show trace information during processing	
-?	—	Show help and usage information	
-h	—	Show help and usage information	
--help	—	Show help and usage information	

Flag	Argument	What it does	When you would use it
<code>--version</code>	—	Show version information	

Gotchas

TODO: operational traps.

See also

`esedbexport`, `esedbinfo`, `AmcacheParser`

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